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CHAPTER 1

SETS, RELATIONS AND FUNCTIONS

1. INTRODUCTION

The concept of set serves as a fundamental part of the present day mathematics. Today this concept is being used in almost every branch of mathematics. Sets are used to define the concepts of relations and functions. The study of geometry, sequences, probability etc. requires theknowledge of sets.

2. SETS AND THEIR REPRESENTATIONS

A set is a well-defined collection of objects.

The following points may be noted :

(i) Objects, elements and members of a set are synonymous terms.

(ii) Sets are usually denoted by capital letters A, B, C, X, Y, Z etc.

(iii) The elements of a set are represented by small letters a, b, c, x, y, z etc.

If a is an element of a set A, we say that " a belongs to A" the Greek symbol $\in$ (epsilon) is used to denote the phrase 'belongs to'. Thus, we write $a \in A$. If 'b' is not an element of a set A, we write $b \notin A$ and read "b does not belong to A".

Thus, in the set V of vowels in the English alphabet, $a \in V$ but $b \notin V$. In the set P of prime factors of 30, $3 \in P$ but $15 \notin P$.

There are two methods of representing a set :

(i) Roster or tabular form (ii) Set-builder form.

(i) In roster form, all the elements of a set are listed, the elements are being separated by commas and are enclosed within braces { }. For example, the set of all even positive integers less than 7 is described in roster form as {2, 4, 6}.

(ii) In set-builder form, all the elements of a set possess a single common property which is not possessed by any element outside the set. For example, in the set {a, e, i, o, u}, all the elements possess a common property, namely, each of them is a vowel in the English alphabet, and no other letter possess this property. Denoting this set by V, we write

$V = \{x : x \text{ is a vowel in English alphabet}\}$

Comparison table :

Statement	Roster form	Set builder form
(1) The set of currencies used in USA, England, Japan, Germany and Russia.	{Dollar, Pound, Yen, Euro, Rouble}	{x : x is the currencies used in USA, England, Japan, Germany and Russia}
(2) The set of Capital of Kerala, Karnataka, Tamilnadu, Andra Pradesh and Gujarat.	{Thiruvananthapuram, Bangalore, Chennai Hyderabad, and Gandhi Nagar}	{x : x is the capitals of Kerala, Karnataka, Tamilnadu, Andra Pradesh and Gujarat}
(3) The set of all distinct letters used in the word student.	{s, t, u, d, e, n}	{x : x is the distinct letters used in the word student}
(4) The set of all the states of India beginning with the letter A.	{Andhra Pradesh, Arunchal Pradesh, Assam}	{x : x is the state of India beginning with the letter A}
(5) The set of six presidents of India since 1980.	{Neelam Sanjeeva Reddy, Gyani Zail Singh, Radha Swami Venkat Raman Dr. Shankar Dayal Sharma, K.R. Naryan A.P.J. Abdul Kalam}	{x : x is the presidents of India since 1980}
(6) The set of all natural numbers between 11 and 15.	{12, 13, 14}	{x : x $\in$ N, 11 < x < 15}

Illustration 1: Write the set $\{x : x \text{ is a positive integer and } x^2 < 40\}$ in the roster form.

Solution : The required numbers are 1, 2, 3, 4, 5, 6. So, the given set in the roster form is {1, 2, 3, 4, 5, 6}.

Illustration 2: Write the set $\left\{\frac{1}{2},\frac{2}{3},\frac{3}{4},\frac{4}{5},\frac{5}{6},\frac{6}{7}\right\}$ in the set-builder form.

Solution : We see that each member in the given set has the numerator one less than the denominator. Also, the numerator begin from 1 and do not exceed 6. Hence, in the set-builder form the given set is $\left\{x : x = \frac{n}{n+1}, \text{ where n is a natural number and } 1 \le n \le 6\right\}$

TYPES OF SETS

1. The Empty set :

A set which does not contain any element is called the empty set or the null set or the void set. According to this definition, *B* is an empty set while *A* is not an empty set. The empty set is denoted by the symbol ϕ or { }

We give below a few examples of empty sets.

(i) Let $A = \{x : 1 < x < 2, x \text{ is a natural number}\}$. Then, A is the empty set, because there is no natural number between 1 and 2.

(ii) $B = \{x : x^2 - 2 = 0 \text{ and } x \text{ is a rational number}\}$. Then, *B* is the empty set because the equation $x^2 - 2 = 0$ is not satisfied by any rational value of *x*.

(iii) $C = \{x : x \text{ is an even prime number greater than 2}\}$. Then *C* is the empty set, because 2 is the only even prime number.

(iv) $D = \{x : x^2 = 4, x \text{ is odd }\}$. Then *D* is the empty set, because the equation $x^2 = 4$ is not satisfied by any odd value of *x*.

2. Finite and Infinite Sets :

A set which is empty or consists of a definite number of elements is called finite otherwise, the set is called infinite.

Consider some examples :

(i) Let *W* be the set of the days of the week. Then, *W* is finite.

(ii) Let S be the set of solutions of the equation $x^2 - 16 = 0$. Then, *S* is finite.

(iii) Let *G* be the set of points on a line. Then *G* is infinite.

When we represent a set in the roster form, we write all the elements of the set within braces { }. It is not possible to write all the elements of an infinite set within braces { } because the numbers of elements of such a set is not finite. So, we represent some infinite set in the roster form by writing a few elements which clearly indicate the structure of the set followed (or preceded) by three dots. For example, {1, 2, 3 . . .} is the set of natural numbers, {1, 3, 5, 7, . . .} is the set of odd natural numbers,

{. . .,–3, –2, –1, 0,1, 2 ,3, . . .} is the set of integers. All these sets are infinite.

3. Equal sets :

Two sets *A* and *B* are said to be equal if they have exactly the same elements and we write $A = B$. Otherwise, the sets are said to be unequal and we write $A \neq B$.

(i) Let $A = \{1, 2, 3, 4\}$ and $B = \{3, 1, 4, 2\}$. Then $A = B$.

(ii) Let A be the set o`f prime numbers less than 6 and *P* the set of prime factors of 30. Then *A* and *P* are equal, since 2, 3 and 5 are the only prime factors of 30 and also these are less than 6.

4. Singleton set :

A set, consisting of a single element is called a singleton set. The sets {0}, {5}, {–7} are singleton sets. $\{x : x + 6 = 0, x \in Z\}$ is a singleton set, because this set contains only integer namely, – 6.

Illustration 3: Which of the following pairs of sets are equal? Justify your answer.

(i) *X*, the set of letters in "ALLOY" and *B*, the set of letters in "LOYAL".

(ii) $A = \{n : n \in Z \text{ and } n^2 \le 4\}$ and $B = \{x : x \in R \text{ and } x^2 - 3x + 2 = 0\}$.

Solution : (i)We have, $X = \{A, L, L, O, Y\}$, $B = \{L, O, Y, A, L\}$. Then *X* and *B* are equal sets as repetition of elements in a set do not change a set. Thus, $X = \{A, L, O, Y\} = B$

(ii)$A = \{-2, -1, 0, 1, 2\}$, $B = \{1, 2\}$. Since $0 \in A$ and $0 \notin B$, *A* and *B* are not equal sets.

Illustration 4: Are the following pair of sets equal ? Give reason.

(i) $A = \{2, 3\}$, $B = \{x : x \text{ is a solution of } x^2 + 5x + 6 = 0\}$

(ii) $A = \{x : x \text{ is a letter in the word FOLLOW}\}$

$B = \{y : y \text{ is a letter in the word WOLF}\}$.

Solution :

We have,

(i) $A = \{2, 3\}$, $B = \{x : x \text{ is a solution of } x^2 + 5x + 6 = 0\}$

Now, $x^2 + 5x + 6 = 0$

$\Rightarrow \quad x^2 + 3x + 2x + 6 = 0$

$\Rightarrow \quad x(x + 3) + 2(x + 3) = 0$

$\Rightarrow \quad (x + 3)(x + 2) = 0 \Rightarrow x = -2, -3$

Therefore, $B = \{-2, -3\}$

Here, we observe that the elements of set A are not exactly the same to that of set B, hence A and B are not equal sets.

(ii) We have, $A = \{x : x \text{ is a letter in the word FOLLOW}\}$

$\Rightarrow \quad A = \{F, O, L, W\}$

And $B = \{x : x \text{ is a letter in the word WOLF}\}$

$\Rightarrow \quad B = \{W, O, L, F\}$

Here, we observe that the elements of both sets are exactly same, hence the sets are equal.

3. SUBSETS

A set A is said to be a subset of a set B if every element of A is also an element of B. In other words, $A \subseteq B$ if whenever $a \in A$, then $a \in B$. It is often convenient to use the symbol "$\Rightarrow$" which means implies. Using this symbol, we can write the definition of subset as follows:

$A \subseteq B$ if $a \in A \Rightarrow a \in B$

We read the above statement as "A is a subset of B if a is an element of A implies that a is also an element of B". If A is not a subset of B, we write $A \not\subset B$. For example :

(i) The set $\mathbb{Q}$ of rational numbers is a subset of the set $\mathbb{R}$ of real numbers, and we write $\mathbb{Q} \subset \mathbb{R}$.

(ii) If A is the set of all divisors of 56 and B the set of all prime divisors of 56, then B is a subset of A and we write $B \subseteq A$.

(iii) Let $A = \{1, 3, 5\}$ and $B = \{x : x \text{ is an odd natural number less than 6}\}$. Then $A \subset B$ and $B \subset A$ and hence $A = B$.

(iv) Let $A = \{a, e, i, o, u\}$ and $B = \{a, b, c, d\}$. Then A is not a subset of B, also B is not a subset of A.

Proper subset : If $A \subseteq B$ and $A \neq B$, then A is called a proper subset of B, written as $A \subset B$.

For example : Let $A = \{x : x \text{ is an even natural number}\}$ and $B = \{x : x \text{ is a natural number}\}$. Then, $A = \{2, 4, 6, 8, ...\}$ and $B = \{1, 2, 3, 4, 5, ...\} \Rightarrow A \subset B$.

Theorems on subsets :

Theorem 1 : Every set is a subset of itself.

Proof : Let A be any set. Then, each element of A is clearly in A. Hence $A \subseteq A$.

Theorem 2 : The empty set is a subset of every set.

Proof : Let A be any set and ϕ be the empty set. In order to show that $\phi \subseteq A$, we must show that every element of ϕ is an element of A also. But, ϕ contains no element, So, every element of ϕ is in A. Hence $\phi \subset A$.

Theorem 3 : The total number of subsets of a finite set containing n element is 2^n.

Proof : Let A be a set of n elements.

The null set is a subset of A containing no element.

$\therefore$ Number of subsets of A containing no element = $1 = {}^nC_0$.

Number of subsets of A containing 1 element = Number of groups of n elements taking 1 at a time = ${}^nC_1$.

Number of subsets of A containing 2 elements = Number of groups of n elements taking 2 at a time = ${}^nC_2$.

Number of subsets of A containing n elements i.e. $A = 1 = {}^nC_n$

Total number of subsets of $A = {}^nC_0 + {}^nC_1 + {}^nC_2 + ... + {}^nC_n = 2^n$ [Using binomial theorem]

Some properties of subsets :

(a) If $A \subseteq B$ and $B \subseteq C$, then $A \subseteq C$.

Let $x \in A \Rightarrow x \in B \quad (\because A \subseteq B)$ and $x \in B \Rightarrow x \in C \quad (\because B \subseteq C) \Rightarrow A \subseteq C$

(b) $A = B$ if and only if $A \subseteq B$ and $B \subseteq A$.

Let $A \subseteq B$ and $B \subseteq A \quad \therefore \; x \in A \Rightarrow x \in B \; (\because A \subseteq B)$ and $x \in B \Rightarrow x \in A \; (\because B \subseteq A)$

$\therefore \quad A = B$

Conversely, Let $A \subseteq B$

$\therefore \quad x \in A \Rightarrow x \in B \ (\because A = B) \ \therefore A \subseteq B$

Similarly, $x \in B \Rightarrow x \in A \ (\because A = B) \quad \therefore B \subseteq A$

Subsets of set of real numbers :

There are many important subsets of $\mathbb{R}$. We give below the names of some of these subsets.

The set of natural numbers $\mathbb{N} = \{1, 2, 3, 4, 5, \ldots\}$

The set of integers $\mathbb{Z} = \{\ldots, -3, -2, -1, 0, 1, 2, 3, \ldots\}$

The set of rational numbers $\mathbb{Q} = \{ x : x = \frac{p}{q}, p, q \in \mathbb{Z} \text{ and } q \neq 0\}$

which is read " $\mathbb{Q}$ is the set of all numbers x such that x equals the quotient $\frac{p}{q}$, where p and q are integers and q is not zero". Members of $\mathbb{Q}$ include –5 (which can be expressed as $-\frac{5}{4}, \frac{5}{7}, 3\frac{1}{2}$ (which can be expresses as $\frac{7}{2}$) and $-\frac{11}{3}$.

The set of irrational numbers, denoted by T, is composed of all other real numbers. Thus $T = \{ x : x \in \mathbb{R}$ and $x \notin \mathbb{Q}\}$, i.e., all real numbers that are not rational. Members of T include $\sqrt{2}$, $\sqrt{5}$ and π.

Some of the obvious relations among these subsets are:

$\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q}, \mathbb{Q} \subset \mathbb{R}, \mathbb{T} \subset \mathbb{R}, \mathbb{N} \not\subset \mathbb{T}$.

Intervals as subsets of $\mathbb{R}$: Let $a, b \in \mathbb{R}$ and $a < b$. Then the set of real numbers $\{y : a < y < b\}$ is called an open interval and is denoted by (a, b). All the points between a and b belong to the open interval (a, b) but a, b themselves do not belong to this interval.

The interval which contains the end points also is called closed interval and is denoted by $[a, b]$. Thus

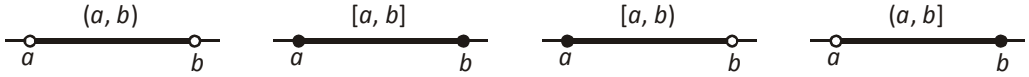
$[a, b] = \{x : a \leq x \leq b\}$

We can also have intervals closed at one end and open at the other, i.e.,

$[a, b) = \{x : a \leq x < b\}$ is an open interval from a to b, including a but excluding b.

$(a, b] = \{ x : a < x \leq b \}$ is an open interval from a to b including b but excluding a.

These sets can be shown by the dark portion of the number line.



The number $(b - a)$ is called the length of any of the intervals (a, b), $[a, b)$, $(a, b]$, $[a, b]$

POWER SET : The collection of all subsets of a set A is called the power set of A. It is denoted by $P(A)$. In $P(A)$, every element is a set. If A has n elements then its power set has 2^n elements

UNIVERSAL SET : If there are some sets under consideration, then there happens to be a set which is a superset of each one of the given sets. Such a set is known as the universal set, denoted by U or ξ.For example,

(i) In the context of human population studies, the universal set consists of all the people in the world.

(ii) Let $A = \{1, 2, 3, 4\}$, $B = \{2, 5, 6\}$, $C = \{1, 3, 7, 8, 9\}$, then

$U = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$ is the universal set.

Illustration 5: Show that : $n\{P\{P(P(\phi))\}\} = 4$

Solution : We have, $P(\phi) = \{\phi\}$

$\therefore P(P(\phi)) = \{\phi, \{\phi\}\} \Rightarrow P\{P(P(\phi))\} = \{\phi, \{\phi\}, \{\{\phi\}\}, \{\phi, \{\phi\}\}\}$

Hence, $n\{P\{P(P(\phi))\}\} = 4$

Illustration 6: Prove that $A \subset \phi \Rightarrow A = \phi$

Solution : We know that, Two sets A and B are equal if an only if $A \subset B$ and $B \subset A$

Also, we know that, $\phi \subset A$ and $A \subset \phi$

$\therefore A = \phi$

Illustration 7: Write the following intervals in set builder form :

(i)(–3, 0) (ii) [6, 12] (iii) (6, 12] (iv) [–23, 5)

Solution : The following intervals are written in set builder form

(i) (–3, 0) is an open interval which excludes both – 3 and 0. So, it can be shown in the form of a set i.e.,

$\{x : x \in \mathbb{R}, -3 < x < 0\}$

(ii) [6, 12] is a closed interval which includes both 6 and 12. So, it can be shown in the set builder form i.e., $\{x : x \in \mathbb{R}, 6 \le x \le 12\}$

(iii) (6, 12] is an interval closed at the second end only i.e. it excludes 6 but includes 12. So it is shown in the set builder form i.e., $\{x : x \in \mathbb{R}, 6 < x \le 12\}$

(iv) [–23, 5) is an interval closed at the first end point but open at the second end point. It means that the interval includes –23 but excludes 5. It is written in the set builder form i.e. $\{x : x \in \mathbb{R}, -23 \le x < 5\}$.

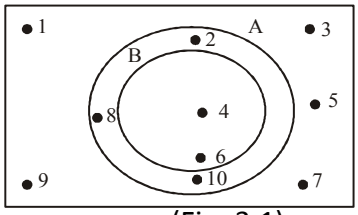
Illustration 8: What universal set would you propose for each of the following :

(i)The set of right triangles. (ii) The set of isosceles triangles.

Solution : **(i)**The universal set for the set of right triangles is set of triangle.

(ii) The universal set for the set of isosceles triangle is set of equilateral triangle or set of triangles.

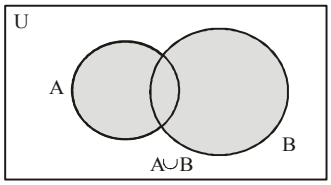
VENN DIAGRAMS : In order to illustrate in a clear and simple way, the ideas involving universal sets, subsets there of, and certain operations on sets, we make use of geometric figures. These figures are called Venn-Diagrams.In Venn diagrams, the elements of the sets are written in their respective circles. In Fig 3.1, $U = \{1, 2, 3, ..., 10\}$ is the universal set of which $A = \{2,4,6,8,10\}$and $B = \{4, 6\}$ are subsets,and also $B \subset A$.



(Fig. 3.1)

OPERATIONS ON SETS

Union of sets : The union of two sets A and B is the set C which consists of all those elements which are either in A or in B (including those which are in both). In symbols, we write $A \cup B = \{ x : x \in A \text{ or } x \in B \}$ The union of two sets can be represented by a Venn diagram as shown in Fig. 3.2



(Fig. 3.2)

The shaded portion in Fig represents $A \cup B$.

Some Properties of the Operation of Union

(i) $A \cup B = B \cup A$ (Commutative law)

(ii) $(A \cup B) \cup C = A \cup (B \cup C)$ (Associative law)

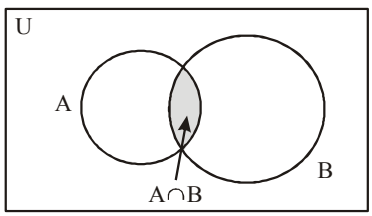
(iii) $A \cup \phi = A$ (Law of identity element, ϕ is the identity of $\cup$)

(iv) $A \cup A = A$ (Idempotent law)

(v) $U \cup A = U$ (Law of U)

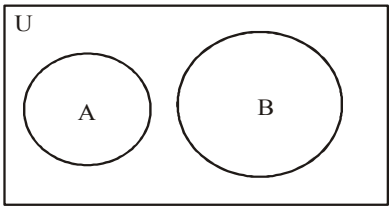
Intersection of sets : The intersection of two sets A and B is the set of all those elements which belong to both A and B. Symbolically, we write $A \cap B = \{x : x \in A \text{ and } x \in B\}$

The shaded portion in Fig. 3.3 indicates the intersection of A and B.



(Fig. 3.3)

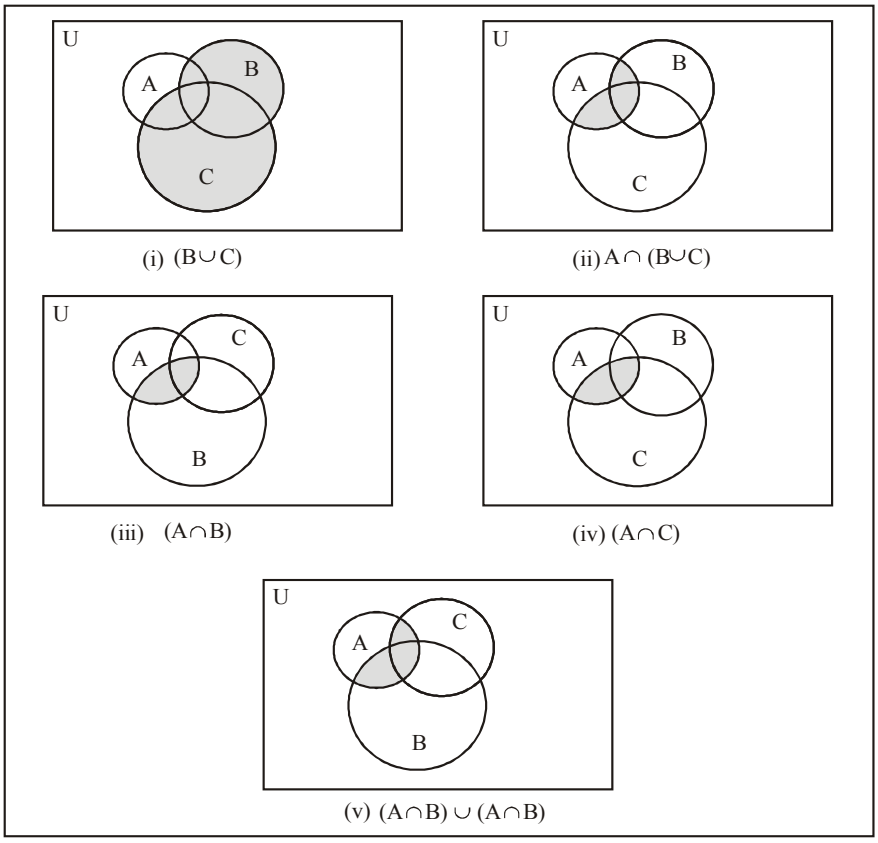
If A and B are two sets such that $A \cap B = \phi$, then A and B are called disjoint sets.
For example, let $A = \{ 2, 4, 6, 8 \}$ and $B = \{ 1, 3, 5, 7 \}$. Then A and B are disjoint sets, because there are no elements which are common to A and B. The disjoint sets can be represented by means of Venn diagram as shown in the Fig. 3.4
In the above diagram, A and B are disjoint sets.



(Fig. 3.4)

Some properties of operation of intersection

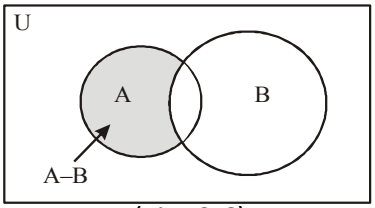
(i) $A \cap B = B \cap A$ (Commutative law).
(ii) $(A \cap B) \cap C = A \cap (B \cap C)$ (Associative law).
(iii) $\phi \cap A = \phi$, $U \cap A = A$ (Law of ϕ and U).
(iv) $A \cap A = A$ (Idempotent law)
(v) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ (Distributive law) i. e., $\cap$ distributes over $\cup$
This can be seen easily from the following Venn diagrams [Figs 3.5 (i) to (v)].



(Fig. 3.5)

Difference of sets :

The difference of the sets A and B in this order is the set of elements which belong to A but not to B. Symbolically, we write $A - B$ and read as " A minus B".
Using the setbuilder notation, we can rewrite the definition of difference



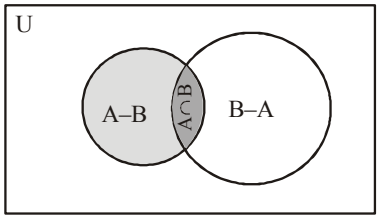
(Fig. 3.6)

as $A - B = \{ x : x \in A \text{ and } x \notin B \}$

The difference of two sets A and B can be represented by Venn diagram as shown in Fig. 3.6

The shaded portion represents the difference of the two sets A and B.

The sets $A - B$, $A \cap B$ and $B - A$ are mutually disjoint sets, i.e., the intersection of any of these two sets is the null set as shown in Fig. 3.7



(Fig. 3.7)

Complement of set :

Let U be the universal set and A a subset of U. Then the complement of A is the set of all elements of U which are not the elements of A. Symbolically, we write A' to denote the complement of A with respect to U. Thus, $A' = \{x : x \in U \text{ and } x \notin A \}$.

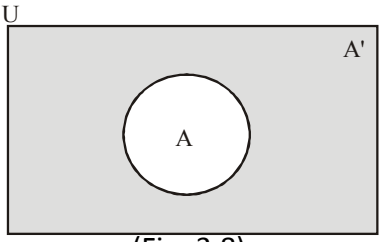
Obviously $A' = U - A$

We note that the complement of a set A can be looked upon, alternatively, as the difference between a universal set U and the set A.

The complement of the union of two sets is the intersection of their complements and the complement of the intersection of two sets is the union of their complements. These are called De Morgan's laws.

These are named after the mathematician De Morgan.

The complement A' of a set A can be represented by a Venn diagram as



(Fig. 3.8)

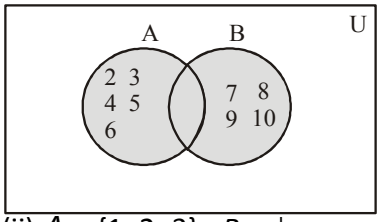
shown in Fig. 3.8 The shaded portion represents the complement of the set A.

Some Properties of Complement Sets :

1. Complement laws: (i) $A \cup A' = U$ (ii) $A \cap A' = \phi$
2. De Morgan's law: (i) $(A \cup B)' = A' \cap B'$ (ii) $(A \cap B)' = A' \cup B'$
3. Law of double complementation : $(A')' = A$
4. Laws of empty set and universal set $\phi' = U$ and $U' = \phi$.
   These laws can be verified by using Venn diagrams.

Illustration 9: Find the union of each of the following pairs of sets :

(i) $A = \{x : x \text{ is a natural number and } 1 < x \leq 6\}$
$B = \{ x : x \text{ is a natural number and } 6 < x \leq 10\}$



(ii) $A = \{1, 2, 3\}$, $B = \phi$

Solution : (i) $A = \{ x : x \text{ is a natural number and } 1 < x \leq 6\}$

$\Rightarrow A = \{2, 3, 4, 5, 6\}$

$B = \{ x : x \text{ is a natural number and } 6 < x \leq 10\}$

$\Rightarrow B = \{7, 8, 9, 10\}$

$\therefore A \cup B = \{2, 3, 4, 5, 6\} \cup \{7, 8, 9, 10\}$

$\Rightarrow$ $A \cup B = \{2, 3, 4, 5, 6, 7, 8, 9, 10\}$

(ii) We have, $A = \{1, 2, 3\}$, $B = \phi$

$\Rightarrow$ $A \cup B = \{1, 2, 3\} \cup \phi$

$\Rightarrow$ $A \cup B = \{1, 2, 3\}$

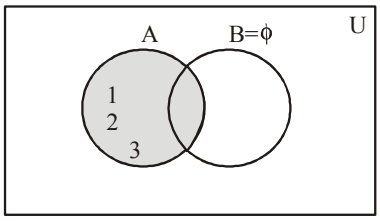


Illustration 10: If $A = \{x : x = 3n, n \in \mathbb{Z}\}$ and $B = \{x : x = 4n, n \in \mathbb{Z}\}$, then find $(A \cap B)$.

Solution : Let $x \in (A \cap B) \Leftrightarrow x \in A$ and $x \in B$

$\Leftrightarrow x$ is a multiple of 3 and x is a multiple of 4.

$\Leftrightarrow x$ is a multiple of 3 and 4 both

$\Leftrightarrow x$ is a multiple of 12

$\Leftrightarrow x = 12n, \ n \in \mathbb{Z}$

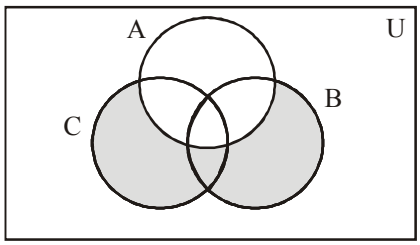
Hence $A \cap B = \{x : x = 12n, n \in \mathbb{Z}\}$

Illustration 11: Shade the following sets :

(i) $A' \cap (B \cup C)$ (ii) $A' \cap (C - B)$

Solution : We have, the given sets shaded in the following figures.

(i) $A' \cap (B \cup C)$ (ii) $A' \cap (C - B)$



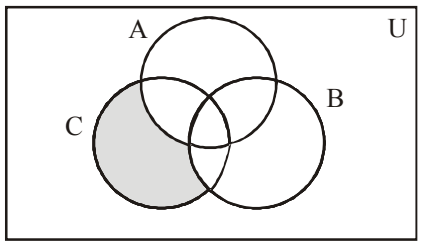


Illustration 12: In a class of 35 students, 24 like to play cricket and 16 like to play football. Also, each student likes to play at least one of the two games. How many students like to play both cricket and football ?

Solution : Let X be the set of students who like to play cricket and Y be the set of students who like to play football. Then $X \cup Y$ is the set of students who like to play at least one game, and $X \cap Y$ is the set of students who like to play both games.

Given $n(X) = 24$, $n(Y) = 16$, $n(X \cup Y) = 35$, $n(X \cap Y) = ?$

Using the formula $n(X \cup Y) = n(X) + n(Y) - n(X \cap Y)$, we get

$35 = 24 + 16 - n(X \cap Y)$

Thus, $n(X \cap Y) = 5$ i.e., 5 students like to play both games.

USEFUL TIPS

1. In roster form, the order in which the elements are listed is immaterial.

2. All infinite sets cannot be described in the roster form. For example, the set of real numbers cannot be described in this form, because the elements of this set do not follow any particular pattern.

3. A set does not change if one or more elements of the set are repeated. For example, the sets $A = \{1, 2, 3\}$ and $B = \{2, 2, 1, 3, 3\}$ are equal, since each element of A is in B and vice-versa. That is why we generally do not repeat any element in describing a set.

4. If A is a subset of the universal set U, then its complement A' is also a subset of U.

5. Union rule for counting : $n(A \cup B) = n(A) + n(B) - n(A \cap B)$

Illustration 13: There are 200 individuals with a skin disorder, 120 had been exposed to the chemical C_1, 50 to chemical C_2, and 30 to both the chemicals C_1 and C_2. Find the number of individuals exposed to (i) Chemical C_1 but not chemical C_2 (ii) Chemical C_2 but not chemical C_1 (iii) Chemical C_1 or chemical C_2

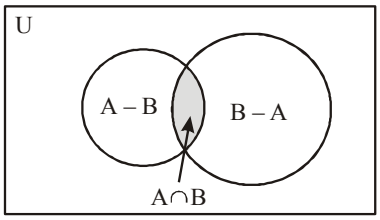
Solution : Let U denote the universal set consisting of individuals suffering from the skin disorder, A denote the set of individuals exposed to the chemical C_1 and B denote the set of individuals exposed to the chemical C_2.

Here $n(U) = 200$, $n(A) = 120$, $n(B) = 50$ and $n(A \cap B) = 30$

(i) From the Venn diagram given in Fig. below, we have $A = (A - B) \cup (A \cap B)$.

$n(A) = n(A - B) + n(A \cap B)$ (Since $(A - B)$ and $A \cap B$ are disjoint.)

or $n(A - B) = n(A) - n(A \cap B) = 120 - 30 = 90$



Hence, the number of individuals exposed to chemical C_1 but not to chemical C_2 is 90.

(ii) From the Fig, we have $B = (B - A) \cup (A \cap B)$.

and so, $n(B) = n(B - A) + n(A \cap B)$ (Since $B - A$ and $A \cap B$ are disjoint.)

or $n(B - A) = n(B) - n(A \cap B) = 50 - 30 = 20$

Thus, the number of individuals exposed to chemical C_2 and not to chemical C_1 is 20.

(iii) The number of individuals exposed either to chemical C_1 or to chemical C_2, i.e.,

$n(A \cup B) = n(A) + n(B) - n(A \cap B) = 120 + 50 - 30 = 140$.

Illustration 14: Let B be a subset of a set A and let $P(A : B) = \{X \in P(A) : B \subset X\}$

(i)Show that $P(A : \phi) = P(A)$

(ii)If $A = \{a, b, c, d\}$ and $B = \{a, b\}$. List all the members of the set $P(A : B)$.

Solution : (i)We have, $P(A : B) = \{X \in P(A) : B \subset X\}$

= Set of all those subsets of A which contain B.

$\therefore P(A : \phi)$ = Set of all those subsets of A which contain ϕ.

= Set of all subsets of set $A = P(A)$

(ii)If $A = \{a, b, c, d\}$ and $B = \{a, b\}$. Then,

$P(A : B)$ = Set of all those subsets of set A which contain B

$= \{a, b, c\}, \{a, b, d\}, \{a, b, c, d\}$

Illustration 15: A college warded 38 medals in football, 15 in basketball and 20 in cricket. If these medals went to a total of 58 men and only three men got medals in all the three sports, how many received medals in exactly two of the three sports ?

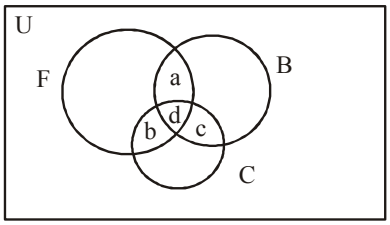
Solution : Let F, B and C denote the set of men who received medals in football, basketball and cricket respectively.

Then $n(F) = 38$, $n(B) = 15$, $n(C) = 20$

$n(F \cup B \cup C) = 58$ and $n(F \cap B \cap C) = 3$

Therefore, $n(F \cup B \cup C) = n(F) + n(B) + n(C)$

$- n(F \cap B) - n(F \cap C) - n(B \cap C) + n(F \cap B \cap C)$,



gives $n(F \cap B) + n(F \cap C) + n(B \cap C) = 18$

Consider the Venn diagram as given in the above figure.

Here, 'a' denotes the number of men who got medals in football and basketball only, 'b' denotes the number of men who got medals in football and cricket only, c denotes the number of men who got medals in basket ball and cricket only and d denotes the number of men who got medal in all the three. Thus, $d = n(F \cap B \cap C) = 3$ and $a + d + b + d + c + d = 18$

Therefore $a + b + c = 9$, which is the number of people who got medals in exactly two of the three sports.

Illustration 16: In a group of 65 people, 40 like cricket, 10 like both cricket and tennis. How many like tennis only and not cricket ? How many like tennis.

Solution : Let A be the set of people who like cricket and B be the set of people who like tennis. Then $(A \cup B) = 65$, $n(A) = 40$, $n(A \cap B) = 10$

We know that, $n(A \cup B) = n(A) + n(B) - n(A \cap B)$

$\Rightarrow$ $65 = 40 + n(B) - 10 \Rightarrow n(B) = 35$

Number of people who like tennis only and not cricket
$= n(B - A)$
Also, $n(B) = n(B - A) + n(A \cap B)$
$35 = n(B - A) + 10$
$n(B - A) = 35 - 10 = 25$

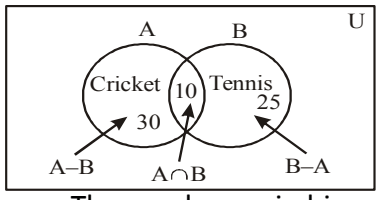
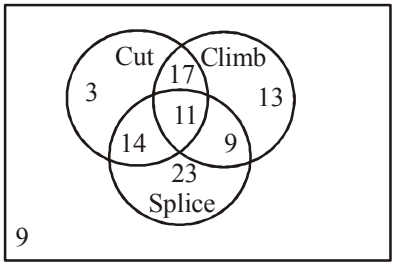


Illustration 17: Jeff Friedman is a section chief for an electric utility company. The employees in his section cut down all trees, climb poles, and splice wire. Friedman reported the following information to the management of the utility.
Of the 100 employees in my section, 45 can cut all trees, 50 can climb poles, 57 can splice wire, 28 can cut trees and climb poles, 20 can climb poles and splice wire, 25 can cut trees and splice wire, 11 can do all three, 9 can't do any of the three (management trainees).

Solution : The data supplied by Friedman lead to the numbers shown in figure. Add the numbers from all the regions to get the total number of Friedman's employees.
9 + 13 + 14 + 23 + 11 + 9 + 17 + 13 = 99
Friedman claimed to have 100 employees, but this data indicate only 99.



The management decided that Friedman didn't qualify as a section chief, and reassigned him as a nightshift meter reader in Guam.

Illustration 18: State which of the following sets are finite or infinite :
(i)$\{x : x \in N$ and $(x - 1)(x - 2) = 0\}$ (ii) $\{x : x \in N$ and $2x = 4\}$
(iii)$\{x : x \in N$ and $2x - 1 = 0\}$ (iv) $\{x : x \in N$ and x is prime$\}$
(v)$\{x : x \in N$ and is odd$\}$

Solution : (i)Given set = {1, 2}. Hence, it is finite.
(ii)Given set = {2}. Hence, it is finite.
(iii)Given set = ϕ. Hence, it is finite.
(iv)The given set is the set of all prime numbers and since set of prime numbers is infinite. Hence the given set is infinite
(v)Since there are infinite number of odd numbers, hence, the given set is infinite.

Illustration 19: Let $A = \{a, e, i, o, u\}$ and $B = \{a, i, u\}$. Show that $A \cup B = A$

Solution : We have, $A \cup B = \{a, e, i, o, u\} = A$.
This example illustrates that union of sets A and its subset B is the set A itself, i.e., if $B \subset A$, then $A \cup B = A$.

Student Task

Q.1 In a survey it was found that 21 people liked product *A*, 26 liked product *B* and 29 liked product *C*. If 14 people liked products *A* and *B*, 12 people liked products *C* and *A*, 14 people liked products *B* and *C* and 8 liked all the three products. How many liked product C only –
(A) 11 (B) 12 (C) 13 (D) 10

Q.2 In a survey of 60 people, it was found that 25 people read newspaper *H*, 26 read newspaper *T*, 26 read newspaper I, 9 read both *H* and I, 11 read both *H* and *T*, 8 read both *T* and I, 3 read all three newspapers. The number of people who read at least one of the newspapers is
(A) 50 (B) 52 (C) 40 (D) 60

Q.3 In a group of students, 100 students know Hindi, 50 know English and 25 know both. Each of the students knows either Hindi or English. How many students are there in the group –
(A) 100 (B) 115 (C) 110 (D) 125

Q.4 In a survey of 600 students in a school, 150 students were found to be taking tea and 225 taking coffee, 100 were taking both tea and coffee. How many students were taking neither tea nor coffee –
(A) 325 (B) 300 (C) 330 (D) 350

Q.5 Let A and B be sets. If $A \cap X = B \cap X = \phi$ and $A \cup X = B \cup X$ for some set X, then
(A) $A = B$. (B) $A \neq B$ (C) $A = A \cap (A \cap X)$ (D) None of these

Q.6 Let A, B, and C be the sets such that $A \cup B = A \cup C$ and $A \cap B = A \cap C$ then
(A) $B \neq C$ (B) $B = C$ (C) $B = B \cap (B \cap C)$ (D) None of these

Q.7 If $A \subset B$ and $B \subset C$, then
(A) $A \subset C$ (B) $A \in C$ (C) $A \not\subset C$ (D) None of these

Q.8 If $A \subset B$ and $x \notin B$, then
(A) $A \subset C$ (B) $x \notin B$ (C) $x \notin A$ (D) None of these

Q.9 If X and Y are two sets such that $n(X) = 17$, $n(Y) = 23$ and $n(X \cup Y) = 38$, then $n(X \cap Y)$
(A) 2 (B) 1 (C) 3 (D) 4

Q.10 If X and Y are two sets such that $X \cup Y$ has 18 elements, X has 8 elements and Y has 15 elements; how many elements does $X \cap Y$ have
(A) 1 (B) 3 (C) 4 (D) 5

Q.11 In a group of 400 people, 250 can speak Hindi and 200 can speak English. How many people can speak both Hindi and English –
(A) 60 (B) 50 (C) 55 (D) 65

ANSWER KEY

1.	(A)	**2.**	(B)	**3.**	(D)	**4.**	(A)	**5.**	(A)	**6.**	(B)	**7.**	(A)
8.	(C)	**9.**	(A)	**10.**	(D)	**11.**	(B)						

4. RELATION

Definition

Let A and B be two non-empty sets, then every subset of $A \times B$ defines a relation from A to B and every relation from A to B is a subset of $A \times B$.

Let $R \subseteq A \times B$ and $(a, b) \in R$. Then, we say that a is related to b by the relation R and write it as $a R b$. If $(a,b) \in R$, we write it as $a R b$.

Example: Let $A = \{1, 2, 5, 8, 9\}$, $B = \{1, 3\}$ we set a relation from A to B as: a R b iff $a \leq b$; $a \in A$, $b \in B$. Then $R = \{(1, 1), (1, 3), (2, 3)\} \subset A \times B$

(1) **Total number of relations :** Let A and B be two non-empty finite sets consisting of m and n elements respectively. Then $A \times B$ consists of mn ordered pairs. So, total number of subset of $A \times B$ is 2^{mn}. Since each subset of $A \times B$ defines relation from A to B, so total number of relations from A to B is 2^{mn}. Among these 2^{mn} relations the void relation ϕ and the universal relation $A \times B$ are trivial relations from A to B.

(2) **Domain and range of a relation :** Let R be a relation from a set A to a set B. Then the set of all first components or coordinates of the ordered pairs belonging to R is called the domain of R, while the set of all second components or coordinates of the ordered pairs in R is called the range of R.

Thus, Dom $(R) = \{a : (a, b) \in R\}$ and Range $(R) = \{b : (a, b) \in R\}$.

It is evident from the definition that the domain of a relation from A to B is a subset of A and its range is a subset of B.

(3) **Relation on a set :** Let A be a non-void set. Then, a relation from A to itself *i.e.* a subset of $A \times A$ is called a relation on set A.

Illustration 20: Let $A = \{1, 2, 3\}$. The total number of distinct relations that can be defined over A is
(a) 2^9 (b) 6 (c) 8 (d) None of these

Solution: (a) $n(A \times A) = n(A).n(A) = 3^2 = 9$

So, the total number of subsets of $A \times A$ is 2^9 and a subset of $A \times A$ is a relation over the set A.

Illustration 21: Let $X = \{1, 2, 3, 4, 5\}$ and $Y = \{1, 3, 5, 7, 9\}$. Which of the following is/are relations from X to Y

(a) $R_1 = \{(x, y) \mid y = 2 + x, x \in X, y \in Y\}$ (b) $R_2 = \{(1,1),(2,1),(3,3),(4,3),(5,5)\}$

(c) $R_3 = \{(1,1),(1,3)(3,5),(3,7),(5,7)\}$ (d) $R_4 = \{(1,3),(2,5),(2,4),(7,9)\}$

Solution: (a,b,c) R_4 is not a relation from X to Y, because (7, 9) does not belong to $X \times Y$.

Illustration 22: Given two finite sets A and B such that $n(A) = 2$, $n(B) = 3$. Then total number of relations from A to B is

(a)4 (b) 8 (c) 64 (d) None of these

Solution: (c)Here $n(A \times B) = 2 \times 3 = 6$

Since every subset of $A \times B$ defines a relation from A to B, number of relation from A to B is equal to number of subsets of $A \times B = 2^6 = 64$ which is given in (c).

Illustration 23: The relation R defined on the set of natural numbers as $\{(a, b) : a$ differs from b by 3$\}$, is given by

(a){(1, 4, (2, 5), (3, 6),.....} (b) {(4, 1), (5, 2), (6, 3),.....}

(c){(1, 3), (2, 6), (3, 9),..} (d) None of these

Solution: (b) $R = \{(a,b) : a, b \in N, a - b = 3\} = \{((n+3), n) : n \in N\} = \{(4,1),(5,2),(6,3).....\}$

INVERSE RELATION

Let A, B be two sets and let R be a relation from a set A to a set B. Then the inverse of R, denoted by R^{-1}, is a relation from B to A and is defined by $R^{-1} = \{b, a) : (a, b) \in R\}$

Clearly $(a, b) \in R \Leftrightarrow (b, a) \in R^{-1}$. Also, Dom ($R$) = Range ($R^{-1}$) and Range ($R$)=Dom ($R^{-1}$)

Example **:**

Let $A = \{a, b, c\}$, $B = \{1, 2, 3\}$ and $R = \{(a, 1), (a, 3), (b, 3), (c, 3)\}$.

Then,

(i) $R^{-1} = \{(1, a), (3, a), (3, b), (3, c)\}$

(ii) Dom (R) = $\{a, b, c\}$ = Range (R^{-1})

(iii) Range (R) = $\{1, 3\}$ = Dom (R^{-1})

Illustration 24: Let $A = \{1, 2, 3\}$, $B = \{1, 3, 5\}$. A relation $R : A \to B$ is defined by $R = \{(1, 3), (1, 5), (2, 1)\}$. Then R^{-1} is defined by

(a){(1,2), (3,1), (1,3), (1,5)} (b) {(1, 2), (3, 1), (2, 1)}

(c){(1, 2), (5, 1), (3, 1)} (d) None of these

Solution: (c) $(x, y) \in R \Leftrightarrow (y, x) \in R^{-1}$ $\therefore$ $R^{-1} = \{(3,1),(5,1),(1,2)\}$

Illustration 25: The relation R is defined on the set of natural numbers as $\{(a, b) : a = 2b\}$. Then R^{-1} is given by

(a){(2, 1), (4, 2), (6, 3).....} (b) {(1, 2), (2, 4), (3, 6)....}

(c)is not defined (d) None of these

Solution: (b) $R = \{(2, 1), (4, 2), (6, 3),.....\}$ So, $R^{-1} = \{(1, 2), (2, 4), (3, 6),.....\}$.

Types of Relations

(1) **Reflexive relation :** A relation R on a set A is said to be reflexive if every element of A is related to itself.

Thus, R is reflexive $\Leftrightarrow (a, a) \in R$ for all $a \in A$.

A relation R on a set A is not reflexive if there exists an element $a \in A$ such that $(a, a) \notin R$.

Example**:** Let $A = \{1, 2, 3\}$ and $R = \{(1, 1); (1, 3)\}$

Then R is not reflexive since $3 \in A$ but $(3, 3) \notin R$

(2) **Symmetric relation :** A relation R on a set A is said to be a symmetric relation *iff*

$(a, b) \in R \Rightarrow (b, a) \in R$ for all $a, b \in A$

i.e. $aRb \Rightarrow bRa$ for all $a, b \in A$.

it should be noted that R is symmetric iff $R^{-1} = R$

Note :-

- The identity and the universal relations on a non-void set are symmetric relations.
- A relation R on a set A is not a symmetric relation if there are at least two elements $a, b \in A$ such that $(a, b) \in R$ but $(b, a) \notin R$.
- A reflexive relation on a set A is not necessarily symmetric.

(3) **Anti-symmetric relation :** Let A be any set. A relation R on set A is said to be an anti-symmetric relation *iff* $(a, b) \in R$ and $(b, a) \in R \Rightarrow a = b$ for all $a, b \in A$.

Thus, if $a \neq b$ then a may be related to b or b may be related to a, but never both.

Example: Let N be the set of natural numbers. A relation $R \subseteq N \times N$ is defined by xRy iff x divides y(*i.e.*, x/y).

Then xRy, $yRx \Rightarrow x$ divides y and y divides $x \Rightarrow x = y$

Note :-

- The identity relation on a set A is an anti-symmetric relation.
- The universal relation on a set A containing at least two elements is not anti-symmetric, because if $a \neq b$ are in A, then a is related to b and b is related to a under the universal relation will imply that $a = b$ but $a \neq b$.

(4) **Transitive relation :** Let A be any set. A relation R on set A is said to be a transitive relation *iff* $(a, b) \in R$ and $(b, c) \in R \Rightarrow (a, c) \in R$ for all $a, b, c \in A$ *i.e.,* aRb and $bRc \Rightarrow aRc$ for all $a, b, c \in A$.

In other words, if a is related to b, b is related to c, then a is related to c.

Transitivity fails only when there exists a, b, c such that $a\ R\ b$, $b\ R\ c$ but $a \not R\ c$.

***Example*:** Consider the set $A = \{1, 2, 3\}$ and the relations

$R_1 = \{(1, 2), (1, 3)\}$; $R_2 = \{(1, 2)\}$; $R_3 = \{(1, 1)\}$; $R_4 = \{(1, 2), (2, 1), (1, 1)\}$

Then R_1, R_2, R_3 are transitive while R_4 is not transitive since in $(2, 1) \in R_4$, $(1, 2) \in R_4$ but $(2, 2) \notin R_4$.

Note :-

- The identity and the universal relations on a non-void sets are transitive.
- The relation 'is congruent to' on the set T of all triangles in a plane is a transitive relation.

(5) **Identity relation :** Let A be a set. Then the relation $I_A = \{(a, a) : a \in A\}$ on A is called the identity relation on A.

In other words, a relation I_A on A is called the identity relation if every element of A is related to itself only. Every identity relation will be reflexive, symmetric and transitive.

***Example* :** On the set = $\{1, 2, 3\}$, $R = \{(1, 1), (2, 2), (3, 3)\}$ is the identity relation on A .

Note :-

- It is interesting to note that every identity relation is reflexive but every reflexive relation need not be an identity relation.
- Also, identity relation is reflexive, symmetric and transitive.

(6) **Equivalence relation :** A relation R on a set A is said to be an equivalence relation on A *iff*

(i) It is reflexive *i.e.* $(a, a) \in R$ for all $a \in A$

(ii) It is symmetric *i.e.* $(a, b) \in R \Rightarrow (b, a) \in R$, for all $a, b \in A$

(iii) It is transitive *i.e.* $(a, b) \in R$ and $(b, c) \in R \Rightarrow (a, c) \in R$ for all $a, b, c \in A$.

Note :-

Congruence modulo (m) : Let m be an arbitrary but fixed integer. Two integers a and b are said to be congruence modulo m if it is divisible by m and we write $a = b$ (mod m).

Thus $a = b$ (mod m) $\Leftrightarrow a - b$ is divisible by m. For example, 18 = 3 (mod 5) because 18 – 3 = 15 which is divisible by 5. Similarly, 3 = 13 (mod 2) because 3 – 13 = –10 which is divisible by 2. But $25 \neq 2$ (mod 4) because 4 is not a divisor of 25 – 3 = 22.

The relation "Congruence modulo m" is an equivalence relation.

Important Tips

- ☞ *If R and S are two equivalence relations on a set A , then $R \cap S$ is also an equivalence relation on A.*
- ☞ *The union of two equivalence relations on a set is not necessarily an equivalence relation on the set.*
- ☞ *The inverse of an equivalence relation is an equivalence relation.*

Illustration 26: Given the relation $R = \{(1, 2), (2, 3)\}$ on the set $A = \{1, 2, 3\}$, the minimum number of ordered pairs which when added to R make it an equivalence relation is

(a) 5 (b) 6 (c) 7 (d) 8

Solution: (c) R is reflexive if it contains (1, 1), (2, 2), (3, 3)

$\therefore (1, 2) \in R, (2, 3) \in R$

R is symmetric if $(2, 1), (3, 2) \in R$. Now,

R will be transitive if $(3, 1); (1, 3) \in R$. Thus, R becomes an equivalence relation by adding (1, 1), (2, 2), (3, 3), (2, 1), (3,2), (1, 3), (3, 1). Hence, the total number of ordered pairs is 7.

Illustration 27: The relation $R = \{(1, 1), (2, 2), (3, 3), (1, 2), (2, 3), (1, 3)\}$ on set $A = \{1, 2, 3\}$ is

(a) Reflexive but not symmetric (b) Reflexive but not transitive

(c) Symmetric and Transitive (d) Neither symmetric nor transitive

Solution: (a) Since $(1, 1); (2, 2); (3, 3) \in R$ therefore R is reflexive. $(1, 2) \in R$ but $(2, 1) \notin R$, therefore R is not symmetric. It can be easily seen that R is transitive.

Illustration 28: Let R be the relation on the set R of all real numbers defined by $a\ R\ b$ iff $|a - b| \leq 1$. Then R is

(a) Reflexive and Symmetric (b) Symmetric only

(c) Transitive only (d) Anti-symmetric only

Solution: (a) $|a - b| = 0 < 1 \ \ \therefore \ \ a\ R\ a \ \ \forall \ \ a \in R$

R is reflexive, Again $a\ R\ b \Rightarrow |a-b| \leq 1 \Rightarrow |b-a| \leq 1 \Rightarrow bRa$

R is symmetric, Again $1R\frac{1}{2}$ and $\frac{1}{2}R1$ but $\frac{1}{2} \neq 1$

R is not anti-symmetric

Further, $1\ R\ 2$ and $2\ R\ 3$ but $1\ \not R\ 3$

$[\because |1 - 3| = 2 > 1]$ R is not transitive.

Illustration 29: The relation "less than" in the set of natural numbers is

(a) Only symmetric (b) Only transitive

(c) Only reflexive (d) Equivalence relation

Solution: (b) Since $x < y, y < z \Rightarrow x < z \nrightarrow x, y, z \in N$

$\therefore xRy, yRz \Rightarrow xRz$, $\therefore$ Relation is transitive , $\therefore\ x < y$ does not give $y < x$, $\therefore$ Relation is not symmetric.

Since $x < x$ does not hold, hence relation is not reflexive.

EXERCISE-1

1. A relation from P to Q is

(a) A universal set of $P \times Q$ (b) $P \times Q$

(c) An equivalent set of $P \times Q$ (d) A subset of $P \times Q$

2. Let R be a relation from a set A to set B, then

(a) $R = A \cup B$ (b) $R = A \cap B$ (c) $R \subseteq A \times B$ (d) $R \subseteq B \times A$

3. Let $A = \{a, b, c\}$ and $B = \{1, 2\}$. Consider a relation R defined from set A to set B. Then R may be is equal to set

(a) A (b) B (c) $A \times B$ (d) $B \times A$

4. Let $n(A) = n$. Then the number of all relations on A is

(a) 2^n (b) $2^{(n)!}$ (c) 2^{n^2} (d) None of these

5. If R is a relation from a finite set A having m elements to a finite set B having n elements, then the number of relations from A to B is

(a) 2^{mn} (b) $2^{mn} - 1$ (c) $2mn$ (d) m^n

6. Let R be a reflexive relation on a finite set A having n-elements, and let there be m ordered pairs in R. Then

(a) $m \geq n$ (b) $m \leq n$ (c) $m = n$ (d) None of these

7. The relation R defined on the set $A = \{1, 2, 3, 4, 5\}$ by $R = \{(x, y) : |x^2 - y^2| < 16\}$ is given by
(a) $\{(1, 1), (2, 1), (3, 1), (4, 1), (2, 3)\}$ (b) $\{(2, 2), (3, 2), (4, 2), (2, 4)\}$
(c) $\{(3, 3), (3, 4), (5, 4), (4, 3), (3, 1)\}$ (d) None of these

8. A relation R is defined from $\{2, 3, 4, 5\}$ to $\{3, 6, 7, 10\}$ by; $xRy \Leftrightarrow x$ is relatively prime to y. Then domain of R is
(a) $\{2, 3, 5\}$ (b) $\{3, 5\}$ (c) $\{2, 3, 4\}$ (d) $\{2, 3, 4, 5\}$

9. Let R be a relation on $\mathbb{N}$ defined by $x + 2y = 8$. Then domain of R is
(a) $\{2, 4, 8\}$ (b) $\{2, 4, 6, 8\}$ (c) $\{2, 4, 6\}$ (d) $\{1, 2, 3, 4\}$

10. If $R = \{(x, y) | x, y \in \mathbb{Z}, x^2 + y^2 \leq 4\}$ is a relation in $\mathbb{Z}$, then domain of R is
(a) $\{0, 1, 2\}$ (b) $\{0, -1, -2\}$ (c) $\{-2, -1, 0, 1, 2\}$ (d) None of these

11. If $A = \{1, 2, 3\}$, $B = \{1, 4, 6, 9\}$ and R is a relation from A to B defined by 'x is greater than y'. The range of R is
(a) $\{1, 4, 6, 9\}$ (b) $\{4, 6, 9\}$ (c) $\{1\}$ (d) None of these

12. R is a relation from $\{11, 12, 13\}$ to $\{8, 10, 12\}$ defined by $y = x - 3$. Then R^{-1} is
(a) $\{(8, 11), (10, 13)\}$ (b) $\{(11, 18), (13, 10)\}$
(c) $\{(10, 13), (8, 11)\}$ (d) None of these

13. Let $A = \{1, 2, 3\}$, $B = \{1, 3, 5\}$. If relation R from A to B is given by $R = \{(1, 3), (2, 5), (3, 3)\}$. Then R^{-1} is
(a) $\{(3, 3), (3, 1), (5, 2)\}$ (b) $\{(1, 3), (2, 5), (3, 3)\}$
(c) $\{(1, 3), (5, 2)\}$ (d) None of these

14. Let R be a reflexive relation on a set A and I be the identity relation on A. Then
(a) $R \subset I$ (b) $I \subset R$ (c) $R = I$ (d) None of these

15. Let R be a relation over the set $\mathbb{N} \times \mathbb{N}$ and it is defined by $(a,b)R(c,d) \Rightarrow a + d = b + c$. Then R is
(a) Reflexive only (b) Symmetric only
(c) Transitive only (d) An equivalence relation

16. An integer m is said to be related to another integer n if m is a multiple of n. Then the relation is
(a) Reflexive and symmetric (b) Reflexive and transitive
(c) Symmetric and transitive (d) Equivalence relation

17. The relation R defined in N as $aRb \Leftrightarrow b$ is divisible by a is
(a) Reflexive but not symmetric (b) Symmetric but not transitive
(c) Symmetric and transitive (d) None of these

18. Let R be a relation on a set A such that $R = R^{-1}$, then R is
(a) Reflexive (b) Symmetric (c) Transitive (d) None of these

19. Let $R = \{(a, a)\}$ be a relation on a set A. Then R is
(a) Symmetric (b) Antisymmetric
(c) Symmetric and antisymmetric (d) Neither symmetric nor anti-symmetric

20. In the set $A = \{1, 2, 3, 4, 5\}$, a relation R is defined by $R = \{(x, y) | x, y \in A \text{ and } x < y\}$. Then R is
(a) Reflexive (b) Symmetric (c) Transitive (d) None of these

ANSWER KEY

1. (d)	**2.** (c)	**3.** (c)	**4.** (c)	**5.** (a)	**6.** (a)	**7.** (d)
8. (d)	**9.** (c)	**10.** (c)	**11.** (c)	**12.** (a)	**13.** (a)	**14.** (b)
15. (d)	**16.** (b)	**17.** (a)	**18.** (b)	**19.** (c)	**20.** (c)	

5. FUNCTIONS

Introduction

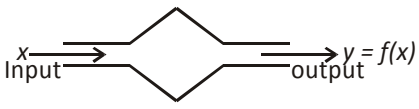
If two variable quantities x and y according to some law are so related that corresponding to each value of x (considered only real), which belongs to set E, there corresponds one and only one finite value of the quantity y (i.e., unique value of y) then y is said to be a functions (single valued) of x, defined by $y = f(x)$ where x is the argument or independent variable and y is a dependent variable defined on the set E.

For example if r be the radius of the sphere and V its volume, then r and V are related by

$$V = \frac{4}{3}\pi r^3 \text{ or } V = f(r)$$

Then we say that the volume V of the sphere is a function of the radius r.

Graphically :



where y is the image of x and x is the pre-image of y under f.

Note :-

1. If to each value of x, which belongs to set E there corresponds one or more than one values of the quantity y. Then y is called the multiple valued function of x defined on the set E.
2. The word FUNCTION is used only for single valued function.

   For Example $y = \sqrt{x}$ is single valued function but $y^2 = x$ is a multiple valued function.

   $\because$ $y^2 = x \Rightarrow y = \pm\sqrt{x}$ for one value of x, y gives two values.

Definition

Let A and B be two non-empty sets and f is a rule which associates each element of A with a unique element of B is called a mapping or function from A to B. If f is a function from A to B, then

we write $f : A \to B$ or $A \xrightarrow{f} B$

which is read as f is a mapping from A to B.

Every function from $A \to B$ satisfies the following conditions :

(i) $f \subset A \times B$

(ii) $\forall a \in A \Rightarrow (a, f(a)) \in f$

(iii) $(a, b) \in f$ and $(a, c) \in f \Rightarrow b = c$

Definition (Graphically)

If any line parallel to y-axis cuts the graph of the functgion at most one point, then it is a single valued function and if it cuts at more than one point then it is a multiple function or it does not form a function.

Illustration 30: Prove that $y = |\sin|x||$ is a function.

Solution : Since any line parallel to y-axis cuts $y = |\sin|x||$ at one and only point. Thus, $y = |\sin|x||$ is a function.

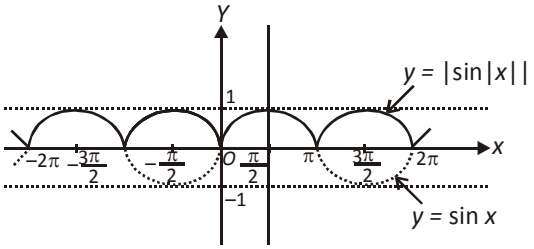
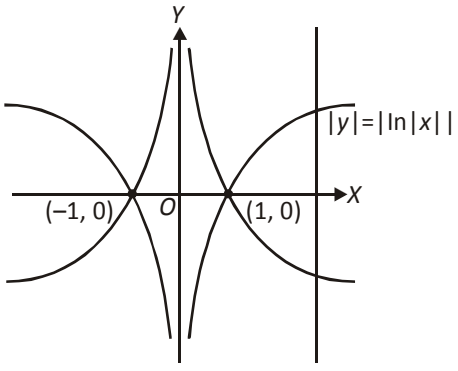


Illustration 31: Prove that $|y| = |\text{In}|x||$ is not a function.

solution :

Since any line parallel to y-axis meets the curve $|y| = |\text{In}|x||$ at two points, Thus, $|y| = |\text{In}|x||$ is not a function.



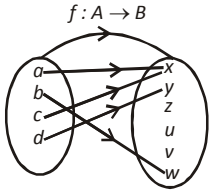
Domain, Co-domain and Range of a Function

Let $f : A \to B$ is a function from A to B, then the set A is called the domain of the function f (denoted by D_f) and the set B is called the Co-domain of the function f (denoted by C_f). The set of all those elements of B which are the images of the elements of set A is called the range of the function f (denoted by R_f).

Domain of $f = D_f = (a : a \in A. (a, f(a)) \in f)$
Range of $f = R_f = \{f(a) : a \in A, f(a) \in B\}$
It should be noted that the range of f is always a subset of Co-domain B.
i.e., $f(A) \subseteq B$

How to find range : First put $y = f(x)$ by suitable substitution, find x in terms of y. Then find all such y for which x is defined i.e., in the domain. Set of these values of y is the range of $f(x)$.

Illustration 32:Find the domain, co-domain and range of the function f which is denoted by the figure below.



Solution : It is clear from the figure
Domain of $f = D_f$ = All elements of set $A = \{a, b, c, d\}$
Co-domain of $f = C_f$ = All elements of set $B = \{x, y, z, u, v, w\}$
Range of $f = R_f$ = All elements of B which are the images of the elements of set $A = \{x, y, w\}$
Also $\{x, y, w\} \subseteq \{x, y, z, u, v, w\}$
i.e., Range $\subseteq$ co-domain.

Formulas for the domain of a function

(i) Domain of $(f(x) \pm g(x))$ = Domain of $f(x) \cap$ Domain of $g(x)$ i.e., $D_{f\pm g} = D_f \cap D_g$

(ii) Domain of $(f(x).g(x))$ = Domain of $f(x) \cap$ Domain of $g(x)$ i.e., $D_{fg} = D_f \cap D_g$

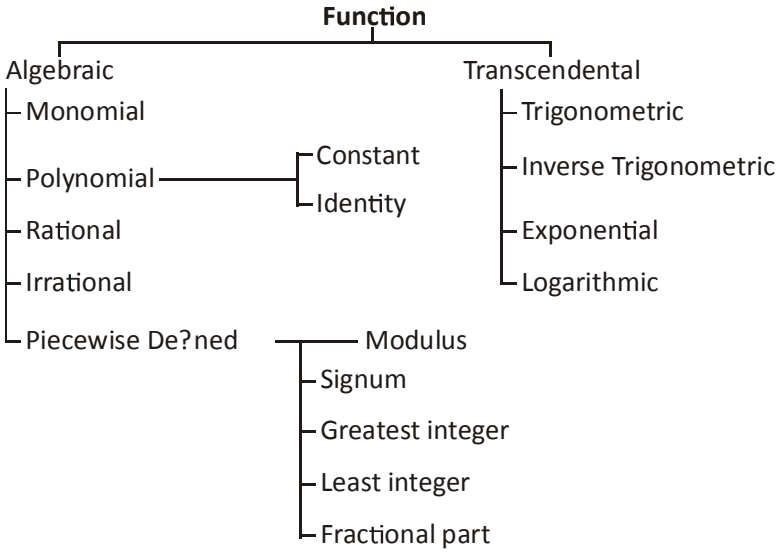
(iii) Domain of $\left(\frac{f(x)}{g(x)}\right)$ = Domain of $f(x) \cap$ Domain of $g(x) \cap \{x : g(x) \neq 0\}$ i.e., $D_{f/g} = D_f \cap D_g \cap \{x : g(x) \neq 0\}$

(iv) Domain of $\sqrt{f(x)}$ = Domain of $f(x) \cap \{x : f(x) \geq 0\}$ i.e., $D_{\sqrt{f}} = D_f \cap \{x : f \geq 0\}$

(v) Domain of $\log_a f(x)$ = Domain of $f(x) \cap \{x : f(x) > 0\}$ i.e., $D_{\log_a f} = D_f \cap \{x : f > 0\}$

(vi) Domain of $(fog)x$ = Domain of $g(x)$
i.e., $D_{fog} = D_g$ [where $(fog)x = f\{g(x)\}$]

6. CLASSIFICATION OF FUNCTIONS



(A) Algebraic functions

An algebraic function consists of a finite number of terms involving powers and roots of the variable x and the operations : addition, substraction, multiplication and division.

e.g., $7x^3 + 2x^2 - 3x + 2$, $5x^2 + 2x^{1/2} + 7$, $\frac{3x^2 - 2x + 5}{7x^2 + 4x - 1}$ etc.

(a) **Monomial functions :** Any function of the form $f(x) = kx^n$ where k is a constant and $n \in W$ is known as monomial function.

i.e., $f(x) = 5x^3$

Note :- The function $f(x) = ax$, $a \neq 0$ is called an odd linear function.

(b) **Polynomial functions :** Any function which is obtained by addition or subtraction of several monomials is called a polynomial whose domain and co-domain both is the set of real numbers. Clearly in this

$f(x) = a_0x^n + a_nx^{n-1} + a_2x^{n-2} + ...+ a_{n-1}x + a_n$ $(a_0 \neq 0)$

where $a_0, a_1, a_2, ...a_{n-1}, a_n \in \mathbb{R}$ and $n \in \mathbb{N}$

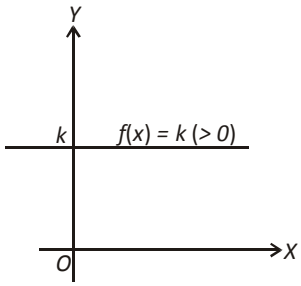
'n' is called the degree of polynomial which is maximum value of the exponents of x.

e.g.	Polynomial	Degree
(i)	$f(x) = 125x^{150} + 22x^{99} + x^{27}$	150
(ii)	$f(x) = 3x^3 + 2x^2 + 3x + 7$	3
(iii)	$f(x) = 2x^2 + 4x - 7$	2
(iv)	$f(x) = 5x + 8$	1
(v)	$f(x) = 4$	0
(vi)	$f(x) = 0$	undefined

1. **Constant function :** The function $f : R \to R$ is called a constant function if a constant $k \in R$ such that $f(x) = k$, $\forall\ k \in R$

$\therefore$ Domain of $f(x) = D_f = R$

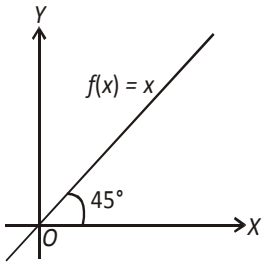
and Range of $f(x) = R_f = k$.



2. **Identity function :** The function $f : R \to R$ is called an identity function if $f(x) = x\ \forall\ x \in R$

$\therefore$ $D_f = R_f = R$

(i.e., output is same as the input)



(c) **Rational function :** A function defined by the quotient of two polynomial functions is called a rational function. Theus $f(x) = \frac{P(x)}{Q(x)}$, where $P(x)$ and $Q(x)$ $(\neq 0)$ are polynomial functions.

e.g.	Rational Functions	Domain
(i)	$\frac{a_0x^n + a_1x^{n-1} + a_2x^{n-2} + ... + a_n}{b_0x^n + b_1x^{n-1} + b_2x^{n-2} + ... + b_n}$	All real numbers except the real roots of the equation $b_0x^n + b_1x^{n-1} + b_2x^{n-2} + ... + b_n = 0$
(ii)	$\frac{5x^2 - 7x + 1}{x^3 - 6x^2 + 11x - 6}$	$\mathbb{R} - \{1, 2, 3\}$
(iii)	$\frac{7x^2 - 4x + 5}{x^2 + 3x + 2}$	$\mathbb{R} - \{-1, -2\}$
(iv)	$\frac{1}{x^3}$	$\mathbb{R} - \{0\}$

(d) Irrational function : An algebraic function or rational function containing one or more radicals (non-integral rational powers of x) is called irrational function.

e.g., $\frac{3x^2 + 2\sqrt{x}}{3x - 2}, 2x^3 - 7x^{1/2} + 5x - 2, \frac{(x^3 - 2x^2 + 7)^{2/3}}{\sqrt{(x+2)}}$

then domains are $x > 0$, $x \neq 2/3$; $x > 0$; $x > -2$

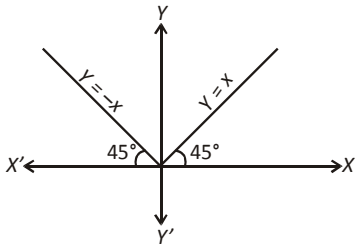
i.e., $(0, \infty) - \left\{\frac{2}{3}\right\}$, $(0, \infty)$; $(-2, \infty)$ respectively.

(e) Piecewise defined functions

(I) Modulus function (or absolute value function)

It is defined as $y = f(x) = |x| = \begin{cases} x; & x \geq 0 \\ -x; & x < 0 \end{cases}$

$D_f = R$, $R_f = [0, \infty)$



Properties of Modulus Function

(i) $|x| \leq a \Rightarrow -a \leq x \leq a$; $(a \geq 0)$

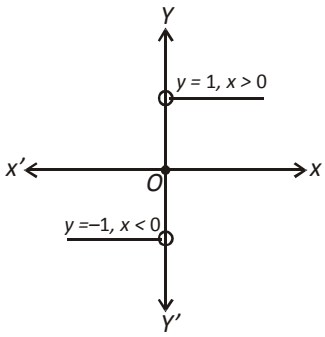
(ii) $|x| \geq a \Rightarrow x \leq -a$ and $x \geq a$; $(a \geq 0)$

(iii) $|x + y| = |x| + |y|$
$\Leftrightarrow x \geq 0$ and $y \geq 0$ or $x \leq 0$ and $y \leq 0$

(iv) $|x - y| = |x| - |y| \Rightarrow x \geq 0$ and $|x| \geq |y|$
or $x \leq 0$ and $y \leq 0$ and $|x| \leq |y|$

(v) $|x \pm y| \leq |x| + |y|$

(vi) $|x \pm y| \geq ||x| - |y||$



(II) Signum function

The signum function f is defined as

$$y = f(x) = \text{Sgn}(x) = \begin{cases} \frac{|x|}{x} \text{ or } \frac{x}{|x|}; & x \neq 0 \\ 0 \quad ; & x = 0 \end{cases} = \begin{cases} -1; x < 0 \\ 0; x = 0 \\ 1; x > 0 \end{cases} \qquad D_f = R, R_f = \{-1, 0, 1\}$$

(III) Greatest Integer Function

$[x]$ denotes the Greatest integer less than or equal to x i.e., $[x] \le x$. It is also known **as floor of x**.

Thus, $[3.5779] = 3$, $[0.89] = 0$, $[3] = 3$

$[-8.7285] = -9$, $[-0.6] = -1$

In general if n is an integer and x is any real number between n and $(n + 1)$,

i.e., $n \le x < n + 1$ then $[x] = n$

$\therefore \quad f(x) = [x]$

For $x \in [-3, -2)$; $y = -3$

For $x \in [-2, -1)$; $y = -2$

For $x \in [-1, 0)$; $y = -1$

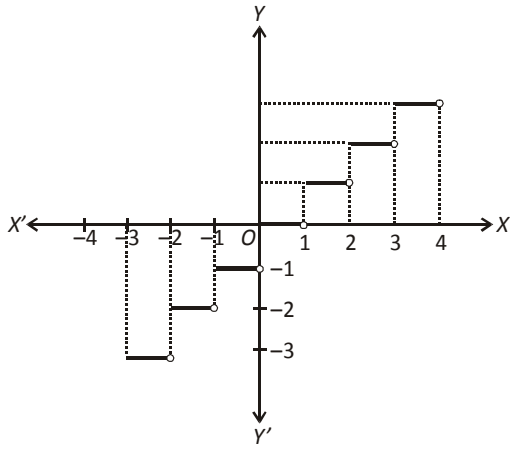
For $x \in [0, 1)$; $y = 0$

For $x \in [1, 2)$; $y = 1$

For $x \in [2, 3)$; $y = 2$

For $x \in [3, 4)$; $y = 3$

$D_f = R$, $R_f = I$



Properties of Greatest Integer Function

(i) $[x + n] = n + [x], n \in I$

(ii) $x = [x] + \{x\}$, $\{x\}$ denotes the fractional part of x.

(iii) $[-x] = -[x], x \in I$

(iv) $[-x] = -[x] - 1, x \notin I$

(v) $[x] \ge n \Rightarrow x \ge n, n \in I$

(vi) $[x] > n \Rightarrow x \ge n + 1, n \in I$

(vii) $[x] \le n \Rightarrow x < n + 1, n \in I$

(viii) $[x] < n \Rightarrow x < n, n \in I$

(ix) $n_2 \le [x] \le n_1 \Rightarrow n_2 \le x < n_1 + 1; n_1, n_2 \in I$

(x) $[x + y] \ge [x] + [y]$

(xi) $\left[\frac{[x]}{n}\right] = \left[\frac{x}{n}\right], n \in N$

(xii) $\left[\frac{n+1}{2}\right] + \left[\frac{n+2}{4}\right] + \left[\frac{n+4}{8}\right] + \left[\frac{n+8}{16}\right] + \ldots = n, n \in N$

(xiii) $[x] + \left[x + \frac{1}{n}\right] + \left[x + \frac{2}{n}\right] + \ldots \left[x + \frac{n-1}{n}\right] = [nx], n \in N$

(IV) Least Integer Function

(x) or $\lceil x \rceil$ denotes the least integer function which is greater than or equal to x. It is also known as **ceiling of x**.

Thus, $(3.578) = 4$, $(0.87) = 1$, $(4) = 4$,

$\lceil -8.239 \rceil = -8, \lceil -0.7 \rceil = 0$

In general if n is an integer and x is any real number between n and $(n + 1)$

i.e., $n < x \le n + 1$ then $(x) = n + 1$

$\therefore \quad f(x) = (x) = \lceil x \rceil$

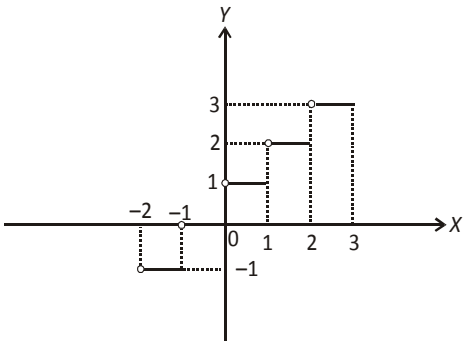
For $x \in (-2, -1]$; $(x) = -1$

For $x \in (-1, 0]$; $(x) = 0$

For $x \in (0, 1]$; $(x) = 1$

For $x \in (1, 2]$; $(x) = 2$

For $x \in (2, 3]$; $(x) = 3$



Properties of Least integer function

(i) $(x + n) = (x) + n, n \in I$

(ii) $x = (x) + \{x\} - 1$, $\{x\}$ denotes the fractional part of x

(iii) $(-x) = -(x), x \in I$

(iv) $(-x) = -(x) + 1, x \notin I$

(v) $(x) \geq n \Rightarrow x > n - 1, n \in I$

(vi) $(x) > n \Rightarrow x > n, n \in I$

(vii) $(x) \leq n \Rightarrow x < n, n \in I$

(viii) $(x) < n \Rightarrow x < n - 1, n \in I$

(ix) $n_2 \leq (x) \leq n_1 \Rightarrow n_2 - 1 < x < n_1; n_1, n_2 \in I$

(x) $(x + y) \leq (x) + (y)$

(xi) $\left(\frac{(x)}{n}\right) = \left(\frac{x}{n}\right), n \in N$

(xii) $\left(\frac{n+1}{2}\right) + \left(\frac{n+2}{4}\right) + \left(\frac{n+4}{8}\right) + \left(\frac{n+8}{16}\right) + \ldots\ldots = 2(n-1), n \in N$

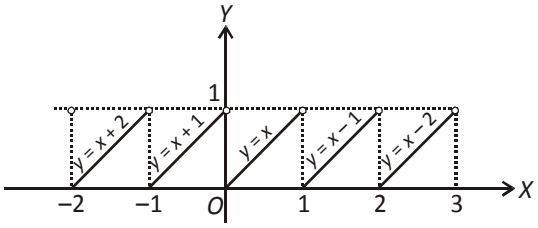
(xiii) $(x) + \left(x + \frac{1}{n}\right) + \left(x + \frac{2}{n}\right) + \ldots\ldots \left(x + \frac{n-1}{n}\right) = (nx) + n - 1, n \in N$

(V) Fractional Part Function

It is denoted as $f(x) = \{x\}$ and defined as

(i) $\{x\} = f$ if $x = n + f$ where $n \in I$ and $0 \leq f < 1$

(ii) $\{x\} = x - [x]$



i.e., $\{2.7\} = 0.7$, $\{3\} = 0$, $\{-3.6\} = 0.4$

Note :-

1. For proper fraction $0 < f < 1$
2. Domain and Range of $\frac{1}{\{x\}}$ are $R - I$ and $(0, 1)$ respectively

Illustration 33: Find the domain of definition of the function $f(x) = \frac{1}{\sqrt{\left|\left[|x| - 1\right]\right| - 5}}$, [.] denotes the greatest integer function.

Solution : Since $f(x) = \frac{1}{\sqrt{\left|\left[|x| - 1\right]\right| - 5}}$

$\Rightarrow f(x)$ is defined when $|[|x| - 1]| - 5 > 0$

Then $[|x| - 1] > 5$ and $[|x| - 1] < -5$ $\Rightarrow |x| - 1 \geq 6$ and $|x| - 1 < -5$

$\Rightarrow |x| \geq 7$ or $|x| < -4$ (impossible) $\Rightarrow x \leq -7$ and $x \geq 7$

$\therefore D_f = (-\infty, -7] \cup [7, \infty)$

Illustration 34: Find the domain of definition of the function $f(x) = \sqrt{\left(\frac{1 - |x|}{2 - |x|}\right)}$

Solution : Since $f(x) = \sqrt{\left(\frac{1-|x|}{2-|x|}\right)}$

$\Rightarrow$ f(x) is defined when $\frac{1-|x|}{2-|x|} \geq 0$

$\Rightarrow \frac{|x|-1}{|x|-2} \geq 0$



$\therefore$ Critical points of $|x|$ are 1 and 2

$\therefore$ From wavy curve method

$|x| \leq 1$ and $|x| > 2$

Then $-1 \leq x \leq 1$ and $x < -2$ and $x > 2$

$\Rightarrow x \in (-\infty, -2) \cup [-1, 1] \cup (2, \infty)$

$\therefore D_f = (-\infty, -2) \cup [-1, 1] \cup (2, \infty)$

Illustration 35: Find the range of the function $f(x) = \frac{e^x - e^{|x|}}{e^x + e^{|x|}}$

Solution : $\because f(x) = \frac{e^x - e^{|x|}}{e^x + e^{|x|}} = \begin{cases} 0 & , x \geq 0 \\ \frac{e^x - e^{-x}}{e^x + e^{-x}} & , x < 0 \end{cases}$

Range of $f(x) = 0$ for $x \geq 0$ ...(1)

and let $y = f(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$ for x < 0

or $\frac{y+1}{1-y} = \frac{2e^x}{2e^{-x}}$ (From componendo and dividendo rule)

$\Rightarrow \frac{y+1}{1-y} = e^{2x} \quad \therefore \quad 0 < \frac{y+1}{1-y} < 1$

$\Rightarrow 0 > \frac{y+1}{y-1} > -1$

$\therefore y \in (-1, 1)$ and $y \in (-\infty, 0) \cup (1, \infty)$ then $y \in (-1, 0)$ ...(2)

$\therefore$ From (1) and (2)

Range of $f(x)$ is $R_f = (-1, 0]$

Illustration 36: Find the range of the function $f(x) = \frac{e^{-x}}{1+[x]}$, [.] denotes the greatest integer function.

Solution : Given $f(x) = \frac{e^{-x}}{1+[x]}$

Let $y = f(x) = \frac{e^{-x}}{1+[x]}$ or $y(1 + [x]) = e^{-x} > 0$

$\Rightarrow y(1 + [x]) > 0$

If $1 + [x] > 0$ then $y > 0$

and If $1 + [x] < 0$ then $y < 0$

Hence $y \in R - \{0\}$

Illustration 37: If [x] denotes the integral part of x for real x, then find the value of

$$\left[\frac{1}{4}\right]+\left[\frac{1}{4}+\frac{1}{200}\right]+\left[\frac{1}{4}+\frac{1}{100}\right]+\left[\frac{1}{4}+\frac{3}{200}\right]+.....+\left[\frac{1}{4}+\frac{199}{200}\right]$$

Solution : The given expression can be written as

$$\left[\frac{1}{4}\right]+\left[\frac{1}{4}+\frac{1}{200}\right]+\left[\frac{1}{4}+\frac{2}{200}\right]+\left[\frac{1}{4}+\frac{3}{200}\right]+.....\left[\frac{1}{4}+\frac{199}{200}\right]=\left[200.\frac{1}{4}\right]=[50]=50$$

Illustration 38: Let [a] denote the larger integer not exceeding the real number a. If x and y satisfy the equations $y = 2[x] + 3$ and $y = 3[x – 2]$ simultaneously, determine [$x + y$].

Solution : We have $y = 2[x] + 3 = 3[x – 2]$

$\Rightarrow 2[x] + 3 = 3[x] – 6 \Rightarrow [x] = 9$

$\therefore y = 2[x] + 3 = 2.9 + 3 = 21$

$\therefore [x + y] = [x + 21] = [x] + 21 = 9 + 21 = 30$

Illustration 39: Let {x} and [x] denote the fractional and integral parts of a real number x respectively. Then solve theequation $4\{x\} = x + [x]$

Solution : $x = [x] + \{x\}$ ...(1)

Substituting the value of x from (1) in the given equation, then

$4\{x\} = [x] + \{x\} + [x]$

or $3\{x\} = 2[x] \Rightarrow \{x\} = \frac{2}{3}[x]$ ...(2)

but$0 \le \{x\} < 1 \Rightarrow 0 \le \frac{2}{3}[x] < 1$ [From (2)]

$\Rightarrow 0 \le [x] < \frac{3}{2} \quad \therefore [x] = 0, 1$

Hence x from (2), $\{x\} = 0, \frac{2}{3}$

Hence from (1) $x = 0 + 0 = 0$ and $x = 1 + \frac{2}{3} = \frac{5}{3}$

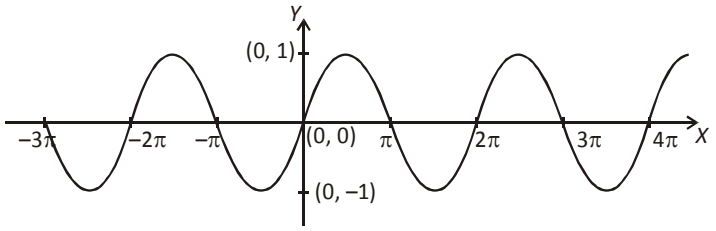
Hence $x = 0, 5/3$

(B) Transcendental functions

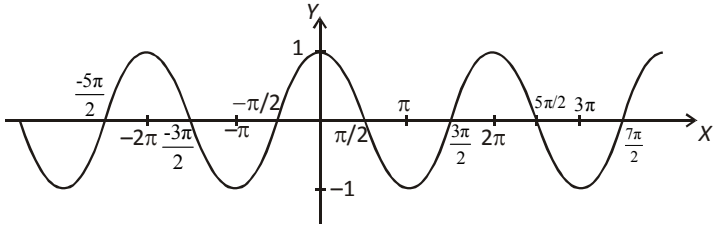
All functions which are not algebraic are called transcendental functions.

(a) Trigonometric Functions

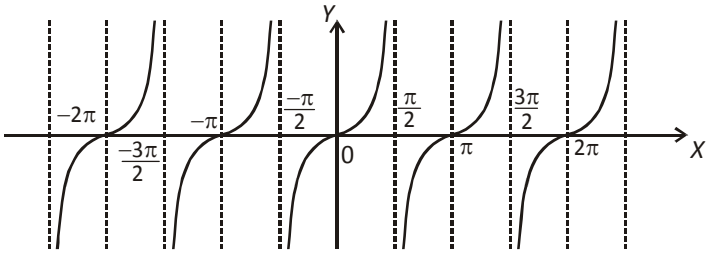
(i) Sine Function : $f(x) = \sin x$ then domain and range of $f(x)$ are R and [–1, 1] respectively.



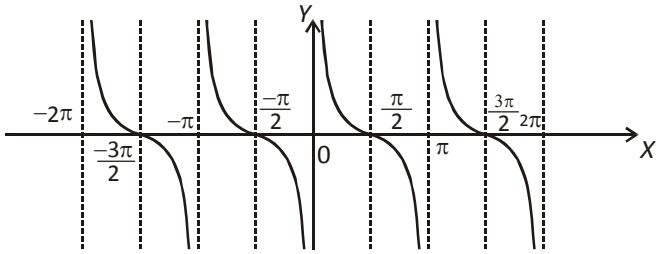
(ii) Cosine Function : $f(x) = \cos x$ then domain and range of $f(x)$ are R and [–1, 1] respectively.



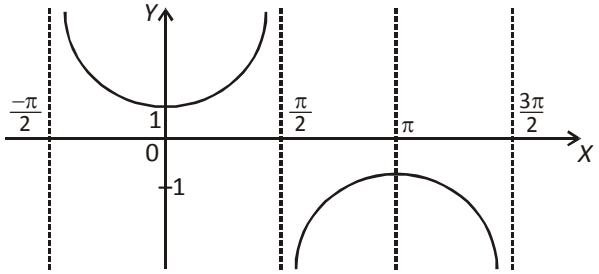
(iii) **Tangent Function :** $f(x) = \tan x$ then domain and range of $f(x)$ are $R - \left\{n\pi + \frac{\pi}{2}, n \in I\right\}$ and R respectively.



(iv) **Cotangent Function :** $f(x) = \cot x$ then domain and range of $f(x)$ are $R - \{n\pi, n \in I\}$ and R respectively.



(v) **Secant Function :** $f(x) = \sec x$ then domain and range of $f(x)$ are $R - \left\{n\pi + \frac{\pi}{2}, n \in I\right\}$ and $R - (-1, 1)$ respectively.



(vi) **Cosecant Function :** $f(x) = \text{cosec } x$ then domain and range of $f(x)$ are $R - \{n\pi, n \in I\}$ and $R - (-1, 1)$ respectively.

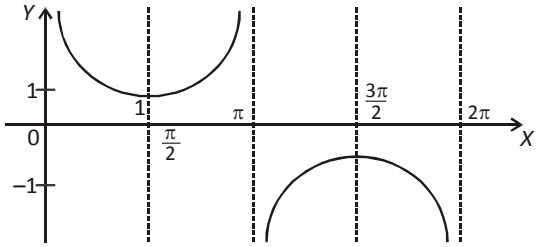


Table of Trigonometric Functions

	Functions	Domain (x)	Range (y)
1.	$y = \sin x$	$x \in R$	$y \in [-1, 1]$
2.	$y = \cos x$	$x \in R$	$y \in [-1, 1]$
3.	$y = \tan x$	$x \in R - \left\{n\pi + \frac{\pi}{2}, n \in I\right\}$	$y \in R$
4.	$y = \cot x$	$x \in R - \{n\pi, n \in I\}$	$y \in R$
5.	$y = \sec x$	$x \in R - \left\{n\pi + \frac{\pi}{2}, n \in I\right\}$	$y \in R - (-1, 1)$
6.	$y = \text{cosec } x$	$x \in R - \{n\pi, n \in I\}$	$y \in R - (-1, 1)$

Illustration 40: Find the domain of definition of the following functions :

(i) $f(x) = \sqrt{\cos(\sin x)}$ **(ii)** $f(x) = \sqrt{\sin(\cos x)}$

Solution : **(i)** Since $f(x) = \sqrt{\cos(\sin x)}$

$f(x)$ is defined when $\cos(\sin x) \geq 0$

but $-1 \leq \sin x \leq 1 \ \forall \ x \in R$

if $\sin x = \theta$, then $\cos\theta \geq 0$

$\therefore \theta$ lies in I and IV quadrant.

$$\Rightarrow \theta \in \left[2n\pi - \frac{\pi}{2}, 2n\pi, + \frac{\pi}{2}\right], n \in I$$

Hence it is clear that $\cos(\sin x) \geq 0 \ \forall \ x \in R$ then $D_f = R$

(ii) Since $f(x) = \sqrt{\sin(\cos x)}$

$f(x)$ is defined when $\sin(\cos x) \geq 0$

$\Rightarrow \cos x \geq \sin^{-1} 0$ $\Rightarrow \cos x \geq 0$

$\therefore x$ lies in I and IV quadrant

$$\therefore 2n\pi - \frac{\pi}{2} \leq x \leq 2n\pi + \frac{\pi}{2}, n \in I \qquad D_f = \left[2n\pi - \frac{\pi}{2}, 2n\pi, + \frac{\pi}{2}\right], n \in I.$$

7. SOME IMPORTANT GRAPHS

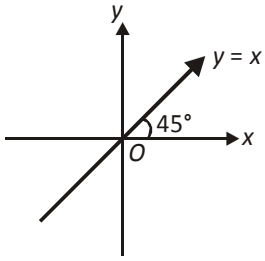
Some of basic polynomial functions are

(i) Identity function/Graph $f(x) = x$

A function f defined by $f(x) = x$ for all $x \in R$, is called the identity function.

Here, $y = x$ clearly represents a straight line passing through the origin and inclined at an angle of 45° with x-axis shown as :

The domain and range of identity function are both equal to R.

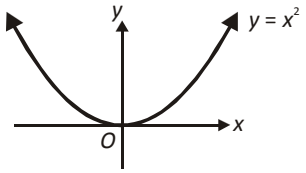


(ii) Graph of $f(x) = x^2$

A function given by $f(x) = x^2$ is called the square function.

The domain of square function is R and its range is $R^+ \cup \{0\}$ or $[0, \infty)$

Clearly $y = x^2$, is a parabola. Since $y = x^2$ is an even function, so its graph is **symmetrical about y-axis,** shown as :

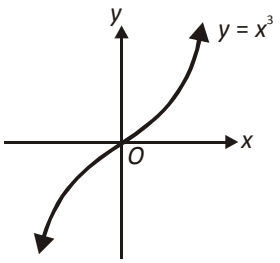


(iii) Graph of $f(x) = x^3$

A function given by $f(x) = x^3$ is called the cube function.

The domain and range of cube are both equal to R.

Since, $y = x^3$ is an old function, so its graph is **symmetrical about opposite quadrant,** i.e., **"origin"**, shown as :



(iv) Graph of $f(x) = x^{2n}; n \in N$

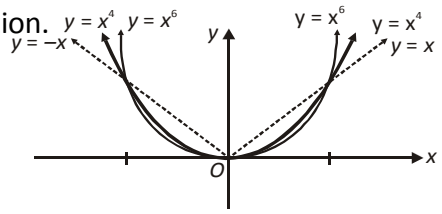
If $n \in N$, then function f given by $f(x) = x^{2n}$ is an even function.

So, its graph is always **symmetrical about y-axis.**

Also, $x^2 > x^2 > x^6 > x^8 > .....$ for all $x \in (-1, 1)$

and $x^2 < x^4 < x^6 < x^8 < .....$ for all $x \in (-\infty, -1) \cup (1, \infty)$

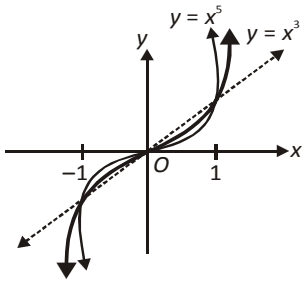
Graphs of $y = x^2$, $y = x^4$, $y = x^6$,....., etc. are shown as :



(v) Graph of $f(x) = x^{2n-1}; n \in N$

If $n \in N$, then the function f given by $f(x) = x^{2n-1}$ is an odd function. So, its graph is **symmetrical about origin or opposite quadrants.**

Here, comparison of values of x, x^3, x^5,.....
for



$x \in (1, \infty)$ $\quad x < x^3 < x^5 < .....$
$x \in (0, 1)$ $\quad x > x^3 > x^5 > .....$
$x \in (-1, 0)$ $\quad x < x^3 < x^5 < .....$
$x \in (-\infty, -1)$ $\quad x > x^3 > x^5 > .....$

Graphs of $f(x) = x$, $f(x) = x^3$, $f(x) = x^5$, ... are shown as in figure.

Rational Expression

A function obtained by dividing a polynomial by another polynomial is called a rational function.

$\Rightarrow$ $\quad f(x) = \dfrac{P(x)}{Q(x)}$ $\quad$ Domain $\in R - \{x \mid Q(x) = 0\}$

i.e., $\quad$ Domain $\in R$ except points for which denominator = 0.

Graphs of Some Simple Rational Funciton

(i) Graph of $f(x) = \dfrac{1}{x}$

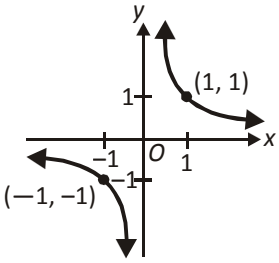
A function defined by $f(x) = \dfrac{1}{x}$ is called the reciprocal function or rectangular hyperbola, with coordinate axis as asymptotes. The domain and range of $f(x) = \dfrac{1}{x}$ is ***R* – {0}.**

Since, $f(x)$ is odd function, so its graph is **symmetrical about opposite quadrants.** Also, we observe

$\lim\limits_{x \to 0^+} f(x) = +\infty$ and $\lim\limits_{x \to 0^-} f(x) = -\infty$

and $\quad$ as $x \to \pm\infty \Rightarrow f(x) \to 0$

Thus, $f(x) = \dfrac{1}{x}$ could be shown as in figure.



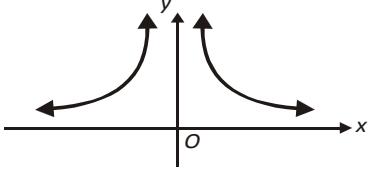
(ii) Graph of $f(x) = \dfrac{1}{x^2}$

Here, $f(x) = \dfrac{1}{x^2}$ is an even function, so its graph is **symmetrical about y-axis.**

Domain of $f(x)$ is $R - \{0\}$ and range $(0, \infty)$.

Also, as $y \to \infty$ $\quad$ as $\lim\limits_{x \to 0^+} f(x)$ $\quad$ or $\quad \lim\limits_{x \to 0^-} f(x)$

and $\quad y \to 0$ $\quad$ as $\lim\limits_{x \to \pm\infty} f(x)$.



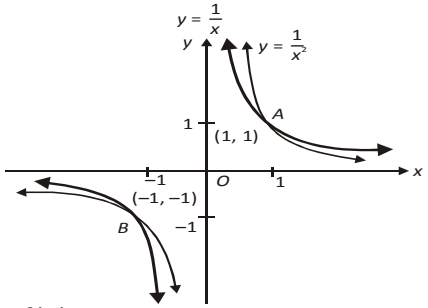
Thus, $f(x) = \dfrac{1}{x^2}$ could be shown as in figure.

(iii) Graph of $f(x) = \dfrac{1}{x^{2n-1}}; n \in N$

Here, $f(x) = \dfrac{1}{x^{2n-1}}$ is an odd function, so its graph is

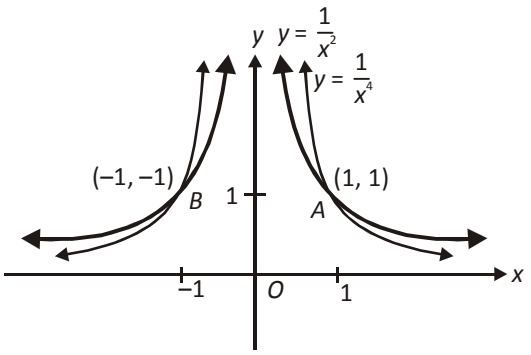
Symmetrical in opposite quadrants.

Also, $y \to \infty$ $\quad$ when $\lim\limits_{x \to 0^+} f(x)$ and $\quad y \to -\infty$ when $\lim\limits_{x \to 0^-} f(x)$



Thus, the graph for $f(x) = \dfrac{1}{x^3}; f(x) = \dfrac{1}{x^5}$, ..., etc. will be similar tothe graph of $f(x) = \dfrac{1}{x}$ **which has asymptotes as coordinates axes,** shown as in figure.

(iv) **Graph of** $f(x)=\frac{1}{x^{2n}};n\in N$



We observe that the function $f(x)=\frac{1}{x^{2n}}$ is an even function, so its graph is **symmetrical about y-axis.**

Also, $y\to\infty$ as $\lim_{x\to0^+} f(x)$ or $\lim_{x\to0^-} f(x)$

and $y\to0$ as $\lim_{x\to-\infty} f(x)$ or $\lim_{x\to+\infty} f(x)$

The values of *y* decrease as the values of *x* increase.

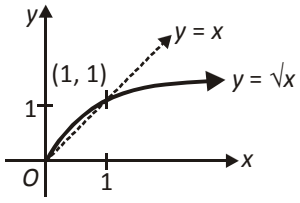
Thus, the graph of $f(x)=\frac{1}{x^2};f(x)=\frac{1}{x^4}$,....., etc. will be similar as graph of $f(x)=\frac{1}{x^2}$, which has **asymptotes as coordinates axis.** Shown as in figure.

Irrational Function

The algebraic function containing terms having non-integral rational powers of *x* are called irrational functions.

(i) **Graph of $f(x)=x^{1/2}$**

Here; $f(x)=\sqrt{x}$ is the portion of the parabola $y^2 = x$, which lies above x-axis.



Domain of $f(x) \in R^+ \cup \{0\}$ or $[0, \infty)$

and range of $f(x) \in R^+ \cup \{0\}$ or $[0, \infty)$

Thus, the graph of $f(x) = x^{1/2}$ is shown as;

Note :- If $f(x) = x^n$ and $g(x) = x^{1/n}$, then $f(x)$ and $g(x)$ inverse of each other.

$\therefore$ $f(x) = x^n$ and $g(x) = x^{1/n}$ is the mirror image about $y = x$.

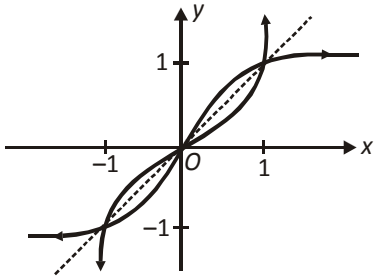
(ii) **Graph of $f(x) = x^{1/3}$**

As discussed above, if $g(x) = x^3$. Then $f(x) = x^{1/3}$ is image of g(x) about $y = x$.

where domain $f(x) \in R$.

and range of $f(x) \in R$.

Thus, the graph of $f(x) = x^{1/3}$ is shown as :



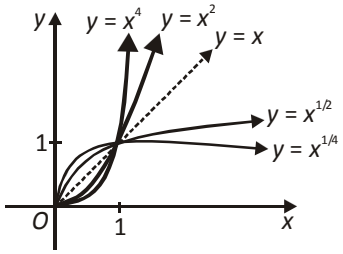
(iii) **Graph of $f(x) = x^{1/2n}$; $n \in N$**

Here, $f(x) = x^{1/2n}$ is defined for all $x \in [0, \infty)$ and the values taken by $f(x)$ are positive.

So, domain and range of $f(x)$ are $[0, \infty)$.

Here, the graph of $f(x) = x^{1/2n}$ is the mirror image of the graph of $f(x) = x^{2n}$ about the line $y = x$, when $x \in [0, \infty)$.

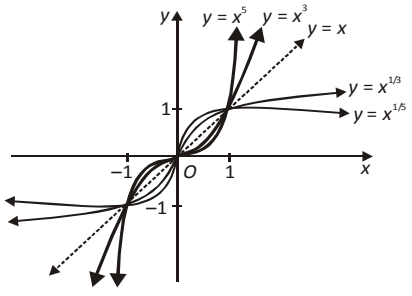
Thus, $f(x) = x^{1/2}$, $f(x) = x^{1/4}$, ... are shown as ;



(iv) **Graph of $f(x) = x^{1/(2n-1)}$, when $n \in N$**

Here, $f(x) = x^{1/(2n-1)}$ is defined for all $x \in R$. So, domain of $f(x) \in R$, and range of $f(x) \in R$. Also the graph of $f(x) = x^{1/(2n-1)}$ about the line $y = x$ when $x \in R$.

Thus, $f(x) = x^{1/3}$, $f(x) = x^{1/5}$,....., are shown as;



MISCELLANEOUS EXAMPLES

Example 1 : If f(x + y + 1) = $(\sqrt{f(x)}+\sqrt{f(y)})^2$ and f(0) = 1 $\forall$ x, $y \in R$.
Determine $f(x)$.

Solution : Given f(x + y + 1) = $(\sqrt{f(x)}+\sqrt{f(y)})^2$

Put $x = y = 0$, $f(1)=2^2$
Again putting $x = 0$, $y = 1$
Then $f(2) = (\sqrt{f(0)}+\sqrt{f(1)})^2$
$= (1 + 2)^2 = 3^2$
and for $x = 1$, $y = 1$

$$f(3)=(\sqrt{f(1)}+\sqrt{f(1)})^2=(2+2)^2=4^2$$

Similarly, $f(x) = (x + 1)^2$

Example 2 : A function f, defined foll all $x, y \in R$ is such that $f(1) = 2$; $f(2) = 8$ and $f(x + y) - kxy = f(x) + 2y^2$, when k is some constant. Find $f(x)$ and show that :

$$f(x+y)f\left(\frac{1}{x+y}\right)=k \text{ for } x+y\neq 0$$

Solution : Given $f(x + y) - kxy = f(x) + 2y^2$
Replacing y by $-x$ then we get

$$f(0) + kx^2 = f(x) + 2x^2$$

$\Rightarrow$ $f(x) = f(0) + kx^2 - 2x^2$ ... (1)
$\therefore f(1) = f(0) + k - 2 = 2$ given ... (2)
and $f(2) = f(0) + 4k - 8 = 8$ (given) ... (3)
Solving (2) and (3) we get

$$k = 4 \text{ and } f(0) = 0 \quad \text{... (4)}$$

$\therefore$ From (1), we get $f(x) = 2x^2$

Also, $f(x+y).f\left(\frac{1}{x+y}\right)=2(x+y)^2.2\left(\frac{1}{x+y}\right)^2$

$= 4 = k$ {From (4)}

Example 3 : Consider a real function $f(x)$ satisfying $2f(xy) = (f(x))^y + (f(y))^x$ for all $x, y \in R$ and $f(1) = a$ where $a \neq 1$. Prove that $(a-1)\sum_{i=1}^{n} f(i)=a^{n+1}-a$

Soltion : Given $2f(xy) = (f(x))^y + (f(y))^x$, replacing y by 1, we get
$2f(x) = f(x) + f(1))^x$
$\Rightarrow f(x) = (f(1))^x = a^x$ ($\because f(1) = a$)

$$\therefore \text{L.H.L.} = (a-1)\sum_{i=1}^{n} f(i)=(a-1)\sum_{i=1}^{n} a^i$$

$= (a - 1)(a + a^2 + a^3 + ... + a^n)$

$$= (a-1)\left\{\frac{a(a^n-1)}{(a-1)}\right\} \quad \text{(If } a > 1\text{)}$$

$= a(a^n - 1) = (a^{n+1} - a)$ = R.H.S.

Example 4 : If $\{x\}$ and $[x]$ represent fractional and integral part of x respectively, then find the value of

$$[x]+\sum_{r=1}^{2000}\frac{\{x+r\}}{2000}$$

Solution : $\because$ $\{x + r\} = \{x\}$

$$\therefore [x]+\sum_{r=1}^{2000}\frac{\{x+r\}}{2000}=[x]+\sum_{r=1}^{2000}\frac{\{x\}}{2000}$$

$$=[x]+\frac{\{x\}}{2000}\sum_{r=1}^{2000}1=[x]+\frac{\{x\}}{2000}\cdot 2000$$

$= [x] + \{x\} = x$

Example 5 : Let $f(x) = (2 \cos x - 1)\,(2 \cos 2x - 1)(2 \cos 2^2x - 1) \ldots (2 \cos 2^{n-1} x - 1)$ (where $n \geq 1$)

Then prove that $f\left(\frac{2\pi k}{2^n \pm 1}\right) = 1 \forall k \in I$

Solution : Since $f(x) = (2 \cos x - 1)\,(2 \cos 2x - 1)\,(2 \cos 2^2x - 1) \ldots (2 \cos 2^{n-1} x - 1)$

$$\therefore f(x)=\frac{(2\cos x+1)(2\cos x-1)(2\cos 2x-1)(2\cos 2^2x-1)\ldots(2\cos 2^{n-1}x-1)}{(2\cos x+1)}$$

$$=\frac{(4\cos^2 x-1)(2\cos 2x-1)(2\cos 2^2x-1)\ldots(2\cos 2^{2n-1}x-1)}{(2\cos x+1)}$$

$$=\frac{(2\cos 2x+1)(2\cos 2x-1)(2\cos 2^2x-1)\ldots(2\cos 2^{n-1}x-1)}{(2\cos x+1)}$$

$$=\frac{(4\cos^2 2x-1)(2\cos 2^2x-1)\ldots(2\cos 2^{n-1}x-1)}{(2\cos x+1)}$$

$$=\frac{(2\cos 2^2x+1)(2\cos 2^2x-1)\ldots(2\cos 2^{n-1}x-1)}{(2\cos x+1)}$$

Similarly $f(x)=\frac{(2\cos^2 2^{n-1}x+1)(2\cos 2^{n-1}x-1)}{(2\cos x+1)}$

$$=\frac{(4\cos^2 2^{n-1}x-1)}{(2\cos x+1)}=\frac{(2\cos 2^n x+1)}{(2\cos x+1)}$$

$$f\left(\frac{2\pi k}{2^n \pm 1}\right)=\frac{2\cos\left(\frac{2^{n+1}\pi k}{2^n \pm 1}\right)+1}{2\cos\left(\frac{2\pi k}{2^n \pm 1}\right)+1}=\frac{2\cos\left(2\pi k \mp \frac{2\pi k}{2^n \pm 1}\right)+1}{2\cos\left(\frac{2\pi k}{2^n \pm 1}\right)+1}$$

$$=\frac{2\cos\left(\frac{2\pi k}{2^n \pm 1}\right)+1}{2\cos\left(\frac{2\pi k}{2^n \pm 1}\right)+1}$$

UNSOLVED EXERCISE

Section-A
(Straight objective type with single option correct)

Topic : 1 Classification of functions , Domain and Range

1. The domain of the function f (x) $= \frac{1}{\sqrt{x^2-3x+2}}$ is

(a) $(-\infty, 1)$ (b) $(-\infty, 1) \cup (2, \infty)$

(c) $(-\infty, 1] \cup [2, \infty)$ (d) $(2, \infty)$

2. If $\log_{1/2}\left(x^2-5x+7\right)>0$, then exhaustive range of values of x is

(a) $(-\infty, 2)\cup(3, \infty)$

(b) (2, 3)

(c) $(-\infty, 1)\cup(1, 2)\cup(2, \infty)$

(d) None of these

3. The range of the function $f(x)=\log_{\sqrt{2}}(2-\log_2(16\sin^2 x+1))$ is –

(a) $(-\infty, 1)$ (b) $(-\infty, 2)$

(c) $(-\infty, 1]$ (d) $(-\infty, 2]$

4. If f(x) is defined on domain [0, 1] then f(2 sinx) is defined on

(a) $\bigcup_{n\in I}\left\{\left[2n\pi, 2n\pi+\frac{\pi}{6}\right]\bigcup\left[2n\pi+\frac{5\pi}{6}, (2n+1)\pi\right]\right\}$

(b) $\bigcup_{n\in I}\left[2n\pi, 2n\pi+\frac{\pi}{6}\right]$

(c) $\bigcup_{n\in I}\left[2n\pi+\frac{5\pi}{6}, (2n+1)\pi\right]$

(d) None of these

5. If $3f(x)+5f\left(\frac{1}{x}\right)=\frac{1}{x}-3, \forall\ x\ (\neq 0)\in R$, then $f(x)$ =

(a) $\frac{1}{14}\left(\frac{3}{x}+5x-6\right)$ (b) $\frac{1}{14}\left(-\frac{3}{x}+5x-6\right)$

(c) $\frac{1}{14}\left(-\frac{3}{x}+5x+6\right)$ (d) None of these

6. The domain of the function $f(x) = \sqrt{\log_{10}\frac{3-x}{x}}$ is

(a) (0, 3/2) (b) (0, 3)

(c) $\left(-\infty, \frac{3}{2}\right]$ (d) $\left(0, \frac{3}{2}\right]$

7. The domain of the function $f(x) = \sqrt{x-\sqrt{1-x^2}}$ is

(a) $\left[-1, -\frac{1}{\sqrt{2}}\right]\cup\left[\frac{1}{\sqrt{2}}, 1\right]$

(b) [–1, 1]

(c) $\left(-\infty, -\frac{1}{2}\right]\cup\left[\frac{1}{\sqrt{2}}, +\infty\right)$

(d) $\left[\frac{1}{\sqrt{2}}, 1\right]$

8. The set of all values of x in $[0, \pi]$ for which the function $\sqrt{1-2\sin 3x}$ takes real values is

(a) $\left[0, \frac{\pi}{6}\right]\cup\left[\frac{5\pi}{6}, \pi\right]$

(b) $\left[0, \frac{\pi}{9}\right]\cup\left[\frac{2\pi}{9}, \frac{7\pi}{9}\right]\cup\left[\frac{8\pi}{9}, \pi\right]$

(c) $\left[0, \frac{\pi}{6}\right]\cup\left[\frac{5\pi}{18}, \frac{13\pi}{18}\right]\cup\left[\frac{17\pi}{18}, \pi\right]$

(d) None of these

9. The range of $f(x) = \cos\frac{\pi[x]}{2}$ is

(a) {0, 1} (b) {– 1, 1}

(c) {–1, 0, 1} (d) [–1, 1]

10. Let $f(x) = \frac{9.3^{2x}-6.3^x+4}{9.3^{2x}+6.3^x+4}$, then range of $f(x)$ is

(a) $\left[\frac{1}{3}, 3\right]$ (b) $\left[\frac{1}{3}, 1\right)$

(c) [0, 2] (d) None of these

11. Let $f(x) = \frac{\sin x}{\sqrt{1+\tan^2 x}} - \frac{\cos x}{\sqrt{1+\cot^2 x}}$ then range of $f(x)$ is

(a) $[-1, 0]$ (b) $[0, 1]$

(c) $[-1, 1]$ (d) None of these

12. If $3f(x) - f\left(\frac{1}{x}\right) = \log x^4$, then $f(e^{-x})$ is

(a) $1 + x$ (b) $1/x$

(c) x (d) $-x$

13. Let $f : [-10, 10] \to R$, where $f(x) = \sin x + [x^2/a]$ and [.] denotes the greatest integer function be an odd function. Then set of values of parameter 'a' is / are

(a) $(-10, 10) - \{0\}$ (b) $(0, 10)$

(c) $[100, \infty)$ (d) $(100, \infty)$

14. The value of the function $f(x) = 3 \sin\left(\sqrt{\frac{\pi^2}{16} - x^2}\right)$ lies in the interval

(a) $\left[-\frac{\pi}{4}, \frac{\pi}{4}\right]$ (b) $\left[0, \frac{3}{\sqrt{2}}\right]$

(c) $(-3, 3)$ (d) None of these

15. The range of the function $y = 3 \sin\sqrt{\frac{\pi^2}{16} - x^2}$ is

(a) $\left[0, \frac{3}{\sqrt{2}}\right]$ (b) $\left[-\frac{3}{\sqrt{2}}, \frac{3}{\sqrt{2}}\right]$

(c) $\left[-\frac{3}{\sqrt{2}}, 0\right]$ (d) None of these

16. The domain of $f(x) = \sqrt{\cos(\sin x)} + \sqrt{\log_x \{x\}}$; {.} denote the fractional part, is

(a) $[1, \pi)$ (b) $(0, 2\pi) - [1, \pi)$

(c) $\left(0, \frac{\pi}{2}\right) - \{1\}$ (d) $(0, 1)$

17. If $f(x) = \frac{2^x + 2^{-x}}{2}$, then $f(x + y).\ f(x - y)$ is equal to

(a) $\frac{1}{2}[f(x + y) + f(x - y)]$

(b) $\frac{1}{2}[f(2x) + f(2y)]$

(c) $\frac{1}{2}[f(x + y) .\ f(x - y)]$

(d) None of these

18. The least value of x for which $f(x) = \frac{\sqrt{x^2 - 4}}{\sin^{-1}(2 - x)}$ is defined is –

(a) $x = 1$ (b) $x = 2$

(c) $x = 3$ (d) Does not exist

19. If $f(x + y) = f(x) + 2y^2 + kxy$ and $f(1) = 2$, $f(2) = 8$, then $f(x)$ is of the form

(a) $2x^2$ (b) $2x^2 + 1$

(c) $2x^2 - 1$ (d) x^2

20. Given that, $f(x)$ is a polynomial in x satisfying the condition $f(x) + f\left(\frac{1}{x}\right) = f(x)\ x\, f\left(\frac{1}{x}\right)$ and $f(5) = 126$, $f(6) =$

(a) 216 (b) 218

(c) 217 (d) –215

21. The range of the function $f(x) = \frac{x^2}{x^4 + 1}$ is

(a) $(0, 1/2)$ (b) $[0, 1/2]$

(c) $[0, \infty)$ (d) $[0, 2]$

22. The range of the function $f(x) = \frac{\sin\left(\pi\left[x^2 + 1\right]\right)}{x^4 + 1}$ where , [ ] is greatest Integer function, is

(a) $[0,\ 1]$ (b) $[-1,\ 1]$

(c) $\{0\}$ (d) none of these

23. If f (x) is defined on (0, 1), then the domain of definition of f (e^x) + f(ln |x|) is

(a) $(-e, -1)$ (b) $(-e, -1) \cup (1, e)$

(c) $(-\infty, -1) \cup (1, \infty)$ (d) $(-e, e)$

24. The value of $\left[\frac{1}{2}\right] + \left[\frac{1}{2} + \frac{1}{100}\right] + \left[\frac{1}{2} + \frac{2}{100}\right] + \ldots + \left[\frac{1}{2} + \frac{99}{100}\right]$ is

(a) 49 (b) 50

(c) 51 (d) 98

25. The domain of definition of f (x) = $\sqrt{\frac{\log_{0.3}|x-2|}{|x|}}$ is

(a) $[1, 2)\cup(2, 3]$ (b) $[1, 3]$
(c) $\mathbb{R} - (1, 3]$ (d) none of these

26. The domain of definition of the function f

$$(x) = \frac{\sqrt{\sin^{-1}x+\sqrt{x^2+1}}+\sqrt{x-[x]+\log x}}{e^{\sqrt{\sin x+\cos x}}+\log\left(\sin\left(\frac{1}{\sqrt{-x^2}}\right)\right)} \text{ is}$$

(a) $(-1, 1)$ (b) $(0, 1)$
(c) $(1, 0)$ (d) none of these

Topic : 2 Composite Function, Even & Odd Function

27. If f(x) is a function that is odd and even simultaneously, then f(C) – f(B) is equal to
(a) 1 (b) – 1
(c) 0 (d) None of these

28. $f(x) = \frac{\cos x}{\left[\frac{x}{\pi}\right]+\frac{1}{2}}$, where x is not an integral multiple of π and [×] denote the greatest integer function, is
(a) An odd function (b) An even function
(c) Neither odd nor even (d) None of these

29. If $f(x) = \frac{1-x}{1+x}$; $x \neq 0$ then $f[f(x)] + f\left[f\left(\frac{1}{x}\right)\right]$
(a) < 2 (b) ≥ 2
(c) $= 2$ (d) None of these

30. If $f(x) = \cos[\pi^2]x + \cos[-\pi^2]x$, where $[x]$ = greatest integer $\leq x$, then
(a) $f(\pi/2) = -1$ (b) $f(\pi) = 1$
(c) $f(-\pi) = -1$ (d) $f(\pi/4) = 2$

31. The function $f(x)=\frac{x}{e^x-1}+\frac{x}{2}+1$ is –
(a) An odd function (b) An even function
(c) Neither an odd nor an even function
(d) A periodic function

32. The function $f(x) = \sec\left[\log(x+\sqrt{1+x^2})\right]$ is
(a) Even (b) Odd
(c) Constant (d) None of these

33. If $f(x) = (a - x^n)^{1/n}$, then $f[f(x)]$ equals to
(a) $x^{1/n}$ (b) x^n
(c) $a - x$ (d) x

34. If $f\left(2x+\frac{y}{8}, 2x-\frac{y}{8}\right)=xy$, then $f(m,n)+f(n,m)=0$
(a) only when m = n (b) only when $m \neq n$
(c) only when $m = -n$ (d) for all m and n

Topic : 3 Periodic Functions

35. If $f(x) = \sin\sqrt{[a]}x$, (where $[a]$ denotes the greatest integer function, has π as it's fundamental period, then
(a) $a = 1$ (b) $a \in [1, 2)$
(c) $a = 9$ (d) $a \in [4, 5)$

36. $f(x) = \sin^4\left(x+\frac{\pi}{6}\right) + \cos^4\left(x+\frac{\pi}{6}\right)$ is-
(a) Not a periodic function
(b) A periodic function with period 2π
(c) A periodic function with period π
(d) A period function with period $\pi/2$

37. If $f(x) . f(y) = f(x) + f(y) + f(xy) - 2$ $\forall x, y \in R$ and if $f(x)$ is not a constant function, then the value of $f(1)$ is
(a) 1 (b) 2
(c) 0 (d) – 1

38. The period of $\tan 3\theta$ is
(a) π (b) $\frac{3\pi}{4}$
(c) $\frac{\pi}{2}$ (d) None of these

39. Period of $f(x)\sin 2\sqrt{x+1}$ is
(a) 2π (b) π
(c) $\pi/2$ (d) None of these

40. If f is a function such that $f(0) = 2$, $f(1) = 3$ and $f(x + 2) = 2f(x) - f(x + 1)$ for every real x then $f(5)$ is
(a) 7 (b) 13
(c) 1 (d) 5

41. If $2f(x)+3f\left(\frac{1}{x}\right)=x^2-1$, then $f(x)$ is
(a) a periodic function (b) an even functioon
(c) an odd function (d) none of these

42. Let $f: R \to R$ be a periodic function such that

$$f(T+x)=1+\left[1-3f(x)+3(f(x))^2-(f(x))^3\right]^{1/3},$$

where T is a fixed positive number, then period of $f(x)$ is
(a) T (b) 2T
(c) 3T (d) none of these

43. The period of the function

$$f(x)=\cos\left(\frac{\pi x}{n!}\right)-\sin\left(\frac{\pi x}{(n+1)!}\right) \text{ is}$$

(a) $2(n+1)!$ (b) $2(n!)$

(c) $(n+1)$ (d) not periodic

44. The fucntion $f(x)=k|\cos x|+k^2|\sin x|+\phi(k)$ has period $\frac{\pi}{2}$ if k is equal to

(a) 1 (b) 2

(c) 3 (d)none of these

Topic : 4 Set Theory and Types of Relations

45. Suppose A_1, A_2,....A_{30} are thirty sets, each with five elements and B_1, B_2,....,B_n are n sets each with three elements. Let

$$\bigcup_{i=1}^{30} A_i = \bigcup_{j=1}^{n} B_j = S$$

If each elements of S belongs to exactly ten of the $A_i's$ and exactly nine of the $B_j's$ then n =

(a)45 (b)35

(c)40 (d) none of these

46. If $A=\{1,3,5,7,9,11,15,17\}$, $B=\{2,4,.....18\}$, and N is the universal set, then $A'\cup((A\cup B)\cap B')$ is

(a) A (b) N

(c) B (d) none of these

47. In a group of 65 pepole, 40 like cricket, 10 like both cricket and tennis. the number of persons liking tennis only and not cricket is

(a) 21 (b) 25

(c) 15 (d) none of these

48. In a group of 1000 pepole, there are 750 pepole who can speak Hindi and 400 who can speak English. then number of persons who can speak Hindi only is

(a) 300 (b) 400

(c) 600 (d) none of these

49. In a statistical investigation of 1,003 families of Kolkata, it was found that 63 families had neither a radio nor a TV, 794 families had a radio and 187 had a TV. The number of families in that group having both a radio and a T.V is

(a) 36 (b) 41

(c) 32 (d) none of these

50. Let R be a reation defined as a R b iff $|a-b|>0$. Then, the relation R is

(a) reflexive (b) symmetric

(c) transitive (d) none of these

51. Let R be a relation defined as a R b iff $1+ab>0$. Then, the relation R is

(a) reflexive (b) symmetric

(c) transitive (d) none of these

52. Let R be a relation defined as a R b iff $|a|\le b$. Then, the relation R is

(a) reflexive (b) symmetric

(c) transitive (d) none of these

53. N is the set of naturral numbers. The relation R is defined on N × N as follows $(a,b)R(c,d)\Leftrightarrow a+d=b+c$. then, R is

(a) reflexive only

(b) symmetric only

(c) transitive only

(d)equivalence relation

54. Out of 800 boys in a school, 224 played cricket, 240 played hockey and 336 played basketball. of the total, 64 played both basketball and hockey; 80 played cricket and basketball and 40 played cricket and hockey; 24 played all the three games. the number of boys who did not play any game is

(a) 160 (b) 240

(c) 216 (d) 128

55. Let $S=\{1,2,3,4,5\}$ and let $A=S\times S$. Define the relation R on A as follows (a,b) R(c,d) if and only if $ad=cb$. Then, R is

(a) reflexive only

(b) symmetric only

(c) transitive only

(d)equivalence relation

56. In a certain town 25% families own a phone and 15% own a car, 65% families own neither a phone nor a car. 2000 families own both a car and a phone. Consider the following statements in this regard:

1. 10% families own both a car and a phone.
2. 35% families own either a car or a phone.
3. 40,000 families live in the town.

Which of the above statemetns are correct?

(a) 1 and 2 (b) 1 and 3

(c) 2 and 3 (d) 1, 2 and 3

57. For real numbers x nad y, define a realtion R, xR y if and only if $x-y+\sqrt{2}$ is an irrational number. Then the relation R is

(a) reflexive (b) symmetric

(c) transitive

(d) an equivalence relagtion

58. If two sets A and B are having 99 elements in common then the number of elements common to each ofthe sets $A\times B$ and $B\times A$ are
(a) 2^{99} (b) 99^2
(c) 100 (d) 18

59. Let $A=\{2,3,4,5,.....,16,17,18\}$. Let $\approx$ be the equivalence relation on A × A defined by (a, b) $\approx$ (c,d) if ad = bc. Then the number of ordered pairs of the equivalence class of (3,2)is
(a) 5 (b) 6
(c) 7 (d) none of these

60. Let U be the universal set and $A\cup B\cup C=U$. Then $\{(A-B)\cup(B-C)\cup(C-A)\}'$ is equal to
(a) $A\cup B\cup C$ (b) $A\cup(B\cap C)$
(c) $A\cap B\cap C$ (d) $A\cap(B\cup C)$

61 The set $(A\cup B\cup C)\cap(A\cap B'\cap C')\cap C'$ is equal to
(a) $A\cap C'$ (b) $A\cap C$
(c) $B\cap C$ (d) none of these

Section-B
(Previous Year Question)

1. Let f be a real valued function defined on the interval (– 1, 1) such that $e^{-x} f(x) = 2 + \int_0^x \sqrt{t^4+1}\cdot dt, \forall x\in(-1,1)$ and let f^{-1} be the inverse function of f. Then $[f^{-1}(2)]'$ is equal to
(a) 1 (b) $\frac{1}{3}$
(c) $\frac{1}{2}$ (d) $\frac{1}{c}$

2. Domain of definition of the function $f(x) = \sqrt{\sin^{-1}(2x)+\frac{\pi}{6}}$ for real valued x, is
(a) $\left[-\frac{1}{4},\frac{1}{2}\right]$ (b) $\left[-\frac{1}{2},\frac{1}{2}\right]$
(c) $\left(-\frac{1}{2},\frac{1}{9}\right)$ (d) $\left[-\frac{1}{4},\frac{1}{4}\right]$

3. Range of the function $f(x) = \frac{x^2+x+2}{x^2+x+1}$; $x\in R$ is
(a) $(1, \infty)$ (b) $\left(1,\frac{11}{7}\right)$
(c) $\left(1,\frac{7}{3}\right]$ (d) $\left(1,\frac{7}{5}\right)$

4. Let $f(x) = (1 + b^2)x^2 + 2bx + 1$ and let m(b) be the minimum value of f (x). As b varies, the range of m(b) is
(a) [0, 1] (b) $\left[0,\frac{1}{2}\right]$
(c) $\left[\frac{1}{2},1\right]$ (d) (0, 1]

5. The domain of definition of $f(x) = \frac{\log_2(x+3)}{x^2+3x+2}$ is
(a) R/{– 1, – 2} (b) (– 2, ∞)
(c) R/{– 1, – 2, – 3} (d) (– 3, ∞)/{– 1, – 2}

6. The domain of definition of the function y(x) is given by the equation $2^x + 2^y = 2$, is
(a) $0 < x \le 1$ (b) $0 \le x \le 1$
(c) $-\infty < x \le 0$ (d) $-\infty < x < 1$

7. Let $f(\theta) = \sin\theta(\sin\theta + \sin 3\theta)$. Then, $f(\theta)$
(a) ≥ 0, only when $\theta \ge 0$
(b) ≤ 0, for all real θ
(c) ≥ 0, for all real θ
(d) ≤ 0, only when $\theta \le 0$

8. The domain of definition of the function $y = \frac{1}{\log_{10}(1-x)} + \sqrt{x+2}$ is
(a) (– 3, – 2) excluding – 2.5
(b) [0, 1] excluding 0.5
(c) (– 2, 1) excluding 0
(d) None of these

9. The function $f: R \to \left[-\frac{1}{2},\frac{1}{2}\right]$ defined as f (x) $= \frac{x}{1+x^2}$ is
(a) invertible
(b) injective but not surjective
(c) surjective but not injective
(d) neither injective nor surjective

10. Let a, b, c∈R. If $f(x) = ax^2 + bx + c$ be such that a + b + c = 3 and f (x + y) = f (x) + f(y) + xy, ∀ x, y∈R, then $\sum_{n=1}^{10} f(n)$ is equal to
(a) 330 (b) 165
(c) 190 (d) 255

11. If $f_k(x) = 1/k(\sin^k x + \cos^k x)$, where x∈R and $k \ge 1$, then $f_4(x) - f_6(x)$ is equal to
(a) 1/6 (b) 1/3
(c) 1/3 (d) 1/12

12. If $f(x) = x^2$ and g(x) = sin x for all x∈R. Then, the set of all x satisfying (*fogogof*)(*x*) =

(*gogof*)(*x*), where (*fog*)(*x*) = *f*(*g*(*x*)), is

(a) $\pm\sqrt{n\pi}, n \in \{0, 1, 2, ...\}$

(b) $\pm\sqrt{n\pi}, n \in \{1, 2, ...\}$

(c) $\pi/2 + 2n\pi$, $n \in \{....., -2, -1, 0, 1, 2, ...\}$

(d) $2n\pi$, $n \in \{...., -2, -1, 0, 1, 2, ...\}$

13. Let $f(x) = \frac{\alpha x}{x+1}$, $x \neq -1$. Then, for what value of α is $f[f(x)] = x$?

(a) $\sqrt{2}$ (b) $-\sqrt{2}$

(c) 1 (d) -1

14. Let $g(x) = 1 + x - [x]$ and $f(x) = \begin{cases} -1, & x < 0 \\ 0, & x = 0 \\ 1, & x > 0 \end{cases}$, then for all x, f [g (x)] is equal to

(a) x (b) 1

(c) f (x) (d) g(x)

15. If $g\{f(x)\} = |\sin x|$ and $f\{g(x)\} = (\sin\sqrt{x})^2$, then

(a) $f(x) = \sin^2 x$, $g(x) = \sqrt{x}$

(b) $f(x) = \sin x$, $g(x) = |x|$

(c) $f(x) = x^2$, $g(x) = \sin\sqrt{x}$

(d) *f* and *g* cannot be determined

16. If $f(x) = \cos(\log x)$, then $f(x)\cdot f(y) - \frac{1}{2}\left[f\left(\frac{x}{y}\right) + f(xy)\right]$ has the value

(a) -1 (b) $\frac{1}{2}$

(c) -2 (d) None of these

17. Let $f(x) = |x - 1|$. Then,

(a) $f(x^2) = \{f(x)\}^2$ (b) $f(x+y) = f(x) + f(y)$

(c) $f(|x|) = |f(x)|$ (d) None of these

18. The function $f: [0, 3] \to [1, 29]$, defined by $f(x) = 2x^3 - 15x^2 + 36x + 1$, is

(a) one-one and onto

(b) onto but not one-one

(c) one-one but not onto

(d) neither one-one nor onto

19. $f(x) = \begin{cases} x, & \text{if } x \text{ is rational} \\ 0, & \text{if } x \text{ is irrational} \end{cases}$,

$g(x) = \begin{cases} 0, & \text{if } x \text{ is rational} \\ x, & \text{if } x \text{ is irrational} \end{cases}$. Then, f – g is

(a) one-one and into

(b) neither one-one nor onto

(c) many one and onto

(d) one-one and onto

20. If $f: [0, \infty) \to [0, \infty)$ and $f(x) = \frac{x}{1+x}$, then *f* is

(a) one-one and onto

(b) one-one but not onto

(c) onto but not one-one

(d) neither one-one nor onto

21. Let function $f: R \to R$ be defined by $f(x) = 2x + \sin x$ for $x \in R$. Then, *f* is

(a) one-to-one and onto

(b) one-to-one but not onto

(c) onto but not one-to-one

(d) neither one-to-one nor onto

22. Let E = {1, 2, 3, 4} and F = {1, 2}. Then, the number of onto functions from E to F is

(a) 14 (b) 16

(c) 12 (d) 8

23. If X and Y are two non-empty sets where $f: X \to Y$, is function is defined such that $f(C) = \{f(x) : x \in C\}$ for $C \subseteq X$ and $f^{-1}(D) = \{x : f(x) \in D\}$ for $D \subseteq Y$, for any $A \subseteq Y$ and $B \subseteq Y$, then

(a) $f^{-1}\{f(A)\} = A$

(b) $f^{-1}\{f(A)\} = A$, only if $f(X) = Y$

(c) $f\{f^{-1}(B)\} = B$, only if $B \subseteq f(x)$

(d) $f\{f^{-1}(B)\} = B$

24. If $f(x) = \sin x + \cos x$, $g(x) = x^2 - 1$, then g {f (x)} is invertible in the domain

(a) $\left[0, \frac{\pi}{2}\right]$ (b) $\left[-\frac{\pi}{4}, \frac{\pi}{4}\right]$

(c) $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ (d) $[0, \pi]$

25. Suppose $f(x) = (x + 1)^2$ for $x \geq -1$. If g(x) is the function whose graph is reflection of the graph of *f* (x) with respect to the line y = x, then g (x) equals:

(a) $-\sqrt{x} - 1, x \geq 0$ (b) $\frac{1}{(x+1)^2}$, $x > -1$

(c) $\sqrt{x+1}, x \geq -1$ (d) $\sqrt{x} - 1, x \geq 0$

27. If $f: [1, \infty) \to [2, \infty)$ is given by $f(x) = x + \frac{1}{x}$, then $f^{-1}(x)$ equals

(a) $\frac{x + \sqrt{x^2 - 4}}{2}$ (b) $\frac{x}{1 + x^2}$

(c) $\frac{x - \sqrt{x^2 - 4}}{2}$ (d) $1 + \sqrt{x^2 - 4}$

28. If the function $f: [1, \infty) \to [1, \infty)$ is defined by $f(x) = 2^{x(x-1)}$, then $f^{-1}(x)$ is

(a) $\left(\frac{1}{2}\right)^{x(x-1)}$ (b) $\frac{1}{2}(1 + \sqrt{1 + 4\log_2 x})$

(c) $\frac{1}{2}(1 - \sqrt{1 + 4\log_2 x})$ (d) not defined

29. If $f(x) = 3x - 5$, then $f^{-1}(x)$

(a) is given by $\frac{1}{3x-5}$

(b) is given by $\frac{x+5}{3}$

(c) does not exist because f is not one-one

(d) does not exist because f is not onto

30. Which of the following functions is periodic?

(a) f (x) = x – [x], where [x] denotes the greatest integer less than or equal to the real number x

(b) f (x) = sin (1/x) for $x \neq 0$, f (0) = 0

(c) f (x) = x cos x

(d) None of the above

31. Let f : (0, 1) $\to$ R be defined by f (x) = $\frac{b-x}{1-bx}$, where b is a constant such that 0 < b < 1. Then,

(a) f is not invertible on (0, 1)

(b) $f \neq f^{-1}$ on (0, 1) and f' (b) = $\frac{1}{f'(0)}$

(c) $f = f^{-1}$ on (0, 1) and f'(b) = $\frac{1}{f'(0)}$

(d) f^{-1} is differentiable on (0, 1)

Section-C
(SCHOOL / BOARD PATTERN)

1. In a survey of 100 students, the number of students studying the various languages were found to be : English only 18, English But not Hindi 23, English but not Hindi 23, English and Sanskrit 8, English 26, Sanskrit 48, Sanskrit and Hindi 8, no language 24. find:
(i) How many students were styding hindi?
(ii) How many students were studying English and Hindi?

2. In a survey it was found that 21 persons liked product P_1 ,26 liked product p_2 and 29 liked product P_3. If 14 persons liked products P_1 and P_2; 12 persons liked product p_3 and P_1 ;14 persons liked product P_2 and P_3 and 8 liked all the three products. Find how many liked product P_3 only.

3. Let $A = \{1,2,3,4,5,6\}$. Let R be a relation on A defined by $R = \{(a,b) : a,b \in A, b$ is exactly divisible by a $\}$
(i) Write R in roster form (ii) find the domain of R (iii) Find the range of R .

4. Let $A = \{1,2,3\}$ and $R = \{(a,b) : |a^2 - b^2| \leq 5, a,b \in A\}$.
Then write R as set of ordered pairs.

5. Let $f : R^+ \to R$, where R^+ is the set of all positive real numbers, be such that $f(x) = \log_e x$. Determine
(i) the image set of the domain of f
(ii) $\{x : f(x) = -2\}$
(iii)whether $f(xy) = f(x) + f(y)$ hold

6 If $f(x) = 3x^4 - 5x^2 + 9$, find $f(x-1)$.

7. Let f be defined by $f(x) = x - 4$ and g be defined by $g(x) = \begin{cases} \frac{x^2-16}{x+4}, x \neq -4 \\ \lambda \quad , x = -4 \end{cases}$ Find λ such that $f(x) = g(x)$ for all x.

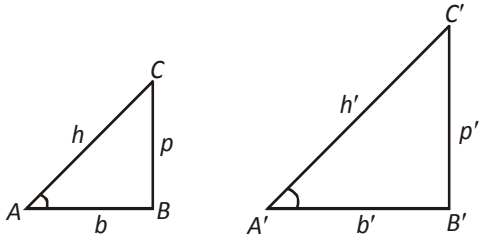
8. Let $f = \left\{\left(x, \frac{x^2}{1+x^2}\right) : x \in R\right\}$ be a function from R into R. Determine the range of f.

9. If $f(x) = \cos\left[\pi^2\right]x + \cos\left[-\pi^2\right]x$, where [x] denotes the greatest integer less than or equal to x, then write the vlaue of $f(\pi)$.

CHAPTER 2 TRIGONOMETRICAL FUNCTION OF ANGLES

1. TRIGONOMETRICAL FUNCTIONS OF ACUTE ANGLES

An angle whose measure is greater than 0° but less than 90° is called an acute angle. Consider a right angled triangle *ABC* with right angle at *B*. Side opposite to right angle is called the hypotenuse, side opposite to angle *A* is called perpendicular for angle *A* and side opposite to third angle is called base for angle *A*.



Any ratio of two of the side depends only on the measure of angle *A*, for if we take a larger or smaller right angle *A′ B′ C′* and *A* = *A′* then.

$\frac{h}{h'} = \frac{b}{b'} = \frac{p}{p'}$, (as these triangle are similar)

Thus any ratio of the lengths of two sides of the triangle is completely determined by angle *A* alone and is independent of the size of the triangle. There are six possible ratios that can be formed from the three sides of a right angled triangle. Each of them has been given a name as follows.

Definitions :

(i) $\sin A = \frac{p}{h}$ (ii) $\cos A = \frac{b}{h}$ (iii) $\tan A = \frac{p}{b}$ (iv) $\cot A = \frac{b}{p}$

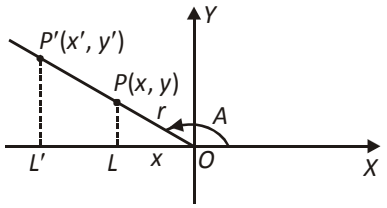
(v) $\sec A = \frac{h}{b}$ (vi) $\operatorname{cosec} A = \frac{h}{p}$

The abbreviations stand for sine, cosine, tangent, cotangent, secant and cosecant of A respectively. These functions of angle *A* are called **trigonometrical functions or trigonometrical ratios.**

Trigonometrical functions of any angle : Let A be a given angle with specified initial ray. We introduce a rectangular coordinate system in the plane with the vertex of angle *A* as the origin and the initial ray of angle *A* as the positive ray of the x-axis. We choose any point *P* on the terminal ray of angle *A*. Let the coordinates of *P* be (*x*, *y*) and its distance from the origin be *r*, then we define.

(i) $\sin A = \frac{y}{r}$ (ii) $\cos A = \frac{x}{r}$ (iii) $\tan A = \frac{y}{x}$

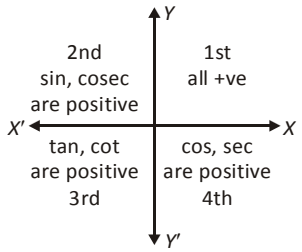
(iv) $\cot A = \frac{x}{y}$ (v) $\sec A = \frac{r}{x}$ (vi) $\operatorname{cosec} A = \frac{r}{y}$



These quantities are functions of the angle *A* alone. They do not depend on the choice of the point *P* and the terminal ray for if we choose a different point *P′* (*x′*, *y′*) on the terminal ray of *A* at a distance *r′* from the origin, it is clear that *x′* and *y′* will have the same sign as those of *x* and *y* respectively and because of similar ΔOPL and $\Delta OP'L'$, $\frac{y'}{r'}, \frac{x'}{r'}, \frac{y'}{x'}$ etc. will be equal to $\frac{y}{r'}, \frac{x}{r'}, \frac{y}{x}$ etc. respectively (*r* being always positive).

Also any trigonometrical function of an angle *A*° is equal to the same trigonometrical function of any angle 360*n* + *A*, where *n* is any integer since all these angles will have the same terminal ray. For example, sin 60° = sin 420° = sin (–300°). After the coordinates system has been introduced,

the plane is divided into four quadrants. An angle is said to be in that quadrant in which its terminal ray lies. For positive acute angles this definition gives the same result as in case of a right angled triangles since x and y are both positive for any point in the first quadrant and consequently are the length of base and perpendicular of the angle A.



(i) Clearly in Ist quadrant sin$A$, cos$A$, tan$A$, cot$A$, sec$A$ and cosec$A$ are all positive as x, y are positive.
(ii) In 2nd quadrant x is negative and y is positive therefore, only sin$A$, cosec$A$ are positive.
(iii) In 3rd quadrant x and y are both negative, therefore only tan$A$ and cot$A$ are positive.
(iv) In 4th quadrant x is positive and y is negative, therefore only cos$A$ and sec$A$ are positive.

Limits of the values of trigonometrical functions :

$\because \quad r = \sqrt{x^2 + y^2}$ $\quad\quad\quad\quad$ $\therefore \quad |x| \le r$ and $|y| \le r$

therefore,

(i) $|\sin A| = \left|\frac{y}{r}\right| = \frac{|y|}{|r|} = \frac{|y|}{r} \le 1$ $\quad\quad$ $\therefore$ $\quad \boxed{-1 \le \sin A \le 1}$

(ii) $|\cos A| = \left|\frac{x}{r}\right| = \frac{|x|}{|r|} = \frac{|x|}{r} \le 1$ $\quad\quad$ $\therefore$ $\quad \boxed{-1 \le \cos A \le 1}$

(iii) $|\sec A| = \left|\frac{r}{x}\right| = \frac{|r|}{|x|} = \frac{|r|}{|x|} \ge 1$ $\quad\quad$ $\therefore$ $\quad \boxed{\sec A \le -1 \text{ or } \sec A \ge 1}$

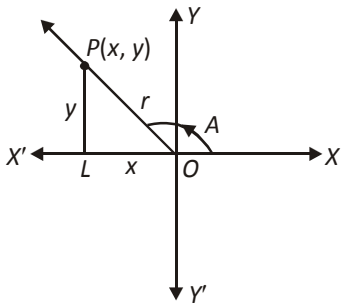
(iv) $|\text{cosec } A| = \left|\frac{r}{y}\right| = \frac{|r|}{|y|} = \frac{r}{|r|} \ge 1$ $\quad\quad$ $\therefore$ $\quad \boxed{\text{cosec } A \le -1 \text{ or cosec } A \ge 1}$

(v) $|\tan A| = \left|\frac{y}{x}\right| = \frac{|y|}{|x|}$ = any positive number $\quad$ $\therefore$ $\quad \boxed{-\infty < \tan A < \infty}$

(vi) $|\cot A| = \left|\frac{x}{y}\right| = \frac{|x|}{|y|}$ = any positive number $\quad$ $\therefore$ $\quad \boxed{-\infty < \cot A < \infty}$

Relation between the trigonometrical ratios of angle :

Let O be the vertex of an angle A, OX its initial ray and OP its terminal ray. Let P (x, y) be a point on the terminal ray.



Let PL be perpendicular to OX'. Let $OP = r$ then

$r^2 = x^2 + y^2$ .... (i)

Now (i) $\sin A = \frac{y}{r}$ (ii) $\cos A = \frac{x}{r}$ (iii) $\tan A = \frac{y}{x}$ (iv) $\cot A = \frac{x}{y}$

(v) $\sec A = \frac{r}{x}$ (vi) $\operatorname{cosec} A = \frac{r}{y}$

Important Results

1. $\sin^2 A + \cos^2 A = 1$
2. $\sin A = \pm\sqrt{1-\cos^2 A}; \cos A = \pm\sqrt{1-\sin^2 A}$
3. $\tan A = \pm\sqrt{\sec^2 A - 1}; \sec A = \pm\sqrt{1+\tan^2 A}$
4. $\cot A = \pm\sqrt{\operatorname{cosec}^2 A - 1}; \operatorname{cosec} A = \pm\sqrt{1+\cot^2 A}$

   Positive or negative sign before the root depends on the quadrant in which A lies
5. $\sin A . \operatorname{cosec} A = 1$; $\cos A . \sec A = 1$; $\tan A . \cot A = 1$

Type 1. Problems based on identity

Illustration 1: Prove that $\sqrt{\frac{1+\sin\theta}{1-\sin\theta}} = \sec\theta + \tan\theta$

Solution : L.H.S. $= \sqrt{\frac{1+\sin\theta}{1-\sin\theta}} = \sqrt{\frac{1+\sin\theta}{1-\sin\theta}.\frac{1+\sin\theta}{1+\sin\theta}}$

$$= \sqrt{\frac{(1+\sin\theta)^2}{1-\sin^2\theta}} = \sqrt{\frac{(1+\sin\theta)^2}{\cos^2\theta}} = \frac{1+\sin\theta}{\cos\theta}$$

$$= \frac{1}{\cos\theta} + \frac{\sin\theta}{\cos\theta} = \sec\theta + \tan\theta$$ **= R.H.S.**

Illustration 2: Show that $(1 + \cot A - \operatorname{cosec} A)(1 + \tan A + \sec A) = 2$

Solution : L.H.S. $= (1 + \cot A - \operatorname{cosec} A)(1 + \tan A + \sec A)$

$$= \left(1 + \frac{\cos A}{\sin A} - \frac{1}{\sin A}\right)\left(1 + \frac{\sin A}{\cos A} + \frac{1}{\cos A}\right)$$

$$= \left(\frac{\sin A + \cos A - 1}{\sin A}\right)\left(\frac{\cos A + \sin A + 1}{\cos A}\right)$$

$$= \frac{(\sin A + \cos A)^2 - 1^2}{\sin A \cos A}$$

$$= \frac{\sin^2 A + \cos^2 A + 2\sin A\cos A - 1}{\sin A\cos A}$$

$$= \frac{2\sin A\cos A}{\sin A\cos A} = 2 = \text{R.H.S.}$$

Illustration 3: Prove that : $\frac{1}{\sec A - \tan A} - \frac{1}{\cos A} = \frac{1}{\cos A} - \frac{1}{\sec A + \tan A}$

Solution : We have to prove that, $\frac{1}{\sec A - \tan A} - \frac{1}{\cos A} = \frac{1}{\cos A} - \frac{1}{\sec A + \tan A}$

or, $\frac{1}{\sec A - \tan A} + \frac{1}{\sec A + \tan A} = \frac{1}{\cos A} + \frac{1}{\cos A} = \frac{2}{\cos A}$ ... (1)

Now L.H.S. of (1) $= \dfrac{1}{\sec A - \tan A} + \dfrac{1}{\sec A + \tan A}$

$= \dfrac{\sec A + \tan A + \sec A - \tan A}{(\sec A - \tan A)(\sec A + \tan A)}$

$= \dfrac{2\sec A}{\sec^2 A - \tan^2 A} = \dfrac{2}{\cos A} =$ R.H.S. of (1)

Illustration 4 **:** Prove that : $3(\sin x - \cos x)^4 + 4(\sin^6 x + \cos^6 x) + 6(\sin x + \cos x)^2 = 13$

Solution : L.H.S. $= 3[(\sin x - \cos x)^2]^2 + 4[(\sin^2 x)^3 + (\cos^2 x)^3]$
$+ 6(\sin^2 x + \cos^2 x + 2\sin x \cos x)$
$= 3(\sin^2 x + \cos^2 x - 2\sin x \cos x)^2 + 4(\sin^2 x + \cos^2 x)(\sin^4 x + \cos^4 x - \sin^2 x \cos^2 x)$
$+ 6(\sin^2 x + \cos^2 x + 2\sin x \cos x)$
$= 3(1 - 2\sin x \cos x)^2 + 4(\sin^4 x + \cos^4 x) - \sin^2 x \cos^2 x$
$+ 6(1 + 2\sin x . \cos x)$
$= 3(1 + 4\sin^2 x \cos^2 x - 4\sin x \cos x) + 4[(\sin^2 x + \cos^2 x)^2 - 2\sin^2 x \cos^2 x - \sin^2 x \cos^2 x]$
$+ 6 + 12 \sin x \cos x$
$= 3 + 12\sin^2 x \cos^2 x - 12\sin x \cos x + 4(1 - 3\sin^2 x \cos^2 x) + 6 + 12\sin x \cos x$
$= 3 + 12\sin^2 x \cos^2 x + 4 - 12\sin^2 x \cos^2 x + 6 = 13$

Illustration 5**:** Prove that : $\sin^8 A - \cos^8 A = (\sin^2 A - \cos^2 A)(1 - 2\sin^2 A \cos^2 A)$

Solution : L.H.S. $= \sin^8 A - \cos^8 A = (\sin^4 A)^2 - (\cos^4 A)^2$
$= (\sin^4 A - \cos^4 A)(\sin^4 A + \cos^4 A)$
$= (\sin^2 A - \cos^2 A)(\sin^2 A + \cos^2 A)$
$[(\sin^2 A + \cos^2 A)^2 - 2\sin^2 A \cos^2 A]$
$= (\sin^2 A - \cos^2 A)(1 - 2\sin^2 A \cos^2 A)$ $[\because \sin^2 A + \cos^2 A = 1]$

Type-II. Problems based on conditional identity :

Illustration 6: If $3\sin\theta + 5\cos\theta = 5$, show that $5\sin\theta - 3\cos\theta = \pm 3$

Solution : Given,
$3\sin\theta + 5\cos\theta = 5$ ... (i)
Let $5\sin\theta - 3\cos\theta = x$ ... (ii)
Squaring and adding we get
$(9\sin^2\theta + 25\cos^2\theta + 30\sin\theta \cos\theta) + (25\sin^2\theta + 9\cos^2\theta - 30\sin\theta \cos\theta) = 25 + x^2$
or, $9(\sin^2\theta + \cos^2\theta) + 25(\sin^2\theta + \cos^2\theta) = 25 + x^2$
or, $34 = 25 + x^2$ or, $x^2 = 9$ $\therefore$ $x = \pm 3$

Note :-
If $x = -3$
From (i), $3\sin\theta + 5\cos\theta = 5$ ... (iii)
and from (ii), $5\sin\theta - 3\cos\theta = -3$ ... (iv)
Multiplying eq. (iii) by 3 and eq. (iv) by 5 and adding, we get
$34\sin\theta = 0$ $\because$ $\sin\theta = 0$ and $\cos\theta = 1$

Illustration 7: If $(\sec A + \tan A)(\sec B + \tan B)(\sec C + \tan C)$
$= (\sec A - \tan A)(\sec B - \tan B)(\sec C - \tan C)$
Prove that the value of each side is ± 1

Solution : Let $(\sec A + \tan A)(\sec B + \tan B)(\sec C + \tan C) = x$ ... (i)
then $(\sec A - \tan A)(\sec B - \tan B)(\sec C - \tan C) = x$ ... (ii)
Multiplying (i) and (ii) we get
$(\sec^2 A - \tan^2 A)(\sec^2 B - \tan^2 B)(\sec^2 C - \tan^2 C) = x^2$
or, $x^2 = 1$ $\because$ $x = \pm 1$ Hence each side $= \pm 1$

Illustration 8: If $\frac{\cos^4 A}{\cos^2 B}+\frac{\sin^4 A}{\sin^2 B}=1$, then prove that

(i) $\sin^4A + \sin^4B = 2\sin^2A \sin^2B$ (ii) $\frac{\cos^4 B}{\cos^2 A}+\frac{\sin^4 B}{\sin^2 A}=1$

Solution : Given, $\frac{\cos^4 A}{\cos^2 B}+\frac{\sin^4 A}{\sin^2 B}=1=\cos^2 A+\sin^2 A$

or, $\frac{\cos^4 A}{\cos^2 B}-\cos^2 A=\sin^2 A-\frac{\sin^4 A}{\sin^2 B}$

or, $\frac{\cos^2 A(\cos^2 A-\cos^2 B)}{\cos^2 B}=\sin^2 A\frac{(\sin^2 B-\sin^2 A)}{\sin^2 B}$

or, $\frac{\cos^2 A}{\cos^2 B}(\cos^2 A-\cos^2 B)=\frac{\sin^2 A}{\sin^2 B}[(1-\cos^2B)-(1-\cos^2A)]$

or, $\frac{\cos^2 A}{\cos^2 B}(\cos^2 A-\cos^2 B)=\frac{\sin^2 A}{\sin^2 B}(\cos^2 A-\cos^2 B)$

or, $(\cos^2 A-\cos^2 B)\left(\frac{\cos^2 A}{\cos^2 B}-\frac{\sin^2 A}{\sin^2 B}\right)=0$

When $\cos^2A - \cos^2B = 0$, $\cos^2A = \cos^2B$ ... (i)

when $\frac{\cos^2 A}{\cos^2 B}-\frac{\sin^2 A}{\sin^2 B}$ = 0, $\cos^2A \sin^2B = \sin^2A \cos^2B$

or, $\cos^2A(1 - \cos^2B) = (1 - \cos^2A) \cos^2B$

or, $\cos^2A - \cos^2A \cos^2B = \cos^2B - \cos^2A \cos^2B$

or, $\cos^2A = \cos^2B$ ... (ii)

Thus in both the cases, $\cos^2 A=\cos^2 B$

$\therefore 1-\sin^2 A=1-\sin^2 B$ or, $\sin^2 A=\sin^2 B$ ... (iii)

(i) L.H.S. $= \sin^4A + \sin^4B = (\sin^2A - \sin^2B)^2 + 2\sin^2A \sin^2B$

$= 2\sin^2A \sin^2B$ = R.H.S. [$\because \sin^2A = \sin^2B$]

(ii) L.H.S. $=\frac{\cos^4 B}{\cos^2 A}+\frac{\sin^4 B}{\sin^2 A}=\frac{\cos^4 B}{\cos^2 B}+\frac{\sin^4 B}{\sin^2 B}$ [From (iii)]

$= \cos^2B + \sin^2B = 1$

Type-III. Problems based on finding the value of trigonometrical ratios when one trigonometrical ratio is known :

Illustration 9: If $\tan^2\theta = 1 - e^2$, prove that $\sec\theta + \tan^3\theta \operatorname{cosec}\theta = (2-e^2)^{\frac{3}{2}}$

Solution : Given, $\tan^2\theta = 1 - e^2$

Now, L.H.S. $= \sec\theta + \tan^3\theta \operatorname{cosec}\theta$

$= \sec\theta\left(1+\tan^3\theta\frac{\operatorname{cosec}\theta}{\sec\theta}\right)$

$= \sec\theta (1 + \tan^3\theta. \cot\theta) = \sec\theta (1 + \tan^2\theta)$

$= \sec\theta \sec^2\theta$

$= \sec^3\theta=(\sec^2\theta)^{\frac{3}{2}}=(1+\tan^2\theta)^{\frac{3}{2}}$

$= (1+1-e^2)^{\frac{3}{2}}=(2-e^2)^{\frac{3}{2}}$

Illustration 10: If $\tan\theta + \sec\theta = 1.5$, find $\sin\theta$, $\tan\theta$ and $\sec\theta$.

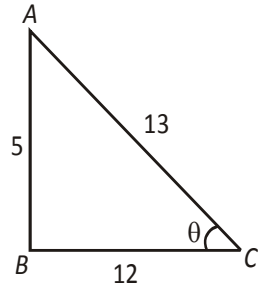
Solution : Given, $\sec\theta + \tan\theta = \frac{3}{2}$ ... (i)

Now, $\sec\theta - \tan\theta = \frac{1}{\sec\theta + \tan\theta} = \frac{2}{3}$ ... (ii)

adding (i) and (ii), we get

$2\sec\theta = \frac{3}{2} + \frac{2}{3} = \frac{13}{6}$

$\therefore \sec\theta = \frac{13}{12} \Rightarrow \tan\theta = \frac{5}{12}$ and $\sin\theta = \frac{5}{13}$



Type-IV. Problems based on elimination :

Illustration 11: Eliminate θ between the equations $a\sec\theta + b\tan\theta + c = 0$ and $p\sec\theta + q\tan\theta + r = 0$.

Solution : Given $a\sec\theta + b\tan\theta + c = 0$ ... (1)

and $p\sec\theta + q\tan\theta + r = 0$ ... (2)

Solving (1) and (2) by cross multiplication method, we have

$$\frac{\sec\theta}{\underset{(i)}{br - qc}} = \frac{\tan\theta}{\underset{(ii)}{pc - ar}} = \frac{1}{\underset{(iii)}{aq - pb}}$$

From (i) and (iii), we get $\sec\theta = \frac{br - qc}{aq - pb}$ ... (3)

From (ii) and (iii), we get $\tan\theta = \frac{pc - ar}{pc - pb}$ ... (4)

$\because \sec^2\theta - \tan^2\theta = 1$

$\therefore \left(\frac{br - qc}{aq - pb}\right)^2 - \left(\frac{pc - ar}{aq - pb}\right)^2 = 1$ or, $(br - qc)^2 - (pc - ar)^2 = (aq - pb)^2$

Illustration 12: If $\text{cosec}\theta - \sin\theta = m$ and $\sec\theta - \cos\theta = n$, eliminate θ.

Solution : Given, $\text{cosec}\theta - \sin\theta = m$ or, $\frac{1}{\sin\theta} - \sin\theta = m$

or, $\frac{1 - \sin^2\theta}{\sin\theta} = m$ or, $\frac{\cos^2\theta}{\sin\theta} = m$ ... (i)

Again $\sec\theta - \cos\theta = n$ or, $\frac{1}{\cos\theta} - \cos\theta = n$

or, $\frac{1 - \cos^2\theta}{\cos\theta} = n$ or, $\frac{\sin^2\theta}{\cos\theta} = n$ ... (ii)

From (i), $\sin\theta = \frac{\cos^2\theta}{m}$ ... (iii)

Putting in (ii), we get $\frac{\cos^4\theta}{m^2\cos\theta} = n$ or, $\cos^3\theta = m^2 n$

$\therefore \cos\theta = (m^2 n)^{\frac{1}{3}}$ or, $\cos^2\theta = (m^2 n)^{\frac{2}{3}}$ ... (iv)

From (iii), $\sin\theta = \frac{\cos^2\theta}{m} = \frac{(m^2 n)^{\frac{2}{3}}}{m} = \frac{m^{\frac{4}{3}} n^{\frac{2}{3}}}{m} = m^{\frac{1}{3}} n^{\frac{2}{3}} = (mn^2)^{\frac{1}{3}}$

$\therefore \quad \sin^2\theta = (mn^2)^{\frac{2}{3}}$ ... (v)

adding (iv) and (v), we get

$(m^2n)^{\frac{2}{3}} + (mn^2)^{\frac{2}{3}} = \cos^2\theta + \sin^2\theta$ or, $(m^2n)^{\frac{2}{3}} + (mn^2)^{\frac{2}{3}} = 1.$

Type V. Problems based on limits between which sinθ, cosθ, secθ, cosecθ, lies or does not lie.

Working Rule : Use the following whichever is applicable.

(i) $-1 \le \sin\theta \le 1$ or, $|\sin\theta| \le 1$ or, $\sin^2\theta \le 1$

(ii) $-1 \le \cos\theta \le 1$ or, $\cos^2\theta \le 1$

(iii) $\sec\theta \le -1$ or $\sec\theta \ge 1$ i.e. $\sec^2\theta \ge 1$

(iv) $\operatorname{cosec}\theta \le -1$ or $\operatorname{cosec}\theta \ge 1$ i.e. $\operatorname{cosec}^2\theta \ge 1$

Illustration 13: Is the equation $\sec^2\theta = \frac{4xy}{(x+y)^2}$ possible for real values of *x* and *y* ?

Solution : Given, $\sec^2\theta = \frac{4xy}{(x+y)^2}$

$\because \quad \sec^2\theta \ge 1 \qquad \therefore \quad \frac{4xy}{(x+y)^2} \ge 1$

or, $(x + y)^2 \le 4xy$ or, $(x + y)^2 - 4xy \le 0$ or, $(x - y)^2 \le 0$

But for real values of *x* and *y*, $(x - y)^2 \not< 0$

$\therefore \quad (x - y)^2 = 0 \qquad \therefore \quad x = y$ Also $x + y \ne 0 \Rightarrow x \ne 0.\ y \ne 0$

$\therefore$ given equation $\sec^2\theta = \frac{4xy}{(x+y)^2}$ is possible for real values of *x* and *y* only when $x = y$ and $x \ne 0$.

Illustration 14: Show that the equation $\sin\theta = x + \frac{1}{x}$ is not possible for any value of θ.

Solution : Given, $\sin\theta = x + \frac{1}{x}$

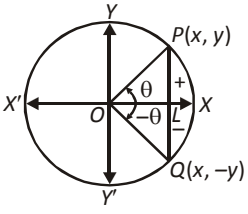
$\therefore \sin^2\theta = x^2 + \frac{1}{x^2} + 2x.\frac{1}{x} = x^2 + \frac{1}{x^2} + 2 \ge 2$

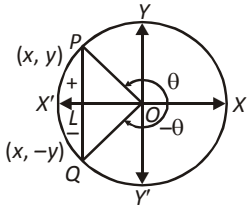
which is not possible since $\sin^2\theta \le 1$

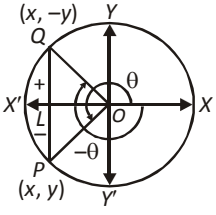
2. TRIGONOMETRICAL FUNCTIONS OF ANGLES OF ARBITRARY MAGNITUDE

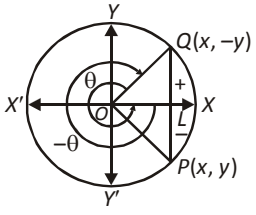
Trigonometrical ratios (functions) of the angles (– θ) in term of those of θ, for all values of θ :

Let the revolving line *OP* starting from the position *OX* rotate in the anticlock wise i.e. positive direction and describe the $\angle XOP = \theta$.









Let another revolving line OQ (= OP) starting from the position OX rotate in the clockwise i.e. negative direction and describe the $\angle XOQ = -\theta$.

Draw $PL \perp OX$. Join QL, then QL will be perpendicular to OX

$\therefore$ In right angled ΔPOL and ΔQOL

$\angle POL = \angle QOL$ and $OP = OQ$

$\therefore$ $\Delta POL = \Delta QOL$

$\therefore$ $PL = QL$ (numerically) $= -QL$ (algebraically)

Let $OP = OQ = r$

Now from ΔQOL,

$$\sin(-\theta) = \frac{QL}{OQ} = \frac{-y}{r} = -\frac{y}{r} = -\frac{PL}{OP} = -\sin\theta$$

$$\cos(-\theta) = \frac{OL}{OQ} = \frac{x}{r} = \frac{OL}{OP} = \cos\theta$$

$$\tan(-\theta) = \frac{QL}{OL} = \frac{-y}{x} = -\frac{y}{x} = -\frac{PL}{OL} = -\tan\theta$$

$$\cot(-\theta) = \frac{OL}{QL} = \frac{x}{-y} = -\frac{x}{y} = -\frac{OL}{PL} = -\cot\theta$$

$$\sec(-\theta) = \frac{OQ}{OL} = \frac{r}{x} = \frac{OP}{OL} = \sec\theta$$

$$\text{cosec}\,(-\theta) = \frac{OQ}{QL} = \frac{r}{-y} = \frac{r}{y} = -\frac{OP}{PL} = -\text{cosec}\theta$$

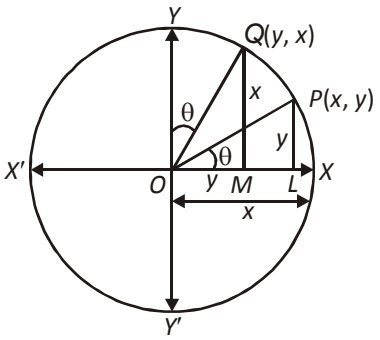
Trigonometrical ratios of the angle $90° \pm \theta$ in terms of θ, for all value of θ :

when θ is in the Ist quadrant.

Let the revolving line OP starting from the position OX and rotating in the positive direction describe the $\angle XOP = \theta$.

Let another revolving line OQ (= OP) starting from the same position OX first coincide with OY so as to describe $\angle XOY = 90°$ and then rotate in the negative direction through an angle θ, then OQ in its final position describes $\angle XOQ = 90° - \theta$.

Draw PL and QM perpendicular to OX.



In right angled ΔQOM

$OP = OQ$

$\therefore$ $\angle POL = \angle OQM = \theta$ [$\because$ $\angle OQM = \angle YOQ$]

$\Delta POL = \Delta QOM$

$\therefore$ $PL = OM$ (algebraically) and $OL = QM$ (algebraically)

Let the co-ordinates of P be (x, y)

then Co-ordinates of Q will be (y, x).

Let $OP = OQ = r$

Now from ΔQOM

$$\sin(90° - \theta) = \frac{QM}{OQ} = \frac{x}{r} = \frac{OL}{OP} = \cos\theta \text{ [from } \Delta POL\text{]}$$

$$\cos(90° - \theta) = \frac{OM}{OQ} = \frac{y}{r} = \frac{PL}{OP} = \sin\theta \text{ [from } \Delta POL\text{]}$$

$$\tan(90° - \theta) = \frac{QM}{OM} = \frac{x}{y} = \frac{OL}{PL} = \cot\theta$$

$$\cot(90° - \theta) = \frac{OM}{QM} = \frac{y}{x} = \frac{PL}{OL} = \tan\theta$$

$$\sec(90^\circ-\theta)=\frac{OQ}{OM}=\frac{r}{y}=\frac{OP}{PL}=\operatorname{cosec}\theta$$

$$\operatorname{cosec}(90^\circ-\theta)=\frac{OQ}{QM}=\frac{r}{x}=\frac{OP}{OL}=\sec\theta$$

Similarly these results can be established when θ lies in the 2nd. 3rd or 4th quadrant.

Trigonometrical ratios of the angle $180^\circ \pm \theta$ in terms of those of θ, for all values of θ.

Let the revolving line *OP* starting from the position *OX* rotate in the positive direction and describe $\angle XOP = \theta$. Let another revolving line *OQ* (= *OP*) starting from the position *OX* rotate in the positive direction through an angle 180° to coincide with *OX*′ and then rotate in the negative direction through an angle θ and in its final position *OQ* describes $\angle XOQ = 180^\circ - \theta$.

In right angled ΔPOL and ΔQOM,

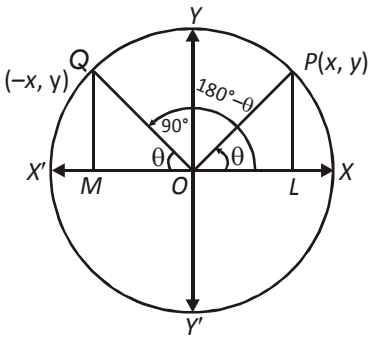
$\angle QOM = \angle POL = \theta$

$OP = OQ$

$\therefore$ $\Delta QOM = \Delta POL$

$\therefore$ $OL = OM$ (numerically) $= -OM$ (algebarically) &

$PL = QM$ (numerically) $= OM$ (algebarically)



Let the co-ordinates of *P* be (*x*, *y*) then the co-ordinates of *Q* will be (–*x*, *y*)

Let $OP = OQ = r$

Now from ΔQOM

$$\sin(180^\circ-\theta)=\frac{QM}{OQ}=\frac{y}{r}=\frac{PL}{OP}=\sin\theta$$

$$\cos(180^\circ-\theta)=\frac{OM}{OQ}=\frac{-x}{r}=-\frac{x}{r}=-\frac{OL}{OP}=-\cos\theta$$

$$\tan(180^\circ-\theta)=\frac{QM}{OM}=\frac{y}{-x}=-\frac{y}{x}=-\frac{PL}{OL}=-\tan\theta$$

$$\cot(180^\circ-\theta)=\frac{OM}{QM}=\frac{-x}{y}=-\frac{x}{y}=-\frac{OL}{PL}=-\cot\theta$$

$$\sec(180^\circ-\theta)=\frac{OQ}{OM}=\frac{r}{-x}=-\frac{r}{x}=-\frac{OP}{OL}=-\sec\theta$$

$$\operatorname{cosec}(180^\circ-\theta)=\frac{OQ}{QM}=\frac{r}{y}=\frac{OP}{PL}=\operatorname{cosec}\theta$$

Step I : Trigonometric function remains unchanged i.e., there is no change in trigonometric function.

sin remains sin, cos remains cos, tan remains tan

Angle associated remains θ.

Step II : Sign is affixed according to the quadrant in which the angle lies.

i.e. for sign, check the quadrant.

B. When trigonometric function is operated on an angle of the form $n\frac{\pi}{2}\pm\theta$, where *n* is an odd integer then

Step I : $\sin \rightleftharpoons \cos$, $\tan \rightleftharpoons \cot$, $\sec \rightleftharpoons \operatorname{cosec}$,

i.e. sine changes to cosine and vice versa etc.

angle associated remains 'θ'

Step II : sign is affixed according to quadrant in which the angle lies.

Note :- sign in always considered on the basis of the operating function.

Illustration 15: Suppose we have to find the values of sin $(3\pi + \theta)$.

Here the angle is of the form $n\pi + \theta$.

Hence the value will be $(+ \sin\theta)$ or $(- \sin\theta)$ according to step I of rule A. Now whether it is $(+ \sin\theta)$ or $(-\sin\theta)$ will be decided by step II of rule A. The quadrant in which $(3\pi + \theta)$ lies is 3rd quadrant and sine funciton is negative in 3rd quadrant. Hence we will select $(- \sin\theta)$.

i.e. $\sin(3\pi + \theta) = -\sin\theta$

Illustration 16: $\sin\left(\frac{3\pi}{2}+\theta\right) = -\cos\theta,$ because the angle is, an odd multiple of $\frac{\pi}{2}+\theta$. Quadrant is 4th and sign of sine in 4th quadrant is negative.

Some values of trigonometrical ratios :

Students are already familiar with the values of sin, cos, tan, cot, sec and cosec of angles 0°, 30°, 45°, 60° and 90° which have been given in the following table.

	0°	30°	45°	60°	90°
sin	0	$\frac{1}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{\sqrt{3}}{2}$	1
cos	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$	0
tan	0	$\frac{1}{\sqrt{3}}$	1	$\sqrt{3}$	undefined
cot	undefined	$\sqrt{3}$	1	$\frac{1}{\sqrt{3}}$	0
sec	1	$\frac{2}{\sqrt{3}}$	$\sqrt{2}$	2	undefined
cosec	undefiend	2	$\sqrt{2}$	$\frac{2}{\sqrt{3}}$	1

Type I. Problems based on values of sin, cos, tan, cot, sec, cosec of angles of different magnitudes :

Working Rule : Use the values from the table.

Illustration 17: Find the value of $3\tan^2 45° - \sin^2 60° - \frac{1}{2}\cot^2 30° + \frac{1}{8}\sec^2 45°$

Solution : $3\tan^2 45° - \sin^2 60° - \frac{1}{2}\cot^2 30° + \frac{1}{8}\sec^2 45°$

$$= 3.1^2 - \left(\frac{\sqrt{3}}{2}\right)^2 - \frac{1}{2}.(\sqrt{3}) + \frac{1}{8}.(\sqrt{2})^2 = 3 - \frac{3}{4} - \frac{3}{2} + \frac{2}{8} = 3 - \frac{3}{4} - \frac{3}{2} + \frac{1}{4}$$

$$= \frac{12-3-6+1}{4} = \frac{4}{4} = 1$$

Illustration 18: Simplify $\frac{\sin 300°\cdot\tan 330°\cdot\sec 420°}{\tan 135°\cdot\sin 210°\cdot\sec 315°}$

Solution : Given expression $= \frac{\sin(2\pi - 60°)\cdot\tan(2\pi - 30°)\cdot\sec(2\pi + 60°)}{\tan(\pi - 45°)\cdot\sin(\pi + 30°)\cdot\sec(2\pi - 45°)}$

$$= \frac{(-\sin 60°)\cdot(-\tan 30°)\cdot\sec 60°}{(-\tan 45°)\cdot(-\sin 30°)\cdot\sec 45°} = \frac{\sin 60°\cdot\tan 30°\cdot\sec 60°}{\tan 45°\cdot\sin 30°\cdot\sec 45°} = \frac{\frac{\sqrt{3}}{2}\cdot\frac{1}{\sqrt{3}}\cdot 2}{1\cdot\frac{1}{2}\cdot\sqrt{2}} = \frac{1}{1/\sqrt{2}} = \sqrt{2}$$

Type II. Problems based on trigonometrical ratios of the angle 90° – θ, 90° + θ, 180° – θ, 180° + θ in terms of θ.

Illustration 19: Show that tan1° tan2° ..... tan89° = 1

Solution : L.H.S. = (tan1° tan89°) (tan2° · tan88°) .....
= [tan1° tan(90° – 1°)]·[tan2° tan(90° – 2°)] .... [tan44° tan(90° – 44°)] tan 45°
= (tan1°·cot1°) (tan2°·cot2°) .... (tan44°·cot44°) tan45°
= 1 [$\because$ tanθ cotθ = 1 and tan 45° = 1]

***Illustration 20*:** Show that : $\sin^2 5° + \sin^2 10° + \sin^2 15° + ... + \sin^2 90° = 9\frac{1}{2}$

Solution : L.H.S. = $(\sin^2 5° + \sin^2 85°) + (\sin^2 10° + \sin^2 80°) + ... + (\sin^2 40° + \sin^2 50°) + \sin^2 45° + \sin^2 90°$
$= (\sin^2 5° + \cos^2 5°) + (\sin^2 10 + \cos^2 10°) + ... + (\sin^2 40° + \cos^2 40°) + \sin^2 45° + \sin^2 90°$

$$= (1 + 1 + 1 + 1 + 1 + 1 + 1 + 1) + \left(\frac{1}{\sqrt{2}}\right)^2 + 1 = 9\frac{1}{2}$$

Illustration 21: $\cos^2\frac{\pi}{16} + \cos^2\frac{3\pi}{16} + \cos^2\frac{5\pi}{16} + \cos^2\frac{7\pi}{16}$

Solution : L.H.S. = $\cos^2\frac{\pi}{16} + \cos^2\frac{3\pi}{16} + \cos^2\left(\frac{\pi}{2} - \frac{3\pi}{16}\right) + \cos^2\left(\frac{\pi}{2} - \frac{\pi}{16}\right)$

$$= \cos^2\frac{\pi}{16} + \cos^2\frac{3\pi}{16} + \sin^2\frac{3\pi}{16} + \sin^2\frac{\pi}{16}$$

$$= \left(\cos^2\frac{\pi}{16} + \sin^2\frac{\pi}{16}\right) + \left(\cos^2\frac{3\pi}{16} + \sin^2\frac{3\pi}{16}\right)$$

= 1 + 1 = 2

Student Task

Q.1 If [ ] represents greater integer function then the sum of all possible values of k = [sin x] + [cos y] is

Q.2 If $\tan\theta = \frac{\theta}{b}$ and $0 < \theta < \pi$, then the value of $\frac{\sqrt{a^2 + b^2}}{|a|}\sin\theta$ is .....

(Assume $b \neq 0$, $a \neq 0$, $a, b \in R$)

Q.3 The equation $\sin^4 x - 2\cos^2 x + k^2 = 0$ is solvable for $k \in R$, the greater value of $\sqrt{2}k$ is

Q.4 Let α be the common positive root of the quadratics.
$x^2 + k_1 x + 11 = 0$, $x^2 - k_2 x - 13 = 0$ and $x^2 - (k_2 - k_1)x + 7 = 0$. Also θ satisfies cosθ + cos2θ + cos 3θ = α, then find the value of sinθ + sin2θ + sin3θ

Q.5 If $A + B = \frac{\pi}{2}$, where A, B > 0 , then maximum vlaue of sin A + Sin B is equal to

Q.6 If $f(x) = (\sin x + \operatorname{cosec} x)^2 + (\cos x + \sec x)^2$, then minimum value of f(x) is

3. PERIODIC FUNCTION

A function $f(x)$ is said to be periodic function of x, if there exists a positive real number T such that $f(x + T) = f(x)$

The smallest value of T is called the period of the function.

If positive value of T is independent of x then $f(x)$ is periodic function and if the value of T depends upon x, then $f(x)$ is non-periodic.

Definition (Graphically) :

A function is said to be periodic if its graph repeat itself after a fixed interval and the width of that interval is called its period.

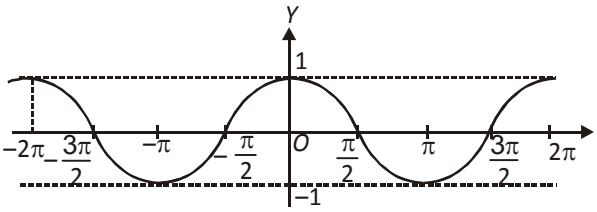
***Illustration 22*:** If cos x is periodic then find its period.

Solution : Let $f(x) = \cos x$. If T is the period, then

$\Rightarrow \cos(x + T) = \cos x \quad \Rightarrow x + T = 2n\pi \pm x$

Put, $x = 0$, then $T = 2n\pi \quad \therefore T = 2\pi, 4\pi, 6\pi, \ldots \quad (\because T > 0)$

Graphically : $\because$ Graph of $f(x) = \cos x$ repeats at an interval 2π.



Thus, $f(x) = \cos x$ is periodic with period 2π.

Note :- Periods of sin x, cos x and tan x are 2π, 2π and π respectively.

Illustration 23: If $\sqrt{(x-[x])}$ is periodic then find its period, where [.] denotes the greatest integer function.

Solution : Let $f(x) = \sqrt{(x-[x])}$

If T is period then $f(x + T) = f(x)$

$\Rightarrow \sqrt{(x+T)-[x+T]} = \sqrt{x-[x]}$

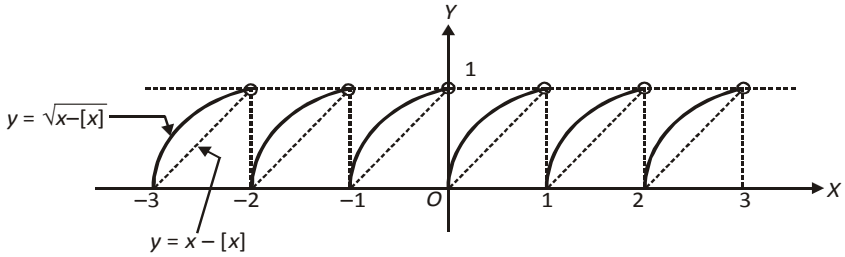
On squaring, we get $x + T - [x + T] = x - [x]$

or $[x + T] - [x] = T \ \forall\, x \in R \quad \Rightarrow T = 1, 2, 3, 4, \ldots$

The smallest value of T is 1

Hence period of $f(x)$ is 1.

Graphically : $\because$ Graph of $f(x) = \sqrt{x-[x]}$ repeats at an interval 1.



Thus, $f(x) = \sqrt{x-[x]}$ is periodic with period 1.

Illustration 24: Let f be a real valued function defined for all real numbers x such that for some fixed $a > 0$.

$$f(x+a) = \frac{1}{2} + \sqrt{f(x) - \{f(x)\}^2} \text{ for all } x$$

Show that the function $f(x)$ is periodic with period $2a$.

Solution : Given $f(x+a)=\frac{1}{2}+\sqrt{f(x)+\{f(x)\}^2}$ ... (1)

Replacing x by $x - a$ and $x + a$ in (1)

then $f(x)=\frac{1}{2}+\sqrt{f(x-a)-\{f(x-a)\}^2}>\frac{1}{2}$

and $f(x+2a)=\frac{1}{2}+\sqrt{f(x+a)-\{f(x+a)\}^2}$ ... (2)

$$=\frac{1}{2}+\sqrt{f(x+a)\{1-f(x+a)\}}$$

$$=\frac{1}{2}+\sqrt{\left(\frac{1}{2}+\sqrt{f(x)-\{f(x)\}^2}\right)\left(\frac{1}{2}-\sqrt{f(x)-\{f(x)\}^2}\right)}$$

$$=\frac{1}{2}+\sqrt{\frac{1}{4}-f(x)+\{f(x)\}^2}$$

$$=\frac{1}{2}+\sqrt{\left(f(x)-\frac{1}{2}\right)^2}$$

$$=\frac{1}{2}+\left|f(x)-\frac{1}{2}\right|$$

$$=\frac{1}{2}+f(x)-\frac{1}{2} \qquad \left\{\because f(x)>\frac{1}{2}\right\}$$

$= f(x)$

$\therefore$ $f(x + 2a) = f(x)$

Hence $f(x)$ is periodic with period $2a$.

Illustration 25: Is $\cos\sqrt{t}$ a periodic function ? If yes, then find the period. If not, then give reasons to explain your answer.

Solution : Let $f(t) = \cos\sqrt{t}$ be periodic with period λ, $\lambda \neq 0$, $\lambda > 0$,

$\therefore f(t + \lambda) = f(t)$

$\cos\sqrt{(t+\lambda)}=\cos\sqrt{t}$

$\Rightarrow \sqrt{(t+\lambda)}=2n\pi\pm\sqrt{t}$ for all $t > 0$

Put $t = 0$, then $\sqrt{\lambda} = 2n\pi$ and put $t = \lambda$, then $\sqrt{2\lambda}=2n\pi\pm\sqrt{\lambda}$

From (1) and (2), $\sqrt{2\lambda}=\sqrt{\lambda}\pm\sqrt{\lambda}$

or $\sqrt{2} = 1 \pm 1$ which is impossible.

Hence $\cos\sqrt{t}$ is non-periodic function.

4. SOME IMPORTANT RESULTS TO FIND PERIOD OF A PERIODIC FUNCTION

(i) $\sin^n x$, $\cos^n x$, $\sec^n x$, $\operatorname{cosec}^n x$ are periodic functions with period 2π or π according as n is odd or even.

(ii) $\tan^n x$, $\cot^n x$ are periodic functions with period π, n is even or odd.

(iii) $|\sin x|$, $|\cos x|$, $|\tan x|$, $|\cot x|$, $|\sec x|$, $|\operatorname{cosec} x|$ are periodic functions with period π.

(iv) $|\sin x| + |\cos x|$, $|\tan x| + |\cot x|$, $|\sec x| + |\operatorname{cosec} x|$ are periodic with period $\pi/2$.

(v) $|\sin x| - |\cos x|$, $|\tan x| - |\sin x|$, $|\cot x| - |\sin x|$, $|\operatorname{cosec} x| - |\sec x|$, $|\cot x| - |\cos x|$, etc. are periodic with period π.

(vi) Algebraic functions $\sqrt{x}$, x^3, $2x^2 + 5x + 4$, ... etc. are not periodic.

(vii) Every constant function is always periodic, with no fundamental period.

(viii) Inverse of a periodic functions does not exist. But in case of trigonometric function since domain and range is restricted and defined then inverse exists.

(ix) If $f(T) = f(0) = f(-T)$ then 'T' is the period.

(x) If $f(x)$ is periodic with period T, then find function $f(ax + b)$ is periodic with period $\frac{T}{|a|}$.

e.g. $\sin 3x$ has period $2\pi/3$, $\cos 2x$ has a period $2\pi/2 = \pi$ and $\tan(-7x + 8)$ has a period $\frac{\pi}{|-7|} = \frac{\pi}{7}$.

(xi) If $f(x)$ is periodic with period T, then $\frac{1}{f(x)}$ and $\sqrt{f(x)}$ are also periodic has a period T.

(xii) If $f_1(x)$ and $f_2(x)$ be two trigonometric functions with periods T_1 and T_2, the $F(x) = k_1 f_1(x) \pm k_2 f_2(x)$ is a periodic function and its period is L.C.M. of T_1 and T_2.
i.e. Period of $F(x)$ = L.C.M. of $\{T_1, T_2\}$

Note :-

1. L.C.M. of $\left(\frac{a}{b}, \frac{c}{d}, \frac{e}{f} \ldots\ldots\right) = \frac{\text{L.C.M of}(a,c,e,\ldots\ldots)}{\text{H.C.F. of}(b,d,f,\ldots\ldots)}$

2. L.C.M. of rational with rational is possible.
i.e. L.C.M. of $\left\{1, \frac{1}{2}, \frac{1}{3}, \ldots, \frac{1}{n}\right\} = \frac{\text{L.C.M. of } \{1,1,1,\ldots,1\}}{\text{H.C.F. of } \{1, 2,3,\ldots,n\}} = \frac{1}{1} = 1$

3. L.C.M. of Irrational with irrational is possible
i.e. L.C.M. of $\left\{2\pi, \frac{\pi}{3}\right\}$ = L.C.M. of $\left\{\frac{2\pi}{1}, \frac{2\pi}{6}\right\} = \frac{\text{L.C.M. of } \{2\pi, 2\pi\}}{\text{H.C.F. of } \{1,6\}} = \frac{2\pi}{1} = 2\pi$

4. L.C.M. of rational with irrational is not possible
i.e. L.C.M. of $\{2\pi, 7\}$ is not possible $\because$ $2\pi \in$ Irrational and $7 \in$ rational
i.e. If $F(x) = 5 \sin x - 7 \sin 8x$
Since, $\sin x$ and $\sin 8x$ are periodic with periods $\frac{2\pi}{1}$ and $\frac{2\pi}{8}$ i.e., 2π and $\frac{\pi}{4}$ respectively, hence
period of $F(x) = \frac{\text{L.C.M. of } \{2\pi, 2\pi\}}{\text{H.C.F. of } \{1, 8\}} = \frac{2\pi}{1} = 2\pi$

5. If $f(x)$ has a period T and $g(x)$ has also period T then it does not mean that $f(x) + g(x)$ must have a period T.
e.g. $f(x) = |\sin x| + |\cos x|$
$\because$ periods of $|\sin x|$ and $|\cos x|$ are π but period of $|\sin x| + |\cos x|$ is $\pi/2$.

6. If $f(x)$ is periodic with period T then $cf(x)$, $f(x \pm c)$, $f(x) \pm c$ are also periodic with period T where c is any constant.

Some Important Results (Fast track results)

1. If x and y are independent variables, then

(i) $f(x + y) = f(x).\ f(y) \Rightarrow f(x) = a^{\lambda x}$, λ is constant.

(ii) $f(X + y) = f(x) + f(y) \Rightarrow f(x) = \lambda x$, λ is constant.

(iii) $f(xy) = f(x) + f(y) \Rightarrow f(x) = \lambda\, \ell n\, x$ or $f(x) = 0$, λ is constant.

(iv) $f(xy) = f(x)\ .\ f(y) \Rightarrow f(x) = x^n$, $n \in R$

2. If $f(x) = a \sin x + b \cos x + c\ \forall\ x \in R$ and a, b, c are constants then

$$c - \sqrt{(a^2 + b^2)} \le f(x) \le c + \sqrt{(a^2 + b^2)}$$

or $f(x) \in [c - \sqrt{(a^2 + b^2)}, c + \sqrt{(a^2 + b^2)}]$.

MISCELLANEOUS EXAMPLES

Example 1 : Find the period of the function
$f(x) = e^{x-[x]+|\cos \pi x|+|\cos 2\pi x|+.....+|\cos n\pi x|}$
where [.] denotes the greatest integer function.

Solution : Given $f(x) = e^{x-[x]+|\cos \pi x|+|\cos 2\pi x|+....+|\cos n\pi x|}$
Since period of $x - [x]$ is 1

Period of $|\cos \pi x| = \frac{\pi}{\pi} = 1$

Period of $|\cos 2\pi x| = \frac{\pi}{2\pi} = \frac{1}{2}$

... ...
... ...

Period of $|\cos n\,\pi\, x| = \frac{\pi}{n\pi} = \frac{1}{n}$

So, period of $f(x)$ will be L.C.M. of all periods so ans = 1

Example 2 : Find the period of the function $f(x) = \sin^4 x + \cos^4 x$

Solution : Since $f(x) = \sin^4 x + \cos^4 x = (\sin^2 x)^2 + (\cos^2 x)^2$

$= (\sin^2 x + \cos^2 x)^2 - 2\sin^2 x \cos^2 x = 1 - \frac{1}{2}\sin^2 2x = 1 - \frac{1}{4}(1 - \cos 4x) = \frac{3}{4} + \frac{1}{4}\cos 4x$

Hence period of $f(x) = \frac{2\pi}{4} = \frac{\pi}{2}$

Example 3 : Find the period of the function $f(x) = \cos(\sin x) + \cos(\cos x)$

Solution : Let T be the period of the function then $f(x + T) = f(x)$
$\Rightarrow \cos(\sin(x + T)) + \cos(\cos(x + T)) = \cos(\sin x) + \cos(\cos x)$
put $x = 0$ then $\cos(\sin T) + \cos(\cos T) = \cos(0) + \cos(1) = \cos(\cos \pi/2) + \cos(\sin \pi/2)$
On comparing we get $T = \pi/2$

Example 4 : If $f(x) = \sin x + \cos ax$ is a periodic function, show that a is rational number.

Solution : Given $f(x) = \sin x + \cos ax$

$\because$ Period of $\sin x = \frac{2\pi}{1}$ and period of $\cos ax = \frac{2\pi}{a}$

Hence period of $f(x)$ = L.C.M. of $\left\{\frac{2\pi}{1}, \frac{2\pi}{a}\right\} = \frac{\text{L.C.M. of } \{2\pi, 2\pi\}}{\text{H.C.F. of } \{1, a\}} = \frac{2\pi}{k}$

where k = H.C.F. of 1 and a

$\therefore \frac{1}{k}$ = integer = q (say), $(\neq 0)$ and $\frac{a}{k}$ = integer = p (say)

$\therefore \frac{a/k}{1/k} = \frac{p}{q} \quad \Rightarrow a = \frac{p}{q} \quad \therefore$ a = rational number

Example 5 : Prove that $\sin(1/x)$, $(x \neq 0)$ is a non-periodic function.

Solution : Let $f(x) = \sin(1/x)$ be periodic with period
T, $T \neq 0$, $T > 0$

$\therefore f(x + T) = f(x) \Rightarrow \sin\left(\frac{1}{x+T}\right) = \sin\left(\frac{1}{x}\right) \Rightarrow \frac{1}{x+T} = n\pi + (-1)^n \frac{1}{x}$ ... (i)

Put $x = T$ and $x = 2T$ in (1), then

$$\frac{1}{2T} = n\pi + (-1)^n \frac{1}{T} \quad \text{... (2)}$$

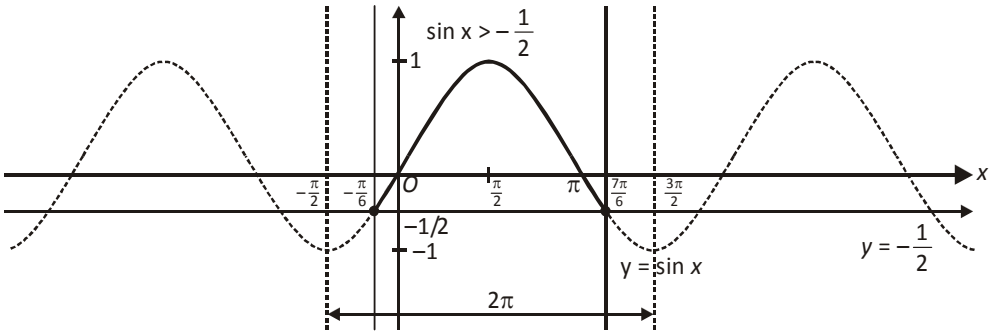
and $$\frac{1}{3T} = n\pi + (-1)^n \frac{1}{2T} \quad \text{... (3)}$$

Subtracting (3) from (2), we get $\frac{1}{6T} = (-1)^n \cdot \frac{1}{2T}$ or $\frac{1}{3} = (-1)^n$ which is impossible.

Hence sin (1/x), ($x \neq 0$) is non-periodic function.

Example 6 : Solve the inequality; $\sin x > -\frac{1}{2}$.

Solution : As the function sin x has least positive period 2π. {That is why it is sufficient to solve inequality of the form $\sin x > a$, $\sin x \geq a$, $\sin x < a$, $\sin x \leq a$ first on the interval of length 2π, and then get the solution set by adding numbers of the form $2\pi n$, $n \in \mathbf{Z}$, to each of the solutions obtained on that interval}. Thus, let us solve this inequality on the interval $\left[-\frac{\pi}{2}, \frac{3\pi}{2}\right]$, where graph of $y = \sin x$ and $y = -\frac{1}{2}$ are taken two curves on x-y plane.



$y = \sin x$ and $y = -\frac{1}{2}$

From above figure, $\sin x > -\frac{\pi}{2}$ when $-\frac{1}{6} < x < \frac{7\pi}{6}$.

Thus, on generalising above solution; $2n\pi - \frac{\pi}{6} < x < 2n\pi + \frac{7\pi}{6}; n \in Z$.

Which implies that those and only those values of x each of which satisfies these two inequalities for a certain $n \in Z$ can serve as solutions to the original inequality.

Example 8 : Solve the inequality : $\cos x \leq -\frac{1}{2}$.

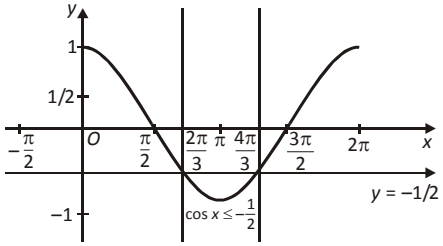
Solution : As discussed in previous example, cos x is periodic with period 2π. So, to check the solution in [0, 2π]. It is clear from figure, $\cos x \leq -\frac{1}{2}$ when; $\frac{2\pi}{3} \leq x \leq \frac{4\pi}{3}$.

On generalising above solution;

$2n\pi + \frac{2\pi}{3} \leq x \leq 2n\pi + \frac{4\pi}{3}; n \in Z$

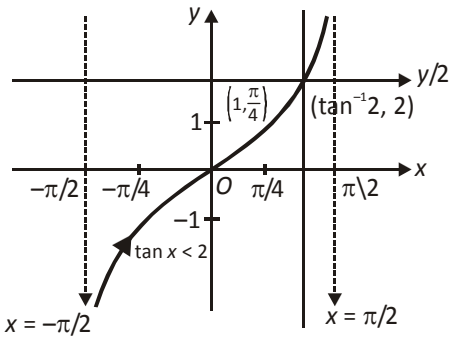
$\therefore$ Solution of $\cos x \leq -\frac{1}{2}$

$\Rightarrow \quad x \in \left[2n\pi + \frac{2\pi}{3}, 2n\pi + \frac{4\pi}{3}\right]; n \in Z.$



Example 9 : Solve the inequality : tan $x < 2$.

Solution : We know tan x is periodic with period π.



x = $\tan^{-1}2$ = arc tan 2

So, to check the solution on the interval $\left(-\frac{\pi}{2},\frac{\pi}{2}\right)$.

It is clear from figure, tan $x < 2$ when;

$-\frac{\pi}{2} < x < \tan^{-1}2$ or $-\frac{\pi}{2} < x < \text{arc tan}2$

$\Rightarrow$ General solution : $2n\pi - \frac{\pi}{2} < x < 2n\pi + \tan^{-1}2$

$\Rightarrow n \in \left(2n\pi - \frac{\pi}{2}, 2n\pi + \text{arc tan}2\right)$

Example 10 : Solve the inequality : $\sin\left(\frac{3x}{2}+\frac{\pi}{12}\right) < \frac{1}{\sqrt{2}}$.

Solution : Here, $\sin\left(\frac{3x}{2}+\frac{\pi}{12}\right) < \frac{1}{\sqrt{2}}$; put $\frac{3x}{2}+\frac{\pi}{12} = t$

$\therefore$ $\sin t < \frac{1}{\sqrt{2}}$, now sin t is periodic with period 2π,

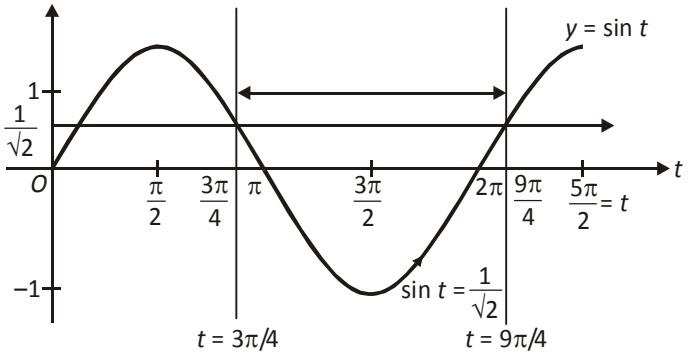
thus to chek on $\left[\frac{\pi}{2},\frac{5\pi}{2}\right]$ or $\left[-\frac{\pi}{2},\frac{3\pi}{2}\right]$

From figure,

$\sin t < \frac{1}{\sqrt{2}}$, when $\frac{3\pi}{4} < t < \frac{9\pi}{4}$.

$\therefore$ general solution

$2n\pi + \frac{3\pi}{4} < t < 2n\pi + \frac{9\pi}{4}; n \in z$



Substituting $t = \frac{3x}{2}+\frac{\pi}{12}$

$2n\pi + \frac{3\pi}{4} < \frac{3x}{2}+\frac{\pi}{12} < 2n\pi + \frac{9\pi}{4} \Rightarrow \frac{4}{9}\pi + \frac{4}{3}\pi n < x < \frac{13}{9}\pi + \frac{4}{3}n\pi; n \in z.$

UNSOLVED EXERCISE

Section-A
(Straight objective type with single option correct)

Topic : 1 Based on Trigonometric Formulae

1. If $\sin\theta$ and $\cos\theta$ are the roots of the equation $ax^2 - bx + c = 0$, then a, b and c satisfy the relation
(a) $a^2 + b^2 + 2ac = 0$ (b) $a^2 - b^2 + 2ac = 0$
(c) $a^2 + c^2 + 2ab = 0$ (d) $a^2 - b^2 - 2ac = 0$

2. If A lies in the second quadrant and $3\tan A + 4 = 0$, the value of $2\cot A - 5\cos A + \sin A$ is equal to –
(a) – 53/10 (b) 23/10
(c) 37/10 (d) 7/10

3. If $3\sin\theta + 5\cos\theta = 5$, then the value of $5\sin\theta - 3\cos\theta$ is equal to
(a) 5 (b) 3
(c) 4 (d) None of these

4. If $\tan\theta = \sqrt{n}$ for some non-square natural number n, then $\sec 2\theta$ is
(a) A rational number
(b) An irrational number
(c) A positive number
(d) None of these

5. If $\tan\theta + \sec\theta = \sqrt{3}$, $0 < \theta < \pi$, then θ is equal to –
(a) $5\pi/6$ (b) $2\pi/3$
(c) $\pi/6$ (d) $\pi/3$

6. If $\tan\theta = a \neq 0$, $\tan 2\theta = b \neq 0$ and $\tan\theta + \tan 2\theta = \tan 3\theta$ then
(a) $a = b$ (b) $ab = 1$
(c) $a + b = 0$ (d) $b = 2a$

7. Out of the following four relations :

(I) $\dfrac{\sin A}{1+\cos A} + \dfrac{\sin A}{1-\cos A} = \dfrac{2}{\sin A}$

(II) $\left(\dfrac{1+\cos A}{\sin A}\right)^2 = \dfrac{1+\cos A}{1-\cos A}$

(III) $\dfrac{\sin 10^\circ}{\cos 80^\circ} = 1$

(IV) $\sin^4 A - \cos^4 A = 1 + \sin^2 A$

the wrong one is –
(a) I (b) II
(c) III (d) IV

8. If $\sin\theta + \operatorname{cosec}\theta = 2$, the value of $\sin^{10}\theta + \operatorname{cosec}^{10}\theta$ is
(a) 2 (b) 1
(c) 0 (d) 4

9. If $x = p\sec\theta$ and $y = q\tan\theta$ then
(a) $x^2 - y^2 = p^2q^2$ (b) $x^2q^2 - y^2p^2 = pq$
(c) $x^2q^2 - y^2p^2 = 1/p^2q^2$ (d) $x^2q^2 - y^2p^2 = p^2q^2$

10. If $\tan\theta + \sin\theta = m$ and $\tan\theta - \sin\theta = n$, the the value of $m^2 - n^2$ is equal to –
(a) $4mn$ (b) $2\sqrt{mn}$
(c) $4\sqrt{mn}$ (d) $2\sqrt{m/n}$

11. A circular wire of radius 7 cm is cut and bend again into an arc of a circle of radius 12 cm. The angle subtended by the arc at the centre is
(a) 50° (b) 210°
(c) 100° (d) 60°

12. If θ satisfies $\sec\theta + \tan\theta = 1$, then which of the following is a root of the equation
$p^2(q^2 - r^2)x^2 + q^2(r^2 - p^2)x + r^2(p^2 - q^2) = 0$
(a) $\sec\theta$ (b) $\cot\theta$
(c) $\tan\theta$ (d) $\operatorname{cosec}\theta$

13. If $a\sin^2\theta + b\cos^2\theta = m$, $b\sin^2\phi + a\cos^2\phi = n$, $a\tan\theta = b\tan\phi$, then which of the following is true ?

(a) $\dfrac{1}{n} - \dfrac{1}{m} = \dfrac{1}{a} - \dfrac{1}{b}$ (b) $\dfrac{1}{n} + \dfrac{1}{m} = \dfrac{1}{a} + \dfrac{1}{b}$
(c) $m^2 + n^2 = a^2 + b^2$ (d) $m^2 - n^2 = a^2 - b^2$

14. If $\cot\alpha + \tan\alpha = m$ and $\dfrac{1}{\cos\alpha} - \cos\alpha = n$, then
(a) $m(mn^2)^{1/3} - n(nm^2)^{1/3} = 1$
(b) $m(m^2n)^{1/3} - n(mn^2)^{1/3} = 1$
(c) $n(mn^2)^{1/3} - m(nm^2)^{1/3} = 1$
(d) $n(m^2n)^{1/3} - m(mn^2)^{1/3} = 1$

15. If $\sin\theta$ and $\cos\theta$ are the roots of the equation $ax^2 - bx + c = 0$, then a, b and c satisfy the relation
(a) $a^2 + b^2 + 2ac = 0$ (B) $a^2 - b^2 + 2ac = 0$
(c) $a^2 + c^2 + 2ab = 0$ (d) $a^2 - b^2 - 2ac = 0$

16. If $\sin x + \sin^2 x = 1$, then the value of $\cos^{12} x + 3\cos^{10} x + 3\cos^8 x + \cos^6 x - 1$ is equal to
(a) 0 (b) 1
(c) –1 (d) 2

17. If $5\cos x + 12\sin x = k$, then number of integral values of k is
(a) 13 (b) 14
(c) 27 (d) 26

Topic : 2 Graph and Conditional Identities

18. If for real values of $x, \cos\theta = x + \frac{1}{x}$, then

(a) θ is an acute angle
(b) θ is a right angle
(c) θ is an obtuse angle
(d) No value of θ is possible

19. Sum of all possible Values of α and β satisfying $3\sin\alpha + 4\cos\beta = 7$ and $0 \le \alpha, \beta \le 4\pi$ is

(a) 7π (b) 8π
(c) 9π (d) 6π

20. If $\sin^2\theta = \frac{4-m^2}{3}, m \in R - (-1,1)$ and $\cot^3\frac{\theta}{2} = \cot\alpha$, then the value of $\cos^{2/3}\alpha + \sin^{2/3}\alpha$ is equal to

(a) $\left(\frac{4}{m}\right)^{2/3}$ (b) $\left(\frac{m}{2}\right)^{2/3}$

(c) $\left(\frac{m}{4}\right)^{2/3}$ (d) $\left(\frac{2}{m}\right)^{2/3}$

21. If $x^2 - 4x + 5 - \text{siny} = 0$, $y \in [0, 2\pi)$, then

(a) x = 1, y = 0 (b) x = 1, y = $\pi/2$
(c) x = 2, y = 0 (d) x = 2, y = $\pi/2$

22. If $a\cos^3\alpha + 3a\cos\alpha\sin^2\alpha = m$ and $a\sin^3\alpha + 3a\cos^2\alpha\sin\alpha = n$, then $(m+n)^{2/3} + (m-n)^{2/3}$ is equal to

(a) $2a^2$ (b) $2a^{1/3}$
(c) $2a^{2/3}$ (d) $2a^3$

23. If $\frac{2\sin\alpha}{1+\cos\alpha+\sin\alpha} = y$, then $\frac{1-\cos\alpha+\sin\alpha}{1+\sin\alpha}$ is equal to

(a) $1/y$ (b) y
(c) $1-y$ (d) $1+y$

Topic : 3 Perodicity and Hight and Distance

24. Period of $\sin 3x + \cos\pi x$ is

(a) $\frac{2\pi}{3}$ (b) 2

(c) $\frac{\pi}{3}$ (d) Does not exist

25. If the fundamental period of $f(x) = \frac{|\sin x| + |\cos x|}{2}$ is T_1 and that of $g(x) = \frac{1}{2}|\sin x| + \frac{1}{3}|\cos x|$ is T_2 then $T_1 + T_2$ is equal to

(a) $3\pi/2$ (b) $\pi/2$
(c) π (d) 2π

26. Let a vertical tower AB have its end A on the level ground. Let C be the mid-point of AB and P be a point on the ground such that AP = 2AB. If $\angle BPC = \beta$, and $\tan\beta$ ισ εθυαλ το

(a) $\frac{6}{7}$ (b) $\frac{1}{4}$

(c) $\frac{2}{9}$ (d) $\frac{4}{9}$

27. A man is walking towards a vertical pillar in a straight path, at a uniform speed. At a certain point A on the path, he observes that the angle of elevation of the top of the pillar is 300. After walking for 10 min from A in the same direction, at a point B, he observes that the angle of elevation of the top of the pillar is 60^0. then, the time taken (in minutes) by him, from B to reach the pillar, is

(a) 6 (b) 10
(c) 20 (d) 5

Topic : 4 Maxima-Minima and Number of Solution

28. For $0 \le \theta \le \frac{\pi}{2}$, the maximum value of $\sin\theta + \cos\theta$ is –

(a) 0 (b) 1
(c) $\sqrt{2}$ (d) 2

29. The number of all possible triplets (a_1, a_2, a_3) such that $a_1 + a_2\cos 2x + a_3\sin^2 x = 0$ for all x is

(a) zero (b) 1
(c) 2 (d) infinite

30. The maximum value of $(7\cos\theta + 24\sin\theta)(7\sin\theta - 24\cos\theta)$ is

(a) 25 (b) 625

(c) $\frac{625}{2}$ (d) $\frac{625}{4}$

31. The number of solutions of the equation $\left(\frac{\pi x}{2\sqrt{3}}\right) = x^2 - 2\sqrt{3}x + 4$

(a) is zero (b) is only one
(c) is only two (d) is greater than 2

32. If $x_i > 0$ for $1 \le i \le n$ and $x_1 + x_2 + \ldots + x_n = \pi$ then the greatest value of the sum $\sin x_1 + \sin x_2 + \ldots + \sin x_n$ is equal to

(a) n (b) 0
(c) $n\sin(\pi/x)$ (d) π

33. Minimum value of $4x^2 - 4x|\sin\theta| - \cos^2\theta$ is equal to

(a) –2 (b) –1
(c) –1/2 (d) 0

Section-B
(Previous Year Question)

1. The expression $\frac{\tan A}{1-\cot A}+\frac{\cot A}{1-\tan A}$ can be written as
   (a) sin A cos A + 1 (b) sec A cosec A + 1
   (c) tan A + cot A (d) sec A + cosec A
2. The number of ordered pairs (α, β), where $\alpha, \beta \in (-\pi, \pi)$ satisfying $\cos(\alpha-\beta)=1$ and $\cos(\alpha+\beta)=\frac{1}{e}$ is
   (a)0 (b)1 (c)2 (d)4
3. Given both θ and ϕ are acute angles and $\sin\theta=\frac{1}{2}$, $\cos\phi=\frac{1}{3}$, then the value of $\theta+\phi$ belongs to
   (a) $\left(\frac{\pi}{3},\frac{\pi}{6}\right]$ (b) $\left(\frac{\pi}{2},\frac{2\pi}{3}\right)$
   (c) $\left(\frac{2\pi}{3},\frac{5\pi}{6}\right]$ (d) $\left(\frac{5\pi}{6},\pi\right]$
4. Which of the following numbers is rational?
   (a) sin 15° (b) cos 15°
   (c) sin 15° cos 15° (d) sin 15° cos 75°
5. $3(\sin x-\cos x)^4+6(\sin x+\cos x)^2+4(\sin^6 x+\cos^6 x)$ equals
   (a)11 (b)12 (c)13 (d)14
6. The value of the expression $\sqrt{3}\operatorname{cosec}20°-\sec 20°$ is equal to
   (a) 2 (b) 2sin 20°/sin 40°
   (c) 4 (d) 4sin 20°/sin40°
7. The expression
   $3\left[\sin^4\left(\frac{3\pi}{2}-\alpha\right)+\sin^4(3\pi+\alpha)\right]-2\left[\sin^6\left(\frac{\pi}{2}+\alpha\right)+\sin^6(5\pi-\alpha)\right]$ is equal to
   (a) 0 (b) 1
   (c) 3 (d) sin 4α + cos 6α
8. $\left(1+\cos\frac{\pi}{8}\right)\left(1+\cos\frac{3\pi}{8}\right)\left(1+\cos\frac{5\pi}{8}\right)\left(1+\cos\frac{7\pi}{8}\right)$ is equal to
   (a) $\frac{1}{2}$ (b) $\cos\frac{\pi}{8}$
   (c) $\frac{1}{8}$ (d) $\frac{1+\sqrt{2}}{2\sqrt{2}}$
9. Given $A=\sin^2\theta+\cos^4\theta$, then for all real values of θ
   (a) $1\le A\le 2$ (b) $\frac{3}{4}\le A\le 1$
   (c) $\frac{13}{16}\le A\le 1$ (d) $\frac{3}{4}\le A\le\frac{13}{16}$
10. If $\tan\theta=-\frac{4}{3}$, then $\sin\theta$ is
    (a) $-\frac{4}{5}$ but not $\frac{4}{5}$ (b) $-\frac{4}{5}$ or $\frac{4}{5}$
    (c) $\frac{4}{5}$ but not $-\frac{4}{5}$ (d) None of the above

Section-C
(School / Board pattern)

1. The angles of a triangle are in A.P. and the number of degrees in the least angle is to the number of degrees in the mean angle as 1 : 120. Find the angles in radians.
2. The number of sides of two regular polygons are 5 : 4 and the difference between their angles is 9^0. Find the number of sides of the polygons.
3. the radius of a circle is 30 cm. Find the length of an arc of this circle, if the length of the chord of the arc is 30 cm.
4. If the arcs of the same length in two circles subtend angles 65^0 and 110^0at the centre, find the ratio of their radii.
5. $\frac{\tan^3\theta}{1+\tan^2\theta}+\frac{\cot^3\theta}{1+\cot^2\theta}=\frac{1-2\sin^2\theta\cos^2\theta}{\sin\theta\cos\theta}$
6. $\frac{(1+\cot\theta+\tan\theta)(\sin\theta-\cos\theta)}{\sec^3\theta-\operatorname{cosec}^3\theta}=\sin^2\theta\cos^2\theta$
7. If $T_n=\sin^n\theta+\cos^n\theta$, prove that
   (i) $\frac{T_3-T_5}{T_1}=\frac{T_5-T_7}{T_3}$ (ii) $2T_6-3T_4+1=0$
   (iii) $6T_{10}-15T_8+10T_6-1=0$
8. If $x=\frac{2\sin\theta}{1+\cos\theta+\sin\theta}$, then prove that $\frac{1-\cos\theta+\sin\theta}{1+\sin\theta}$ is also equal to x.
9. Prove that
   $\cos 570^0\sin 510^0+\sin(-330^0)\cos(-390^0)=0$
10. Prove that
    $\frac{\cos(2\pi+\theta)\operatorname{cosec}(2\pi+\theta)\tan(\pi/2+\theta)}{\sec(\pi/2+\theta)\cos\theta\cot(\pi+\theta)}=1$

CHAPTER 3 TRANSFORMATION FORMULAE

1. TRIGONOMETRIC RATIO OF COMPOUND ANGLES

Trigonometric functions of sum or difference of two angles :

(a) $\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$

(b) $\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$

(c) $\sin^2 A - \sin^2 B = \cos^2 B - \cos^2 A = \sin(A + B).\sin(A - B)$

(d) $\cos^2 A - \sin^2 B = \cos^2 B - \sin^2 A = \cos(A + B).\cos(A - B)$

(e) $\tan(A \pm B) = \dfrac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$ where $A \neq n\pi + \dfrac{\pi}{2}$, $B \neq n\pi + \dfrac{\pi}{2}$ and $A \pm B \neq n\pi + \dfrac{\pi}{2}$

(f) $\cot(A \pm B) = \dfrac{\cot A \cot B \mp 1}{\cot B \pm \cot A}$ where $A \neq n\pi$, $B \neq n\pi$ and $A + B \neq n\pi$

(g) $\tan(A + B + C) = \dfrac{\tan A + \tan B + \tan C - \tan A \tan B \tan C}{1 - \tan A \tan B - \tan B \tan C - \tan C \tan A}$

(h) $\cot(A + B + C) = \dfrac{\cot A + \cot B + \cot C - \cot A \cot B \cot C}{1 - \cot A \cot B - \cot B \cot C - \cot C \cot A}$

(i) $\sin(A + B + C) = \sin A \cos B \cos C + \cos A \sin B \cos C + \cos A \cos B \sin C - \sin A \sin B \sin C$
or,
$\sin(A + B + C) = \cos A \cos B \cos C (\tan A + \tan B + \tan C - \tan A \tan B \tan C)$

(j) $\cos(A + B + C) = \cos A \cos B \cos C - \sin A \sin B \cos C - \sin A \cos B \sin C - \cos A \sin B \sin C$
or,
$\cos(A + B + C) = \cos A \cos B \cos C (1 - \tan A \tan B - \tan B \tan C - \tan C \tan A)$

(h) $\tan(A_1 + A_2 + \ldots\ldots + A_n) = \dfrac{S_1 - S_3 + S_5 - S_7 + \ldots\ldots}{1 - S_2 + S_4 - S_6 + \ldots\ldots}$, where

$S_1 = \tan A_1 + \tan A_2 + \ldots\ldots + \tan A_n$ = the sum of the tangents of the separate angles,
$S_2 = \tan A_1 \tan A_2 + \tan A_2 \tan A_3 + \ldots\ldots$ = the sum of the tangents taken two at a time,
$S_3 = \tan A_1 \tan A_2 \tan A_3 + \tan A_2 \tan A_3 \tan A_4 + \ldots\ldots$ = the sum of the tangents taken three at a time, and so on.

Illustration 1: Calculate :

(i)sin 15°, cos 15°, tan 15°, cot 15° (ii) tan 22.5°, cot 22.5° (iii) tan 7.5°, cot 7.5°

Solution : (i)$\sin 15° = \sin(45° - 30°) = \sin 45° \cos 30° - \cos 45° \sin 30° \Rightarrow \sin 15° = \dfrac{\sqrt{3} - 1}{2\sqrt{2}}$

$\cos 15° = \cos(45° - 30°) = \cos 45° \cos 30° + \sin 45° \sin 30° \Rightarrow \cos 15° = \dfrac{\sqrt{3} + 1}{2\sqrt{2}}$

$\tan 15° = \dfrac{1 - \cos 30°}{\sin 30°}$ $\left(\text{using } \tan\dfrac{\theta}{2} = \dfrac{1 - \cos\theta}{\sin\theta}\right)$

$\Rightarrow \tan 15° = 2 - \sqrt{3}$ and also, $\cot 15° = \dfrac{1}{\tan 15°} = 2 + \sqrt{3}$

(ii) $\tan 22\dfrac{1}{2}° = \dfrac{1 - \cos 15°}{\sin 15°} = \dfrac{1 - \dfrac{\sqrt{3} + 1}{2\sqrt{2}}}{\dfrac{\sqrt{3} - 1}{2\sqrt{2}}} = \dfrac{2\sqrt{2} - \sqrt{3} - 1}{\sqrt{3} - 1} \Rightarrow \tan 7\dfrac{1}{2}° = (\sqrt{3} - \sqrt{2})(\sqrt{2} - 1)$

and also, $\cot 7\frac{1}{2}^\circ = \frac{1}{\tan 7\frac{1}{2}^\circ} = (\sqrt{3}+\sqrt{2})(\sqrt{2}+1)$

Illustration 2: If $\tan\alpha - \tan\beta = x$ and $\cot\beta - \cot\alpha = y$, prove that $\cot(\alpha-\beta) = \frac{x+y}{xy}\left(\alpha \neq \beta \neq \frac{\pi}{4}\right)$.

Solution : $\text{RHS} = \frac{x+y}{xy} = \frac{(\tan\alpha-\tan\beta)+(\cot\beta-\cot\alpha)}{(\tan\alpha-\tan\beta)(\cot\beta-\cot\alpha)}$

$$= \frac{\frac{\cot\beta-\cot\alpha}{\cot\beta\cdot\cot\alpha}+(\cot\beta-\cot\alpha)}{\frac{\cot\beta-\cot\alpha}{\cot\beta\cdot\cot\alpha}(\cot\beta-\cot\alpha)}$$

$$= \frac{(\cot\beta-\cot\alpha)\left[\frac{1}{\cot\alpha\cdot\cot\beta}+1\right]}{(\cot\beta-\cot\alpha)\left[\frac{\cot\beta-\cot\alpha}{\cot\beta\cdot\cot\alpha}\right]} \qquad [\alpha \neq \beta \neq 45°]$$

$$= \frac{1+\cot\alpha\cdot\cot\beta}{\cot\beta-\cot\alpha} = \cot(\alpha-\beta) = \text{LHS}$$

2. TRANSFORMATION FORMULAE

Factorisation of the sum or difference of two sines or cosines :

(a) $\sin C + \sin D = 2\sin\frac{C+D}{2}\cos\frac{C-D}{2}$ (b) $\sin C - \sin D = 2\cos\frac{C+D}{2}\sin\frac{C-D}{2}$

(c) $\cos C + \cos D = 2\cos\frac{C+D}{2}\cos\frac{C-D}{2}$ (d) $\cos C - \cos D = -2\sin\frac{C+D}{2}\sin\frac{C-D}{2}$

Transformation of products into sum or difference of sines & cosines :

(a) $2\sin A\cos B = \sin(A+B)+\sin(A-B)$ (b) $2\cos A\sin B = \sin(A+B)-\sin(A-B)$

(c) $2\cos A\cos B = \cos(A+B)+\cos(A-B)$ (d) $2\sin A\sin B = \cos(A-B)-\cos(A+B)$

Illustration 3: Show that : $\frac{\sin 2x-\sin 3x+\sin 4x}{\cos 2x-\cos 3x+\cos 4x} = \tan 3x$

Solution : $\text{LHS} = \frac{\sin 2x-\sin 3x+\sin 4x}{\cos 2x-\cos 3x+\cos 4x}$

$$= \frac{(\sin 2x+\sin 4x)-\sin 3x}{(\cos 2x+\cos 4x)-\cos 3x} \qquad \left[\text{note that } \frac{2x+4x}{2} = 3x\right]$$

$$= \frac{2\sin\left(\frac{2x+4x}{2}\right)\cos\left(\frac{2x-4x}{2}\right)-\sin 3x}{2\cos\left(\frac{2x+4x}{2}\right)\cos\left(\frac{2x-4x}{2}\right)-\cos 3x} = \frac{2\sin 3x\cos(-x)-\sin 3x}{2\cos 3x\cos(-x)-\cos 3x}$$

$$= \frac{\sin 3x[2\cos x-1]}{\cos 3x[2\cos x-1]} \quad [\because \cos(-\theta) = \cos\theta]$$

$= \tan 3x = \text{RHS}$

Illustration 4: Show that : $\dfrac{\cos 8A\cos 5A-\cos 12A\cos 9A}{\sin 8A\cos 5A+\cos 12A\sin 9A}=\tan 4A$

Solution : $\text{LHS}=\dfrac{2\cos 8A\cos 5A-2\cos 12A\cos 9A}{2\sin 8A\cos 5a+2\cos 12A\sin 9A}$ [Multiply and divide by 2]

$$=\frac{[\cos(8A+5A)+\cos(8A-5A)]-[\cos(12A+9A)+\cos\{12A-9A)]}{[\sin(8A+5A)+\sin(8A-5A)]+[\sin(12A+9A)+\sin(12A-9A)]}$$

$$=\frac{\cos 13A+\cos 3A-\cos 21A-\cos 3A}{\sin 13A+\sin 3A+\sin 21A-\sin 3A}=\frac{\cos 13A-\cos 21A}{\sin 13A+\sin 21A}=\frac{-2\sin\frac{13A+21A}{2}\sin\frac{13A-21A}{2}}{2\sin\frac{13A+21A}{2}\cos\frac{13A-21A}{2}}$$

$$=\frac{-2\sin 17A\sin(-4A)}{2\sin 17A\cos(-4A)}=\frac{\sin 4A}{\cos 4A}=\tan 4A=\text{RHS}$$

Illustration 5: If $m\tan(\theta-30°)=n\tan(\theta+120°)$, then show that : $\cos 2\theta=\dfrac{m+n}{2(m-n)}$

Solution : $\dfrac{\tan(\theta+120°)}{\tan(\theta-30°)}=\dfrac{m}{n}$

$$\Rightarrow \quad \frac{\sin(\theta+120°)\cos(\theta-30°)}{\cos(\theta+120°)\sin(\theta-30°)}=\frac{m}{n}$$

$$\Rightarrow \quad \frac{\sin(\theta+120°)\cos(\theta-30°)-\cos(\theta+120°)\sin(\theta-30°)}{\sin(\theta+120°)\cos(\theta-30°)+\cos(\theta+12°)\sin(\theta-30°)}=\frac{m-n}{m+n}$$ [Apply C and D]

$$\Rightarrow \quad \frac{\sin[(\theta+120°)-(\theta-30°)]}{\sin[(\theta+120°)+(\theta-30°)]}=\frac{m-n}{m+n} \Rightarrow \quad \frac{1}{2}\,(m+n)=(m-n)\cos 2\theta$$

$$\Rightarrow \cos 2\theta=\frac{m+n}{2(m-n)}$$

Illustration 6: If $\tan\beta=\dfrac{\tan\alpha+\tan\gamma}{1+\tan\alpha\tan\gamma}$, prove that : $\sin 2\beta=\dfrac{\sin 2\alpha+\sin 2\gamma}{1+\sin 2\alpha\sin 2\gamma}$.

Solution : We have $\tan\beta$ in terms of α and γ, so that we have to express $\sin 2\beta$ in terms of α, γ.

Hence we will start with $\sin 2\beta=\dfrac{2\tan\beta}{1+\tan^2\beta}$ and substitute for $\tan\beta$ in R.H.S. Also, as the final expression does not contain $\tan\alpha$ and $\tan\gamma$, so express $\tan\beta$ in terms of sine and cosine.

$$\tan\beta=\frac{\sin\alpha\cos\gamma+\cos\alpha\sin\gamma}{\cos\alpha\cos\gamma+\sin\alpha\sin\gamma}=\frac{\sin(\alpha+\gamma)}{\cos(\alpha-\gamma)}$$

$$\text{Now}\sin 2\beta=\frac{2\tan\beta}{1+\tan^2\beta} \Rightarrow \sin 2\beta=\frac{2\frac{\sin(\alpha+\gamma)}{\cos(\alpha-\gamma)}}{1+\frac{\sin^2(\alpha+\gamma)}{\cos^2(\alpha-\gamma)}}=\frac{2\sin(\alpha+\gamma)\cos(\alpha-\gamma)}{\cos^2(\alpha-\gamma)+\sin^2(\alpha+\gamma)}$$

$$=\frac{\sin[\overline{\alpha+\gamma}+\overline{\alpha-\gamma}]+\sin[\overline{\alpha+\gamma}-\overline{\alpha-\gamma}]}{1+\sin^2(\alpha+\gamma)-\sin^2(\alpha-\gamma)}=\frac{\sin 2\alpha+\sin 2\gamma}{1+\sin[\overline{\alpha+\gamma}+\overline{\alpha-\gamma}]\sin[\overline{\alpha+\gamma}-\overline{\alpha-\gamma}]}$$

$$\Rightarrow \sin 2\beta=\frac{\sin 2\alpha+\sin 2\gamma}{1+\sin 2\alpha\sin 2\gamma}$$

Illustration 7: If $a \sin^2 x + b \cos^2 x = c$, b $\sin^2 y + a \cos^2 y = d$ and $a \tan x = b \tan y$, then $\frac{a^2}{b^2}$ is

(A) $\frac{(b-c)(d-b)}{(a-d)(c-d)}$ (B) $\frac{(a-d)(c-a)}{(b-c)(d-b)}$ (C) $\frac{(d-a)(c-a)}{(b-c)(d-b)}$ (D) $\frac{(b-c)(b-d)}{(a-c)(a-d)}$

Solution : (B) We have to find the value of $\frac{a^2}{b^2}$, i.e. $\frac{\tan^2 y}{\tan^2 x}$ [$\because a \tan x = b \tan y$]

Given that : $a \sin^2 x + b \cos^2 x = c \Rightarrow a \sin^2 x + b (1 - \sin^2 x) = c$

$\Rightarrow \sin^2 x (a - b) = c - b \Rightarrow \sin^2 x = \frac{c-b}{a-b}$

$\therefore \cos^2 x = 1 - \sin^2 x \Rightarrow \cos^2 x = \frac{a-c}{a-b}$

So, $\tan^2 x = \frac{c-b}{a-c} = \frac{b-c}{c-a}$ ..... (i)

Similarly, we can find $\tan^2 y$

$\Rightarrow \tan^2 y = \frac{a-d}{d-b}$ [Replace c by d, b by a, a by b in (i)]

$\Rightarrow \frac{a^2}{b^2} = \frac{\tan^2 y}{\tan^2 x} = \frac{a-d}{d-b} \times \frac{c-a}{b-c}$

Illustration 8: If $x = \sin(\alpha - \beta) . \sin(\gamma - \delta)$; $y = \sin(\beta - \gamma) . \sin(\alpha - \delta)$ and $z = \sin(\gamma - \alpha) . \sin(\beta - \delta)$, then

(A) $x + y + z = 0$ (B) $x + y - z = 0$ (C) $y + z - x = 0$ (D) None of these

Solution : **(A)** Consider $x = \sin(\alpha - \beta) \sin(\gamma - \delta)$, multiply both side by '2'. We get :

$\because 2x = 2 \sin(\alpha - \beta) \sin(\gamma - \delta)$

$2x = \cos(\alpha - \beta - \gamma + \delta) - \cos(\alpha - \beta + \gamma - \delta)$ .....(i)

And similarly

$2y = \cos(\beta - \gamma - \alpha + \delta) - \cos(\beta - \gamma + \alpha - \delta)$ .....(ii)

$2z = \cos(\gamma - \alpha - \beta + \delta) - \cos(\gamma - \alpha + \beta - \delta)$ .....(iii)

Now adding at (i), (ii) and (iii) to get :

$2x + 2y + 2z = \cos(\alpha - \beta - \gamma - \delta) - \cos(\alpha - \beta + \gamma - \delta) + \cos(\beta - \gamma - \alpha - \delta) - \cos(\beta - \gamma - \alpha + \delta) + \cos(\gamma - \alpha - \beta + \delta) - \cos(\gamma - \alpha + \beta - \delta)$

As $\cos(\alpha - \beta - \gamma + \delta) = \cos(\gamma - \alpha + \beta - \delta)$

$\cos(\beta - \gamma - \alpha + \delta) = \cos(\alpha - \beta + \gamma - \delta)$

$\cos(\gamma - \alpha - \beta + \delta) = \cos(\beta - \gamma - \alpha - \delta)$ [As $\cos(-\theta) = \cos\theta$]

$\Rightarrow 2x + 2y + 2z = 0 \Rightarrow x + y + z = 0$

Illustration 9: If $A + B = \frac{\pi}{3}$ and $\cos A + \cos B = 1$, then

(A) $|\cos A - \cos B| = \sqrt{\frac{2}{3}}$ (B) $|\cos A - \cos B| = \frac{1}{2\sqrt{3}}$

(C) $|\cos A - \cos B| = \sqrt{\frac{3}{2}}$ (D) $|\cos A - \cos B| = \frac{1}{2}$

Solution : **(A)** Consider L.H.S. : $2\cos\frac{A+B}{2}\cos\frac{A-B}{2} = 1$

$\Rightarrow \quad 2\times\cos\frac{\pi}{6}.\cos\frac{A-B}{2}=1$ As $A+B=\frac{\pi}{3}$

$=2\times\frac{\sqrt{3}}{2}.\cos\frac{A-B}{2}=1$ $\cos\frac{A-B}{2}=\frac{1}{\sqrt{3}}$

Now consider the choice '*A*'

$$|\cos A-\cos B|=\left|-2\sin\frac{A+B}{2}\sin\frac{A-B}{2}\right|$$

$=2\sin\frac{A+B}{2}\sin\frac{A-B}{2}$ $=\sin\frac{\pi}{6}.\sqrt{1-\cos^2\frac{A-B}{2}}$ [As $A+B=\frac{\pi}{3}$]

$=2\times\frac{1}{2}\times\sqrt{1-\frac{1}{3}}=\sqrt{\frac{2}{3}}$

Illustration 10: Prove that $\sqrt{3}$ cosec 20° – sec 20° = 4.

Solution : Consider L.H.S. $=\frac{\sqrt{3}}{\sin20°}-\frac{1}{\cos20°}=\frac{\sqrt{3}\cos20°-\sin20°}{\sin20°\cdot\cos20°}$

Multiply and divided by

$=\frac{4\left(\frac{\sqrt{3}}{2}\cos20°-\frac{1}{2}\sin20°\right)}{2\sin20°\cos20°}$ $=4.\frac{(\sin60°\cdot\cos20°-\cos60°\cdot\sin20°)}{\sin40°}$

$=4\frac{\sin(60°-20°)}{\sin40°}=4.\frac{\sin40°}{\sin40°}=4=\text{R.H.S.}$

Student Task

1. If α, β be two distinct angles satisfying the equation $a\cos\theta + b\sin\theta = c$, show that : $\cos(\alpha+\beta)=\frac{a^2-b^2}{a^2+b^2}$.
2. If $\tan(A+B)=m$ and $\tan(A-B)=n$, then find value of $\tan 2A$ and $\tan 2B$.
3. Show that : $\cos A+\cos B+\cos C+\cos(A+B+C)=4\cos\frac{B+C}{2}\cos\frac{C+A}{2}\cos\frac{A+B}{2}$.
4. If $\sec(\theta+\alpha)+\sec(\theta-\alpha)=2\sec\theta$, prove that $\cos\theta=\pm\sqrt{2}\cos\frac{\alpha}{2}$.
5. Prove : $\frac{\sin7x+\sin3x}{\cos7x+\cos3x}=\tan5x$.

3. MULTIPLE AND SUB-MULTIPLE ANGLES

(a) $\sin 2A = 2\sin A\cos A$; $\sin\theta = 2\sin\frac{\theta}{2}\cos\frac{\theta}{2}$

(b) $\cos 2A = \cos^2A - \sin^2A = 2\cos^2A - 1 = 1 - 2\sin^2A$; $2\cos^2\frac{\theta}{2} = 1+\cos\theta$, $2\sin^2\frac{\theta}{2} = 1-\cos\theta$.

(c) $\tan2A=\frac{2\tan A}{1+\tan^2A}$; $\tan\theta=\frac{2\tan\frac{\theta}{2}}{1-\tan^2\frac{\theta}{2}}$

(d) $\sin 2A = \frac{2\tan A}{1+\tan^2 A}$; $\cos 2A = \frac{1-\tan^2 A}{1+\tan^2 A}$

(e) $\sin 3A = 3\sin A - 4\sin^3 A$

(f) $\cos 3A = 4\cos^3 A - 3\cos A$

(g) $\tan 3A = \frac{3\tan A - \tan^3 A}{1-3\tan^2 A}$

General Formulae :

(a) $\tan A \pm \tan B = \frac{\sin(A \pm B)}{\cos A \cos B}$ where, $A, B \neq n\pi + \frac{\pi}{2}$, $n \in \mathbb{Z}$

(b) $\cot A \pm \cot B = \frac{\sin(B \pm A)}{\sin A \sin B}$ where, $A, B \neq n\pi$, $n \in \mathbb{Z}$

(c) $1 \pm \tan A \cdot \tan B = \frac{\cos(A \mp B)}{\cos A \cos B}$ where $A, B \neq n\pi + \frac{\pi}{2}$, $n \in \mathbb{Z}$

(d) $1 \pm \cot A \cdot \cot B = \pm\frac{\cos(A \mp B)}{\sin A \sin B}$ where, $A \,.\, B \neq n\pi$, $n \in \mathbb{Z}$

(e) $\frac{1-\cos\theta}{\sin\theta} = \tan\frac{\theta}{2} = \frac{\sin\theta}{1+\cos\theta}$ where $\theta \neq 2n\pi$, $n \in \mathbb{Z}$

(f) $\frac{1+\cos\theta}{\sin\theta} = \cot\frac{\theta}{2}$, where $\theta \neq (2n+1)\pi$, $n \in \mathbb{Z}$

(g) $\frac{1-\cos\theta}{1+\cos\theta} = \tan^2\frac{\theta}{2}$, where $\theta \neq (2n+1)\pi$, $n \in \mathbb{Z}$

(h) $\frac{1+\cos\theta}{1-\cos\theta} = \cot^2\frac{\theta}{2}$, where $\theta \neq 2n\pi$, $n \in \mathbb{Z}$

(i) $\tan\left(\frac{\pi}{4}+\theta\right) = \frac{1+\tan\theta}{1-\tan\theta} = \frac{\cos\theta+\sin\theta}{\cos\theta-\sin\theta} = \frac{1+\sin 2\theta}{\cos 2\theta}$

(j) $\tan\left(\frac{\pi}{4}-\theta\right) = \frac{1-\tan\theta}{1+\tan\theta} = \frac{\cos\theta-\sin\theta}{\cos\theta+\sin\theta} = \frac{1-\sin 2\theta}{\cos 2\theta}$

Important trigonometric ratios :

(a) $\sin n\pi = 0$; $\cos n\pi = (-1)^n$; $\tan n\pi = 0$, where $n \in \mathbb{Z}$

(b) $\sin 15° \text{ or } \sin\frac{\pi}{12} = \frac{\sqrt{3}-1}{2\sqrt{2}} = \cos 75° \text{ or } \cos\frac{5\pi}{12}$

$\cos 15° \text{ or } \cos\frac{\pi}{12} = \frac{\sqrt{3}+1}{2\sqrt{2}} = \sin 75° \text{ or } \sin\frac{5\pi}{12}$

$\tan 15° = \frac{\sqrt{3}-1}{\sqrt{3}+1} = 2-\sqrt{3} = \cot 75°$; $\tan 75° = \frac{\sqrt{3}+1}{\sqrt{3}-1} = 2+\sqrt{3} = \cot 15°$

(c) $\sin\frac{\pi}{10}$ or $\sin 18° = \frac{\sqrt{5}-1}{4}$ & $\cos 36°$ or $\cos\frac{\pi}{5} = \frac{\sqrt{5}+1}{4}$

(d) $\sin 22\frac{1°}{2} = \frac{1}{2}\left(\sqrt{2-\sqrt{2}}\right)$ (e) $\cos 22\frac{1°}{2} = \frac{1}{2}\left(\sqrt{2+\sqrt{2}}\right)$

(f) $\tan 22\frac{1°}{2} = \sqrt{2}-1$ (g) $\cot 22\frac{1°}{2} = \sqrt{2}+1$

(h) $\cos 18^\circ = \frac{\sqrt{10+2\sqrt{5}}}{4} = \sin 72^\circ$ (i) $\sin 36^\circ = \frac{\sqrt{10-2\sqrt{5}}}{4} = \cos 54^\circ$

(j) $\sin 9^\circ = \frac{\sqrt{3+\sqrt{5}}-\sqrt{5-\sqrt{5}}}{4} = \cos 81^\circ$ (k) $\cos 9^\circ = \frac{\sqrt{3+\sqrt{5}}+\sqrt{5-\sqrt{5}}}{4} = \sin 81^\circ$

(l) $\cos 36^\circ - \cos 72^\circ = \frac{1}{2}$ (m) $\cos 36^\circ \cos 72^\circ = \frac{1}{4}$

(n) $\sin\theta \sin(60^\circ - \theta) \sin(60^\circ + \theta) = 1/4 \sin 3\theta$

(o) $\cos\theta \cos(60^\circ - \theta) \cos(60^\circ + \theta) = 1/4 \cos 3\theta$

(p) $\tan\theta \tan(60^\circ - \theta) \tan(60^\circ + \theta) = \tan 3\theta$

(q) $\sin\frac{A}{2} + \cos\frac{A}{2} = \pm\sqrt{1+\sin A}$ (r) $\sin\frac{A}{2} - \cos\frac{A}{2} = \pm\sqrt{1-\sin A}$

Illustration 11: Show that :

(i) $\frac{1-\cos\theta}{\sin\theta} = \tan\frac{\theta}{2}$ (ii) $\frac{\sin\theta}{1+\cos\theta} = \tan\frac{\theta}{2}$

(iii) $(\cos\theta + \sin\theta)^2 = 1 + \sin 2\theta$ (iv) $(\cos\theta - \sin\theta)^2 = 1 - \sin 2\theta$

(v) $\tan\left(\frac{\pi}{4}+\theta\right) = \frac{1+\tan\theta}{1-\tan\theta} = \frac{\cos\theta+\sin\theta}{\cos\theta-\sin\theta}$ (vi) $\cot\theta - \tan\theta = 2\cot 2\theta$

$\tan\left(\frac{\pi}{4}-\theta\right) = \frac{1-\tan\theta}{1+\tan\theta} = \frac{\cos\theta-\sin\theta}{\cos\theta+\sin\theta}$

Solution : (i) $\text{LHS} = \frac{2\sin^2\theta/2}{2\sin\theta/2\cos\theta/2} = \tan\theta/2$

(ii) $\text{LHS} = \frac{2\sin\theta/2\cos\theta/2}{2\cos^2\theta/2} = \tan\theta/2$

(iii)& (iv) expand LHS to get answer.

(v)expand using tan ($A + B$), tan ($A - B$) and tan $\frac{\pi}{4} = 1$

(vi) $\text{LHS} = \frac{\cos\theta}{\sin\theta} - \frac{\sin\theta}{\cos\theta} = \frac{\cos^2\theta-\sin^2\theta}{\sin\theta\cos\theta} = \frac{2\cos 2\theta}{\sin 2\theta} = 2\cot 2\theta$

Illustration 12: If $2\tan\alpha = 3\tan\beta$ then show that : $\tan(\alpha-\beta) = \frac{\sin 2\beta}{5-\cos 2\beta}$

Solution : We have to express $\tan(\alpha - \beta)$ in terms of β only. Starting with standard result of $\tan(\alpha - \beta)$ and substituting for $\tan\alpha = 3/2 \tan\beta$ in R.H.S. we have :

$\Rightarrow \tan(\alpha-\beta) = \frac{\tan\alpha-\tan\beta}{1+\tan\alpha\tan\beta} = \frac{3/2\tan\beta-\tan\beta}{1+3/2\tan^2\beta}$

$\Rightarrow \tan(\alpha-\beta) = \frac{\tan\beta}{2+3\tan^2\beta} = \frac{\sin\beta\cos\beta}{2\cos^2\beta+3\sin^2\beta}$

$= \frac{2\sin\beta\cos\beta}{4\cos^2\beta+6\sin^2\beta} = \frac{\sin 2\beta}{2(1+\cos 2\beta)+3(1-\cos 2\beta)} \Rightarrow \tan(\alpha-\beta) = \frac{\sin 2\beta}{5-\cos 2\beta}$

Range of trigonometric expression :

$E = a \sin\theta + b \cos\theta$

$E = \sqrt{a^2+b^2} \sin(\theta + \alpha)$, where $\tan\alpha = \frac{b}{a}$ $= \sqrt{a^2+b^2} \cos(\theta - \beta)$, where $\tan\beta = \frac{a}{b}$

Hence for any real value of $\theta, -\sqrt{a^2+b^2} \le E \le \sqrt{a^2+b^2}$

Some Important formulae

1. For any natural number n,
   (a) $\sin nA = {}^nC_1(\cos A)^{n-1}.\sin A - {}^nC_3(\cos A)^{n-3}(\sin A)^3 + {}^nC_5(\cos A)^{n-5}(\sin A)^5 - \ldots\ldots$
   (b) $\cos nA = {}^nC_0(\cos A)^n - {}^nC_2 \cdot (\cos A)^{n-2}\sin A + {}^nC_4 \cdot (\cos A)^{n-4}\sin^4 A - \ldots\ldots$
   (c) $\tan nA = \frac{{}^nC_1\tan A - {}^nC_3\tan^3 A + {}^nC_5\tan^5 A - \ldots\ldots}{1 - {}^nC_2\tan^2 + {}^nC_4\tan^4 A - \ldots\ldots\ldots}$
2. $\sqrt{2+\sqrt{2+\sqrt{2+\sqrt{2+\ldots\ldots+\sqrt{2+2\cos 2^n A}}}}} = 2\cos A, \ \forall\ n \in \mathbb{N}$, where there are n square root sign on L.H.S.
3. $\tan A + 2\tan 2A + 2^2\tan 2^2A + \ldots\ldots + 2^n\tan 2^nA = \cot A - 2^{n+1}\cot 2^{n+1}A \ \forall\ n \in \mathbb{N}$.

Illustration 13: Find set of all possible values of α in $[-\pi, \pi]$ such that $\sqrt{\frac{1-\sin\alpha}{1+\sin\alpha}}$ is equal to ($\sec\alpha - \tan\alpha$).

Solutoin : Clearly $\alpha \neq \pm\frac{\pi}{2}$

as, $\sec\alpha - \tan\alpha = \frac{1-\sin\alpha}{\cos\alpha}$ .....(i)

and $\sqrt{\frac{1-\sin\alpha}{1+\sin\alpha}} = \sqrt{\frac{(1-\sin\alpha)^2}{\cos^2\alpha}} = \left|\frac{1-\sin\alpha}{\cos\alpha}\right| = \frac{1-\sin\alpha}{|\cos\alpha|}$ .....(ii) As [$1 - \sin\alpha$ is always + ve]

From (i) and (ii) two expressions are equal only if $\alpha > 0$, i.e., $-\pi/2 < \alpha < \pi/2$

$\therefore \sqrt{\frac{1-\sin\alpha}{1+\sin\alpha}}$ and $\sec\alpha - \tan\alpha$ are equal where $\alpha = \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

Illustration 14: Find the value of, $\frac{\sqrt{1+\sin 2A} + \sqrt{1-\sin 2A}}{\sqrt{1+\sin 2A} - \sqrt{1-\sin 2A}}$; where $|\tan A| < 1$ and $|A|$ is acute.

Solution : As $|\tan A| < 1 \Rightarrow \frac{-\pi}{4} < A < \frac{\pi}{4}$ [As $-\pi/4 < A < \pi/4$ and in this interval $\cos A > \sin A$]

$\Rightarrow \frac{\sqrt{(\cos A + \sin A)^2} + \sqrt{(\cos A - \sin A)^2}}{\sqrt{(\cos A + \sin A)^2} - \sqrt{(\cos A - \sin A)^2}} \Rightarrow \frac{2\cos A}{2\sin A} \Rightarrow \cot A$

$\Rightarrow \frac{|\cos A + \sin A| + |\cos A - \sin A|}{|\cos A + \sin A| - |\cos A - \sin A|}$ $\therefore \frac{\sqrt{1+\sin 2A} + \sqrt{1-\sin 2A}}{\sqrt{1+\sin 2A} - \sqrt{1-\sin 2A}} = \cot A$

$\Rightarrow \frac{\cos A + \sin A + \cos A - \sin A}{\cos A + \sin A - (\cos A - \sin A)}$ when $|\tan A| < 1$ and $|A|$ is acute.

Illustration 15: If $\frac{x}{\cos\theta} = \frac{y}{\cos\left(\theta + \frac{2\pi}{3}\right)} = \frac{z}{\cos\left(\theta - \frac{2\pi}{3}\right)}$, then $x + y + z$ is equal to :

(A) -1 (B) 1 (C) 0 (D) None of these

Solution : (C) Given $\frac{x}{\cos\theta}=\frac{y}{\cos\left(\theta+\frac{2\pi}{3}\right)}=\frac{z}{\cos\left(\theta-\frac{2\pi}{3}\right)}=\lambda$ (say)

$$\Rightarrow x+y+z=\lambda\left\{\cos\theta+\cos\left(\theta+\frac{2\pi}{3}\right)+\cos\left(\theta-\frac{2\pi}{3}\right)\right\}=\lambda\left\{\cos\theta+2\cos\theta\cos\frac{2\pi}{3}\right\}=0$$

Illustration 16: Match the column :

Column-I		Column-II	
(A)	$\cos^2 52\frac{1^\circ}{2}-\sin^2 22\frac{1^\circ}{2}$	p.	1
(B)	$\sin^2\frac{3\pi}{5}+\cos^2\frac{4\pi}{5}$	q.	$\frac{3-\sqrt{3}}{4\sqrt{2}}$
(C)	sin 24° + cos 6°	r.	$\frac{3}{4}$
(D)	$\sin^2 50^\circ+\cos^2 130^\circ$	s.	$\frac{\sqrt{15}+\sqrt{3}}{4}$

Solution : A → (q); B → (r); C → (s); D → (p)

(i) $\cos^2 52\frac{1}{2}-\sin^2 22\frac{1}{2}=\cos 75^\circ\cos 30^\circ=\frac{\sqrt{3}-1}{2\sqrt{2}}\times\frac{\sqrt{3}}{2}=\frac{3-\sqrt{3}}{4\sqrt{2}}$

(ii) $\sin^2\frac{3\pi}{5}+\cos^2\frac{4\pi}{5}=\left(\frac{\sqrt{5}-1}{4}\right)^2+\left(\frac{\sqrt{5}+1}{4}\right)^2=\frac{3}{4}$

(iii) sin 24° + cos 6° = 2 sin 54 cos 30 = $\frac{\sqrt{15}+\sqrt{3}}{4}$

(iv) $\sin^2 50^\circ+\cos^2 130^\circ=1$

EXERCISE

Q.1 Prove that $\cos^2\theta+\sec^2\theta$ can never be less than 2 for all values of θ.

Q.2 1f 7 (cosec θ – 1) = 3 cotθ, show that 3 (cosec θ + 1) = 7 cot θ.

Q.3 Show that : $\frac{1+\cos ec^2\alpha\tan^2\theta}{1+\cos ec^2\beta\tan^2\theta}=\frac{1+\cot^2\alpha\sin^2\theta}{1+\cot^2\beta\sin^2\theta}$.

Q.4 If $\frac{\cos A}{\cos B}=n$, $\frac{\sin A}{\sin B}=m$, show that :

(a) $(m^2-n^2)\sin^2 B=1-n^2$ (b) $(n^2-m^2)\sin^2 A=m^2(n^2-1)$

Q.5 If $\tan\theta=\frac{x\sin\phi}{1-x\cos\phi}$ and tan $\phi=\frac{y\sin\theta}{1-y\cos\theta}$. Show that $x\sin\phi=y\sin\theta$.

Q.6 If $p\tan\alpha=\alpha\tan p\alpha$, show that $\frac{\sin^2 p\alpha}{\sin^2\alpha}=\frac{p^2}{1+(p^2-1)\sin^2\alpha}$.

Q.7 If $x=y\tan\theta-x\tan^2\theta$ and $2x\sin\alpha=y\cos\alpha-2x\cos\alpha\tan\theta$. Show that $\cot\alpha=\tan 2\theta$.

Q.8 Eliminate θ from the following :

(i) $a\cot\theta+b\operatorname{cosec}\theta=x^2$ and $b\cot\theta+a\operatorname{cosec}\theta=y^2$

(ii) $\cos\theta+\sin\theta=m$ and $\sec\theta+\operatorname{cosec}\theta=n$.

Q.9 If $\sin\alpha\sin\beta-\cos\alpha\cos\beta+1=0$, prove that $1+\cot\alpha\tan\beta=0$.

Q.10 If $\alpha + \beta = \theta$ and $\sin\alpha = k \sin\beta$, show that $\tan\alpha = \dfrac{k\sin\theta}{1+k\cos\theta}$ and $\tan\beta = \dfrac{\sin\theta}{k+\cos\theta}$.

Q.11 If $\tan A \tan B = \sqrt{\dfrac{a-b}{a+b}}$, show that : $(a - b\cos 2A)(a - b\cos 2B) = a^2 - b^2$.

Q.12 If $\sin x + \sin y = a$, $\cos x + \cos y = b$, prove that $\sin(x + y) = \dfrac{2ab}{a^2+b^2}$.

Q.13 Show that :

(i) $\sec\left(\dfrac{\pi}{4}+\theta\right)\sec\left(\dfrac{\pi}{4}-\theta\right) = 2\sec 2\theta$ (ii) $2\cos\theta = \sqrt{2+\sqrt{2+2\cos 4\theta}}$

Q.14 Prove the following :

(i) $\sin 10° \sin 50° \sin 70° = 1/8$ (ii) $\sin 20° \sin 40° \sin 80° = \sqrt{3}/8$

(iii) $\cos 15° - \sin 15° = 1/\sqrt{2}$ (iv) $\dfrac{\cos 9° + \sin 9°}{\cos 9° - \sin 9°} = \tan 54°$

(v) $4 \sin \pi/10 \sin 13\pi/10 = -1$.

Q.15 If $\sec\theta = \sqrt{2}$ and $\dfrac{3\pi}{2} < \theta < 2\pi$, find the value of $\dfrac{1+\tan\theta+\operatorname{cosec}\theta}{1+\cos\theta-\operatorname{cosec}\theta}$.

4. MAXIMUM & MINIMUM VALUES OF TRIGONOMETRICAL EXPRESSIONS

In this section, we shall discuss problems on finding the maximum and minimum values of various trigonometrical expression.

As discussed that $-1 \le \sin x \le 1$, $-1 \le \cos x \le 1$, $-\infty < \tan x < \infty$, $|\sec x| \ge 1$ and $|\operatorname{cosec} x| \ge 1$.

Consider the expression $a\cos\theta \pm b\sin\theta$, where θ is a variable.

Let $y = a\cos\theta \pm b\sin\theta$

Further, let $a = r\cos\alpha$ and $b = r\sin\alpha$. Then, $r = \sqrt{a^2+b^2}$ and $\tan\alpha = \dfrac{b}{a}$

$\therefore$ $y = r\cos\alpha\cos\theta \pm r\sin\alpha\sin\theta$

$\Rightarrow$ $y = r\cos(\theta \mp \alpha)$

We know that $-1 \le \cos(\theta \mp \alpha) \le 1$ for all θ

$\Rightarrow$ $-r \le r\cos(\theta \mp \alpha) \le r$ for all θ

$\Rightarrow$ $-\sqrt{a^2+b^2} \le y \le \sqrt{a^2+b^2}$ for all θ

$\Rightarrow$ $-\sqrt{a^2+b^2} \le a\cos\theta \pm b\sin\theta \le \sqrt{a^2+b^2}$ for all θ

If follows from the above discussion that $-\sqrt{a^2+b^2}$ and $\sqrt{a^2+b^2}$ are minimum and maximum values of $a\cos\theta \pm b\sin\theta$ for varying values of θ.

Note :- Above results can also be derived taking $a = r\sin\alpha$ and $b = r\cos\alpha$.

Important : The maximum and minimum values of $a\cos\theta \pm b\sin\theta + c$ are

$c + \sqrt{a^2+b^2}$ and $c - \sqrt{a^2+b^2}$, respectively.

i.e., $c - \sqrt{a^2+b^2} \le a\cos\theta \pm b\sin\theta + c \le c + \sqrt{a^2+b^2}$.

Illustration 17: Find the maximum and minimum values of :

(i)$\sin\theta + \cos\theta$ (ii) $\sqrt{3}\sin\theta - \cos\theta$ (iii) $5\sin\theta + 12\cos\theta + 7$

Solution : Given expressions are in the form of :

$a\sin\theta + b\cos\theta$

Express this in terms of one T-ratio by dividing and multiplying by $(a^2 + b^2)^{1/2}$

(i) $\sin\theta + \cos\theta = 1 \cdot \sin\theta + 1 \cdot \cos\theta$

$$= \sqrt{2}\left(\frac{1}{\sqrt{2}}\sin\theta + \frac{1}{\sqrt{2}}\cos\theta\right)$$

$$= \sqrt{2}\left(\sin\theta\cos\frac{\pi}{4} + \cos\theta\sin\frac{\pi}{4}\right) = \sqrt{2}\sin\left(\theta + \frac{\pi}{4}\right)$$

Now sine of an angle must be between –1 and 1.

$$\Rightarrow \quad -1 \le \sin\left(\theta + \frac{\pi}{4}\right) \le 1 \quad \Rightarrow \quad -\sqrt{2} \le \sqrt{2}\sin\left(\theta + \frac{\pi}{4}\right) \le \sqrt{2}$$

So maximum value of $\sin\theta + \cos\theta$ is $\sqrt{2}$ and minimum value of $\sin\theta + \cos\theta$ is $-\sqrt{2}$.

(ii) $$\sqrt{3}\sin\theta - \cos\theta = 2\left(\frac{\sqrt{3}}{2}\sin\theta - \frac{1}{2}\cos\theta\right)$$

[Multiplying and divide by $\sqrt{(\sqrt{3})^2 + (1)^2}$]

$$= 2\left(\sin\theta\cos\frac{\pi}{6} - \cos\theta\sin\frac{\pi}{6}\right)$$

$$= 2\sin\left(\theta - \frac{\pi}{6}\right)$$

as $$-1 \le \sin\left(\theta - \frac{\pi}{6}\right) \le 1 \quad \Rightarrow \quad -2 \le 2\sin\left(\theta - \frac{\pi}{6}\right) \le 2$$

So maximum value is 2 and minimum value is –2.

(iii) Consider $5\sin\theta + 12\cos\theta = 13\ [5/13\sin\theta + 12/13\cos\theta]$

[Multiplying and divide by $\sqrt{5^2 + 12^2}$]

Construct a triangle with sides, 5, 12, 13. If α is an angle of triangle,

then $\cos\alpha = 5/13$, $\sin\alpha = 12/13$,

$5\sin\theta + 12\cos\theta = 13\ [\sin\theta\cos\alpha + \cos\theta\sin\alpha]$

$5\sin\theta + 12\cos\theta + 7 = 13\ [\sin(\theta + \alpha)] + 7$

as $-1 \le \sin(\theta + \alpha) \le 1$

$-13 + 7 \le 13\sin(\theta + \alpha) + 7 \le 13 + 7$

So maximum value is 20 and minimum value is –6.

Illustration 18 (i) Prove that $5\cos x + 3\cos(x + \pi/3) + 3$ lies between –4 and 10.

(ii)Show that whatever be the value of θ, the expression $a\sin^2\theta + b\sin\theta\cos\theta + \cos^2\theta$ lies between $\frac{a+c}{2} - \frac{1}{2}\sqrt{b^2 + (a-c)^2}$ and $\frac{a+c}{2} + \frac{1}{2}\sqrt{b^2 + (a-c)^2}$.

Solution : (i) $$5\cos x + 3\left(\cos x\cos\frac{\pi}{3} - \sin x\sin\frac{\pi}{3}\right) + 3$$

$$= \cos x\left(5 + \frac{3}{2}\right) - \sin x.\frac{3\sqrt{3}}{2} + 3$$

$$= \frac{13}{2}\cos x - \frac{3\sqrt{3}}{2}\sin x + 3 = \sqrt{\frac{169}{4} + \frac{27}{4}}\left\{\frac{13/2}{\sqrt{\frac{169}{4} + \frac{27}{4}}}\cos x - \frac{3\sqrt{3}/2}{\sqrt{\frac{169}{4} + \frac{27}{4}}}\sin x\right\} + 3$$

[Multiplying and divide by $\sqrt{\left(\frac{13}{2}\right)^2+\left(\frac{3\sqrt{3}}{2}\right)^2}$]

$= 7(\cos\alpha\cos x - \sin\alpha\sin x) + 3$

$= 7\cos(\alpha + x) + 3$

As $-1 \le \cos(\alpha + x) \le 1$

i.e., $-7 + 3 \le 7\cos(\alpha + x) + 3 \le 7 + 3$

i.e., $-4 \le 7\cos(\alpha + x) + 3 \le 10$

(ii) Let $f(\theta) = a\sin^2\theta + b\sin\theta\cos\theta + c\cos^2\theta$

$$=\frac{a(1-\cos2\theta)}{2}+\frac{b}{2}\sin2\theta+\frac{c(1+\cos2\theta)}{2}=\left(\frac{c}{2}-\frac{a}{2}\right)\cos2\theta+\frac{b}{2}\sin2\theta+\frac{a}{2}+\frac{c}{2}$$

$$=\frac{1}{2}(a+c)+\frac{1}{2}\left(\sqrt{b^2+(a-c)^2}\right)(\sin2\theta\cos\alpha-\cos2\theta\sin\alpha)=\frac{1}{2}(a+c)+\frac{\sqrt{b^2(a-c)^2}}{2}\sin(2\theta-\alpha)$$

As $-1 \le \sin(2\theta - \alpha) \le 1$

$$\therefore \quad \left\{\frac{a+c}{2}\right\}-\frac{\sqrt{b^2+(a-c)^2}}{2}\le f(\theta)\le\left\{\frac{a+c}{2}\right\}+\frac{\sqrt{b^2+(a-c)^2}}{2}$$

Illustration 19: Find the maximum and minimum values of $\sin^6 x + \cos^6 x$.

Solution : If the function contains only even powers of sine and cosine of the same angle, use the following properties

(i) $\sin^2 x + \cos^2 x = 1$ (ii) $2\sin x\cos x = \sin 2x$.

Let $f(x) = \sin^6 x + \cos^6 x$

$= (\sin^2 x)^3 + (\cos^2 x)^3$

$= (\sin^2 x + \cos^2 x)^3 - 3\sin^2 x\cos^2 x(\sin^2 x + \cos^2 x)$

$= 1 - 3\sin^2 x\cos^2 x = 1 - \frac{3}{4}(\sin 2x)^2$

When $(\sin 2x)^2$ is minimum i.e. 0, then y will be maximum and when $(\sin 2x)^2$ is maximum i.e. 1, then y will be minimum.

Hence $f(x)_{max} = 1-\frac{3}{4}\times0 = 1$ and $f(x)_{min} = 1-\frac{3}{4}\times1 = \frac{1}{4}$.

Student Task

1. Show that minimum and the maximum values of expression :

   $\cos\theta(\sin\theta+\sqrt{\sin^2\theta+\sin^2\alpha})$ are $-\sqrt{1+\sin^2\alpha}$ and $\sqrt{1+\sin^2\alpha}$, where α is a constant.

2. Find the maximum and minimum value of the following :

   (i) $3\sin 2x + 4\cos 2x + 3$ (ii) $6\sin x\cos x + 4\cos 2x$ (iii) $\frac{\sin 3x}{\sin x}, x \ne \pi,$

3. Find the maximum value of $1+\sin\left(\frac{\pi}{4}+\theta\right)+2\cos\left(\frac{\pi}{4}-\theta\right)$ for all real values of θ.

4. Find the minimum value of $\cos 2\theta + \cos\theta$ for all real values of θ.

5. If $A = \cos^2\theta + \sin^4\theta$, then for all values of θ,

   (A) $1 \le A \le 2$ (B) $\frac{13}{16} \le A \le 1$ (C) $\frac{3}{4} \le A \le \frac{13}{16}$ (D) $\frac{3}{4} \le A \le 1$

5. TRIGONOMETRIC SERIES IN WHICH ANGLES ARE IN A.P.

TYPE-1 Problems based on finding the sum of series of sines or cosines whose angles are in A.P.
Following two results are very useful in solving such problems.

Result I : $\sin\alpha + \sin(\alpha+\beta) + \sin(\alpha+2\beta) + ... + \sin(\alpha+\overline{n-1}\,\beta) = \dfrac{\sin\frac{n\beta}{2}}{\sin\frac{\beta}{2}}.\sin\left[\alpha+(n-1)\dfrac{\beta}{2}\right]$

Result II : $\cos\alpha + \cos(\alpha+\beta) + \cos(\alpha+2\beta) + ... + \cos(\alpha+\overline{n-1}\,\beta) = \dfrac{\sin\frac{n\beta}{2}}{\sin\frac{\beta}{2}}.\cos\left[\alpha+(n-1)\dfrac{\beta}{2}\right]$

Proof for Result 1 :

Let $S = \sin\alpha + \sin(\alpha+\beta) + \sin(\alpha+2\beta) + ... + \sin(\alpha+\overline{n-1}\,\beta)$

Here angles ae in A.P. and common difference of angles = β

Note :- In a series of sine and cosine whenever angles are in A.P. and power of sine and cosine is one

then we multiply each term by $2\sin\left(\dfrac{\text{common difference of angles}}{2}\right)$, then express each term as a differene of two terms and add.

So, multiply both sides by $2\sin\frac{\beta}{2}$, we get :

$$S.2\sin\frac{\beta}{2} = 2\sin\alpha\sin\frac{\beta}{2} + 2\sin(\alpha+\beta).\sin\frac{\beta}{2} + ... + 2\sin(\alpha+\overline{n-1}\,\beta)\sin\frac{\beta}{2} \quad .....(i)$$

Now, first term above R.H.S. can be written as $2\sin\alpha.\sin\frac{\beta}{2} = \cos\left(\alpha-\frac{\beta}{2}\right) - \cos\left(\alpha+\frac{\beta}{2}\right)$

Similarly, the second term R.H.S. can be written as $2\sin(\alpha+\beta)\sin\frac{\beta}{2} = \cos\left(\alpha+\frac{\beta}{2}\right) - \cos\left(\alpha+\frac{3\beta}{2}\right)$

... ...
... ...

and also the last term of R.H.S. can be written as

$$2\sin(\alpha+\overline{n-1}\,\beta).\sin\frac{\beta}{2} = \cos\left[\alpha+(2n-3)\frac{\beta}{2}\right] - \cos\left[\alpha+(2n-1)\frac{\beta}{2}\right]$$

Now, adding above all terms, we get R.H. S. of (1) as $\cos\left(\alpha-\frac{\beta}{2}\right) - \cos\left[\alpha+(2n-1)\frac{\beta}{2}\right]$.

From (i), we get : $2\sin\frac{\beta}{2}.S = 2\sin\left[\alpha+(n-1)\frac{\beta}{2}\right].\sin\frac{n\beta}{2} \Rightarrow S = \dfrac{\sin\frac{n\beta}{2}}{\sin\frac{\beta}{2}}.\sin\left[\alpha+(n-1)\frac{\beta}{2}\right]$

Students are advised to prove the result – II themselves.

***Illustration 20*:** Show that : $\cos 2\pi/7 + \cos 4\pi/7 + \cos 6\pi/7 = -1/2$.

Solution : $\text{L.H.S.} = \dfrac{2\sin\frac{\pi}{7}\left(\cos\frac{2\pi}{7}+\cos\frac{4\pi}{7}+\cos\frac{6\pi}{7}\right)}{2\sin\frac{\pi}{7}}$

$$= \frac{1}{2\sin\frac{\pi}{7}}\left[\left(\sin\frac{3\pi}{7}-\sin\frac{\pi}{7}\right)+\left(\sin\frac{5\pi}{7}-\sin\frac{3\pi}{7}\right)+\left(\sin\frac{7\pi}{7}-\sin\frac{5\pi}{7}\right)\right] = \frac{\sin\pi-\sin\frac{\pi}{7}}{2\sin\frac{\pi}{7}} = -\frac{1}{2}$$

Alternative Method :

We can also use the relation :

$$\cos a + \cos(a+d) + \ldots\ldots \cos(a+\overline{n-1}\,d) = \frac{\sin(nd/2)}{\sin(d/2)}\cos\left(\frac{2a+\overline{n-1}\,d}{2}\right)$$

[where d is common difference of AP]

$$\Rightarrow \quad \text{L.H.S.} = \frac{\sin 3\left(\frac{2\pi/7}{2}\right)}{\sin\frac{2\pi/7}{2}}\cos\left(\frac{\frac{4\pi}{7}+2\left(\frac{2\pi}{7}\right)}{2}\right)$$

$$= \frac{\sin\frac{3\pi}{7}.\cos\frac{4\pi}{7}}{\sin\frac{\pi}{7}} = \frac{\cos\left(\frac{\pi}{2}-\frac{3\pi}{7}\right)\sin\left(\frac{\pi}{2}-\frac{4\pi}{7}\right)}{\sin\frac{\pi}{7}} = \frac{\cos\frac{\pi}{14}\sin\left(\frac{-\pi}{14}\right)}{\sin\frac{\pi}{7}} = \frac{-1}{2}\frac{\sin\frac{\pi}{7}}{\sin\frac{\pi}{7}} = \frac{-1}{2}$$

TYPE - 2 If angles are in A.P. and sum of the first and the last angles is π or $\pi/2$. Then we will use the following working rule and power of sine and cosine may or may not be one.

Working Rule :

If angles are in A.P. and sum of the first and last angles in $\pi/2$ or π etc., then group equidistant terms from both ends, express last angle in terms of the first in each group and simplify.

Illustration 21: Prove that : $\cos^4\frac{\pi}{4}+\cos^4\frac{3\pi}{8}+\cos^4\frac{5\pi}{8}+\cos^4\frac{7\pi}{8}=\frac{3}{2}$.

Solution : $\text{L.H.S.} = \left(\cos^4\frac{\pi}{8}+\cos^4\frac{7\pi}{8}\right)+\left(\cos^4\frac{3\pi}{8}+\cos^4\frac{5\pi}{8}\right)$

$$= \left[\cos^4\frac{\pi}{8}+\cos^4\left(\pi-\frac{\pi}{8}\right)\right]+\left[\cos^4\frac{3\pi}{8}+\cos^4\left(\pi-\frac{3\pi}{8}\right)\right] = 2\cos^4\frac{\pi}{8}+2\cos^4\frac{3\pi}{8}$$

$$= 2\left[\left(\cos^2\frac{\pi}{8}\right)^2+\left(\cos^2\frac{3\pi}{8}\right)^2\right] = 2\left[\left(\frac{1+\cos\frac{\pi}{4}}{2}\right)^2+\left(\frac{1+\cos\frac{3\pi}{4}}{2}\right)^2\right]$$

$$= \frac{1}{2}\left[\left(1+\frac{1}{\sqrt{2}}\right)^2\left(1-\frac{1}{\sqrt{2}}\right)^2\right] = \frac{1}{2}\left[1+\frac{1}{2}+1+\frac{1}{2}\right] = \frac{3}{2}$$

TYPE - 3 If angles are in G.P. having common ratio 2 or 1/2, terms of sine or cosine are in power one and in product.

Illustration 22: $\cos A \cos 2A \cos 4A \cos 8A \ldots\ldots \cos 2^{n-1}A = \frac{1}{2^n \sin A}\sin(2^n A)$.

Solution : Multiply above and below by $2^n \sin A$

$$\therefore \quad \text{L.H.S.} = \frac{2^{n-1}}{2^n \sin A}\ [2 \sin A \cos A \cos 2A \cos 4A \ldots\ldots \cos 2^{n-1}A]$$

$= \frac{2^{n-2}}{2^n \sin A}$ [2 sin 2A cos 2A cos 4A ..... cos $2^{n-1} A$]

$= \frac{1}{2^n \sin A}[2\sin 2^{n-1} A \cos 2^{n-1} A]$ $= \frac{1}{2^n \sin A}\sin(2 . 2^{n-1} A) = \frac{\sin(2^n A)}{2^n \sin A}$

***Illustration 23*:** (i) cos 20° cos 40° cos 60° cos 80° = 1/16.

(ii) sin 10° sin 30° sin 50° sin 70° = 1/16.

Solution : (i)In given expression : $\cos 60° = \frac{1}{2}$ and $n = 3$

$\Rightarrow \text{L.H.S.} = \frac{1}{2}\left[\frac{1}{2^3 \sin 2\theta}\right]\sin(2^3 . 20°) = \frac{1}{16} \times \frac{1}{\sin 20°} . \sin 160° = \frac{1}{16 \sin 20°} . \sin(180 - 20°)$

$= \frac{1}{16}\left(\frac{\sin 20°}{\sin 20°}\right) = \frac{1}{16}$

(ii) By complementry rule

sin θ = cos (90 – θ)

The given expression reduces to cos 20° cos 40° cos 60° cos 80°

Which is same as above illustration.

TYPE - 4 If angle are in A.P., terms in sine or cosine having power one, and in product and sum of the first and last angles is not $\pi/2$ or π etc. then

Working Rule :

(i) Change the last angle using formula for $\pi - \theta$ etc.

(ii) Rearrange the terms in ascending order of angles.

(iii) Group the terms in two parts : One part consisting of terms having angles in G.P. and the other part consisting of remaining parts.

(iv) Simplify

Illustration 24: Prove that : $2\cos\frac{\pi}{13}\cos\frac{9\pi}{13} + \cos\frac{3\pi}{13} + \cos\frac{5\pi}{13} = 0$

Solution : $\text{L.H.S.} = 2\cos\frac{\pi}{13}\cos\frac{9\pi}{13} + 2\cos\frac{4\pi}{13}\cos\frac{\pi}{13}$

$= 2\cos\frac{\pi}{13}\left[\cos\frac{9\pi}{13} + \cos\frac{4\pi}{13}\right]$ $= 2\cos\frac{\pi}{13} . 2\cos\frac{\pi}{2}\cos\frac{5\pi}{26} = 0$ [as cos ($\pi/2$) = 0]

Illustration 25: $\sin^2 12° + \sin^2 21° + \sin^2 39° + \sin^2 48° = 1 + \sin^2 9° + \sin^2 18°$

Solution : $\sin^2 12° + \sin^2 21° + \sin^2 39° + \sin^2 48° - \sin^2 9° - \sin^2 18° = 1$

$\sin^2 12° + \sin^2 21^{\circ} + (\sin^2 39° - \sin^2 9°) + (\sin^2 48° - \sin^2 18°) = 1$

Multiplying both sides by 2,

$2 \sin^2 12° + 2 \sin^2 21° + 2 (\sin^2 39° - \sin^2 9°) + 2 (\sin^2 48° - \sin^2 18°) = 2$

By using $2 \sin^2 \theta = 1 - \cos 2\theta$ and $\sin^2 \theta - \sin^2 \phi = \sin(\theta + \phi) . \sin(\theta - \phi)$

$\Rightarrow 1 - \cos 24° + 1 - \cos 42° + 2 \sin 48° \sin 30° + 2 \sin 66° \sin 30° = 2$

As $\sin 30° = \frac{1}{2}$ and by complementary rule sin 66° = cos 24° ; sin 48° = cos 42°.

Illustration 26: Show that : sin 6° sin 42° sin 66° sin 78° = 1/16.

Solution : Note that (66 + 6)/2 = 36 and (66 – 6)/2 = 30. Hence sin 6° and sin 66° should be combined.

L.H.S. = 1/4 [2 sin 6° sin 66°] [2 sin 42° sin 78°]

= 1/4 [cos (6° – 66°) – cos (6° + 66°)][cos (42° – 78°) – cos (42° + 78°)]

= 1/4 [cos 60° – cos 72°][cos 36° – cos 120°]

Substituting the values, we get : L.H.S. = $\frac{1}{4}\left(\frac{1}{2} - \frac{\sqrt{5}-1}{4}\right)\left(\frac{\sqrt{5}+1}{4} + \frac{1}{2}\right)$

$$= \frac{1}{4}\left(\frac{2-\sqrt{5}+1}{4}\right)\left(\frac{\sqrt{5}+1+2}{4}\right) = \frac{1}{64}\left(3-\sqrt{5}\right)\left(3+\sqrt{5}\right) = \frac{1}{16} = \text{R.H.S.}$$

Illustration 27: Prove that : sin 20 sin 40 sin 80 = $\sqrt{3}/8$ (where angles are in degrees)

Solution : Consider L.H.S. = sin 20 sin 40 sin 80 = sin 20 sin (60 – 20) sin (60 + 20)

$$= \frac{1}{4}\sin 3(20) = \frac{1}{4}\sin 60 = \frac{\sqrt{3}}{8}$$

***Illustration 28*:** Prove that :

(i) $\sin\frac{\pi}{14}\sin\frac{3\pi}{14}\sin\frac{5\pi}{14}\sin\frac{7\pi}{14} = \frac{1}{8}$ (ii) $\sin\frac{\pi}{14}\sin\frac{3\pi}{14}\sin\frac{5\pi}{14}\sin\frac{7\pi}{14}\sin\frac{9\pi}{14}\sin\frac{11\pi}{14} = \frac{1}{64}$

Solution : (i) $\sin\frac{\pi}{14} = \sin\left(\frac{\pi}{2}-\frac{6\pi}{14}\right) = \cos\frac{6\pi}{14} = \cos\left(\pi-\frac{8\pi}{14}\right) = -\cos\frac{8\pi}{14}$

$\sin\frac{3\pi}{14} = \sin\left(\frac{\pi}{2}-\frac{4\pi}{14}\right) = \cos\frac{4\pi}{14}$; $\sin\frac{5\pi}{14} = \sin\left(\frac{\pi}{2}-\frac{2\pi}{14}\right) = \cos\frac{2\pi}{14}$

$\therefore$ L.H.S. $= -\cos\frac{2\pi}{14}\cos\frac{4\pi}{14}\cos\frac{8\pi}{14}$

$= -\frac{1}{2^3\sin A}.\sin(2^3 A)$ [where $A = \frac{2\pi}{14}$]

$= -\frac{1}{8\sin\frac{\pi}{7}}\sin\frac{8\pi}{7}$ $= -\frac{1}{8\sin\frac{\pi}{7}}\sin\left(\pi+\frac{\pi}{7}\right) = -\frac{1}{8}(-1) = \frac{1}{8}$ [sin (π + θ) = – sin θ]

(ii) In the given expression $\sin\frac{7\pi}{14} = \sin\frac{\pi}{2} = 1$ and sum of first and last angle is π

$\Rightarrow$ Terms are equidistant from beginning and end.

$\Rightarrow$ $\left(\sin\frac{\pi}{14}\sin\frac{3\pi}{14}\sin\frac{5\pi}{14}\sin\frac{7\pi}{14}\right)^2 = \left(\frac{1}{8}\right)^2 = \frac{1}{64}.$

Illustration 29: Prove that : (i) $\left(1+\cos\frac{\pi}{10}\right)\left(1+\cos\frac{3\pi}{10}\right)\left(1+\cos\frac{7\pi}{10}\right)\left(1+\cos\frac{9\pi}{10}\right) = \frac{1}{16}$

(ii) $\left(1+\cos\frac{\pi}{8}\right)\left(1+\cos\frac{3\pi}{8}\right)\left(1+\cos\frac{5\pi}{8}\right)\left(1+\cos\frac{7\pi}{8}\right) = \frac{1}{8}$

(iii) cos 60° cos 36° cos 42° cos 78° = 1/16.

Solution : (i)By supplementary rule : cos (π – A) = – cos A

$$\text{L.H.S.} = \left(1+\cos\frac{\pi}{10}\right)\left(1+\cos\frac{3\pi}{10}\right)\left(1-\cos\frac{3\pi}{10}\right)\left(1-\cos\frac{\pi}{10}\right)$$

$$= \left(1-\cos^2\frac{\pi}{10}\right)\left(1-\cos^2\frac{3\pi}{10}\right) = \sin^2 18°\sin^2 54°$$

$$= \left(\frac{\sqrt{5}-1}{4}.\frac{\sqrt{5}+1}{4}\right)^2\frac{1}{16}$$

(ii) $\cos\frac{3\pi}{8}=\cos\left\{\frac{\pi}{2}-\frac{\pi}{8}\right\}=\sin\frac{\pi}{8}.$

$\cos\frac{5\pi}{8}=\cos\left\{\frac{\pi}{2}+\frac{\pi}{8}\right\}=-\sin\frac{\pi}{8};$ $\cos\frac{7\pi}{8}=\cos\left\{\pi-\frac{\pi}{8}\right\}=-\cos\frac{\pi}{8}$

$\therefore$ L.H.S. = $(1+\cos\pi/8)(1+\sin\pi/8)(1-\sin\pi/8)(1-\cos\pi/8)$
$=(1-\cos^2\pi/8)(1-\sin^2\pi/8)$

$=\frac{1}{4}\left[\sin\frac{\pi}{4}\right]^2=\frac{1}{4}\left\{\frac{1}{\sqrt{2}}\right\}^2=\frac{1}{8}$

(iii) Consider L.H.S. :

$=\frac{1}{2}.\frac{\sqrt{5}+1}{4}.\frac{1}{2}.(2\cos42°\cos78°)$ $=\frac{1}{16}(\sqrt{5}+1)(\cos120°+\cos36°)$

$=\frac{1}{16}(\sqrt{5}+1).\frac{\sqrt{5}-1}{4}=\frac{1}{16}.\frac{5-1}{4}=\frac{1}{16}$

Student Task

Prove that :

1. $\sin x \sin y \sin(x-y)+\sin y \sin z \sin(y-z)+\sin z \cdot \sin x \sin(z-x)+\sin(x-y)\sin(y-z)\sin(z-x)=0$
2. If $\tan\theta=\frac{1-\cos\phi}{\sin\phi}$ then prove that $\tan 2\theta=\tan 2\phi$
3. Prove that : $\frac{\sin\theta+\sin3\theta+\sin5\theta+.....+\sin2013\theta}{\cos\theta+\cos3\theta+\cos5\theta+.....+\cos2013\theta}=\tan1007\theta$
4. If $\sin\theta = k\sin(\theta+2\alpha)$, prove that $\tan(\theta+\alpha)=\frac{1+k}{1-k}\tan\alpha$
5. If $\cos5\theta=0$, then prove that $16\cos^4\theta-20\cos^2\theta+5=0$ where $0<\theta<\frac{\pi}{2}$
6. If $3\tan\alpha = 4\tan\beta$, then prove that $\tan(\alpha-\beta)=\frac{\sin2\beta}{7-\cos2\beta}$
7. Let $\alpha+\beta+\gamma=\pi$, then show that $\tan(\alpha+\beta-\gamma)+\tan(\beta+\gamma-\alpha)+\tan(\gamma+\alpha-\beta)=\tan(\alpha+\beta-\gamma)\tan(\beta+\gamma-\alpha)\tan(\gamma+\alpha-\beta)$
8. If $A+B+C=2S$, then show that $\cos^2S+\cos^2(S-A)+\cos^2(S-B)+\cos^2(S-C)=2+2\cos A\cos B\cos C$
9. If $0<\theta<\pi$, prove that $\cot\frac{\theta}{2}\geq1+\cot\theta$
10. In ΔABC, if $\cos A\cos B\cos(A-B)+\cos B\cos C\cos(B-C)+\cos C\cos A\cos(C-A)+2\cos A\cos B\cos C=1$ then prove that ΔABC is an equilateral triangle.
11. If $A+B+C=\pi$, then prove that $(\tan A+\tan B+\tan C)(\cot A+\cot B+\cot C)=1+\sec A\sec B\sec C$
12. If $\cos\alpha+\cos\beta+\cos\gamma=\sin\alpha+\sin\beta+\sin\gamma=0$ then prove that
    (a) $\cos3\alpha+\cos3\beta+\cos3\gamma=3\cos(\alpha+\beta+\gamma)$
    (b) $\cos2\alpha+\cos2\beta+\cos2\gamma=0$
    (c) $\cos(\alpha+\beta)+\cos(\beta+\gamma)+\cos(\gamma+\alpha)=0$

6. CONDITIONAL IDENTITIES

TYPE - I Problems based on transformation of the plus form of sine or cosine in product form :

Working Rule :

(i) Simplify the terms containing A and B using the formula of $\sin C \pm \sin D$ and $\cos C \pm \cos D$ whichever applicable.

(ii) Simplify the term containing C by using $\sin 2\theta = 2 \sin\theta \cos\theta$ or $\cos 2\theta = 2\cos^2\theta - 1$ or $\cos 2\theta = 1 - 2\sin^2\theta$. and write $A + B$ in terms of C and then take out the common factor (which will be a term containing C).

(iii) The transform C in $A + B$ (not in the factor which has been taken common).

(iv) Then simplify to get the desired result.

Illustration 30: If $A + B + C = \pi$, then show that :

(i) $\sin 2A + \sin 2B + \sin 2C = 4 \sin A \sin B \sin C$

(ii) $\cos 2A + \cos 2B + \cos 2C = -1 - 4 \cos A \cos B \cos C$

(iii) $\cos A + \cos B + \cos C = 1 + 4 \sin A/2 \sin B/2 \sin C/2$

(iv) $\sin A + \sin B + \sin C = 4 \cos A/2 \cos B/2 \cos C/2$

Solution : (i) L.H.S. $= \sin 2A + \sin 2B + \sin 2C$

$= 2 \sin (A + B) \cos (A - B) + \sin 2C$

$= 2 \sin (\pi - C) \cos (A - B) + \sin 2C$

$= 2 \sin C \cos (A - B) + 2 \sin C \cos C$

$= 2 \sin C [\cos (A - B) + \cos C]$

Convert the expression with in brackets in A and B only.

$= 2 \sin C [\cos (A - B) + \cos (\pi - A - B)]$

$= 2 \sin C [\cos (A - B) - \cos (A + B)]$

$= 2 \sin C (2 \sin A \sin B)$

$= 4 \sin A \sin B \sin C$ = R.H.S.

(ii) L.H.S. $= \cos 2A + \cos 2B + \cos 2C$

$= 2 \cos (A + B) \cos (A - B) + \cos 2C$

$= -2 \cos C \cos (A - B) + 2\cos^2 C - 1$

$= -2 \cos C [\cos (A - B) - \cos C] - 1$

$= -2 \cos C [\cos (A - B) + \cos (A + B)] - 1$

$= -1 - 4 \cos A \cos B \cos C$ = RHS

(iii) LHS $= \cos A + \cos B + \cos C$

$$= 2\cos\frac{A+B}{2}\cos\frac{A-B}{2} + \cos C \quad = 2\cos\left(\frac{\pi}{2} - \frac{C}{2}\right)\cos\frac{A-B}{2} + \cos C$$

$$= 2\sin\frac{C}{2}\cos\frac{A-B}{2} + \left(1 - 2\sin^2\frac{C}{2}\right) \qquad \left[\text{Express } \cos C \text{ in terms of } \sin\frac{C}{2}\right]$$

$$= 2\sin\frac{C}{2}\left[\cos\frac{A-B}{2} - \sin\frac{C}{2}\right] + 1$$

$$= 2\sin\frac{C}{2}\left[\cos\frac{A-B}{2} - \cos\frac{A+B}{2}\right] + 1$$

$$= 2\sin\frac{C}{2}\left[2\sin\frac{A}{2}\sin\frac{B}{2}\right] + 1 \quad = 1 + 4\sin\frac{A}{2}\sin\frac{B}{2}\sin\frac{C}{2} = \text{RHS}$$

(iv)Try On The Same Pattern. (As in part 3 above)

TYPE - II Problems involving squares of sines or cosines of angles.

Working Rule :

(i) First, rearrange the terms in L.H.S. so that either $\sin^2 A - \sin^2 B = \sin (A + B) . \sin (A - B)$ or $\cos^2 A - \sin^2 B = \cos (A + B) . \cos (A - B)$ can be applied.

(ii) Simplify the term containing C and then take out the common factor (which will be a term containing C).

(iii) Then transform C in $A + B$ (not in the factor which has been taken common).

(iv) Then simplify to get the desired result.

***Illustration 31*:** If $A + B + C = \pi$, then show that :

(i) $\sin^2 A + \sin^2 B - \sin^2 C = 2 \sin A \sin B \cos C$

(ii) $\cos^2 A/2 + \cos^2 B/2 + \cos^2 C/2 = 2 + 2 \sin A/2 \sin B/2 \sin C/2$.

(iii) $\sin^2 A + \sin^2 B + \sin^2 C = 2 + 2 \cos A \cos B \cos C$

Solution :(i) Starting from L.H.S. :

$= \sin^2 A + \sin^2 B - \sin^2 C$

$= \sin^2 A + \sin(\pi - A) \sin(B - C)$

$= \sin^2 A + \sin(\pi - A) \sin(B - C)$

$= \sin A [\sin A + \sin(B - C)]$

$= \sin A [\sin[\pi - (B + C)] + \sin(B - C)]$

$= \sin A [\sin(B + C) + \sin(B - C)]$

$= \sin A [2 \sin B \cos C] = 2 \sin A \sin B \cos C$

(ii) L.H.S. $= \cos^2 A/2 + (1 - \sin^2 B/2) + \cos^2 C/2$

$= 1 + (\cos^2 A/2 - \sin^2 B/2) + \cos^2 C/2$

$= 1 + \cos(A + B)/2 \cos(A - B)/2 + \cos^2 C/2$

$= 1 + \sin C/2 \cos(A - B)/2 + 1 - \sin^2 C/2$

$= 2 + \sin C/2 [\cos(A - B)/2 - \sin C/2]$

$= 2 + \sin C/2 [\cos(A - B)/2 - \cos(A + B)/2]$

$= 2 + 2 \sin C/2 \sin A/2 \sin B/2$

(iii) L.H.S. $= \sin^2 A + \sin^2 B + \sin^2 C$

$= 1 - (\cos^2 A - \sin^2 B) + \sin^2 C$

$= 1 - \cos(A + B) \cos(A - B) + \sin^2 C$

$= 1 + \cos C \cos(A - B) + 1 - \cos^2 C$

$= 2 + \cos C [\cos(A - B) - \cos C]$

$= 2 + \cos C [\cos(A - B) + \cos(A + B)]$

$= 2 + 2 \cos C \cos A \cos B$

TYPE - III Problems involving only tangents or cotangents of angles.

Working Rule :

(i) Write the given relation between angles keeping two angles on one side and remaining on the other side.

(ii) Take tangent or cotangent of both sides and simplify to get the desired result.

Illustration 32: In a ΔABC, prove that $\tan A + \tan B + \tan C = \tan A \tan B \tan C$.

Solution : In ΔABC, $\therefore \quad A + B + C = \pi \quad \therefore \quad A + B = \pi - C$

$\Rightarrow \quad \tan(A + B) = \tan(\pi - C)$ [By taking tan on both side]

$\Rightarrow \quad \dfrac{\tan A + \tan B}{1 - \tan A \tan B} = -\tan C$ [$\because \tan(\pi - C) = -\tan C$]

$\Rightarrow \quad \tan A + \tan B = -\tan C + \tan A \tan B \tan C \quad \Rightarrow \tan A + \tan B + \tan C = \tan A \tan B \tan C$

Illustration 33: If $A + B + C = \pi$, prove that : $\tan\dfrac{A}{2}\tan\dfrac{B}{2} + \tan\dfrac{B}{2}\tan\dfrac{C}{2} + \tan\dfrac{C}{2}\tan\dfrac{A}{2} = 1.$

Solution : $\because A + B + C = \pi \quad \therefore \quad \dfrac{A}{2} + \dfrac{B}{2} + \dfrac{C}{2} = \dfrac{\pi}{2}$ or, $\dfrac{A}{2} + \dfrac{B}{2} = \dfrac{\pi}{2} - \dfrac{C}{2}$

$\therefore \tan\left(\dfrac{A}{2} + \dfrac{B}{2}\right) = \tan\left(\dfrac{\pi}{2} - \dfrac{C}{2}\right) = \cot\dfrac{C}{2}$ [By taking tan on both side] $\Rightarrow \dfrac{\tan\frac{A}{2} + \tan\frac{B}{2}}{1 - \tan\frac{A}{2}\tan\frac{B}{2}} = \dfrac{1}{\tan\frac{C}{2}}$

$\Rightarrow \quad \tan\dfrac{A}{2}\tan\dfrac{C}{2} + \tan\dfrac{B}{2}\tan\dfrac{C}{2} = 1 - \tan\dfrac{A}{2}\tan\dfrac{B}{2} \quad \Rightarrow \quad \tan\dfrac{A}{2}\tan\dfrac{B}{2} + \tan\dfrac{B}{2}\tan\dfrac{C}{2} + \tan\dfrac{C}{2}\tan\dfrac{A}{2} = 1.$

Illustration 34: If $A + B + C = \pi$, prove that cot B cot C + cot C cot A + cot A cot B = 1.

Solution : $\because$ $A + B + C = \pi$ $\therefore$ $A + B = \pi - C$

$\Rightarrow$ cot $(A + B)$ = cot $(\pi - C)$ [By taking cot on both side]

$\Rightarrow$ $\dfrac{\cot A.\cot B - 1}{\cot B + \cot A} = -\cot C$

$\Rightarrow$ cot A cot $B - 1 = -$ cot B cot $C -$ cot C cot A

$\Rightarrow$ cot A cot B + cot B cot C + cot C cot $A = 1$

TYPE - IV Miscellaneous Type

Working Rule :

(i) If it is given that $x + y + z = xyz$ or, $xy + yz + zx = 1$ put $x = \tan A$, $y = \tan B$ and $z = \tan C$ in it and simplify it to find the value of $A + B + C$.

(ii) Then put tan A, tan B, tan C in place of x, y, z respectively in the L.H.S. of the given question and simplify it to get the desired identity using the value of $A + B + C$.

Illustration 35: If $xy + yz + zx = 1$, prove that : $\dfrac{x}{1-x^2} + \dfrac{y}{1-y^2} + \dfrac{z}{1-z^2} = \dfrac{4xyz}{(1-x^2)(1-y^2)(1-z^2)}$

Solution : Let x =tan A, y = tan B, z = tan C

Given, $xy + yz + zx = 1$

$\therefore$ tan A tan B + tan B tan C + tan C tan $A = 1$

$\Rightarrow$ tan C (tan A + tan B) = 1– tan A tan B

$\Rightarrow$ $\dfrac{\tan A + \tan B}{1 - \tan A \tan B} = \dfrac{1}{\tan C} = \cot C = \tan\left(\dfrac{\pi}{2} - C\right) \Rightarrow \tan (A + B) = \tan\left(\dfrac{\pi}{2} - C\right)$

$\therefore$ $A + B = \dfrac{\pi}{2} - C$ [Taking principal value]

$\Rightarrow$ $A + B + C = \dfrac{\pi}{2}$ ..... (i)

Now, L.H.S. = $\dfrac{x}{1-x^2} + \dfrac{y}{1-y^2} + \dfrac{z}{1-z^2}$

$= \dfrac{\tan A}{1 - \tan^2 A} + \dfrac{\tan B}{1 - \tan^2 B} + \dfrac{\tan C}{1 - \tan^2 C} = \dfrac{1}{2}\left[\dfrac{2\tan A}{1 - \tan^2 A} + \dfrac{2\tan B}{1 - \tan^2 B} + \dfrac{2\tan C}{1 - \tan^2 C}\right]$

$= \dfrac{1}{2}$ (tan $2A$ + tan $2B$ + tan $2C$) .....(ii)

Now from (i), $A + B + C = \dfrac{\pi}{2}$ $\Rightarrow$ $2A + 2B + 2C = \pi$

or, $2A + 2B = \pi - 2C$ $\Rightarrow$ tan $(2A + 2B)$ = tan $(\pi - 2C)$

or, $\dfrac{\tan 2A + \tan 2B}{1 - \tan 2A \tan 2B} = -\tan 2C$ $\Rightarrow$ tan $2A$ + tan $2B = -$ tan $2C$ + tan $2A$ tan $2B$ tan $2C$

or, tan $2A$ + tan $2C$ + tan $2B$ = tan $2A$ tan $2B$ tan $2C$ ..... (iii)

From (iii), L.H.S. = $\dfrac{1}{2}$ (tan $2A$ + tan $2B$ + tan $2C$)

$= \dfrac{1}{2}$ (tan $2A$ tan $2B$ tan $2C$) [From (iii)]

$= \dfrac{1}{2} \cdot \dfrac{2\tan A}{1 - \tan^2 A} \cdot \dfrac{2\tan B}{1 - \tan^2 B} \cdot \dfrac{2\tan C}{1 - \tan^2 C}$

$= \dfrac{1}{2} \cdot \dfrac{2x}{1-x^2} \cdot \dfrac{2y}{1-y^2} \cdot \dfrac{2z}{1-z^2} = \dfrac{4yxz}{(1-x^2)(1-y^2)(1-z^2)}$

Student Task

Q.1 If A + B + C = π, then $\frac{\cot A+\cot B}{\tan A+\tan B}+\frac{\cot B+\cot C}{\tan B+\tan C}+\frac{\cot C+\cot A}{\tan C+\tan A}=$

(a) –1 (b) 0 (c) 1 (d) 3

Q.2 If A + B + C = π and $\sin^2\frac{A}{2}+\sin^2\frac{B}{2}-\sin^2\frac{C}{2}=\lambda-2\cos\frac{A}{2}\cos\frac{B}{2}\sin\frac{C}{2}$ then λ =

(a) –1 (b) 0 (c) 1 (d) 3

Q.3 If A + B + C = π and cos A + cos B – cos C = $\lambda+\mu\cos\frac{A}{2}\cos\frac{B}{2}\sin\frac{C}{2}$ then $\lambda+\mu$ =

(a) –1 (b) 0 (c) 1 (d) 3

Q.4 If A + B + C = π then sin 2A + sin 2B + sin 2C =

(a) 4 cos A cos B cos C (b) 4 sin A sin B sin C

(c) 4 cos A cos B sin C (d) 4 sin A sin B cos C

Q.5 If x + y + z = xyz, then $\frac{2x}{1-x^2}+\frac{2y}{1-y^2}+\frac{2z}{1-z^2}-\frac{2x}{1-x^2}\cdot\frac{2y}{1-y^2}\cdot\frac{2z}{1-z^2}=$

(a) 0 (b) 1 (c) 3 (d) –1

Q.6 If $A + B + C = \pi$, then minimum value of $\tan^2\frac{A}{2}+\tan^2\frac{B}{2}+\tan^2\frac{C}{2}$ is

(a) 0 (b) 1 (c) –1 (d) 2

Q.7 If $A + B + C = \pi$, then minimum value of $\cot^2 A + \cot^2 B + \cot^2 C$ is

(a) 0 (b) 1 (c) –1 (d) 2

Q.8 In ΔABC, the minimum value of $\frac{\Sigma\cot^2(A/2)\cot^2(B/2)}{\prod\cot^2(A/2)}$ is

(a) 1 (b) 2 (c) 3 (d) Non existent

Q.9 If in triangle ABC, $\cos^2 A + \cos^2 B - \cos^2 C = 1$, then

(a) $\angle A = \pi/2$ (b) $\angle B = \pi/2$ (c) $\angle C = \pi/2$ (d) None of these

Answers

1. (c) 2. (c) 3. (d) 4. (c) 5. (a) 6. (b) 7. (b)
8. (a) 9. (c)

MISCELLANEOUS EXAMPLES

Example 1 : If $\cos\theta=\frac{\cos\alpha-\cos\beta}{1-\cos\alpha\cos\beta}$, then show that $\tan\frac{\theta}{2}=\pm\tan\frac{\alpha}{2}\cot\frac{\beta}{2}$.

Solution : Manipulating the given condition as follows :

$$\frac{1}{\cos\theta}=\frac{1-\cos\alpha\cos\beta}{\cos\alpha-\cos\beta}$$

$$\Rightarrow \quad \frac{1-\cos\theta}{1+\cos\theta}=\frac{(1-\cos\alpha\cos\beta)-(\cos\alpha-\cos\beta)}{(1-\cos\alpha\cos\beta)+(\cos\alpha-\cos\beta)} \qquad \text{[Apply 'C' and 'D']}$$

$$\Rightarrow \quad \tan^2\frac{\theta}{2}=\frac{(1-\cos\alpha)+\cos\beta(1-\cos\alpha)}{(1+\cos\alpha)-\cos\beta(1+\cos\alpha)}=\frac{(1-\cos\alpha)(1+\cos\beta)}{(1+\cos\alpha)(1-\cos\beta)}\Rightarrow\tan^2\frac{\theta}{2}=\frac{\frac{1-\cos\alpha}{1+\cos\alpha}}{\frac{1-\cos\beta}{1+\cos\beta}}=\frac{\tan^2\frac{\alpha}{2}}{\tan^2\frac{\beta}{2}}$$

$$\Rightarrow \quad \tan^2\frac{\theta}{2}=\tan^2\frac{\alpha}{2}.\cot^2\frac{\beta}{2} \quad \Rightarrow \quad \tan\frac{\theta}{2}=\pm\tan\frac{\alpha}{2}.\cot\frac{\beta}{2}$$

Example 2 : If $\tan\frac{\theta}{2}=\sqrt{\frac{1-e}{1+e}}\tan\frac{\phi}{2}$, show that $\cos\phi = \frac{\cos\theta-e}{1-e\cos\theta}$

Solution : We have to find $\cos\phi$ in terms of e and $\cos\theta$, so try to convert $\tan\theta/2$ to $\cos\phi$.

$$\tan^2\frac{\theta}{2}=\frac{1-e}{1+e}\tan^2\frac{\phi}{2}$$

$$\Rightarrow \tan^2\frac{\phi}{2}=\frac{1+e}{1-e}\tan^2\frac{\theta}{2}=\frac{1+e}{1-e}\left(\frac{1-\cos\theta}{1+\cos\theta}\right)\Rightarrow \frac{\tan^2\frac{\phi}{2}}{1}=\frac{1+e-\cos\theta-e\cos\theta}{1-e+\cos\theta-e\cos\theta}$$

$$\Rightarrow\frac{1-\tan^2\phi/2}{1+\tan^2\phi/2}=\frac{(1-e+\cos\theta-e\cos\theta)-(1+e-\cos\theta-e\cos\theta)}{(1-e+\cos\theta-e\cos\theta)+(1+e-\cos\theta-e\cos\theta)} \qquad \text{[Apply C and D]}$$

$$\Rightarrow\cos\phi=\frac{-2e+2\cos\theta}{2-2e\cos\theta}=\frac{\cos\theta-e}{1-e\cos\theta}$$

Example 3 : If $x + y + z = 0$, show that $\cot(x+z-y)\cot(x+y-z)+\cot(x+y-z)\cot(y+z-x)+\cot(y+z-x)\cot(z+x-y)=1$.

Solution : Let $A = x + y - z$; $B = z + x - y$; $C = y + z - x$

$\therefore \quad A + B + C = x + y + z = 0 \Rightarrow A + B = -C$

Taking cot on both sides we get :

$$\cot(A+B)=\cot(-C)\Rightarrow \frac{\cot A.\cot B-1}{\cot A+\cot B}=-\cot C$$

$\Rightarrow \cot A \cdot \cot B + \cot B \cot C + \cot C \cot A = 1$

Putting the value of A, B and C, we get :

$\cot(x+y-z)\cot(z+x-y)+\cot(z+x-y)\cot(y+z-x)+\cot(y+z-x)\cot(x+y-z)=1$.

Example 4 : If $\alpha + \beta = 90°$ and $\beta + \gamma = \alpha$ then prove that $\tan\alpha = \tan\beta + 2\tan\gamma$.

Solution : $\alpha + \beta = 90°$

Taking tan on both sides

$$\text{Now } \tan(\alpha+\beta)=\frac{\tan\alpha+\tan\beta}{1-\tan\alpha\tan\beta}=\tan 90°=\frac{1}{0}$$

$\Rightarrow \quad \tan\alpha\tan\beta = 1$

Now $\tan\alpha = \tan(\beta+\gamma)$

$$= \frac{\tan\beta+\tan\gamma}{1-\tan\beta\tan\gamma}$$

$\Rightarrow \tan\alpha - \tan\alpha\tan\beta\tan\gamma = \tan\beta + \tan\gamma$

$\Rightarrow \tan\alpha - \tan\gamma = \tan\beta + \tan\gamma$

$\Rightarrow \tan\alpha = \tan\beta + 2\tan\gamma$

Example 5 : If $\sin^3 x \cdot \sin 3x = \sum_{m=0}^{n} c_m \cdot \cos mx$ is an identity in x. Where c_m's are constants, then find the value of n.

Solution : Here, $\sin^3 x \cdot \sin 3x = \frac{3\sin x-\sin 3x}{4}.\sin 3x$ also,

$$\sum_{m=0}^{n} c_m \cdot \cos mx = c_0 + c_1\cos x + c_2\cos 2x + \ldots\ldots + c_n\cos nx \qquad \ldots\ldots(ii)$$

$$= \frac{3}{8}(2\sin x \sin 3x) - \frac{1}{4}(\sin^2 3x)$$

$$= \frac{3}{8}(\cos 2x - \cos 4x) - \frac{1}{8}(1 - \cos 6x)$$

$$= -\frac{1}{8} + \frac{3}{8}\cos 2x - \frac{3}{8}\cos 4x + \frac{1}{8}\cos 6x \quad \text{..... (i)}$$

On comparing (i) and (ii) we get,

$n = 6$

Example 6 : For any real θ, the maximum value of $\cos^2(\cos\theta) + \sin^2(\sin\theta)$ is

(A) 1 (B) $1 + \sin^2 1$ (C) $1 + \cos^2 1$ (D) Does not exist

Solution : (B)

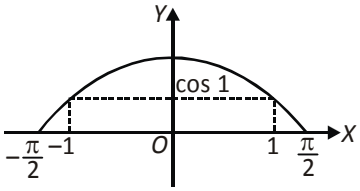
Let $f(\theta) = \cos^2(\cos\theta) + \sin^2(\sin\theta)$

$\because$ $-1 \le \cos\theta \le 1$ and $-1 \le \sin\theta \le 1$

$\therefore$ $\cos 1 \le \cos(\cos\theta) \le 1$ and $-1 \le \sin\theta \le 1$

$\therefore$ $\cos^2 1 \le \cos^2(\cos\theta) \le 1$ ..... (i)

and $0 \le \sin^2(\sin\theta) \le \sin^2 1$ .....(ii)



Now combining (i) and (ii), max. value of $\cos^2(\cos\theta) = 1$ at $\theta = \pi/2$ and max. value of $\sin^2(\sin\theta) = \sin^2 1$ at $\theta = \pi/2$

$\Rightarrow$ Maximum value of $f(\theta) = 1 + \sin^2 1$ at $\theta = \pi/2$

Example 7 : Show that : $\cos\frac{2\pi}{7}.\cos\frac{4\pi}{7}.\cos\frac{6\pi}{7} = \frac{1}{8}$

Solution : Let $y = \cos\frac{2\pi}{7}.\cos\frac{4\pi}{7}.\cos\frac{6\pi}{7}$

$$= \cos\frac{2\pi}{7}.\cos\frac{4\pi}{7}.\cos\left(\pi - \frac{\pi}{7}\right)$$

$$= -\cos\frac{\pi}{7}.\cos\frac{2\pi}{7}.\cos\frac{4\pi}{7}$$

[Here there is only one group consisting of terms having angles in G.P. So, we multiply both sides by 2 sin $\pi/7$ and simplify]

$\therefore$ $2y\sin\frac{\pi}{7} = -\left(2\sin\frac{\pi}{7}\cos\frac{\pi}{7}\right)\cos\frac{2\pi}{7}\cos\frac{4\pi}{7}$

$\Rightarrow$ $2y\sin\frac{\pi}{7} = -\sin\frac{2\pi}{7}\cos\frac{2\pi}{7}\cos\frac{4\pi}{7}$

$\Rightarrow$ $4y\sin\frac{\pi}{7} = -\left(2\sin\frac{2\pi}{7}\cos\frac{2\pi}{7}\right).\cos\frac{4\pi}{7}$ [Multiplying both side by 2]

$\Rightarrow$ $4y\sin\frac{\pi}{7} = -\sin\frac{4\pi}{7}\cos\frac{4\pi}{7}$

$\Rightarrow$ $8y\sin\frac{\pi}{7} = -2\sin\frac{4\pi}{7}.\cos\frac{4\pi}{7}$ [Multiplying both side by 2]

$\Rightarrow$ $8y\sin\frac{\pi}{7} = -\sin\frac{8\pi}{7} = -\sin\left(\pi + \frac{\pi}{7}\right)$

$\Rightarrow$ $8y\sin\frac{\pi}{7} = +\sin\frac{\pi}{7}$ $\Rightarrow$ $y = \frac{1}{8}$

Example 8 : For a positive integer n, let

$$f_n(\theta)=\left(\tan\frac{\theta}{2}\right)(1+\sec\theta)(1+\sec 2\theta)(1+\sec 4\theta).....(1+\sec 2^n\theta).$$ Then :

(A) $f_2\left(\frac{\pi}{16}\right)=1$ (B) $f_3\left(\frac{\pi}{32}\right)=1$ (C) $f_4\left(\frac{\pi}{64}\right)=1$ (D) $f_5\left(\frac{\pi}{128}\right)=1$

Solution : We have

$$\left(\tan\frac{\theta}{2}\right)(1+\sec\theta)=\frac{\sin(\theta/2)}{\cos(\theta/2)}\left[1+\frac{1}{\cos\theta}\right]=\frac{\sin(\theta/2)}{\cos(\theta/2)}.\frac{2\cos^2(\theta/2)}{\cos\theta}=\frac{2\sin(\theta/2)\cos(\theta/2)}{\cos\theta}=\frac{\sin\theta}{\cos\theta}=\tan\theta$$

Thus, $f_n(\theta)=\left(\tan\frac{\theta}{2}\right)(1+\sec\theta)(1+\sec 2\theta)(1+\sec 4\theta).....(1+\sec 2^n\theta)$

= (tan θ) (1 + sec 2θ) (1 + sec 4θ) ..... (1 + sec 2^n θ)
= (tan 2θ) (1 + sec 4θ) ..... (1 + sec 2^n θ)
= (tan 4θ) (1 + sec 8θ) ..... (1 + sec 2^n θ)
= .....
= tan (2^n θ)

Now, $f_2\left(\frac{\pi}{16}\right)=\tan\left(2^2\frac{\pi}{16}\right)=\tan\left(\frac{\pi}{4}\right)=1$

$f_3\left(\frac{\pi}{32}\right)=\tan\left(2^3\frac{\pi}{32}\right)=\tan\left(\frac{\pi}{4}\right)=1$

$f_4\left(\frac{\pi}{64}\right)=\tan\left(2^4\frac{\pi}{64}\right)=\tan\left(\frac{\pi}{4}\right)=1$

and $f_5\left(\frac{\pi}{128}\right)=\tan\left(2^5\frac{\pi}{128}\right)=\tan\left(\frac{\pi}{4}\right)=1$

Example 9 : If $\alpha+\beta+\gamma=\pi$, show that tan ($\beta+\gamma-\alpha$) + tan ($\gamma+\alpha-\beta$) + tan ($\alpha+\beta-\gamma$) = tan ($\beta+\gamma-\alpha$) tan ($\gamma+\alpha-\beta$) tan ($\alpha+\beta-\gamma$).

Solution : Let $\beta+\gamma-\alpha=A, \gamma+\alpha-\beta=B,$ and $\alpha+\beta-\gamma=C$

Now, $A+B+C=\beta+\gamma-\alpha+\gamma+\alpha-\beta+\alpha+\beta-\gamma$

or, $A+B+C=\alpha+\beta+\gamma=\pi$ [As $\alpha+\beta+\gamma=\pi$]

or, $A+B=\pi-C$

$\therefore$ tan ($A+B$) = tan ($\pi-C$)

or, $\frac{\tan A+\tan B}{1-\tan A\tan B}=-\tan C$ [As tan ($\pi-C$) = – tan C]

or, tan A + tan B = – tan C + tan A tan B tan C or, tan A + tan B + tan C = tan A tan B tan C

Putting the values of A, B and C, we get :

tan ($\beta+\gamma-\alpha$) + tan ($\gamma+\alpha-\beta$) + tan ($\alpha+\beta-\gamma$) = tan ($\beta+\gamma-\alpha$) tan ($\gamma+\alpha-\beta$) tan ($\alpha+\beta-\gamma$).

Example 10 : If $A+B+C=\pi$, prove that (cot B + cot C) (cot C + cot A) (cot A + cot B) = cosec A cosec B cosec C.

Solution : $\cot B+\cot C=\frac{\cos B}{\sin B}+\frac{\cos C}{\sin C}=\frac{\cos B\sin C+\sin B\cos C}{\sin B\sin C}=\frac{\sin(B+C)}{\sin B\sin C}$

$=\frac{\sin(\pi-A)}{\sin B\sin C}=\frac{\sin A}{\sin B\sin C}$ ..... (i)

Similarly, cot C + cot A = $\frac{\sin B}{\sin A \sin C}$ .....(ii)

and $\cot A + \cot B = \frac{\sin C}{\sin A \sin B}$ .....(iii)

Multiplying (i), (ii), and (iii), we get :

$$\text{L.H.S.} = \frac{\sin A}{\sin B \sin C} . \frac{\sin B}{\sin C \sin A} . \frac{\sin c}{\sin A \sin B}$$

= cosec *A* cosec *B* cosec *C*

UNSOLVED EXERCISE

Section-A
(Straight objective type with single option correct)

Topic : 1 Based on Trigonomatric Formule

1. If $\tan\frac{\pi}{9}$, *x* and $\tan\frac{5\pi}{18}$ are in A.P. and $\tan\frac{\pi}{9}$, *y* and $\tan\frac{7\pi}{18}$ are also in A.P., then
(a) $x = y$ (b) $x = 2y$
(c) $2x = y$ (d) $x + y = \frac{\pi}{2}$

2. If $\cos 28^o + \sin 28^o = k^3$, then $\cos 17^o$ is equal to
(a) $\frac{k^3}{\sqrt{2}}$ (b) $-\frac{k^3}{\sqrt{2}}$
(c) $\pm\frac{k^3}{\sqrt{2}}$ (d) None of these

3. If $x^2 - 2\cos\alpha\, x + 1 = 0$ and $y^2 - 2\cos\beta\, y + 1 = 0$ then $2\cos(\alpha + \beta)$ is equal to
(a) $\frac{x}{y} + \frac{y}{x}$ (b) $\left(x + \frac{1}{y}\right)\left(y + \frac{1}{x}\right)$
(c) $xy + \frac{1}{xy}$ (d) None of these

4. The value of $\cos\frac{2\pi}{7}\cos\frac{4\pi}{7}\cos\frac{6\pi}{7}$ is
(a) –1/4 (b) 1/4
(c) –1/8 (d) 1/8

5. The value of $64\sqrt{3}\sin\frac{\pi}{48}\cos\frac{\pi}{48}\cos\frac{\pi}{24}\cos\frac{\pi}{12}\cos\frac{\pi}{6}$ is
(a) 8 (b) 6
(c) 4 (d) 12

6. The expression S = sec 11° · sec 19° – 2 cot 71° reduces to
(a) 2 cot 11° (b) tan 19°
(c) 2 tan 11° (d) $\frac{1}{2}$ tan 19°

7. The average of the numbers *n* sin *n*°, for n = 2, 4, 6, ..., 180, is
(a) tan 1° (b) sin 1°
(c) cot 1° (d) cos 1°

8. Let *a* and *b* real numbers such that sin a + sin b = $\frac{\sqrt{2}}{2}$ cos a + cos b = $\frac{\sqrt{6}}{2}$ Then (a + b) =
(a) $\frac{\pi}{6}$ (b) $\frac{\pi}{4}$
(c) $\frac{\pi}{3}$ (d) $\frac{\pi}{2}$

9. $\frac{3 + \cot 76^o \cot 16^o}{\cot 76^o + \cot 16^o}$ is equal to
(a) tan 16° (b) cot 76°
(c) tan 44° (d) cot 44°

10. The numerical value of tan 9° – tan 27° – tan 63° + tan 81° is equal to
(a) 2 (b) 4
(c) 6 (d) 0

11. The value of sin 55° – sin 19° + sin 53° – sin 17° is always equal to
(a) cos 1° (b) sin 1°
(c) tan 1° (d) cot 1°

12. If $\frac{1 + \sin 2x}{1 - \sin 2x} = \cot^2(a + x)$, $\forall x \in \mathbb{R} - \left\{n\pi + \frac{\pi}{4}\right\}$, $n \in \mathbb{N}$, then 'a' is equal to
(a) $\frac{\pi}{4}$ (b) $\frac{3\pi}{4}$
(c) $\frac{\pi}{2}$ (d) $\frac{\pi}{3}$

13. If $\frac{\cos x}{a}=\frac{\sin x}{b}$, then |a cos 2x+b sin 2x| is equal to

(a) $|a|$ (b) $\frac{a^2}{|b|}$

(c) $\frac{b^2}{|a|}$ (d) $\frac{a^2}{b}$

14. If $\frac{\sec^4 x}{\alpha}+\frac{\tan^4 x}{\beta}=\frac{1}{\alpha+\beta}$, then which of the following statements is correct?

(a) $\sin^2 x = \alpha/\beta$ (b) $\sin^2 x = -\beta/\alpha$
(c) $\sin^2 x = -\alpha/\beta$ (d) $\sin^2 x = \beta/\alpha$

15. If $\theta \in \left(\frac{\pi}{4}, \frac{5\pi}{8}\right)$ then the value of $\sqrt{\cos^2 2\theta - 4\sin\theta\cos\theta\cos 2\theta + \sin^2 2\theta}$ is equal to

(a) $\sin 2\theta - \cos 2\theta$ (b) $\cos 2\theta - \sin 2\theta$
(c) $|\sin 2\theta| - |\cos 2\theta|$ (d) Either (1) and (2)

16 If $x \sec\theta = 1 - y \tan\theta$ and $x^2 \sec^2\theta = 5 + y^2 \tan^2\theta$, then

(a) $x^2 = \frac{9y^2}{y^2+4}$ (b) $x^2 = \frac{4y^2}{9-y^2}$

(c) $y^2 = \frac{4x^2}{9+x^2}$ (d) $y^2 = \frac{9x^2}{x^2+4}$

17. If $x = a\cos^3\theta \sin^2\theta$, $y = a\sin^3\theta\cos^2\theta$ and $\frac{(x^2+y^2)^p}{(xy)^q}$ $(p, q \in \mathbb{N})$ is independent of θ, then

(a) $p = 4, q = 4$ (b) $p = 5, q = 5$
(c) $q = 4, p = 5$ (d) $q = 5, p = 4$

18. Which of the following numbers is/are rational?

(a) $\sin 15^\circ$ (b) $\cos 15^\circ$
(c) $\sin 15^\circ \cos 15^\circ$ (d) $\sin 15^\circ \cos 75^\circ$

19. Let $\frac{3\pi}{4} < x < \pi$ then $\sqrt{2\cot\alpha + \frac{1}{\sin^2\alpha}}$ is

(a) $\frac{\sin\alpha - \cos\alpha}{\sin\alpha}$ (b) $\frac{\sin\alpha + \cos\alpha}{\sin\alpha}$

(c) $\frac{-\sin\alpha + \cos\alpha}{\sin\alpha}$ (d) $\frac{-\sin\alpha - \cos\alpha}{\sin\alpha}$

Topic : 2 Graph & Conditional Identites

20. If $\left(\cos^2 x + \frac{1}{\cos^2 x}\right)$ + (1 + $\tan^2 2y$) (3 + sin 3z) = 4, then y can take values equal to

(a) $\frac{\pi}{4}$ (b) $\frac{\pi}{3}$

(c) $\frac{\pi}{2}$ (d) None of these

21. If the sum of the series $\frac{1}{\cos\theta-\cos 3\theta}+\frac{1}{\cos\theta-\cos 5\theta}+\frac{1}{\cos\theta-\cos 7\theta}+ \ldots+\frac{1}{\cos\theta-\cos(2n+1)\theta}$ is equal to $\frac{\cot\theta-\cot(n+1)\theta}{2^\lambda \sin\theta}$ then λ =

(a) 1 (b) 2
(c) –1 (d) 0

22. If $\alpha, \beta, \gamma, \delta$ are the smallest positive angles in ascending order of magnitude which have their sines equal to the positive quantity K, then the value of $4\sin\frac{\alpha}{2} + 3\sin\frac{\beta}{2} + 2\sin\frac{\gamma}{2} + \sin\frac{\delta}{2}$ is equal to

(a) $2\sqrt{1-K}$ (b) $2\sqrt{1+K}$

(c) $\frac{\sqrt{1+K}}{2}$ (d) $\sqrt{1+k}$

23. The number of nonnegative integral values of λ for which the equation 3 sin x + 11 cos x = λ has a solution is

(a) 14 (b) 12
(c) 10 (d) 23

24. If $\frac{\cos x}{a}=\frac{\cos(x+\theta)}{b}=\frac{\cos(x+2\theta)}{c}=\frac{\cos(x+3\theta)}{d}$, then $\frac{a+c}{b+d}$ is equal to

(a) $\frac{a}{d}$ (b) $\frac{c}{b}$

(c) $\frac{b}{c}$ (d) $\frac{d}{a}$

25. Let $0^\circ < \theta < 45^\circ$, arrange $t_1 = (\tan\theta)^{\tan\theta}$, $t_2 = (\tan\theta)^{\cot\theta}$, $t_3 = (\cot\theta)^{\tan\theta}$, $t_4 = (\cot\theta)^{\cot\theta}$ in decreasing order then

(a) $t_1 > t_2 > t_3 > t_4$ (b) $t_4 > t_3 > t_1 > t_2$
(c) $t_4 > t_1 > t_2 > t_3$ (d) $t_4 > t_3 > t_2 > t_1$

26. The equation $4\sin^2 x + 4\sin x + a^2 - 3 = 0$ possesses a solution if a belongs to the interval

(a) (–1, 3) (b) (–3, 1)
(c) (–2, 2) (d) R – (–2, 2)

27. If $|\sin x + \cos x| = |\sin x| + |\cos x|$, then x belong to the quadrant

(a) I or III (b) II or IV
(c) I or II (d) III or IV

28. The smallest positive root of the equation $\tan x - x = 0$ lies in

(a) $\left(0, \frac{\pi}{2}\right)$ (b) $\left(\frac{\pi}{2}, \pi\right)$
(c) $\left(\pi, \frac{3\pi}{2}\right)$ (d) $\left(\frac{3\pi}{2}, 2\pi\right)$

29. The equation $k \cos x - 3 \sin x = k + 1$ is solvable only if k belongs to the interval
(a) $[4, +\infty]$ (b) $[-4, 4]$
(c) $(-\infty, 4]$ (d) None of these

30. If $\sin\theta$ and $\cos\theta$ are the roots of equation $ax^2 - bx + c = 0$, then a, b and c satisfy the relation
(a) $a^2 + b^2 + 2ac = 0$ (b) $a^2 - b^2 + 2ac = 0$
(c) $a^2 + c^2 + 2ab = 0$ (d) $a^2 - b^2 - 2ac = 0$

31. The number of integral values of k for which the equation $7 \cos x + 5 \sin x = 2k + 1$ has a solutions is
(a) 4 (b) 8
(c) 10 (d) 12

32. If $\frac{\cos\theta}{p} = \frac{\sin\theta}{q}$, then $\frac{p}{\sec 2\theta} + \frac{q}{\operatorname{cosec} 2\theta}$ is
(a) p (b) 0
(c) qp (d) None of these

33. $\tan 9° - (4\cos^2 9° - 3)(4\cos^2 27° - 3) =$
(a) 0 (b) 1
(c) 2 (d) 3

Topic : 3 Maxima & Minima

34. If $a \sin x + b \cos(x + \theta) + b \cos(x - \theta) = d$, then the minimum value of $|\cos\theta|$ is equal to
(a) $\frac{1}{2|b|}\sqrt{d^2 - a^2}$ (b) $\frac{1}{2|a|}\sqrt{d^2 - a^2}$
(c) $\frac{1}{2|d|}\sqrt{d^2 - a^2}$ (d) None of these

35. Maximum value of $a \sin^2\theta + b \sin\theta \cos\theta + c \cos^2\theta$ is equal to $k\left[(a + c) + \sqrt{b^2 + (c - a)^2}\right]$ $\forall \theta \in \mathbb{R}$ then $k =$
(a) 1 (b) 2
(c) $\frac{1}{2}$ (d) $\frac{3}{2}$

36. For what and only what values of α lying between 0 and π is the inequality $\sin\alpha \cdot \cos^3\alpha > \sin^3\alpha \cdot \cos\alpha$ valid?
(a) $\alpha \in \left(0, \frac{\pi}{4}\right)$ (b) $\alpha \in \left(0, \frac{\pi}{2}\right)$
(c) $\alpha \in \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$ (d) $\alpha \in \left(\frac{\pi}{2}, \frac{3\pi}{4}\right)$

37. The least value of $\left(\sqrt{3} + \sqrt{2}\right)\cos x + \left(\sqrt{3} - \sqrt{2}\right)\sin x$ is
(a) $\sqrt{7}$ (b) $-\sqrt{10}$
(c) $-\sqrt{13}$ (d) $\sqrt{5}$

38. Given both θ and ϕ are acute angles and $\sin\theta = \frac{1}{2}$, $\cos\phi = \frac{1}{3}$, then the value of $\theta + \phi$ belongs to
(a) $\left(\frac{\pi}{3}, \frac{\pi}{2}\right]$ (b) $\left(\frac{\pi}{2}, \frac{2\pi}{3}\right]$
(c) $\left(\frac{2\pi}{3}, \frac{5\pi}{6}\right]$ (d) $\left(\frac{5\pi}{6}, \pi\right]$

39. For $x \in R$, the difference between the greatest and the least value of $y = \frac{x}{x^2 + 1}$ is
(a) 1 (b) 2
(c) 3 (d) 1/2

Section-B
(Previous Year Question)

1. Let $f : (-1, 1) \to R$ be such that $f(\cos 4\theta) = \frac{2}{2 - \sec^2\theta}$ for $\theta \in \left(0, \frac{\pi}{4}\right) \cup \left(\frac{\pi}{4}, \frac{\pi}{2}\right)$. Then, the value(s) of $f\left(\frac{1}{3}\right)$ is/are
(a) $1 - \sqrt{\frac{3}{2}}$ (b) $1 + \sqrt{\frac{3}{2}}$
(c) $1 - \sqrt{\frac{2}{3}}$ (d) $1 + \sqrt{\frac{2}{3}}$

2. For $0 < \theta < \frac{\pi}{2}$, the solution(s) of $\sum_{m=1}^{6} \operatorname{cosec}\left(\theta + \frac{(m-1)\pi}{4}\right)\operatorname{cosec}\left(\theta + \frac{m\pi}{4}\right) = 4\sqrt{2}$ is/are
(a) $\frac{\pi}{4}$ (b) $\frac{\pi}{6}$
(c) $\frac{\pi}{12}$ (d) $\frac{5\pi}{12}$

3. If $\frac{\sin^4 x}{2}+\frac{\cos^4 x}{3}=\frac{1}{5}$, then

(a) $\tan^2 x=\frac{2}{3}$ (b) $\frac{\sin^8 x}{8}+\frac{\cos^8 x}{27}=\frac{1}{125}$

(c) $\tan^2 x=\frac{1}{3}$ (d) $\frac{\sin^8 x}{8}+\frac{\cos^8 x}{27}=\frac{2}{125}$

4. For a positive integer n, let $f_n(\theta)=\left(\tan\frac{\theta}{2}\right)(1+\sec\theta)(1+\sec 2\theta)(1+\sec 2^2\theta)\ldots.(1+\sec 2^n\theta)$, then

(a) $f_2\left(\frac{\pi}{16}\right)=1$ (b) $f_3\left(\frac{\pi}{32}\right)=1$

(c) $f_4\left(\frac{\pi}{64}\right)=1$ (d) $f_5\left(\frac{\pi}{128}\right)=1$

5. If $\alpha+\beta=\frac{\pi}{2}$ and $\beta+\gamma=\alpha$, then $\tan\alpha$ equals

(a) $2(\tan\beta+\tan\gamma)$ (b) $\tan\beta+\tan\gamma$

(c) $\tan\beta+2\tan\gamma$ (d) $2\tan\beta+\tan\gamma$

6. If $\alpha+\beta+\gamma=2\pi$, then

(a) $\tan\frac{\alpha}{2}+\tan\frac{\beta}{2}+\tan\frac{\gamma}{2}=\tan\frac{\alpha}{2}\tan\frac{\beta}{2}\tan\frac{\gamma}{2}$

(b) $\tan\frac{\alpha}{2}\tan\frac{\beta}{2}+\tan\frac{\beta}{2}\tan\frac{\gamma}{2}+\tan\frac{\gamma}{2}\tan\frac{\alpha}{2}=1$

(c) $\tan\frac{\alpha}{2}+\tan\frac{\beta}{2}+\tan\frac{\gamma}{2}=-\tan\frac{\alpha}{2}\tan\frac{\beta}{2}\tan\frac{\gamma}{2}$

(d) None of the above

7. Let $\theta\in\left(0,\frac{\pi}{4}\right)$ and $t_1=(\tan\theta)^{\tan\theta}$, $t_2=(\tan\theta)^{\cot\theta}$, $t_3=(\cot\theta)^{\tan\theta}$ and $t_4=(\cot\theta)^{\cot\theta}$, then

(a) $t_1>t_2>t_3>t_4$ (b) $t_4>t_3>t_1>t_2$

(c) $t_3>t_1>t_2>t_4$ (d) $t_2>t_3>t_1>t_4$

8. Let a vertical tower AB have its end A on the level ground. Let C be the mid-point of AB and P be a point on the ground such that AP = 2AB. If $\angle BPC=\beta$, then $\tan\beta$ is equal to

(a) $\frac{6}{7}$ (b) $\frac{1}{4}$

(c) $\frac{2}{9}$ (d) $\frac{4}{9}$

9. A man is walking towards a vertical pillar in a straight path, at a uniform speed. At a certain point A on the path, he observes that the angle of elevation of the top of the pillar is 30°. After walking for 10 min from A in the same direction, at a point B, he observes that the angle of elevation of the top of the pillar is 60°. Then, the time taken (in minutes) by him, from B to reach the pillar, is

(a) 6 (b) 10

(c) 20 (d) 5

Section-C

(School / Board pattern)

1. Write the value of $\frac{2\left(\sin^6\theta+\cos^6\theta\right)-3}{\left(\sin^4\theta+\cos^4\theta\right)+1}$

2. Prove that : $\frac{\tan^2 2\theta-\tan^2\theta}{1-\tan^2 2\theta\tan^2\theta}=\tan 3\theta\tan\theta$

3. Prove that : $\frac{1}{\sin(x-a)\cos(x-b)}=\frac{\cot(x-a)+\tan(x-b)}{\cos(a-b)}$

4. If angle θ is divided into two parts such that the tangents of one part is λ times the tangent of other, and ϕ is theri difference, then show that $\sin\theta=\frac{\lambda+1}{\lambda-1}\sin\phi$.

5. If $\sin 2A=\lambda\sin 2B$, Prove that: $\frac{\tan(A+B)}{\tan(A-B)}=\frac{\lambda+1}{\lambda-1}$

6. If $\cos(A+B)\sin(C-D)=\cos(A-B)\sin(C+D)$, prove that $\tan A\tan B\tan C+\tan D=0$

7. If $\cos A=m\cos B$, tehn write the value of $\cot\frac{A+B}{2}\cot\frac{A-B}{2}$.

8. If $\cos\theta=\frac{\cos\alpha+\cos\beta}{1+\cos\alpha\cos\beta}$, prove that $\tan\frac{\theta}{2}=\pm\tan\frac{\alpha}{2}\tan\frac{\beta}{2}$

9. Prove that: $\left(1+\cos\frac{\pi}{10}\right)\left(1+\cos\frac{3\pi}{10}\right)\left(1+\cos\frac{7\pi}{10}\right)\left(1+\cos\frac{9\pi}{10}\right)=\frac{1}{16}$

10. The value of $\sin^2\left(\frac{\pi}{18}\right)+\sin^2\left(\frac{\pi}{9}\right)+\sin^2\left(\frac{7\pi}{18}\right)+\sin^2\left(\frac{4\pi}{9}\right)$ is

CHAPTER 4 TRIGONOMETRIC EQUATIONS

1. SOME BASIC RESULTS

(i) If $\sin\theta = 0$, then $\theta = 0, \pm\pi, \pm 2\pi, ...$
$\Rightarrow$ $\theta = n\pi$ where $n \in I$.

(ii) If $\cos\theta = 0$, then $\theta = \pm\pi/2, \pm 3\pi/2$ .....
$\Rightarrow$ $\theta = (2n + 1)\pi/2$ where $n \in I$.

(iii) If $\tan\theta = 0$ then, $\theta = 0, \pm\pi, \pm 2\pi, ...$
$\Rightarrow$ $\theta = n\pi$ where $n \in I$.

(iv) If $\sin\theta = 1$ then, $\theta = \frac{\pi}{2}, \frac{5\pi}{2}, \frac{9\pi}{2}, ...$

$\Rightarrow$ $\theta = (4n+1)\frac{\pi}{2}$ where $n \in I$.

(v) If $\cos\theta = 1$ then, $\theta = 0, 2\pi, 4\pi, ...$
$\Rightarrow$ $\theta = 2n\pi$ where $n \in I$.

(vi) If $\tan\theta = 1$ then, $\theta = \frac{\pi}{4}, \frac{5\pi}{4}, \frac{9\pi}{4}, ...$

$\Rightarrow$ $\theta = (4n+1)\frac{\pi}{4}$ where $n \in I$.

Illustration 1: Find the values of x lying between 0 and 2π and satisfying the equation : $\sin x + \sin 3x = 0$.

Solution : The given equation is $\sin x + \sin 3x = 0$

$$2\sin\frac{x+3x}{2}\cos\frac{x-3x}{2} = 0$$

$\Rightarrow 2\sin 2x \cos x = 0$

Either : $\sin 2x = 0$ or $\cos x = 0$

$\Rightarrow$ $2x = n\pi$, or $x = (2n + 1)\pi/2$, [where, $n \in I$]

$\Rightarrow$ $x = n\pi/2$, or $x = (2n + 1)\pi/2$, [where, $n \in I$]

This is the general solution of the equation. To get particular solution satisfying $0 < x < 2\pi$, we will substitute integral values of n.

(i) $n = 0 \Rightarrow x = 0, \pi/2$ (ii) $n = 1 \Rightarrow x = \pi/2, 3\pi/2$

(iii) $n = 2 \Rightarrow x = \pi, 5\pi/2$ (iv) $n = 3 \Rightarrow x = 3\pi/2, 7\pi/2$

Hence for $0 < x < 2\pi$, the solution is $x = \pi/2, \pi, 3\pi/2$.

SOME MORE BASIC RESULTS

(i) If $\sin\theta = \sin\alpha$
then $\theta = n\pi + (-1)^n\alpha$, where $n \in I$

(ii) If $\cos\theta = \cos\alpha$
then $\theta = 2n\pi \pm \alpha$, where $n \in I$

(iii) If $\tan\theta = \tan\alpha$
then $\theta = n\pi + \alpha$, where $n \in I$

(iv) If $\tan^2\theta = \tan^2\alpha$
then $\theta = n\pi \pm \alpha$, where $n \in I$

(v) If $\cos^2\theta = \cos^2\alpha$
then $\theta = n\pi \pm \alpha$, where $n \in I$

(vi) If $\sin^2\theta = \sin^2\alpha$
then $\theta = n\pi \pm \alpha$, where $n \in I$

***Illustration 2*:** Find the values of θ satisfying $\sin\theta = \sin\alpha$.

Solution : $\sin\theta = \sin\alpha$

$\Rightarrow \sin\theta - \sin\alpha = 0 \Rightarrow 2\cos\frac{\theta+\alpha}{2}\sin\frac{\theta-\alpha}{2} = 0$

Either : $\cos\frac{\theta+\alpha}{2} = 0$ or $\sin\frac{\theta-\alpha}{2} = 0$

$\Rightarrow \frac{\theta+\alpha}{2} = (2\ell+1)\frac{\pi}{2}$ or $\frac{\theta-\alpha}{2} = n\pi$ [where ℓ, n are integers]

$\theta = (2\ell + 1)\pi - \alpha$ or $\theta = 2n\pi + \alpha$

$\theta = (\text{odd no.})\pi - \alpha$ or $\theta = (\text{even no.})\pi + \alpha$

$\theta = n\pi + (-1)^n\alpha$, $n \in I$. [where $n \in I$]

Illustration 3: Find the values of θ satisfying $\cos\theta = \cos\alpha$ in the interval $0 \le \theta \le \pi$.

Solution : $\cos\theta = \cos\alpha$

$\Rightarrow \cos\theta - \cos\alpha = 0 \Rightarrow -2\sin\frac{\theta+\alpha}{2}\sin\frac{\theta-\alpha}{2} = 0$

Either $\sin\frac{\theta+\alpha}{2} = 0$

or $\sin\frac{\theta-\alpha}{2} = 0 \Rightarrow \frac{\theta+\alpha}{2} = n\pi$

or $\frac{\theta-\alpha}{2} = n\pi$ [where, $n \in I$]

$\Rightarrow \theta = 2n\pi - \alpha$

or $\theta = 2n\pi + \alpha$ [where, $n \in I$]

Combining the two values, we get :

$\theta = 2n\pi \pm \alpha$ $n \in I$

Illustration 4: Find the values of θ satisfying the equation : $\tan\theta = \tan\alpha$.

Solution : $\tan\theta = \tan\alpha$

$\Rightarrow \frac{\sin\theta}{\cos\alpha} = \frac{\sin\alpha}{\cos\alpha} \Rightarrow \sin\theta\cos\alpha - \cos\theta\sin\alpha = 0$

$\Rightarrow \sin(\theta - \alpha) = 0 \Rightarrow \theta - \alpha = n\pi, n \in I \Rightarrow \theta = n\pi + \alpha, n \in I$

Note :- The following

(i) $\sin\theta = \sin\alpha \Rightarrow \theta = n\pi + (-1)^n\alpha$, $n \in I$

(ii) $\cos\theta = \cos\alpha \Rightarrow \theta = 2n\pi \pm \alpha$, $n \in I$

(iii) $\tan\theta = \tan\alpha \Rightarrow \theta = n\pi + \alpha$, $n \in I$

Important : Every trigonometric equation should be manipulated so that it reduces to any of the above results.

Student Task

Q.1 The solution of the equation $e^{\sin x} - e^{-\sin x} - 4 = 0$ is

(a) $x = 0$ (b) $x = \sin^{-1}[\log(2 - \sqrt{5})]$

(c) No real solution (d) None of these

Q.2 The number of real solution of the equation $\sin(e^x) = 5^x + 5^{-x}$ is

(a) 0 (b) 1 (c) 2 (d) Infinitely many

Q.3 The equation $\sin^4 x - (k+2)\sin^2 x - (k+3) = 0$ possesses a solution if

(a) $k > -3$ (b) $k < -2$ (c) $-3 \le k \le -2$ (d) k is any (+ve) value

Q.4 In interval $\left[\frac{-\pi}{2}, \frac{\pi}{2}\right]$, the equation $\log_{\sin\theta}(\cos^2\theta) = 2$ has

(a) No solution (b) A unique solution

(c) Two solution (d) Infinitely many solution

Q.5 If max $\{5\sin\theta + 3\sin(\theta - \alpha)\} = 7$, then the set of possible value of α is $\theta \in R$

(a) $\left\{x : x = 2n\pi \pm \frac{\pi}{3}, n \in I\right\}$ (b) $\left\{x : x = 2n\pi \pm \frac{2\pi}{3}, n \in I\right\}$

(c) $\left[\frac{\pi}{3}, \frac{2\pi}{3}\right]$ (d) None of these

Q.6 Total number of solutions of $\sin x = \frac{|x|}{10}$ is equal to

(a) 4 (b) 6 (c) 7 (d) None of these

Q.7 If $4\sin^2 x - 8\sin x + 3 \le 0$, $0 \le x \le 2\pi$, then the solution set for x is

(a) $\left[0, \frac{\pi}{6}\right]$ (b) $\left[0, \frac{5\pi}{6}\right]$ (c) $\left[\frac{5\pi}{6}, 2\pi\right]$ (d) $\left[\frac{\pi}{6}, \frac{5\pi}{6}\right]$

Q.8 The general solution of the equation $\frac{1 - \sin x + ... + (-1)^n \sin^n x + ...}{1 + \sin x + ... + \sin^n x + ...} = \frac{1 - \cos 2x}{1 + \cos 2x}$ is

(a) $(-1)^n\left(\frac{\pi}{3}\right) + n\pi$ (b) $(-1)^n\left(\frac{\pi}{6}\right) + n\pi$

(c) $(-1)^{n+1}\left(\frac{\pi}{6}\right) + n\pi$ (d) $(-1)^{n-1}\left(\frac{\pi}{3}\right) + n\pi, n \in I$

Answers

1. (c) **2.** (a) **3.** (c) **4.** (b) **5.** (a) **6.** (b) **7.** (d) **8.** (b)

2. HOW TO SOLVE TRIGONOMETRIC EQUATIONS

While solving equations following points must be kept in mind.

(i) Squaring should be avoided as far as possible, if squaring is done check for the extraneous roots.

(ii) Never cancel equal terms containing 'unknown or variable' on two sides which are in product. It may cause root loss.

(iii) The answer should not contain such values of θ which make any of the terms undefined.

(iv) Domain should not change. If it changes, necessary correction must be made.

(v) Check that denominator is not zero at any stage while solving equations.

TYPE-I

Problems based on method of factorisation :

Working Rule :

Step I. Take all the terms to L.H.S. i.e. make R.H.S. zero, then factorise L.H.S.

Step II. Equate each factor to zero and solve.

Step III. Check for root loss and extraneous roots.

Illustration 5: Solve the equation $\cos x + \cos 2x + \cos 4x = 0$.

Solution : $\cos x + (\cos 2x + \cos 4x) = 0$

$\Rightarrow \cos x + 2\cos 3x \cos x = 0$

$\Rightarrow \cos x (1 + 2\cos 3x) = 0$

$\Rightarrow \cos x = 0$

OR $1 + 2\cos 3x = 0$

$\Rightarrow \cos x = 0$

or $\cos 3x = -1/2 = \cos 2\pi/3$

$\Rightarrow$ $x = (2n + 1)\,\pi/2$

or $3x = 2n\pi \pm 2\pi/3$ [where, $n \in I$]

$\Rightarrow$ $x = (2n + 1)\,\pi/2$,

or $x = 2n\pi/3 \pm 2\pi/9$, [where, $n \in I$]

This is the general solution of the equation. To get particular solution satisfying $0 \le x \le \pi$, we will substitute integral values of n.

(i) $n = 0 \Rightarrow x = \pi/2, \pm 2\pi/9$

(ii) $n = 1 \Rightarrow x = 3\pi/2, 8\pi/9, 4\pi/9$

(iii) $n = 2 \Rightarrow x = 5\pi/2, 14\pi/9, 10\pi/9$ (greater than π)

(iv) $n = -1 \Rightarrow x = -\pi/2, -2\pi/3 \pm 2\pi/9$ (less than 0)

Hence the values for $0 \le x \le \pi$ are $x = \pi/2, 2\pi/9, 4\pi/9, 8\pi/9$.

***Illustration 6*:** Solve the equation $\sin x = \cos 4x$ for $0 \le x \le \pi$.

Solution : $\sin x = \cos 4x$

$\Rightarrow$ $\cos 4x = \cos(\pi/2 - x)$

$\Rightarrow$ $4x = 2n\pi \pm (\pi/2 - x)$

Either $4x = 2n\pi + \pi/2 - x$

or $4x = 2n\pi - \pi/2 + x$

$\Rightarrow$ $x = 2n\pi/5 + \pi/10$

or $x = 2n\pi/3 - \pi/6$

This is the general solution of the equation. To get particular solution satisfying $0 \le x \le \pi$, we will substitute integral values of n.

(i) $n = 0 \Rightarrow x = \pi/10, -\pi/6$

(ii) $n = 1 \Rightarrow x = \pi/2$

(iii) $n = 2 \Rightarrow x = 9\pi/10, 7\pi/6$

(iv) $n = 3 \Rightarrow x = 13\pi/10, 11\pi/6$ (greater than π)

(v) $n = -1 \Rightarrow x = -3\pi/10, -5\pi/6$ (less than 0)

Hence the required solution for $0 \le x \le \pi$ is $x = \pi/10, \pi/2, 9\pi/10$.

Illustration 7: Find the values of x satisfying $3 - 2\cos x - 4\sin x - \cos 2x + \sin 2x = 0$

Solution : Put $\cos 2x = 2\cos^2 x - 1$

$\Rightarrow$ $3 - 2\cos x - 4\sin x - (2\cos^2 x - 1) + \sin 2x = 0$

$\Rightarrow$ $(4 - 4\sin x) - 2\cos^2 x - 2\cos x + \sin 2x = 0$

$\Rightarrow$ $4(1 - \sin x) - 2(1 - \sin^2 x) - 2\cos x\,(1 - \sin x) = 0$

Either $\sin x = 1$ or $\sin x + \cos x = 1$

$\Rightarrow$ $\sin x = \sin \pi/2$ or $\sqrt{2}\cos(x - \pi/4) = 1$

$\Rightarrow$ $x = n\pi + (-1)^n\,\pi/2$ or $x - \pi/4 = 2n\pi \pm \pi/4$ [where, $n \in I$]

$\Rightarrow$ $x = n\pi + (-1)^n\,\pi/2$ or $x = 2n\pi \pm \pi/4 + \pi/4$ [where, $n \in I$]

$\Rightarrow$ $x = n\pi + (-1)^n\,\pi/2$ or $x = 2n\pi,\ 2n\pi + \pi/2$

Combining the two, we get : $x = 2n\pi,\ 2n\pi + \pi/2$.

Illustration 8: Solve for x : $\sec 4x - \sec 2x = 2$; in $-\pi \le x \le \pi$.

Solution : $\sec 4x - \sec 2x = 2$

$\Rightarrow$ $\dfrac{1}{\cos 4x} - \dfrac{1}{\cos 2x} = 2$

$\Rightarrow$ $\cos 2x - \cos 4x = 2\cos 2x \cos 4x$

$\Rightarrow$ $\cos 2x - \cos 4x = \cos 6x + \cos 2x$

$\Rightarrow$ $\cos 6x + \cos 4x = 0 \Rightarrow 2\ \cos x.\cos x = 0$

Either $\cos 5x = 0$ or $\cos x = 0$

$\Rightarrow$ $5x = 2n\pi \pm \pi/2$

or $x = 2n\pi \pm \pi/2$ [where, $n \in I$]

$\Rightarrow$ $x = 2n\pi/5 \pm \pi/10$

or $x = 2n\pi \pm \pi/2$ [where, $n \in I$]

This is the general solution of the equation. To get particular solution satisfying $-\pi \le x \le \pi$, we will substitute integral values of n.

$n = 0 \quad \Rightarrow x = \pm \pi/10$

$n = \pm 1 \quad \Rightarrow x = \pm \pi/2, -\pi/2, -3\pi/10$

Consider $x = 2n\pi \pm \pi/2$; $n \in \mathbb{Z}$

$n = 0 \quad \Rightarrow x = \pm \pi/2$

These are the only values of x in $[-\pi, \pi]$.

Illustration 9: Solve: fourθ: $\cos\theta \cos 2\theta \cos 3\theta = 1/4$.

Solution : $4 \cos\theta \cos 2\theta \cos 3\theta = 1$

or, $(2 \cos 3\theta \cos\theta)(2 \cos 2\theta) = 1$

or, $(\cos 4\theta + \cos 2\theta)(2 \cos 2\theta) - 1 = 0$

or, $2 \cos 4\theta \cos 2\theta + 2\cos^2 2\theta - 1 = 0$

or, $2\cos 4\theta \cos 2\theta + \cos 4\theta = 0$

or, $\cos 4\theta [2 \cos 2\theta + 1] = 0$

$\Rightarrow$ Either $\cos 4\theta = 0$, or $2 \cos 2\theta + 1 = 0$.

$\Rightarrow \quad 4\theta = (2n+1)\frac{\pi}{2}$, h $\in$ z

or $\quad \cos 2\theta = -\frac{1}{2} = \cos\frac{2\pi}{3}$

$\Rightarrow \quad \theta = (2n+1)\frac{\pi}{8}; n \in \mathbb{Z}$

or $\quad 2\theta = 2n\pi \pm \frac{2\pi}{3}; n \in \mathbb{Z}$

$\Rightarrow \quad \theta = (2n+1)\frac{\pi}{8}$ or $\theta = n\pi \pm \frac{\pi}{3}; n \in \mathbb{Z}$ [where, $n \in z$]

Hence,

$\theta = (2n+1)\frac{\pi}{8}, n\pi \pm \frac{\pi}{3}$ where $n = 0, \pm 1, \pm 2, ...$

Illustration 10: Find all solutions of the equation, $4 \cos^2 x \sin x - 2 \sin^2 x = 3 \sin x$.

Solution : $4 \cos^2 x \sin x - 2 \sin^2 x - 3 \sin x = 0 \quad \Rightarrow \sin x [4 \cos^2 x - 2 \sin x - 3] = 0$

$\Rightarrow \sin x [4 - 4 \sin^2 x - 2 \sin x - 3] = 0 \quad \Rightarrow \sin x = 0$ or $4 \sin^2 x + 2 \sin x - 1 = 0$

If $\sin x = 0, x = n\pi$ or If $4 \sin^2 x + 2 \sin x - 1 = 0$

$$\sin x = \frac{-2 \pm \sqrt{4 - 4.4(-1)}}{2.4} = \frac{-1 \pm \sqrt{5}}{4}$$

when $\sin x = \frac{-1+\sqrt{5}}{4} \quad \Rightarrow \quad \sin x = \sin\frac{\pi}{10}$ [As $\sin 18° = \frac{\sqrt{5}-1}{4}$ and $18° = \frac{\pi}{10}$]

$\Rightarrow \quad x = n\pi + (-1)^n \frac{\pi}{10}; n \in I$

When $\sin x = \frac{-1-\sqrt{5}}{4}, \sin x = -\left(\frac{\sqrt{5}+1}{4}\right) = -\cos 36° \quad \left[\text{As } \cos 36° = \frac{\sqrt{5}+1}{4}\right]$

or, $\sin x = -\sin 54°$

or, $\sin x = \sin\left(-\frac{3\pi}{10}\right)$

$\Rightarrow \quad x = n\pi + (-1)^n \left(-\frac{3\pi}{10}\right)$

$\therefore \quad x = n\pi + (-1)^n \frac{\pi}{10}, \quad n\pi + (-1)^n \left(-\frac{3\pi}{10}\right); n \in I$

TYPE-II

Equations of the form $a\cos\theta + b\sin\theta = c$

Working Rule :

(i) Divide by $\sqrt{a^2+b^2}$ on both sides and get $\frac{a}{\sqrt{a^2+b^2}}\cos\theta + \frac{b}{\sqrt{a^2+b^2}}\sin\theta = \frac{c}{\sqrt{a^2+b^2}}$.

(ii) Write L.H.S. as $\sqrt{a^2+b^2}\cos(\theta-\alpha)$ [where $\cos\alpha = \frac{a}{\sqrt{a^2+b^2}}$ and $\sin\alpha\frac{b}{\sqrt{a^2+b^2}}$]

(iii) Simplify the equation.

Illustration 11: Solve the equation $\sqrt{3}\sin x + \cos x = 1$ in the interval $0 \le x \le 2\pi$.

Solution : $\sqrt{3}\sin x + \cos x = 1$

$\Rightarrow$ $2(\sqrt{3}/2\sin x + 1/2\cos x) = 1$

[Multiplying and divide by $\sqrt{(\sqrt{3})^2+(1)^2}$]

$\Rightarrow$ $2(\cos\pi/3\cos x + \sin\pi/3\sin x) = 1$

$\Rightarrow$ $2\cos(x - \pi/3) = 1$

$\Rightarrow$ $\cos(x - \pi/3) = \cos\pi/3$

$\Rightarrow$ $x - \pi/3 = 2n\pi \pm \pi/3$

$\Rightarrow$ $x = 2n\pi + 2\pi/3,\ x = 2n\pi$ [where, $n \in I$]

This is the general solution of the equation. To get particular solution satisfying $0 \le x \le 2\pi$ we will substitute integral values of n.

(i) $n = 0 \Rightarrow x = 0$ or $2\pi/3$

(ii) $n = 1 \Rightarrow x = 2\pi + 2\pi/3$ or 2π

(iii) $n = 2 \Rightarrow x = 4\pi + 2\pi/3$

or 4p (greater than π)

(iv) $n = -1 \Rightarrow x = -4\pi/3$

or -2π (less than 0)

Hence the required values of x are $0, 2\pi/3, 2\pi$.

Illustration 12: Solve the equation : $\sin x + \cos x = \sin 2x - 1$.

Solution : Let $t = \sin x + \cos x$

$\Rightarrow$ $t^2 = 1 + 2\sin x\cos x$

$\Rightarrow$ $\sin 2x = t^2 - 1$

Hence the given equation is $t = (t^2 - 1) - 1$

$\Rightarrow$ $t^2 - t - 2 = 0$

Solving the equation to get :

$(t - 2)(t + 1) = 0$

$\Rightarrow$ $t = 2$ or $t = -1$

$\Rightarrow$ $\sin x + \cos x = 2$

or $\sin x + \cos x = -1$

$\Rightarrow$ $\sqrt{2}\cos(x - \pi/4) = 2$

or $\sqrt{2}\cos(x - \pi/4) = -1$

$\Rightarrow$ $\cos(x - \pi/4) = \sqrt{2}$

or $\cos(x - \pi/4) = -\frac{1}{\sqrt{2}}$

As $-1 \le \cos\theta \le 1$, $\cos(x - \pi/4) = \sqrt{2}$ is impossible

$\Rightarrow$ $\cos(x - \pi/4) = -\frac{1}{\sqrt{2}}$ is the only possibility

$\Rightarrow$ $\cos(x - \pi/4) = \cos(\pi - \pi/4)$

$\Rightarrow$ $x - \pi/4 = 2n\pi \pm 3\pi/4$ [where, $n \in I$]

$\Rightarrow$ $x = 2n\pi \pm 3\pi/4 + \pi/4$ is the general solution

Illustration 13: Solve the equation $\sin^4 x + \cos^4 x = \frac{7}{2} \sin x \cos x$.

Solution : $\sin^4 x + \cos^4 x = 7/2 \sin x \cos x$

$\Rightarrow \quad (\sin^2 x + \cos^2 x)^2 - 2 \sin^2 x \cos^2 x = \frac{7}{2} \sin x \cos x$

Let $t = 2 \sin x \cos x = \sin 2x$

$\Rightarrow \quad 1 - 2/4\, t^2 = 7/4\, t$

$\Rightarrow \quad 2t^2 + 7t - 4 = 0$

$\Rightarrow \quad (2t - 1)(t + 4) = 0$

$\Rightarrow \quad t = 1/2 \quad$ or $\quad t = -4$

$\Rightarrow \quad \sin 2x = 1/2 \quad$ or $\quad \sin 2x = -4$ (impossible)

$\Rightarrow \quad \sin 2x = \sin \pi/6$

$\Rightarrow \quad 2x = n\pi + (-1)^n \pi/6,\ n \in I$

$x = n\pi/2 + (-1)^n \pi/12$ is the general solution

3. TRIGONOMETRIC INEQUALITY

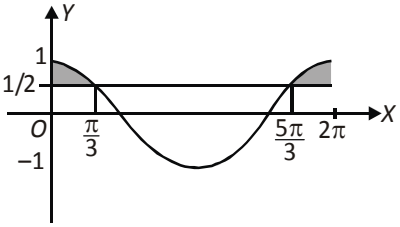
Working Rule :

While solving inequations involving trigonometric functions, it is best to use graphs of trigonometric functions

Illustration 14 **:** Solve : $\cos x \geq \frac{1}{2}$.

Solution : We construct the graph $y_1 = \cos x$, $y_2 = + \frac{1}{2}$.

Now, on this graph we want those values of x for which graph of $y = \cos x$ is more than the graph of $y = \frac{1}{2}$ which is shown by colour part of the graph



We get solution as $\left[0, \frac{\pi}{3}\right] \cup \left[\frac{5\pi}{3}, 2\pi\right]$ or general solution is

$2n\pi \leq x \leq \frac{\pi}{3} + 2n\pi$ and $2n\pi + \frac{5\pi}{3} \leq x \leq 2\pi + 2n\pi;\ n \in \mathbb{Z}$

Illustration 15: Solve: the inequality $\sin x + \cos 2x > 1$ if $0 \leq x \leq \pi/2$.

Solution : Let $\sin x = t \quad \Rightarrow \cos 2x = 1 - 2t^2$

$\Rightarrow$ The inequality is :

$t + 1 - 2t^2 > 1$

$\Rightarrow \quad 2t^2 - t < 0$

$\Rightarrow \quad t(2t - 1) < 0$

$\Rightarrow \quad (t - 0)(t - 1/2) < 0$

$\Rightarrow \quad 0 < t < 1/2$

$\Rightarrow \quad 0 < \sin x < 1/2$

In $0 \leq x \leq \pi/2$, this means than $0 \leq x \leq \pi/6$ is the solution.

Illustration 16: Solve $\tan^2 x - (1 + \sqrt{3})\tan x + \sqrt{3} < 0$.

Solution : The given inequality is $(\tan x - 1)(\tan x - \sqrt{3}) < 0$

$\Rightarrow 1 < \tan x < \sqrt{3} \Rightarrow \frac{\pi}{4} < x < \frac{\pi}{3}$

Since the tangent function repeats after an interval of length π, so the general solution is

$n\pi + \frac{\pi}{4} < x < n\pi + \frac{\pi}{3}$; $n \in \mathbb{Z}$

4. SIMULTANEOUS EQUATIONS

Two equations in one variable.

Working Rule :

(i) If two equations in one unknown (say x) are given, then solve given equations separately for x lying between 0 and 2π.

(ii) Select the value of x for which both the equations are satisfied.

(iii) Add $2n\pi$ to get general solution.

Illustration 17: Solve $\cos x = \frac{1}{\sqrt{2}}$, and $\tan x = -1$.

Solution : $\cos x = \frac{1}{\sqrt{2}} \Rightarrow x = \frac{\pi}{4}, \frac{7\pi}{4}$ in $(0, 2\pi)$

$\tan x = -1 \Rightarrow x = \frac{3\pi}{4}, \frac{7\pi}{4}$ in $(0, 2\pi)$

$\therefore$ Common value of $x = \frac{7\pi}{4}$

Now for general solution add a factor of $2n\pi$.

$\therefore$ General solution is $x = 2n\pi + \frac{7\pi}{4}$; $n \in \mathbb{Z}$

5. TWO EQUATIONS IN TWO VARIABLE

Working Rule :

Step I. Find $A + B$ and $A - B$ between 0 and 2π from the two given equations. Impose the condition that A and B must be smallest and positive, since A and B are positive angles, $A + B > A - B$. Find $A + B$ and $A - B$ accordingly.

Step II. Solve the two equation to get A and B.

Step III. Then generalised it with the help of common period. i.e. adding a factor of np.

Illustration 18: If $\tan(A - B) = 1$, $\sec(A + B) = 2/\sqrt{3}$, calculate the smallest positive values and the most general values of A and B.

Solution : Smallest Positive Values

Let $A, B \in (0, 2\pi)$

$\Rightarrow (A + B) > (A - B)$

Now $\tan(A - B) = 1 \Rightarrow (A - B) = \pi/4, 5\pi/4$

$\sec(A + B) = 3/\sqrt{3} \Rightarrow (A + B) = \pi/6, 11\pi/6$

As $(A + B) > (A - B)$, there are two possibilities :

(i) $A - B = \pi/4$ and $A + B = 11\pi/6$

(ii) $A - B = 5\pi/4$ and $A + B = 11\pi/6$

From (i), we get : $A = \frac{25\pi}{24}$ and $B = \frac{19\pi}{24}$

From (ii), we get : $A = \frac{37\pi}{24}$ and $B = \frac{7\pi}{24}$

General Values

$\tan(A-B)=1 \Rightarrow A-B=n\pi+\pi/4\ ;\ n\in\mathbb{Z}$

$\sec(A+B)=\frac{2}{\sqrt{3}} \Rightarrow A+B=2k\pi\pm\frac{\pi}{6}\ ;\ n\in\mathbb{Z}$

Taking $A-B=n\pi+\frac{\pi}{4}$ and $A+B=2k\pi+\frac{\pi}{6}$ we get :

$A=\frac{(2k+n)\pi}{2}+\frac{5\pi}{24}$ and $B=\frac{(2k-n)\pi}{2}-\frac{\pi}{24}$

Taking $A-B=n\pi+\frac{\pi}{4}$ and $A+B=2k\pi-\frac{\pi}{6}$ we get :

$A=\frac{(2k+n)}{2}+\frac{\pi}{24}$ and $B=\frac{(2k-n)}{2}+\frac{5\pi}{24}$

MISCELLANEOUS TYPE :

(i) Whenever the terms on the two sides (L.H.S. and R.H.S.) of the equation are the different nature, We use inequality method for testing whether the equation has any real solution or not.

Working Rule :

(i) Let y = each side of the equation and break the equation in two parts.

(ii) From the inequality for y taking L.H.S. of the equation and also for R.H.S. of the equation. If there is any value of y satisfying both the inequalities, then the equation will have real solution and if there is no such y, the equation will have no real solution.

Illustration 19: Show that the equation $2\cos^2\left(\frac{x}{2}\right)\sin^2 x=x^2+x^{-2}$ for $0<x\le\frac{\pi}{2}$ has no real solution.

Solution : Let $y_1=2\cos^2\left(\frac{x}{2}\right)\sin^2 x$ ..... (i)

and $y_2=x^2+x^{-2}$ ..... (ii)

From (i), $y_1=2\cos^2\frac{x}{2}.\sin^2 x$

$=(<2)\times(\le 1)\ <2$

i.e. $y_1<2$ ..... (iii)

From (ii), $y_2=x^2+x^{-2}=x^2+\frac{1}{x^2}=\left(x-\frac{1}{x}\right)^2+2\ge 2$ $[\because x>0]$

i.e. $y_2\ge 2$ ..... (iv)

No value of y can be obtained satisfying (iii) and (iv) simultaneously,

$\Rightarrow$ no real solution of the equation exists.

(ii) Whenever the equation contains power terms.

Working Rule :

(i) Equate the base if possible.

(ii) If not possible to equate the base then take log both side.

Illustration 20: Find the values of x in $(-\pi,\pi)$ which satisfy the equation

$8^{1+|\cos x|+|\cos^2 x|+|\cos^3 x|+.....\text{to infinity}}=4^3.$

Solution : $8^{1+|\cos x|+|\cos^2 x|+|\cos^3 x|+.....\text{to infinity}}=8^2$

$\Rightarrow$ $1+|\cos x|+|\cos^2 x|+|\cos^3 x|+.....$ to infinity $=2$

This is an infinite geometric series with first term 1 and common ratio $|\cos x|$.

$\Rightarrow$ $\frac{1}{1-|\cos x|}=2$

[By using sum of infinite geometric series with first term a and common ratio r is $\frac{a}{1-r}$ if $|r| < 1$]

or, $1-|\cos x| = \frac{1}{2} \Rightarrow |\cos x| = \frac{1}{2} \Rightarrow \cos x = \pm\frac{1}{2}$

When $\cos x = \frac{1}{2} = \cos\frac{\pi}{3}, x = 2n\pi \pm \frac{\pi}{3}$

When $\cos x = -\frac{1}{2} = \cos\frac{2\pi}{3}, x = 2n\pi \pm \frac{2\pi}{3}$

The value of x in the given interval $(-\pi, \pi) = \pm\frac{\pi}{3}, \pm\frac{2\pi}{3}$.

***Illustration 21*:** Solve : $|\cos x|^{\sin^2 x - \frac{3}{2}\sin x + \frac{1}{2}} = 1$.

Solution : $|\cos x|^{\sin^2 x - \frac{3}{2}\sin x + \frac{1}{2}} = 1$

Taking log of both sides, we get :

$$\left(\sin^2 x - \frac{3}{2}\sin x + \frac{1}{2}\right)\log|\cos x| = \log 1 = 0$$

If $\sin^2 x - \frac{3}{2}\sin x + \frac{1}{2} = 0$

$2\sin^2 x - 3\sin x + 1 = 0$

or, $(\sin x - 1)(2\sin x - 1) = 0$

When $\sin x - 1 = 0$, $\sin x = 1 \Rightarrow |\cos x| = 0$

$\Rightarrow$ no solution as log0 is not defined.

When $2\sin x - 1 = 0, \sin x = \frac{1}{2} = \sin\frac{\pi}{6}$.

$\Rightarrow x = n\pi + (-1)^n \frac{\pi}{6}$

If $\log|\cos x| = 0$, $|\cos x| = 1 \Rightarrow \cos x = \pm 1$

Taking positive sign,

$\cos x = 1 = \cos 0 \Rightarrow x = n\pi$

Taking negative sign,

$\cos x = -1 = \cos\pi \Rightarrow x = 2n\pi \pm \pi$

$\therefore$ $x = 2n\pi, 2n\pi \pm \pi, n\pi + (-1)^n \frac{\pi}{6}$.

TYPE-VI

Whenever terms are in sin, cos in power 1, all terms connected with plus sign and number of terms in L.H.S. is equal to the number in R.H.S. [with (+) or (–) sign] then each term must have its exteremum value.

Working Rule :

In such problems each term will be (+1) when the number in R.H.S. is (+)ve and each term will be (–1) when the number in R.H.S. is (–)ve.

Illustration 22: Solve the equation $\sin 6x + \cos 4x + 2 = 0$.

Solution : $\sin 6x + \cos 4x + 2 = 0$ or $\sin 6x + \cos 4x = -2$ ..... (1)

$\Rightarrow \sin 6x = -1$ and $\cos 4x = -1$ both satisfied simultaneously

Now, $\sin 6x = -1 = \sin\frac{3\pi}{2} \Rightarrow 6x = 2n\pi + \frac{3\pi}{2} \Rightarrow x = \frac{n\pi}{3} + \frac{\pi}{4}$, where $n \in I$.

$\Rightarrow$ Values of x between 0 and 2π are $\frac{\pi}{4}, \frac{7\pi}{12}, \frac{11\pi}{12}, \frac{5\pi}{4}, \frac{19\pi}{12}, \frac{23\pi}{12}$

Also, $\cos 4x = -1 = \cos\pi \Rightarrow 4x = 2n\pi + \pi$

$\therefore \quad x = \frac{n\pi}{2} + \frac{\pi}{4}$, where $n \in I$.

$\Rightarrow$ Values of x between 0 and 2π are $\frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$

Hence, values of x lying between 0 and 2π satisfying both the equations are $\frac{\pi}{4}, \frac{5\pi}{4}$

$\therefore$ General solution will be given by

$x = 2n\pi + \frac{\pi}{4}$ or, $2n\pi + \frac{5\pi}{4}$ [where $n \in I$]

i.e. $x = 2n\pi + \frac{\pi}{4}$ or, $(2n+1)\pi + \frac{\pi}{4}$

Combining these two results, to get :

$x = m\pi + \frac{\pi}{4}$, [where $n \in I$]

Illustration 23: Solve the equation $\cos^7 x + \sin^4 x = 1$.

Solution : $\because \cos^7 x \le \cos^2 x$ ..... (i)

and $\sin^4 x \le \sin^2 x$ ..... (ii)

$\therefore \quad \cos^7 x + \sin^4 x \le \cos^2 x + \sin^2 x$

or, $\cos^7 x + \sin^4 x \le 1$.

Clearly L.H.S. of equation does not exceed unity and is equal to 1 only when equality occurs in both (i) and (ii).

i.e. Equation is satisfied only when

$\cos^7 x = \cos^2 x$ ..... (iii)

and $\sin^4 x = \sin^2 x$ ..... (iv)

From (iii), $\cos^7 x = \cos^2 x$

or, $\cos^2 x (\cos^5 x - 1) = 0$

$\Rightarrow$ either : $\cos x = 0$ or $\cos x = 1$

$\Rightarrow \quad x = (2n+1)\frac{\pi}{2}$ or $x = 2n\pi$

Clearly (iv) is also satisfied for these values of x.

[When $\cos x = 0$, $\sin x = \pm 1$ and when $\cos x = 1$. $\sin x = 0$]

Hence the required solutions are $x = (2n+1)\frac{\pi}{2}$ or, $2n\pi$. where $n \in I$.

EXERCISE

Q.1 In ΔABC, the line joining the circumcentre to the incentre is parallel to BC, then $\cos B + \cos C$ is equal to

(a) $\frac{3}{2}$ (b) 1 (c) $\frac{3}{4}$ (d) $\frac{1}{2}$

Q.2 The general solution of $\tan\theta + \tan 4\theta + \tan 7\theta = \tan\theta \tan 4\theta . \tan 7\theta$ is

(a) $\theta = \frac{n\pi}{4}$ (b) $\theta = \frac{n\pi}{12}$ (c) $\theta = \frac{n\pi}{6}$ (d) None of these

Q.3 If $\frac{\sin^3\theta - \cos^3\theta}{\sin\theta - \cos\theta} - \frac{\cos\theta}{\sqrt{1+\cot^2\theta}} - 2\tan\theta.\cot\theta = -1$ where $\theta \in [0, 2\pi]$ then

(a) $\theta \in \left(0, \frac{\pi}{2}\right) - \left\{\frac{\pi}{4}\right\}$ (b) $\theta \in \left(\frac{\pi}{2}, \pi\right) - \left\{\frac{3\pi}{4}\right\}$

(c) $\theta \in \left(\pi, \frac{3\pi}{2}\right) - \left\{\frac{5\pi}{4}\right\}$ (d) $\theta \in (0, \pi) - \left\{\frac{\pi}{4}, \frac{\pi}{2}\right\}$

Q.4 Total number of integral values of 'n' so that $\sin x (\sin x + \cos x) = n$ has at least one solution, is
(a) 2 (b) 1 (c) 3 (d) Zero

Q.5 The number of roots of the equation $x + 2\tan x = \frac{\pi}{2}$ in the interval $[0, 2\pi]$ is
(a) 1 (b) 2 (c) 3 (d) Infinite

Q.6 The set of value of x four which $\sin x \cos^3 x > \cos x . \sin^3 x$, $0 \le x \le \pi$, is
(a) $(0, \pi)$ (b) $\left(0, \frac{\pi}{4}\right)$ (c) $\left(\frac{\pi}{4}, \pi\right)$ (d) none of these

Q.7 If $\frac{\sin 3\theta}{2\cos 2\theta + 1} = \frac{1}{2}$ then
(a) $\theta = n\pi + \frac{\pi}{6}; n \in I$ (b) $\theta = n\pi - \frac{\pi}{6}; n \in I$
(c) $\theta = n\pi + (-1)^n \frac{\pi}{6}; n \in I$ (d) $\theta = n\pi - \frac{\pi}{6}; n \in I$

Q.8 If $\frac{\cos 3\theta}{2\cos 2\theta - 1} = \frac{1}{2}$ then
(a) $\theta = n\pi + \frac{\pi}{3}$ (b) $\theta = 2n\pi \pm \frac{\pi}{3}$ (c) $\theta = 2n\pi \pm \frac{\pi}{6}$ (d) $\theta = 2n\pi + \frac{\pi}{6}$

Q.9 Solution of $4\sin^2\theta + \sqrt{3} = 2(1 + \sqrt{3})\sin\theta$ is
(a) $\theta = n\pi + (-1)^n \frac{\pi}{6}$ or $\theta = n\pi + (-1)^n \frac{\pi}{3}$ where $n \in I$
(b) $\theta = 2n\pi + \frac{\pi}{3}; n \in I$
(c) $\theta = n\pi + \frac{\pi}{6}; n \in I$ (d) $\theta = 2n\pi - \frac{\pi}{6}; n \in I$

Q.10 The solution of the equation $\frac{\sin^3 \frac{x}{2} - \cos^3 \frac{x}{2}}{2 + \sin x} = \frac{\cos x}{3}$ is
(a) $x = 2n\pi + \frac{\pi}{2}; n \in I$ (b) $x = 2n\pi + \frac{\pi}{4}; n \in I$
(c) $x = n\pi + \frac{\pi}{4}; n \in I$ (d) None of these

Q.11 The general solution of the equation $\sin^{100} x - \cos^{100} x = 1$ is
(a) $2n\pi + \frac{\pi}{2}, n \in I$ (b) $n\pi + \frac{\pi}{2}, n \in I$ (c) $n\pi - \frac{\pi}{2}, n \in I$ (d) none of these

Q.12 The solution of the inequality $\log_{1/2} \sin x < \log_{1/2} \cos x$ in $[0, 2\pi]$ is
(a) $x \in \left(0, \frac{\pi}{4}\right)$ (b) $x \in \left(-\frac{\pi}{4}, 0\right)$ (c) $x \in \left(0, \frac{\pi}{2}\right)$ (d) none of these

Q.13 If $0 < \theta < 2\pi$ and $2\sin^2\theta - 5\sin\theta + 2 > 0$, then θ lies in the interval
(a) $\left(0, \frac{\pi}{6}\right) \cup \left(\frac{5\pi}{6}, 2\pi\right)$ (b) $\left(\frac{\pi}{6}, \frac{5\pi}{6}\right)$
(c) $(0, \pi)$ (d) $\left(\frac{\pi}{2}, 2\pi\right)$

Q.14 The smallest positive root of the equation $\tan x - x = 0$ lies on

(a) $\left(0,\frac{\pi}{2}\right)$ (b) $\left(\frac{\pi}{2},\pi\right)$ (c) $\left(\pi,\frac{3\pi}{2}\right)$ (d) $\left(\frac{3\pi}{2},2\pi\right)$

Q.15 The number of solution(s) equation $\sin\frac{x}{4}+2\cos\frac{x-2\pi}{3}=3$ is/are

(a) 1 (b) 0 (c) 2 (d) ∞

Answers

1. (b) **2.** (b) **3.** (d) **4.** (a) **5.** (c) **6.** (b) **7.** (c)
8. (b) **9.** (a) **10.** (a) **11.** (b) **12.** (a) **13.** (a) **14.** (c)
15. (b)

MISCELLANEOUS EXAMPLES

Example 1 : (a) The general solution of the equation $(\sqrt{3}-1)\sin\theta+(\sqrt{3}+1)\cos\theta=2$ is –

(a) $2n\pi\pm\frac{\pi}{4}+\frac{\pi}{12}; n\in I$ (b) $n\pi+(-1)^n\frac{\pi}{4}+\frac{\pi}{12}; n\in I$

(c) $2n\pi\pm\frac{\pi}{4}-\frac{\pi}{12}; n\in I$ (d) $n\pi+(-1)^n\frac{\pi}{4}-\frac{\pi}{12}; n\in I$

(b) The values of x between 0 and 2π, which satisfy the equation $\sin x\sqrt{8\cos^2 x}=1$ are in A.P. with common difference

(a)$\pi/4$ (b)$\pi/8$ (c) $3\pi/8$ (d) $5\pi/8$

Solution : (a)

(A) Dividing by $\sqrt{(\sqrt{3}-1)^2+(\sqrt{3}+1)^2}=2\sqrt{2}$

$$\frac{\sqrt{3}-1}{2\sqrt{2}}\sin\theta+\frac{\sqrt{3}+1}{2\sqrt{2}}\cos\theta=\frac{1}{\sqrt{2}},\sin\frac{\pi}{12}=\frac{\sqrt{3}-1}{2\sqrt{2}},\cos\frac{\pi}{12}=\frac{\sqrt{3}+1}{2\sqrt{2}}$$

$$\sin\frac{\pi}{12}\sin\theta+\cos\frac{\pi}{12}\cos\theta=\frac{1}{\sqrt{2}}$$

$$\cos\left(\theta-\frac{\pi}{12}\right)=\frac{1}{\sqrt{2}}=\cos\left(2n\pi\pm\frac{\pi}{4}\right)$$

$$\Rightarrow\ \theta-\frac{\pi}{12}=2n\pi\pm\frac{\pi}{4}\ \Rightarrow\ \theta=2n\pi\pm\frac{\pi}{4}+\frac{\pi}{12}; n\in I$$

(b)(A) We have $\sin x\sqrt{8\cos^2 x}=1$

$\Rightarrow$ sinx |cosx| = $\frac{1}{2\sqrt{2}}$

Case –I : when cosx > 0

In this case sinx cosx = $\frac{1}{2\sqrt{2}}$ $\Rightarrow\ \sin 2x=+\frac{1}{\sqrt{2}}\Rightarrow 2x=\frac{\pi}{4},\frac{3\pi}{4},\frac{9\pi}{4},\frac{13\pi}{4}$

$$\Rightarrow\ x=\frac{\pi}{8},\frac{3\pi}{8},\frac{9\pi}{8},\frac{13\pi}{8}$$

As x lies between 0 and 2π and cos x > 0, $x = \frac{\pi}{8}, \frac{3\pi}{8}$

Case–II : When cosx < 0 .

In this case sinx |cosx| = $\frac{1}{2\sqrt{2}}$ $\Rightarrow$ $\sin x \cos x = -\frac{1}{2\sqrt{2}}$ or $\sin 2x = -\frac{1}{\sqrt{2}}$

$\Rightarrow x = \frac{5\pi}{8}, \frac{7\pi}{8}, \frac{13\pi}{8}, \frac{15\pi}{8}$ $\Rightarrow$ $x = \frac{5\pi}{8}, \frac{7\pi}{8}$ as cos x < 0

Thus the values of x satisfying the given equation which lie between 0 and 2π are

$\frac{\pi}{8}, \frac{3\pi}{8}, \frac{5\pi}{8}, \frac{7\pi}{8}$

These are in A.P. with common difference $\pi/4$.

Example 2 : Show that : $2^{\sin x} + 2^{\cos x} \geq 2^{1-\frac{1}{\sqrt{2}}}$ for all real *x*.

Solution : Clearly, $2^{\sin x}$ and $2^{\cos x}$ are positive, so when we apply A.M. $\geq$ G.M.

$$\therefore \frac{2^{\sin x} + 2^{\cos x}}{2} \geq \sqrt{2^{\sin x}.2^{\cos x}} = \sqrt{2^{\sin x + \cos x}} \quad \text{..... (1)}$$

As we know, $\sin x + \cos x \geq -\sqrt{2}$ [Using $-\sqrt{1^2+1^2} \leq \sin x + \cos x \leq \sqrt{1^2+1^2}$]

$\Rightarrow \sqrt{2^{\sin x + \cos x}} \geq 2^{-\sqrt{2}/2}$ $\Rightarrow$ $\sqrt{2^{\sin x + \cos x}} \geq 2^{-1/\sqrt{2}}$ ..... (ii)

From (i) and (ii) ;

$$\frac{2^{\sin x} + 2^{\cos x}}{2} \geq \sqrt{2^{\sin x + \cos x}} \geq 2^{-1/\sqrt{2}}$$

$$\Rightarrow 2^{\sin x} + 2^{\cos x} \geq 2.2^{-1/\sqrt{2}}$$

$\Rightarrow 2^{\sin x} + 2^{\cos x} \geq 2^{1-1/\sqrt{2}}$ for all real values of *x*.

Example 3 : Let $0 < A, B < \pi/2$ satisfying the Equation $3 \sin^2 A + 2 \sin^2 B = 1$ and $3 \sin 2A = 2 \sin 2B$, then prove that : $A + 2B = \pi/2$.

Solution : From the second equation, we have

$\sin 2B = \frac{3}{2} \sin 2A$ ..... (i)

and from the first equation,

$3 \sin^2 A = 1 - 2 \sin^2 B = \cos 2B$ ..... (ii)

Now, consider the identity

$\cos (A + 2B) = \cos A \cdot \cos 2B - \sin A \cdot \sin 2B$

$= 3 \cos A \cdot \sin^2 A - \frac{3}{2} \sin A \cdot \sin 2A$

$= 3 \cos A \cdot \sin^2 A - 3 \sin^2 A \cos A = 0$

$\cos (A + 2B) = 0$

$\therefore \cos (A + 2B) = 0$ ..... (iii)

Given that $0 < A < \pi/2$ and $0 < B < \pi/2$

$\Rightarrow 0 < A + 2B < \pi + \pi/2$

$\Rightarrow 0 < (A + 2B) < 3\pi/2$ ..... (iv)

$\therefore$ From (iii) and (iv),

$A + 2B = \frac{\pi}{2}$

Example 4 : If $a_{r+1}=\sqrt{\frac{1}{2}(1+a_r)}$, prove that $\cos\left(\frac{\sqrt{1-a_0^2}}{a_1.a_2.a_3.....\text{to}\ \infty}\right)=a_0$.

Solution : Let, $a_0=\cos\theta$, then $a_{r+1}=\sqrt{\frac{1}{2}(1+a_r)}$ gives

$$a_1=\sqrt{\frac{1}{2}(1+a_0)}=\sqrt{\frac{1}{2}(1+\cos\theta)}=\cos\frac{\theta}{2}$$

$$a_2=\sqrt{\frac{1}{2}(1+a_1)}=\sqrt{\frac{1}{2}\left(1+\cos\frac{\theta}{2}\right)}=\cos\frac{\theta}{2^2}\quad[\text{When, } r=1]$$

$$a_3=\sqrt{\frac{1}{2}(1+a_2)}=\sqrt{\frac{1}{2}\left(1+\cos\frac{\theta}{2^2}\right)}=\cos\frac{\theta}{2^3}$$

$$\vdots\qquad\qquad\vdots\qquad\qquad\vdots$$

$$a_n=\sqrt{\frac{1}{2}(1+a_{n-1})}=\sqrt{\frac{1}{2}\left(1+\cos\frac{\theta}{2^{n-1}}\right)}=\cos\frac{\theta}{2^n}$$

$\therefore\quad a_1\,.\,a_2\,.\,a_3\,.....\,a_n$

$$\Rightarrow\quad \cos\frac{\theta}{2}.\cos\frac{\theta}{2^2}.\cos\frac{\theta}{2^2}.....\cos\frac{\theta}{2^n}$$

$$\Rightarrow\quad \frac{\cos\frac{\theta}{2}.\cos\frac{\theta}{2^2}.\cos\frac{\theta}{2^3}....\cos\frac{\theta}{2^n}.2\sin\frac{\theta}{2^n}}{2\sin\frac{\theta}{2^n}}\qquad\left\{\text{Multiply dislide by }\sin\frac{\theta}{2n}\right\}$$

$$\Rightarrow\quad \frac{\cos\frac{\theta}{2}.\cos\frac{\theta}{2^2}....\cos\frac{\theta}{2^{n-1}}.\sin\frac{\theta}{2^{n-1}}}{2\sin\frac{\theta}{2^n}}\qquad\{\because\ 2\sin\alpha\,.\,\cos\alpha=\sin 2\alpha\}$$

$$\Rightarrow\quad \frac{\cos\frac{\theta}{2}.\cos\frac{\theta}{2^2}...\cos\frac{\theta}{2^{n-2}}.\sin\frac{\theta}{2^{n-2}}}{2^2\sin\frac{\theta}{2^n}}=\qquad[\text{Multiply divide by 2}]$$

$$\vdots\qquad\qquad\vdots\qquad\qquad\vdots$$

$$=\frac{\cos\frac{\theta}{2^{n-1(n-1)}}.\sin\frac{\theta}{2^{n-(n-1)}}}{2^{n-1}.\sin\frac{\theta}{2^2}}=\frac{\cos\frac{\theta}{2}.\sin\frac{\theta}{2}}{2^{n-1}.\sin\frac{\theta}{2^n}}=\frac{\sin\theta}{2^n.\sin\theta/2^n}[\text{As } n\to\infty,2^n\to\infty,\frac{\theta}{2^n}\to 0\Rightarrow\underset{n\to\infty}{\text{Limit}}\frac{\sin\frac{\theta}{2^n}}{\frac{\theta}{2^n}}=1]$$

$$\therefore\quad a_1.a_2.....to\ \infty=\lim_{n\to\infty}\frac{\sin\theta}{2^n.\sin\theta/2^n}=\lim_{n\to\infty}\frac{\sin\theta}{\theta.\frac{\sin(\theta/2^n)}{(\theta/2^n)}}=\frac{\sin\theta}{\theta}$$

$$\therefore\quad \frac{\sqrt{1-a_0^2}}{a_1a_2a_3....\text{to}\ \infty}=\frac{\sqrt{1-\cos^2\theta}}{\frac{\sin\theta}{\theta}}=\theta$$

$$\therefore \cos\left(\frac{\frac{\sqrt{1-a_0^2}}{a_1}}{a_1a_2a_3 \text{ to } \infty}\right) = \cos\theta = a_0.$$

Example 5 : Evaluate : $\sum_{r=2}^{n} \sin r\alpha$, where $(n + 2)\ \alpha = 2\pi$ (without using formula).

Solution : Let $S = \sum_{r=2}^{n} \sin r\alpha = \sin 2\alpha + \sin 3\alpha + \sin 4\alpha + ... + \sin n\alpha$

Multiplying both side by $2 \sin \frac{\alpha}{2}$

$$\therefore 2\sin\frac{\alpha}{2}.S = 2\sin\frac{\alpha}{2}.\sin 2\alpha + 2\sin\frac{\alpha}{2}.\sin 3\alpha + 2\sin\frac{\alpha}{2}.\sin 4\alpha + ... + \sin\frac{\alpha}{2}.\sin n\alpha$$

$$= \left\{\cos\frac{3\alpha}{2} - \cos\frac{5\alpha}{2}\right\} + \left\{\cos\frac{5\alpha}{2} - \cos\frac{7\alpha}{2}\right\} + ... + \left\{\cos\left(n-\frac{1}{2}\right)\alpha - \cos\left(n+\frac{1}{2}\right)\alpha\right\}$$

$= \cos\frac{3\alpha}{2} - \cos\left(n+\frac{1}{2}\right)\alpha$ [All terms get cancelled axcept first and last term]

$$= \sin\left(\frac{\frac{3\alpha}{2} + \left(n+\frac{1}{2}\right)\alpha}{2}\right)\sin\left(\frac{\left(n+\frac{1}{2}\right)\alpha - \frac{3\alpha}{2}}{2}\right)$$

$$= 2\sin\frac{(n+2)\alpha}{2}.\sin\frac{(n-1)\alpha}{2}$$

$$\Rightarrow S = \frac{\sin\frac{(n-1)\alpha}{2}}{\sin\frac{\alpha}{2}}.\sin\frac{(n+2)\alpha}{2} = \frac{\sin\frac{(n-1)\alpha}{2}}{\sin\frac{\alpha}{2}}.\sin\frac{2\pi}{2} = 0$$ [As $(n + 2)\ \alpha = 2\pi$]

Example 6 : Find all the values of θ satisfying the equation, $\sin 7\theta = \sin\theta + \sin 3\theta$ such that $0 \le \theta \le \pi$.

Solution : $\sin 7\theta = \sin\theta + \sin 3\theta$

$\therefore (\sin 7\theta - \sin\theta) - \sin 3\theta = 0$

$\therefore 2 \sin 3\theta \cdot \cos 4\theta - \sin 3\theta = 0$

$\therefore \sin 3\theta\ (2 \cos 4\theta - 1) = 0$ $\Rightarrow$ Either : $\sin 3\theta = 0$ or $\cos 4\theta = \frac{1}{2}$

i.e., $3\theta = n\pi,\ n \in I$ or $4\theta = 2n\pi \pm \frac{\pi}{3}, n \in I$

$$\Rightarrow \theta = \frac{n\pi}{3} \quad \text{or} \quad \theta = \frac{n\pi}{2} \pm \frac{\pi}{12}$$

Put $n = 0$, $\theta = 0$ or $\theta = \pm\frac{\pi}{12}, \left(-\frac{\pi}{12} \text{rejected}\right)$

Put n = 1, $\theta = \frac{\pi}{3}$ or $\theta = \frac{\pi}{2} + \frac{\pi}{12}, \frac{\pi}{2} - \frac{\pi}{12}$, $\theta = \frac{7\pi}{12}, \frac{5\pi}{12}$

Put n = 2, $\theta = \frac{2\pi}{3}$ or $\theta = \pi \pm \frac{\pi}{12}\left(\pi + \frac{\pi}{12}\text{rejected}\right)$

Put n = 3, $\theta = \pi$

$\therefore$ Solutions are ;

$0, \frac{\pi}{12}, \frac{\pi}{3}, \frac{5\pi}{12}, \frac{7\pi}{12}, \frac{2\pi}{3}, \frac{11\pi}{12}$ and π

Example 7 : Solve the equation $\sin 3x + \cos 2x = 1$.

Solution : $\sin 3x + \cos 2x = 1$

$\Rightarrow (3 \sin x - 4 \sin^3 x) + (1 - 2 \sin^2 x) = 1$

i.e., $-4 \sin^3 x - 2 \sin^2 x + 3 \sin x = 0$

$\Rightarrow \sin x (4 \sin^2 x + 2 \sin x - 3) = 0$

$\Rightarrow \sin x = 0$ or $x = \frac{-2 \pm \sqrt{4+48}}{8}$

i.e., $\sin x = 0$ or $\sin x = \frac{-1 \pm \sqrt{13}}{4}$

$\Rightarrow x = n\pi$; $n \in \mathbb{Z}$

or $x = n\pi + (-1)^n \sin^{-1}\left(\frac{-1+\sqrt{13}}{4}\right)\left(\sin x = \frac{-1-\sqrt{13}}{4}\text{is rejected as it is} < -1\right)$

Example 8 : Solve the equation : $1 + 2\text{cosec } x = \frac{-\sec^2 \frac{x}{2}}{2}$, where, $x \neq n\pi$.

Solution : Here $1 + 2\text{cosec } x = \frac{-\sec^2 \frac{x}{2}}{2}$

$\Rightarrow 1 + \frac{2}{\sin x} = \frac{-[1 + \tan^2 x/2]}{2}$

$\therefore 2\left[\frac{2\tan\left(\frac{x}{2}\right)}{1+\tan^2\left(\frac{x}{2}\right)} + 2\right] = -\left[1 + \tan^2 \frac{x}{2}\right].\left(\frac{2\tan\frac{x}{2}}{1+\tan^2\frac{x}{2}}\right)$

Put $\tan\frac{x}{2} = t, 2\left[\frac{2t}{1+t^2} + 2\right] = -(1+t^2)\left(\frac{2t}{1+t^2}\right)$

$\therefore 4t^2 + 4t + 2t(1 + t^2) + 4 = 0 \quad \Rightarrow \quad 2t^3 + 4t^2 + 6t + 4 = 0$

$\Rightarrow t^3 + 2t^2 + 3t + 2 = 0 \quad \Rightarrow \quad (t + 1)(t^2 + t + 2) = 0$

$\Rightarrow t = -1 \quad (\because \ t^2 + t + 2 \neq 0)$

or $\tan\frac{x}{2} = t = -1 \quad \therefore \quad \frac{x}{2} = n\pi - \frac{\pi}{4}, n \in \text{I}$

$\therefore \ x = 2n\pi - \frac{\pi}{2}, n \in \text{I}$ is the required solution.

Example 9 : Solve for real x and y.

$\cot^2(x+y) + \tan^2(x+y) + y^2 + 2y - 1 = 0$

Solution : Here, $\cot^2(x+y) + \tan^2(x+y) + y^2 + 2y - 1 = 0$

$\Rightarrow \{1 + \cot^2(x+y)\} + \{1 + \tan^2(x+y)\} + y^2 + 2y - 3 = 0$ [By adding and subtracting 2]

i.e., $\operatorname{cosec}^2(x+y) + \sec^2(x+y) + (y+1)^2 - 4 = 0$

i.e., $\dfrac{1}{\sin^2(x+y).\cos^2(x+y)} - 4 + (y+1)^2 = 0$

$4[\operatorname{cosec}^2(2x+2y) - 1] + (y+1)^2 = 0$ .... (i)

We know that $\{\operatorname{cosec}^2\theta - 1\} \geq 0$ always

$\{\operatorname{cosec}^2(2x+2y) - 1\} \geq 0$; also $(y+1)^2 \geq 0$ for all real y.

$\therefore$ L.H.S. of (i) is always ≥ 0.

$\therefore$ Equality is possible if and only if both the terms,

$4\{\operatorname{cosec}^2(2x+2y) - 1\}$ and $(y+1)^2$ are zero simultaneously.

$\Rightarrow (y+1)^2 = 0$ and $\sin^2(2x+2y) = 1$

$\therefore y = -1$ and $\cos^2(2x+2y) = 0$

$\therefore y = -1$ and $\cos^2(2x-2) = 0$

i.e., $y = -1$ and $2(x-1) = (2n+1)\dfrac{\pi}{2}$, where $n \in I$

i.e, $x - 1 = (2n+1)\dfrac{\pi}{4}, n \in I$

i.e, $y = -1$ and $x = (2n+1)\dfrac{\pi}{4} + 1$, $n \in I$ are the possible solutions.

Example 10 : Find all the solutions of the equation : $(\sin x + \cos x)\sin 2x = a(\sin^3 x + \cos^3 x)$ located between $\dfrac{\pi}{2}$ and π and for which values of 'a' does this equation have at most one solution satisfying the condition $\dfrac{\pi}{2} \leq x \leq \pi$.

Solution : $(\sin x + \cos x)\sin 2x = a(\sin x + \cos x)[\sin^2 x + \cos^2 x - \sin x \cos x]$

$$\Rightarrow (\sin x + \cos x)\left[\sin 2x - a\left(1 - \frac{1}{2}\sin 2x\right)\right] = 0 \quad \text{.... (i)}$$

Either : $\sin x + \cos x = 0 \Rightarrow \tan x = -1 = \tan\dfrac{3\pi}{4}$ OR $\sin 2x - a\left(1 - \dfrac{1}{2}\sin 2x\right) = 0$

$\therefore x = \dfrac{3\pi}{4}$

Hence there is always at least one root lying in $\pi/2$ and π for any value of the parameter a.

or $2\sin 2x - 2a + a\sin 2x = 0$ [From (i)]

$$\Rightarrow \sin 2x = \frac{2a}{2+a} \quad \text{..... (ii)}$$

Since $\dfrac{\pi}{2} < x < \pi$ or $\pi < 2x < 2\pi$

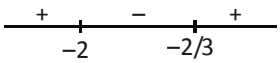
$\Rightarrow -1 \leq \sin 2x < 0$ ..... (iii)

Now, from (ii) and (iii), we have

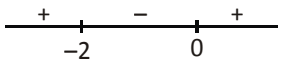
$-1 \leq \dfrac{2a}{2+a} < 0$, where $a \neq -2$

$-1 \le \frac{2a}{2+a}$ and $\frac{2a}{2+a} < 0$

$0 \le \frac{2a}{2+a} + 1$ and $a(2 + a) < 0$



$a < -2$ or $a \ge -2/3$ ..... (A)



$-2 < a \le 0$ ..... (B)

And from (A) and (B)

$$-\frac{2}{3} \le a \le 0$$

Hence for every value of 'a' satisfying the condition $-\frac{2}{3} \le a \le 0$ the equation, sin $2x = \frac{2a}{2+a}$ has the roots lying between $\pi/2$ and π.

Now we have to find the solution of the equation

$$\sin 2x = \frac{2a}{2+a} \text{ where } -\frac{2}{3} \le a \le 0 \text{ and } \frac{\pi}{2} < x < \pi$$

$\therefore \quad \pi < 2x < 2\pi$

Case : I $\pi < 2x < \frac{3\pi}{2}$

$0 < 2x - \pi < \pi/2$

$$\sin(2x) = \frac{2a}{2+a}$$

$\Rightarrow \quad \sin(2x - \pi) = -\frac{2a}{2+a}$

$$x = \frac{\pi}{2} - \frac{1}{2}\sin^{-1}\left(\frac{2a}{2+a}\right)$$

Case : II $\frac{3\pi}{2} < 2x < 2\pi$

$$-\frac{\pi}{2} < 2x - 2\pi < 0$$

Since, $\sin 2x = \frac{2a}{2+a}$

$$\sin(2x - 2\pi) = \frac{2a}{2+a}$$

$$2x - 2\pi = \sin^{-1}\left(\frac{2a}{2+a}\right)$$

$$x = \pi + \frac{1}{2}\sin^{-1}\left(\frac{2a}{2+a}\right)$$

Thus, $x = \left\{ \begin{array}{l} \frac{\pi}{2} - \frac{1}{2}\sin^{-1}\left(\frac{2a}{2+a}\right), \pi < 2x < 3\pi/2 \\ \pi + \frac{1}{2}\sin^{-1}\left(\frac{2a}{2+a}\right), \frac{3\pi}{2} < 2x < 2\pi \end{array} \right\}$ For $a \in \left[\frac{-2}{3}, 0\right]$

Student Task

Solve the following Trigonometric equations :

1. $\cos 2\theta = \left(\sqrt{2}+1\right)\left(\cos\theta - 1/\sqrt{2}\right)$.
2. $\tan p\theta = \cot q\theta$
3. $\cos\theta + \sin\theta = \cos 2\theta + \sin 2\theta$
4. $\sin\theta + \sin 5\theta = \sin 3\theta$ where $0 \le \theta \le \pi$
5. $\sin^3 x + \cos x \sin x + \cos^3 x = 1$
6. $\cos 3x \cos^3 x + \sin 3x \sin^3 x = 0$
7. $\sin 2x + \sin x + \cos 2x + \cos x + 1 = 0$
8. If $y + \cos x = \sin x$ has a real solution, then :

   (A) $y \le -\sqrt{2}$ (B) $-\sqrt{2} \le y \le \sqrt{2}$ (C) $y > \sqrt{2}$ (D) None of these
9. The equation $a\sin x + \cos 2x = 2a - 7$ possesses a solution if :

   (A) $a > 6$ (B) $2 \le a \le 6$ (C) $a > 2$ (D) None of these
10. The equaiton $3(\sin x + \cos x) - 2(\sin^3 x + \cos^3 x) = 8$ has :

    (A) No solutions (B) One solution (C) Two solutions (D) Infinite
11. If $2\cos x < \sqrt{3}$ and $x \in [-\pi, \pi]$ then the solution set for x is :

    (A) $\left[-\pi, -\frac{\pi}{6}\right) \cup \left(\frac{\pi}{6}, \pi\right]$ (B) $\left(-\frac{\pi}{6}, \frac{\pi}{6}\right)$

    (C) $\left[-\pi, -\frac{\pi}{6}\right] \cup \left[\frac{\pi}{6}, \pi\right]$ (D) None of these
12. If $2\sin x + 1 \ge 0$ and $x \in [0, 2\pi]$ then solution set for x is :

    (A) $\left[0, \frac{7\pi}{6}\right]$ (B) $\left[0, \frac{7\pi}{6}\right] \cup \left[\frac{11\pi}{6}, 2\pi\right]$

    (C) $\left[\frac{11\pi}{6}, 2\pi\right]$ (D) None of these
13. The most general solutionof the equation $\sqrt{3}\cos x + \sin x = \sqrt{2}$ is

    (A) $2n\pi + \frac{5\pi}{12}$ (B) $2n\pi \pm \frac{\pi}{4} + \frac{\pi}{6}$ (C) $2n\pi - \frac{\pi}{12}$ (D) None of these

UNSOLVED EXERCISE

Section-A
(Straight objective type with single option correct)

Topic : 1 General Solution

1. If $\sin^6\theta + \cos^6\theta + k\cos^2 2\theta = 1$, then k is equal to

   (a) $\frac{1}{2}\tan^2 2\theta$ (b) $\frac{1}{4}\tan^2 2\theta$

   (c) $4\cot^2 2\theta$ (d) $\frac{3}{4}\tan^2 2\theta$
2. The general solution of the equation $8\tan^2\frac{x}{2} = 1 + \sec x$ is

   (a) $x = 2n\pi \pm \cos^{-1}\left(\frac{-1}{3}\right); n \in I$

   (b) $x = 2n\pi \pm \frac{\pi}{6}; n \in I$

   (c) $x = 2n\pi \pm \cos^{-1}\left(\frac{1}{3}\right); n \in I$

   (d) None of these
3. Let $2\sin^2 x + 3\sin x - 2 > 0$ and $x^2 - x - 2 < 0$ (x is measured in radian). Then 'x' lies in the interval

   (a) $\left(\frac{\pi}{6}, \frac{5\pi}{6}\right)$ (b) $\left(-1, \frac{5\pi}{6}\right)$

   (c) $(-1, 2)$ (d) $\left(\frac{\pi}{6}, 2\right)$

4. The number of all possible triplets (a_1, a_2, a_3) such that $a_1 + a_2 \cos 2x + a_3 \sin^2 x = 0$ for all x is
(a) 0 (b) 1
(c) 3 (d) Infinite

5. The most general values of 'x' for which $\sin x + \cos x = \min_{a \in R} \{1, a^2 - 4a + 6\}$ are given by
(a) $2n\pi$ (b) $2n\pi + \frac{\pi}{2}$
(c) $n\pi + (-1)^n \frac{\pi}{4} - \frac{\pi}{4}$ (d) None of these

6. If $F(x) = \sin x + \cos x$, then the most general solution of $f(x) = \left[F\left(\frac{\pi}{10}\right)\right]$ are (where [x] is the greatest integer less than or equal to 'x')
(a) $2n\pi + \frac{\pi}{2}$ (b) $n\pi, n \in I$
(c) $2n\pi + \frac{2\pi}{3}, n \in I$ (d) None of these

7. **Statement-1 :** If $\cos(\pi \sin x) = \sin(\pi \cos x)$, then $\sin 2x$ is equal to $\frac{3}{4}$ or $\frac{-3}{4}$.
Statement-2 : If $\cos(\pi \sin x) = \sin(\pi \cos x)$, then $\sin 2x \in \left(\frac{3}{4}, 1\right] \cup \left[-1, \frac{-3}{4}\right)$.
Statement-3 : $\cos(\pi \sin x) = \sin(\pi \cos x) \Rightarrow x = \frac{1}{2}$ which of the following statements is true ?
(a) TFT (b) FTT
(c) TTT (d) TFF

8. **Statement-1 :** The solution set (x, y) for which $\sin x \sin y = 1$ contains perfect square multiple of $\frac{\pi}{2}$.
Statement-2 : The solutions set (x, y) of $\sin x \sin y = 1$ is given by $\left((4n+1)\frac{\pi}{2}, (4m+1)\frac{\pi}{2}\right), m, n \in Z$.
Statement-3 : The solution set (x, y) of $\sin x \sin y = 1$ is given by $\left((4n-1)\frac{\pi}{2}, (4m-1)\frac{\pi}{2}\right), m, n \in Z$. Which of the following options is correct ?
(a) TTT (b) FFF
(c) FTT (d) FTF

9. The number of non-zero solutions of the equation $\cos\frac{(\sqrt{2}+1)x}{2}\cos\frac{(\sqrt{2}-1)x}{2} = 1$ is
(a) 1 (b) 2
(c) 0 (d) 3

10. The solution of the equation $12 \tan 2x + \frac{\sqrt{10}}{\cos x} + 1 = 0$ is
(a) $n\pi - \tan^{-1} 3, n \in I$
(b) $n\pi - \tan^{-1} 3$, where is an even integer
(c) $n\pi - \tan^{-1} 3$, where n is an odd integer
(d) none of these

11. Solution of the equation $\sec\left(\frac{\pi}{4} + \theta\right) + \sec\left(\frac{\pi}{4} - \theta\right) = 2\sqrt{2}$ is
(a) $\theta = \frac{n\pi}{3}, n \in I$ or $\theta = n\pi$
(b) $\theta = \frac{2n\pi}{3}, n \in I$ or $\theta = 2n\pi$
(c) $\theta = \frac{n\pi}{6}, n \in I$ or $\theta = n\pi$
(d) $\theta = \frac{n\pi}{2}, n \in I$ or $\theta = n\pi$

12. If $\frac{\sin^3\theta - \cos^3\theta}{\sin\theta - \cos\theta} - \frac{\cos\theta}{\sqrt{1+\cot^2\theta}} - 2\tan\theta\cot\theta = -1$, $\theta \in [0, 2\pi]$, then
(a) $\theta \in \left(0, \frac{\pi}{2}\right) - \left\{\frac{\pi}{4}\right\}$ (b) $\theta \in \left(\frac{\pi}{2}, \pi\right) - \left\{\frac{3\pi}{4}\right\}$
(c) $\theta \in \left(\pi, \frac{3\pi}{2}\right) - \left\{\frac{5\pi}{4}\right\}$ (d) $\theta \in (0, \pi) - \left\{\frac{\pi}{4}, \frac{\pi}{2}\right\}$

13. If $\tan^4 x + \tan^4 y + 2\cot^2 x \cot^2 y = 3 + \sin^2(x + y)$, then

(a) $x = y = n\pi \pm \frac{\pi}{6}, n \in Z$

(b) $x = y = n\pi \pm \frac{\pi}{4}, n \in Z$

(c) $x = y = n\pi \pm \frac{\pi}{3}, n \in Z$

(d) None of these

14. Number of solutions of $\cot 3\theta + 6\,\text{cosec}\,2\theta - 8\,\text{cosec}^3 2\theta = 0$ is

(a) 0 (b) 1
(c) 2 (d) ∞

15. From the identitiy $\sin 3x = 3\sin x - 4\sin^3 x$, it follows that if x is real and $|x| < 1$, then

(a) $3x - 4x^3 > 1$
(b) $3x - 4x^3 \le 1$
(c) $3x - 4x^3 < 1$
(d) Nothing can be said about $3x - 4x^3$

16. Solution of the equation $4\cot 2\theta = \cot^2\theta - \tan^2\theta$ is

(a) $\theta = n\pi \pm \frac{\pi}{2}$ (b) $\theta = n\pi \pm \frac{\pi}{3}$

(c) $\theta = n\pi \pm \frac{\pi}{4}$ (d) None of these

17. A set of values of x satisfying the equation $\cos^2\left(\frac{1}{2}px\right) + \cos^2\left(\frac{1}{2}qx\right) = 0$ form an A.P. with common difference

(a) $\frac{2\pi}{p+q}$ (b) $\frac{2\pi}{p-q}$

(c) $\frac{\pi}{p+q}$ (d) None of these

18. The equation $(\cos p - 1)x^2 + (\cos p)x + \sin p = 0$, where x is a variable, has real roots. Then the interval of p may be any of the following

(a) $(0, 2\pi)$ (b) $(-\pi, 0)$

(c) $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ (d) $(0, \pi)$

19. The solution set of the system of equations: $x + y = \frac{2\pi}{3}, \cos x + \cos y = \frac{3}{2}$, where x and y are real is

(a) a finite non-empty set
(b) nul set
(c) ∞
(d) none of these

20. Let α and β be the angles satisfying $\cos(\alpha - \beta) = 1$ and $\cos(\alpha + \beta) = \frac{1}{e}$, where $\alpha, \beta \in [-\pi, \pi]$

Total number of ordered pairs of the type (α, β) satisfying these conditions is/are

(a) 0 (b) 1
(c) 2 (d) 4

21. General solution of the equation $(\sqrt{3} - 1)\sin\theta + (\sqrt{3} + 1)\cos\theta = 2$ is

(a) $2n\pi \pm \frac{\pi}{4} + \frac{\pi}{12}$ (b) $n\pi + (-1)^n \frac{\pi}{4} + \frac{\pi}{12}$

(c) $2n\pi \pm \frac{\pi}{4} - \frac{\pi}{12}$ (d) $n\pi + (-1)^n \frac{\pi}{4} - \frac{\pi}{12}$

22. The set of values of x for which $\frac{\tan 3x - \tan 2x}{1 + \tan 3x \tan 2x} = 1$ is

(a) ϕ

(b) $\left\{\frac{\pi}{4}\right\}$

(c) $\left\{n\pi + \frac{\pi}{4}, n = 1,2,3,\ldots\right\}$

(d) $\left\{2n\pi + \frac{\pi}{4}, n = 1,2,3,\ldots\right\}$

23. If $\sin 5x + \sin 3x + \sin x = 0$, then the value of x other than zero, lying between $0 \le x \le \frac{\pi}{2}$ is

(a) $\frac{\pi}{6}$ (b) $\frac{\pi}{12}$

(c) $\frac{\pi}{3}$ (d) $\frac{\pi}{4}$

24. The values of θ satisfying $\sin 7\theta = \sin 4\theta - \sin\theta$ and $0 < \theta < \frac{\pi}{2}$ are

(a) $\frac{\pi}{9}, \frac{\pi}{4}$ (b) $\frac{\pi}{3}, \frac{\pi}{9}$

(c) $\frac{\pi}{6}, \frac{\pi}{9}$ (d) $\frac{\pi}{3}, \frac{\pi}{4}$

25. If $\cos 3x + \sin\left(2x - \frac{7\pi}{6}\right) = -2$, then $x =$

(a) $\frac{\pi}{3}(6k + 1)$ (b) $\frac{\pi}{3}(6k - 1)$

(c) $\frac{\pi}{3}(2k + 1)$ (d) None of these

26. The solution of $\tan 2\theta \tan \theta = 1$ is

(a) $\frac{\pi}{3}$ (b) $(6n \pm 1)\frac{\pi}{6}$

(c) $(4n \pm 1)\frac{\pi}{6}$ (d) $(2n + \pi)\frac{\pi}{6}$

27. If $\sin A = \sin B$, $\cos A = \cos B$, then the value of A in terms of B is

(a) $n\pi + B$ (b) $n\pi + (-1)^n B$

(c) $2n\pi + B$ (d) $2n\pi - B$

28. If $5 \cos 2\theta + 2 \cos^2 \frac{\theta}{2} + 1 = 0$, $-\pi < \theta < \pi$, then $\theta =$

(a) $\frac{\pi}{3}$ (b) $\frac{\pi}{3}, \cos^{-1}(3/5)$

(c) $\cos^{-1}(3/5)$ (d) $\frac{\pi}{3}, \pi - \cos^{-1}(3/5)$

29. The solution set of the equation $4 \sin\theta \cos\theta - 2\cos\theta - 2\sqrt{3}\sin\theta + \sqrt{3} = 0$ in the interval $(0, 2\pi)$ is

(a) $\left\{\frac{3\pi}{4}, \frac{7\pi}{4}\right\}$ (b) $\left\{\frac{\pi}{3}, \frac{5\pi}{3}\right\}$

(c) $\left\{\frac{3\pi}{4}, \frac{7\pi}{4}, \frac{\pi}{3}, \frac{5\pi}{3}\right\}$ (d) $\left\{\frac{\pi}{6}, \frac{5\pi}{6}, \frac{11\pi}{6}\right\}$

30. The most general solution of $\tan\theta = -1$, $\cos\theta = \frac{1}{\sqrt{2}}$ is

(a) $n\pi + \frac{7\pi}{4}$ (b) $n\pi + (-1)^n \frac{7\pi}{4}$

(c) $2n\pi + \frac{7\pi}{4}$ (d) None of these

31. The general solution of : $\sin^2\theta \sec\theta + \sqrt{3}\tan\theta = 0$ is

(a) $\theta = n\pi + (-1)^{n+1}\frac{\pi}{3}$, $\theta = n\pi$, $n \in Z$

(b) $\theta = n\pi$, $n \in Z$

(c) $\theta = n\pi + (-1)^{n+1}\frac{\pi}{3}$, $n \in Z$

(d) $\theta = \frac{n\pi}{2}, n \in Z$

32. The equation $3^{\sin 2x + 2\cos^2 x} + 3^{1 - \sin 2x + 2\sin^2 x} = 28$ is satisfied for the values of x given by

(a) $\cos x = 0$ (b) $\tan x = -1$

(c) $\tan x = 1$ (d) None of these

Topic : 2 Solving Equation with graph

33. Number of solutions of the given equation $\tan x + \sec x = 2\cos x$ lying in the interval $[0, 2\pi]$ is

(a) 0 (b) 1

(c) 2 (d) 3

34. The number of points of intersection of the two curves $y = 2\sin x$ and $y = 5x^2 + 2x + 3$ is

(a) 0 (b) 1

(c) 2 (d) ∞

35. If $0 < x < \pi/2$ and $\sin^n x + \cos^n x \geq 1$ then

(a) $n \in [2, \infty)$ (b) $(-\infty, 2]$

(c) $n \in [-1, 1]$ (d) None of these

36. Value of 'x' and 'y' satisfying the equation $\sin^7 y = |x^3 - x^2 - 9x + 9| + |x^3 - 4x - x^2 + 4| + \sec^2 2y + \cos^4 y$ are

(a) $x = 1, y = n\pi$ (b) $x = 1, y = 2n\pi + \frac{\pi}{2}$

(c) $x = 1\ y = 2n\pi$ (d) None of these

37. Total number of solution of $\sin\{x\} = \cos\{x\}$, where $\{.\}$ denotes the fractional part, in $[0, 2\pi]$ is equal to

(a) 5 (b) 6

(c) 7 (d) None of these

38. If $a, b \in [0, \pi]$ and the equation $x^2 + 4 + 3\sin(ax + b) - 2x = 0$ has at least one solution, then the value of $(a + b)$ can be

(a) $\frac{7\pi}{2}$ (b) $\frac{3\pi}{2}$

(c) $\frac{9\pi}{2}$ (d) None of these

39. If $1 - \sin x = \frac{\sqrt{3}}{2}\left|x - \frac{\pi}{2}\right| + a$ has no solution, where $a \in R^+$, then

(a) $a \in R^+$ (b) $a > \frac{3}{2} - \frac{\pi}{\sqrt{3}}$

(c) $a \in \left(0, \frac{3}{2} + \frac{\pi}{\sqrt{3}}\right)$ (d) None of these

40. A solution of the equation $(1 - \tan\theta)(1 + \tan\theta)\sec^2\theta + 2^{\tan^2\theta} = 0$, where θ lies in the interval $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ is given by

(a) $\theta = 0$ (b) $\theta = \frac{\pi}{3}$ or $-\frac{\pi}{3}$

(c) $\theta = -\frac{\pi}{6}$ (d) $\theta = \frac{\pi}{6}$

41. The equation $2\sin^2 x - (p + 3)\sin x + 2p - 2 = 0$ possesses a real solution if

(a) $0 \le p \le 1$ (b) $-1 \le p \le 3$
(c) $4 \le p \le 6$ (d) $p \ge 6$

42. If $0 < \alpha, \beta < \frac{\pi}{2}$ and $\cos\alpha \cos\beta \cos(\alpha+\beta) = -\frac{1}{8}$, then

(a) $\alpha = \frac{\pi}{6}, \beta = \frac{\pi}{3}$ (b) $\alpha = \frac{\pi}{3}, \beta = \frac{\pi}{3}$

(c) $\alpha = \frac{\pi}{3}, \beta = \frac{\pi}{6}$ (d) None of these

43. Solutions of the equation $|\cos x| = 2[x]$ are (where [ ] = G.I.F.)
(a) 2 (b) 1
(c) 3 (d) None of these

43. If $1 + \sin\theta + \sin^2\theta + \ldots$ to $\infty = 4 + 2\sqrt{3}$, $0 < \theta < \pi$, $\theta \ne \frac{\pi}{2}$, then θ =

(a) $\frac{\pi}{6}$ (b) $\frac{\pi}{3}$

(c) $\frac{\pi}{3}$ or $\frac{\pi}{6}$ (d) $\frac{\pi}{3}$ or $\frac{2\pi}{3}$

44. The equation $\sin^6 x + \cos^6 x = \lambda$, has a solution if

(a) $\lambda \in \left[\frac{1}{2}, 1\right]$ (b) $\lambda \in \left[\frac{1}{4}, 1\right]$

(c) $\lambda \in [-1, 1]$ (d) $\lambda \in \left[0, \frac{1}{2}\right]$

Topic : 3 Problems Based on Maximum and Minimum

45. The number of solution of the equation : $\cos(\pi\sqrt{x-4}).\cos\pi\sqrt{x} = 1$ is
(a) > 2 (b) 2
(c) 1 (d) 0

46. If the mapping of $f(x) = ax + b$, $a < 0$ maps [–1, 1] onto [0, 2], then for all values of θ, $A = \cos^2\theta + \sin^4\theta$ is

(a) $F\left(\frac{1}{4}\right) \le A \le F(0)$ (b) $F(0) \le A \le F(-2)$

(c) $F\left(\frac{1}{3}\right) \le A \le F(-2)$ (d) None of these

47. **Statement-1 :** If $\cos x - \cos\alpha = \sin x - \sin\alpha$, then minimum value of $|x + \alpha|$ is $\frac{\pi}{2}$.

Statement-2 : The number of possible integral value of p such that $\sqrt{3}\sin x + \sqrt{2}\cos x = k$ has a solution is 5.

Statement-3 : The number of possible solutions of x such that $\sin^8 x + \cos^8 x = 1$ is always finite. Which of the following options is true ?
(a) TTF (b) FTF
(c) TTT (d) TFF

48. The most general solution of the equation $2^{\sin x} + 2^{\cos x} = 2^{1-1/\sqrt{2}}$ are

(a) $n\pi - \frac{\pi}{4}$ (b) $n\pi + \frac{\pi}{4}$

(c) $n\pi + (-1)^n \frac{\pi}{4}$ (d) $2n\pi \pm (-1)^n \frac{\pi}{4}$

49. The general solution of the equation $\left(\sin\frac{\pi}{6}\right)^{\cot 3x + \cot x} = 2^{\cot x\left[\sin^2(2\pi - x) - \cos(\pi - x)\sin\left(\frac{\pi}{2} + x\right)\right]}$ is

(a) $n\pi \pm \alpha$, where $\tan\alpha = \frac{\sqrt{7}}{5}$

(b) $n\pi \pm \alpha$, where $\tan\alpha = \sqrt{\frac{7}{5}}$

(c) $n\pi \pm \frac{\pi}{4}$
(d) None of these

50. For $x \in (-\pi, \pi)$, the value of x for which the equation $(\sqrt{3}\sin x + \cos x)^{\sqrt{\sqrt{3}\sin 2x - \cos 2x + 2}} = 4$ is satisfied

(a) $\frac{\pi}{2}$ (b) $\frac{\pi}{4}$

(c) $\frac{\pi}{3}$ (d) $\frac{\pi}{6}$

51 The solution set of the inequality $\cos^2\theta < \frac{1}{2}$ is

(a) $\left\{\theta : (8n+1)\frac{\pi}{4} < \theta < (8n+3)\frac{\pi}{4}, n \in Z\right\}$

(b) $\left\{\theta : (8n-1)\frac{\pi}{4} < \theta < (8n-3)\frac{\pi}{4}, n \in Z\right\}$

(c) $\left\{\theta : (4n+1)\frac{\pi}{4} < \theta < (4n+3)\frac{\pi}{4}, n \in Z\right\}$

(d) None of these

Section-B
(Previous Year Question)

1. If $5(\tan^2 x - \cos^2 x) = 2\cos 2x + 9$, then the value of $\cos 4x$ is

(a) $-\frac{3}{5}$ (b) $\frac{1}{3}$

(c) $\frac{2}{9}$ (d) $-\frac{7}{9}$

2. If $0 \le x < 2\pi$, then the number of real values of x, which satisfy the equation $\cos x + \cos 2x + \cos 3x + \cos 4x = 0$, is

(a) 3 (b) 5

(c) 7 (d) 9

3. Let $S = \left\{x \in (-\pi, \pi); x \neq 0, \pm\frac{\pi}{2}\right\}$. The sum of all distinct solutions of the equation $\sqrt{3}\sec x + \text{cosec}\, x + 2(\tan x - \cot x) = 0$ in the set S is equal to

(a) $-\frac{7\pi}{9}$ (b) $-\frac{2\pi}{9}$

(c) 0 (d) $\frac{5\pi}{9}$

4. If $P = \{\theta : \sin\theta - \cos\theta = \sqrt{2}\cos\theta\}$ and $Q = \{\theta : \sin\theta + \cos\theta = \sqrt{2}\sin\theta\}$ be two sets. Then,

(a) $P \subset Q$ and $Q - P \neq \phi$

(b) $Q \not\subset P$

(c) $P \not\subset Q$ (d) $P = Q$

5. Let *n* be an odd integer. If $\sin n\theta = \sum_{r=0}^{n} b_r \sin^r\theta$, for every value of θ, then

(a) $b_0 = 1, b_1 = 3$ (b) $b_0 = 0, b_1 = n$

(c) $b_0 = -1, b_1 = n$ (d) $b_0 = 0, b_1 = n^2 - 3n + 3$

6. The general value of θ satisfying the equation $2\sin^2\theta - 3\sin\theta - 2 = 0$, is

(a) $n\pi + (-1)^n\frac{\pi}{6}$ (b) $n\pi + (-1)^n\frac{\pi}{2}$

(c) $n\pi + (-1)^n\frac{5\pi}{6}$ (d) $n\pi + (-1)^n\frac{7\pi}{6}$

7. In a ΔABC, angle A is greater than angle B. If the measures of angles A and B satisfy the equation $3\sin x - 4\sin^3 x - k = 0$, $0 < k < 1$, then the measure of angle C is

(a) $\frac{\pi}{3}$ (b) $\frac{\pi}{2}$

(c) $\frac{2\pi}{3}$ (d) $\frac{5\pi}{6}$

8. The general solution of $\sin x - 3\sin 2x + \sin 3x = \cos x - 3\cos 2x + \cos 3x$ is

(a) $n\pi + \frac{\pi}{8}$ (b) $\frac{n\pi}{2} + \frac{\pi}{8}$

(c) $(-1)^n\frac{n\pi}{2} + \frac{\pi}{8}$ (d) $2n\pi + \cos^{-1}\frac{3}{2}$

9. The general solution of the trigonometric equation $\sin x + \cos x = 1$ is given by

(a) $x = 2n\pi$; $n = 0, \pm 1, \pm 2, \ldots$

(b) $x = 2n\pi + \pi/2$, $n = 0, \pm 1, \pm 2, \ldots$

(c) $x = n\pi + (-1)^n\frac{\pi}{4} - \frac{\pi}{4}$; $n = 0, \pm 1, \pm 2, \ldots$

(d) None of the above

10. The equation $2\cos^2\left(\frac{x}{2}\right)\sin^2 x = x^2 + x^{-2}, x \le \frac{\pi}{9}$ has

(a) no real solution

(b) one real solution

(c) more than one real solution

(d) none of the above

11. Let α and β be non zero real numbers such that $2(\cos\beta - \cos\alpha) + \cos\alpha\cos\beta = 1$. Then which of the following is/are true?

(a) $\sqrt{3}\tan\left(\frac{\alpha}{2}\right) - \tan\left(\frac{\beta}{2}\right) = 0$

(b) $\tan\left(\frac{\alpha}{2}\right) - \sqrt{3}\tan\left(\frac{\beta}{2}\right) = 0$

(c) $\tan\left(\frac{\alpha}{2}\right) + \sqrt{3}\tan\left(\frac{\beta}{2}\right) = 0$

(d) $\sqrt{3}\tan\left(\frac{\alpha}{2}\right) + \tan\left(\frac{\beta}{2}\right) = 0$

12. The values of θ lying between $\theta = 0$ and $\theta = \pi/2$ and satisfying the equation

$$\begin{vmatrix} 1+\sin^2\theta & \cos^2\theta & 4\sin 4\theta \\ \sin^2\theta & 1+\cos^2\theta & 4\sin 4\theta \\ \sin^2\theta & \cos^2\theta & 1+4\sin 4\theta \end{vmatrix} = 0, \text{ is}$$

(a) $7\pi/24$ (b) $5\pi/24$

(c) $11\pi/24$ (d) $\pi/24$

13. The set of values of θ satisfying inequation $2\sin^2\theta - 5\sin\theta + 2 > 0$, where $0 < \theta < 2\pi$, is

(a) $\left(0, \frac{\pi}{6}\right) \cup \left(\frac{5\pi}{6}, 2\pi\right)$ (b) $\left[0, \frac{\pi}{6}\right] \cup \left[\frac{5\pi}{6}, 2\pi\right]$

(c) $\left[0, \frac{\pi}{3}\right] \cup \left[\frac{2\pi}{3}, 2\pi\right]$ (d) None of the above

14. Let $\theta, \phi \in [0, 2\pi]$ be such that $2\cos\theta(1 - \sin\phi) = \sin^2\theta\left(\tan\frac{\theta}{2} + \cot\frac{\theta}{2}\right)\cos\phi - 1$, $\tan(2\pi -$

$\theta) > 0$ and $-1 < \sin\theta < -\frac{\sqrt{3}}{2}$. Then ϕ cannot satisfy

(a) $0 < \phi < \frac{\pi}{2}$ (b) $\frac{\pi}{2} < \phi < \frac{4\pi}{3}$

(c) $\frac{4\pi}{3} < \phi < \frac{3\pi}{2}$ (d) $\frac{3\pi}{2} < \phi < 2\pi$

15. For $x \in (0, \pi)$, the equation $\sin x + 2\sin 2x - \sin 3x = 3$ has
(a) infinitely many solutions
(b) three solutions
(c) one solution
(d) no solution

16. The number of solutions of the pair of equations $2\sin^2\theta - \cos 2\theta = 0$ and $2\cos^2\theta - 3\sin\theta = 0$ in the interval $[0, 2\pi]$ is
(a) 0 (b) 1
(c) 2 (d) 4

17. The number of integral values of *k* for which the equation $7\cos x + 5\sin x = 2k + 1$ has a solution, is
(a) 4 (b) 8
(c) 10 (d) 12

18. The number of values of x in the interval $[0, 5\pi]$ satisfying the equation $3\sin^2 x - 7\sin x + 2 = 0$ is
(a) 0 (b) 5
(c) 6 (d) 10

19. Number of solutions of the equation $\tan x + \sec x = 2\cos x$ lying in the interval $[0, 2\pi]$ is
(a) 0 (b) 1
(c) 2 (d) 3

20. The number of solutions of the equation $\sin(e^x) = 5^x + 5^{-x}$ is
(a) 0 (b) 1
(c) 2 (d) infinitely many

21. The equation $(\cos p - 1)x^2 + (\cos p)x + \sin p = 0$ in the variable x, has real roots. Then, p can take any value in the interval
(a) $(0, 2\pi)$ (b) $(-\pi, 0)$
(c) $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ (d) $(0, \pi)$

22. The smallest positive root of the equation $\tan x - x = 0$ lies in

(a) $\left(0, \frac{\pi}{2}\right)$ (b) $\left(\frac{\pi}{2}, \pi\right)$

(c) $\left(\pi, \frac{3\pi}{2}\right)$ (d) $\left(\frac{3\pi}{2}, 2\pi\right)$

23. The number of all possible triplets (a_1, a_2, a_3) such that $a_1 + a_2\cos(2x) + a_3\sin^2(x) = 0$, $\forall$ x is
(a) 0 (b) 1
(c) 3 (d) ∞

24. $\sec^2\theta = \frac{4xy}{(x+y)^2}$ is true if and only if
(a) $x = y \neq 0$ (b) $x = y, x \neq 0$
(c) $x = y$ (d) $x \neq 0, y \neq 0$

25. For $0 < \phi < \pi/2$, if $x = \sum_{n=0}^{\infty}\cos^{2n}\phi$, $y = \sum_{n=0}^{\infty}\sin^{2n}\phi$, $z = \sum_{n=0}^{\infty}\cos^{2n}\phi\sin^{2n}\phi$, then
(a) $xyz = xz + y$ (b) $xyz = xy + z$
(c) $xyz = x + y + z$ (d) $xyz = yz + x$

Section-C
(School / Board pattern)

1. Solve the Equation (General solution): $\cos\theta + \cos 3\theta - 2\cos 2\theta = 0$
2. Solve the Equation $\tan\theta + \tan 2\theta + \tan\theta\tan 2\theta = 1$
3. Sole the Equation $2\sin^2 x + \sin^2 2x = 2$
4. Solve the Equation $4\sin x\sin 2x\sin 4x = \sin 3x$
5. Solve the Equation $\sin\theta + \sin 2\theta + \sin 3\theta + \sin 4\theta = 0$
6. The general solution of the equation $7\cos^2\theta + 3\sin^2\theta = 4$ is
7. The general value of x satisfying the equation $\sqrt{3}\sin x + \cos x = \sqrt{3}$ is given by
8. The smallest positive angle which satisfies the equation $2\sin^2\theta + \sqrt{3}\cos\theta + 1 = 0$ is
9. If $\sqrt{3}\cos\theta + \sin\theta = \sqrt{2},$ then general value of θ is
10. The number of values of x in the interval $[0, 5\pi]$ satisfying the equation $3\sin^2 x - 7\sin x + 2 = 0$ is

ANSWER KEY

CHAPTER-01 (SETS, RELATIONS AND FUNCTIONS)

SECTION-A (Straight objective type questions)

1. (b)	2. (b)	3. (d)	4. (a)	5. (b)	6. (d)	7. (d)
8. (c)	9. (c)	10. (d)	11. (c)	12. (d)	13. (d)	14. (b)
15. (a)	16. (d)	17. (b)	18. (d)	19. (a)	20. (c)	21. (b)
22. (c)	23. (a)	24. (b)	25. (a)	26. (d)	27. (c)	28. (a)
29. (b)	30. (a)	31. (b)	32. (b)	33. (d)	34. (d)	35. (d)
36. (d)	37. (b)	38. (d)	39. (d)	40. (b)	41. (b)	42. (b)
43. (a)	44. (a)	45. (a)	46. (b)	47. (b)	48. (c)	49. (b)
50. (b)	51. (a,b)	52. (c)	53. (d)	54. (a)	55. (d)	56. (c)
57. (a)	58. (b)	59. (b)	60. (a,c)	61. (c)		

SECTION-B (Previous Years Questions)

1. (b)	2. (a)	3. (c)	4. (d)	5. (d)	6. (d)	7. (c)
8. (c)	9. (a, d)	10. (c)	11. (a)	12. (d)	13. (b)	14. (d)
15. (b)	16. (a)	17. (d)	18. (d)	19. (a, b, c)	20. (a,c)	21. (b, c)
22. (a, d)	23. (b)	24. (d)	25. (b)	26. (a)	27. (a)	28. (a, b, c)
29. (c)	30. (b)	31. (d)	32. (a)	33. (b)	34. (b)	35. (a)
36. (b)						

SECTION-C(School/Board level type questions)

1. (i) 18 (ii) 3 2. 11

3. (i) $R=\{(1,1)(1,2),(1,3)(1,4)(1,6)(2,2)(2,4)(2,6)(3,3)(3,6)(4,4)(5,5)(6,6)\}$

   (ii) $Df=\{1,2,3,4,5,6\}$ (iii) Range $R=\{1,2,3,4,5,6\}$

4. $\{(1,1)(2,2)(3,3),(1,2)(2,1)(2,3)(3,2)\}$

5. (i) R (ii) $\{e^{-2}\}$ (iii) Yes 6. $3x^4-12x^3+13x^2-2x+7$

7. $\lambda=-8$ 8. $Rf\in[0,1)$ 9. 0 10. $\frac{x}{x-1}$ 11. 0

CHAPTER-02 (TRIGONOMETRICAL FUNCTIONS OF ANGLES)

SECTION-A (Straight objective type questions)

1. (b)	2. (b)	3. (b)	4. (a)	5. (c)	6. (c)	7. (d)
8. (a)	9. (d)	10. (c)	11. (b)	12. (a)	13. (b)	14. (a)
15. (b)	16. (a)	17. (c)	18. (d)	19. (c)	20. (d)	21. (d)
22. (c)	23. (b)	24. (d)	25. (a)	26. (c)	27. (d)	28. (c)
29. (d)	30. (c)	31. (b)	32. (c)	33. (b)		

SECTION-B (Previous Years Questions)

1. (b)	2. (b)	3. (b)	4. (c)	5. (c)	6. (c)	7. (b)
8. (c)	9. (b)	10. (b)				

SECTION-C (School/Board level type questions)

1. $\pi/360,\pi/3,239\pi/360$ 2. 10,8 3. 10π 4. 22:13

CHAPTER-03 (TRANSFORMATION FORMULAE)

SECTION-A (Straight objective type questions)

1. (c)	2. (a)	3. (c)	4. (d)	5. (b)	6. (c)	7. (c)
8. (c)	9. (d)	10. (b)	11. (d)	12. (a)	13. (a)	14. (b)
15. (a)	16. (a)	17. (c)	18. (c)	19. (d)	20. (c)	21. (a)
22. (b)	23. (b)	24. (c)	25. (b)	26. (c)	27. (a)	28. (c)
29. (c)	30. (b)	31. (b)	32. (a)	33. (a)	34. (a)	35. (c)
36. (a)	37. (b)	38. (b)	39. (a)			

SECTION-B (Previous Years Question)

1. (a,b)	2. (c,d)	3. (a,b)	4. (a,b,c,d)	5. (c)	6. (a)	7. (b)
8. (c)	9. (d)					

CHAPTER-04 (TRIGONOMETRIC EQUATIONS)

SECTION-A (Straight objective type questions)

1. (d)	2. (c)	3. (d)	4. (b)	5. (c)	6. (d)	7. (d)
8. (a)	9. (c)	10. (c)	11. (b)	12. (d)	13. (b)	14. (a)
15. (b)	16. (c)	17. (d)	18. (d)	19. (c)	20. (c)	21. (a)
22. (a)	23. (c)	24. (a)	25. (a)	26. (b)	27. (c)	28. (d)
29. (d)	30. (c)	31. (b)	32. (b)	33. (c)	34. (a)	35. (a)
36. (b)	37. (b)	38. (b)	39. (a)	40. (b)	41. (b)	42. (b)
43. (d)	44. (a)	45. (b)	46. (c)	47. (c)	48. (a)	49. (b)
50. (b)	51. (c)	52. (c)				

SECTION-B (Previous Years Questions)

1. (d)	2. (c)	3. (c)	4. (d)	5. (b)	6. (d)	7. (c)
8. (b)	9. (c)	10. (a)	11. (b,c)	12. (a,c)	13. (a)	14. (a,c,d)
15. (d)	16. (c)	17. (b)	18. (c)	19. (c)	20. (a)	21. (d)
22. (c)	23. (d)	24. (a,b)	25. (b,c)			